UM12059

MACsec core

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User manual
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Document information

Information	Content
Keywords	macsec, memory, rx frame, tx frame, register, security
Abstract	This user manual describes the register set and usage of the MACsec core, used in several Ethernet products.

1 Introduction

This user manual describes the register set and usage of the MACsec core, used in several Ethernet products. The manual describes the full feature set of the MACsec core, because different products might use different configurations. Please refer to the product data sheet for the available feature set, e.g. number of supported secure channels, as well as the procedure for accessing the MACsec core.

2 MACsec Memory Map/Register Definitions register descriptions

The following section provides the memory map for each register within the MACsec.

The per SC configuration is done by selecting the RX SC via Section 2.9 and the TX SC via Section 2.11. Then all the context associated with the provided RX/TX SC index is available in the Skyblue memory map.

The per SC configuration for SAK and SAH registers is done by selecting the RX SC via Section 2.10 and the TX SC via Section 2.12.

2.1 MACsec memory map

MACsec base address: 0h

2.1.1 Global_configuration_and_Status memory map

Offset	Register	Width (In bits)	Access	Reset value
3F8h	MACsec IP Block Revision 1 register (MACSEC_IP_REV_1)	32	RO	0A80_0400h
3FCh	MACsec IP Block Revision 2 register (MACSEC_IP_REV_2)	32	RO	0001_0000h
400h	MACsec Event Register (MACSEC_EVR)	32	W1C	0000_0000h
404h	MACsec Event Enable Register (MACSEC_EVER)	32	R/W	0000_0000h
40Ch	MACsec Error Register (MACSEC_ERR)	32	W1C	0000_0000h
410h	MACsec Error Enable Register (MACSEC_ERER)	32	R/W	0000_0000h
440h	MACsec Memory ECC Error Capture Register (MACSEC_MEEC)	32	W1C	0000_0000h
444h	MACsec Idle status Register (MACSEC_STATUS)	32	RO	0000_0001h

2.1.2 MACSec_RX_Keys memory map

Offset	Register	Width (In bits)	Access	Reset value
700h	RX Security Association A key 1 of 8 (MACSEC_RXSAKA1Q)	32	R/W	0000_0000h
704h	RX Security Association A key 2 of 8 (MACSEC_RXSAKA2Q)	32	R/W	0000_0000h
708h	RX Security Association A key 3 of 8 (MACSEC_RXSAKA3Q)	32	R/W	0000_0000h
70Ch	RX Security Association A key 4 of 8 (MACSEC_RXSAKA4Q)	32	R/W	0000_0000h
710h	RX Security Association A key 5 of 8 (MACSEC_RXSAKA5Q)	32	R/W	0000_0000h
714h	RX Security Association A key 6 of 8 (MACSEC_RXSAKA6Q)	32	R/W	0000_0000h
718h	RX Security Association A key 7 of 8 (MACSEC_RXSAKA7Q)	32	R/W	0000_0000h
71Ch	RX Security Association A key 8 of 8 (MACSEC_RXSAKA8Q)	32	R/W	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
720h	RX Security Association A Short Secure Channel Identifier (MACSEC_RXSAASSCI)	32	R/W	0000_0000h
724h	RX Security Association A SALT 1 of 3 (MACSEC_RXSAASALT1T)	32	R/W	0000_0000h
728h	RX Security Association A SALT 2 of 3 (MACSEC_RXSAASALT2T)	32	R/W	0000_0000h
72Ch	RX Security Association A SALT 3 of 3 (MACSEC_RXSAASALT3T)	32	R/W	0000_0000h
740h	RX Security Association B key 1 of 8 (MACSEC_RXSAKB1Q)	32	R/W	0000_0000h
744h	RX Security Association B key 2 of 8 (MACSEC_RXSAKB2Q)	32	R/W	0000_0000h
748h	RX Security Association B key 3 of 8 (MACSEC_RXSAKB3Q)	32	R/W	0000_0000h
74Ch	RX Security Association B key 4 of 8 (MACSEC_RXSAKB4Q)	32	R/W	0000_0000h
750h	RX Security Association B key 5 of 8 (MACSEC_RXSAKB5Q)	32	R/W	0000_0000h
754h	RX Security Association B key 6 of 8 (MACSEC_RXSAKB6Q)	32	R/W	0000_0000h
758h	RX Security Association B key 7 of 8 (MACSEC_RXSAKB7Q)	32	R/W	0000_0000h
75Ch	RX Security Association B key 8 of 8 (MACSEC_RXSAKB8Q)	32	R/W	0000_0000h
760h	RX Security Association B Short Secure Channel Identifier (MACSEC_RXSABSSCI)	32	R/W	0000_0000h
764h	RX Security Association B SALT 1 of 3 (MACSEC_RXSABSALT1T)	32	R/W	0000_0000h
768h	RX Security Association B SALT 2 of 3 (MACSEC_RXSABSALT2T)	32	R/W	0000_0000h
76Ch	RX Security Association B SALT 3 of 3 (MACSEC_RXSABSALT3T)	32	R/W	0000_0000h

2.1.3 MACSec_TX_Keys memory map

Offset	Register	Width (In bits)	Access	Reset value
780h	TX Security Association A key 1 of 8 (MACSEC_TXSAKA1Q)	32	R/W	0000_0000h
784h	TX Security Association A key 2 of 8 (MACSEC_TXSAKA2Q)	32	R/W	0000_0000h
788h	TX Security Association A key 3 of 8 (MACSEC_TXSAKA3Q)	32	R/W	0000_0000h
78Ch	TX Security Association A key 4 of 8 (MACSEC_TXSAKA4Q)	32	R/W	0000_0000h
790h	TX Security Association A key 5 of 8 (MACSEC_TXSAKA5Q)	32	R/W	0000_0000h
794h	TX Security Association A key 6 of 8 (MACSEC_TXSAKA6Q)	32	R/W	0000_0000h
798h	TX Security Association A key 7 of 8 (MACSEC_TXSAKA7Q)	32	R/W	0000_0000h
79Ch	TX Security Association A key 8 of 8 (MACSEC_TXSAKA8Q)	32	R/W	0000_0000h
7A0h	TX Security Association A Short Secure Channel Identifier (MACSEC_TXSAASSCI)	32	R/W	0000_0000h
7A4h	TX Security Association A SALT 1 of 3 (MACSEC_TXSAASALT1T)	32	R/W	0000_0000h
7A8h	TX Security Association A SALT 2 of 3 (MACSEC_TXSAASALT2T)	32	R/W	0000_0000h
7ACh	TX Security Association A SALT 3 of 3 (MACSEC_TXSAASALT3T)	32	R/W	0000_0000h
7C0h	TX Security Association A key 1 of 8 (MACSEC_TXSAKB1Q)	32	R/W	0000_0000h
7C4h	TX Security Association B key 2 of 8 (MACSEC_TXSAKB2Q)	32	R/W	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
7C8h	TX Security Association B key 3 of 8 (MACSEC_TXSAKB3Q)	32	R/W	0000_0000h
7CCh	TX Security Association B key 4 of 8 (MACSEC_TXSAKB4Q)	32	R/W	0000_0000h
7D0h	TX Security Association B key 5 of 8 (MACSEC_TXSAKB5Q)	32	R/W	0000_0000h
7D4h	TX Security Association B key 6 of 8 (MACSEC_TXSAKB6Q)	32	R/W	0000_0000h
7D8h	TX Security Association B key 7 of 8 (MACSEC_TXSAKB7Q)	32	R/W	0000_0000h
7DCh	TX Security Association B key 8 of 8 (MACSEC_TXSAKB8Q)	32	R/W	0000_0000h
7E0h	TX Security Association B Short Secure Channel Identifier (MACSEC_TXSABSSCI)	32	R/W	0000_0000h
7E4h	TX Security Association B SALT 1 of 3 (MACSEC_TXSABSALT1T)	32	R/W	0000_0000h
7E8h	TX Security Association B SALT 2 of 3 (MACSEC_TXSABSALT2T)	32	R/W	0000_0000h
7ECh	TX Security Association B SALT 3 of 3 (MACSEC_TXSABSALT3T)	32	R/W	0000_0000h

2.1.4 Global statistics

The following statistics are maintained globally for the cases where an SCI can not be resolved (no SecTag, KaY frames, bad SecTag, or SCI not found) or filtered to the uncontrolled port. For other scenarios where the SCI is resolved, the per SC counts are used instead.

Offset	Register	Width (In bits)	Access	Reset value
600h	ifInOctetsCp first half Statistic (MACSEC_IFIOCP1HS)	32	R/W	0000_0000h
604h	ifInOctetsCp second half Statistic (MACSEC_IFIOCP2HS)	32	R/W	0000_0000h
608h	ifInUcastPktsCp Statistic (MACSEC_IFIUPCPS)	32	R/W	0000_0000h
610h	ifInOctetsUp first half Statistic (MACSEC_IFIOUP1HS)	32	R/W	0000_0000h
614h	ifInOctetsUp second half Statistic (MACSEC_IFIOUP2HS)	32	R/W	0000_0000h
618h	ifInUcastPktsUp Statistic (MACSEC_IFIUPUPS)	32	R/W	0000_0000h
620h	ifInMulticastPktsCp Statistic (MACSEC_IFIMPCPS)	32	R/W	0000_0000h
624h	ifInBroadcastPktsCp Statistic (MACSEC_IFIBPCPS)	32	R/W	0000_0000h
628h	ifInMulticastPktsUp Statistic (MACSEC_IFIMPUPS)	32	R/W	0000_0000h
62Ch	ifInBroadcastPktsUp Statistic (MACSEC_IFIBPUPS)	32	R/W	0000_0000h
630h	InPktsWithoutTag Statistic (MACSEC_INPWTS)	32	R/W	0000_0000h
634h	InPktsKaY Statistic (MACSEC_INPKAYS)	32	R/W	0000_0000h
638h	InPktsBadTag Statistic (MACSEC_INPBTS)	32	R/W	0000_0000h
63Ch	InPktsSCINotFound Statistic (MACSEC_IPSNFS)	32	R/W	0000_0000h
640h	InPktsUnsupportedEC Statistic (MACSEC_IPUECS)	32	R/W	0000_0000h
644h	nPktsEponSingleCopyBroadcast Statistic (MACSEC_IPESCBS)	32	R/W	0000_0000h
648h	InPktsTooLong Statistic (MACSEC_IPTLS)	32	R/W	0000_0000h
64Ch	InPktsFilteredUp Statistic (MACSEC_IPFUS)	32	R/W	0000_0000h
650h	InPktsDiscardedUp Statistic (MACSEC_IPDUS)	32	R/W	0000_0000h

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2.1.5 Global statistics for the TX path

The following statistics are maintained globally for the TX path.

Offset	Register	Width (In bits)	Access	Reset value
680h	OutPktsDiscarded Statistic (MACSEC_OPDS)	32	R/W	0000_0000h

2.1.6 MACsec internal registers

The following sections provide detailed descriptions of the MACsec internal registers. Reserved fields must be written with zeros when a register is written.

Offset	Register	Width (In bits)	Access	Reset value
0h	MACsec configuration (MACSEC_CFG)	32	R/W	0000_0000h
4h	MACsec EtherType (MACSEC_ET)	32	R/W	0000_88E5h
8h	MACsec Configured TPID1 (MACSEC_CTPID1)	32	R/W	0000_0000h
Ch	MACsec Configured TPID2 (MACSEC_CTPID2)	32	R/W	0000_0000h
10h	MACsec VLAN Classification Control (MACSEC_VCLCTR)	32	R/W	0000_0000h
40h	Maximum Frame Length (MACSEC_MFL)	32	R/W	0000_0000h
44h	TX Packet Number exhaustion threshold (MACSEC_TPNET)	32	R/W	0000_0000h
80h	RX SC access select (MACSEC_RXSCA)	32	R/W	0000_0000h
84h	RX SC Key access select (MACSEC_RXSCKA)	32	R/W	0000_0000h
C0h	TX SC access select (MACSEC_TXSCA)	32	R/W	0000_0000h
C4h	TX SC access select (MACSEC_TXSCKA)	32	R/W	0000_0000h

2.1.7 Secured Channel of the SecY/SC

The following statistics are kept for the each of the 1 to 64 Secured Channel of the SecY/SC. The statistics such as InPktsOK, InPktsInvalid, InPktsNotValid, InNotUsingSA, InPktsUnusedSA are not kept since they can be aggregated by software using per SA statistics.

Offset	Register	Width (In bits)	Access	Reset value
150h	RX Secure Channel InPktsUnchecked Statistic (MACSEC_RXSCIPUS)	32	R/W	0000_0000h
154h	RX Secure Channel InPktsDelayed Statistic (MACSEC_RXSCIPDS)	32	R/W	0000_0000h
158h	RX Secure Channel InPktsLate Statistic (MACSEC_RXSCIPLS)	32	R/W	0000_0000h
160h - 16Ch	RX AN a InNotUsingSA Statistic (MACSEC_RXAN0INUSS - MACSEC_RXAN3INUSS)	32	R/W	0000_0000h
170h - 17Ch	RX AN a InPktsUnusedSA Statistic (MACSEC_RXAN0IPUSS - MACSEC_RXAN3IPUSS)	32	R/W	0000_0000h
180h	RX Security Association A configuration and status (MACSEC_RXSAACS)	32	R/W	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
184h	RX Security Association A nextPN (MACSEC_RXSAANPN)	32	R/W	0000_0000h
188h	RX Security Association Extended A nextPN (MACSEC_RXSAAXNPN)	32	R/W	0000_0000h
18Ch	RX Security Association A lowestPN (MACSEC_RXSAALPN)	32	R/W	0000_0000h
190h	RX Security Association Extended A lowestPN (MACSEC_RXSAAXLPN)	32	R/W	0000_0000h
194h	RX Security Association A InPktsOK Statistic (MACSEC_RXSAAIPOS)	32	R/W	0000_0000h
1A0h	RX Security Association A hash 1 of 4 (MACSEC_RXSAHA1Q)	32	R/W	0000_0000h
1A4h	RX Security Association A hash 2 of 4 (MACSEC_RXSAHA2Q)	32	R/W	0000_0000h
1A8h	RX Security Association A hash 3 of 4 (MACSEC_RXSAHA3Q)	32	R/W	0000_0000h
1ACh	RX Security Association A hash 4 of 4 (MACSEC_RXSAHA4Q)	32	R/W	0000_0000h
1B0h	RX Security Association A InPktsInvalid Statistic (MACSEC_RXSAAIPIS)	32	R/W	0000_0000h
1B4h	RX Security Association A InPktsNotValid Statistic (MACSEC_RXSAAIPNVS)	32	R/W	0000_0000h
1C0h	RX Security Association B configuration and status (MACSEC_RXSABCS)	32	R/W	0000_0000h
1C4h	RX Security Association B nextPN (MACSEC_RXSABNPN)	32	R/W	0000_0000h
1C8h	RX Security Association Extended B nextPN (MACSEC_RXSABXNPN)	32	R/W	0000_0000h
1CCh	RX Security Association B lowestPN (MACSEC_RXSABLPN)	32	R/W	0000_0000h
1D0h	RX Security Association Extended B lowestPN (MACSEC_RXSABXLPN)	32	R/W	0000_0000h
1D4h	RX Security Association B InPktsOK Statistic (MACSEC_RXSABIPOS)	32	R/W	0000_0000h
1E0h	RX Security Association B hash 1 of 4 (MACSEC_RXSAHB1Q)	32	R/W	0000_0000h
1E4h	RX Security Association B hash 2 of 4 (MACSEC_RXSAHB2Q)	32	R/W	0000_0000h
1E8h	RX Security Association B hash 3 of 4 (MACSEC_RXSAHB3Q)	32	R/W	0000_0000h
1ECh	RX Security Association B hash 4 of 4 (MACSEC_RXSAHB4Q)	32	R/W	0000_0000h
1F0h	RX Security Association B InPktsInvalid Statistic (MACSEC_RXSABIPIS)	32	R/W	0000_0000h
1F4h	RX Security Association B InPktsNotValid Statistic (MACSEC_RXSABIPNVS)	32	R/W	0000_0000h

2.1.8 RX configuration, status, and statistic registers

The RX configuration, status and statistic registers are used to access per SC statistics, configure the RX SCI, configure the RX SAs, and access the per SA statistics. The accessed RX SC is configured via the RX SC access select; see MACSEC_RXSCA. For storing the RX SAKs, the RX SC selection is done via memory map.

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Offset	Register	Width (In bits)	Access	Reset value
100h	RX Secure Channel Identifier first half (MACSEC_RXSCI1H)	32	R/W	0000_0000h
104h	RX Secure Channel Identifier second half (MACSEC_RXSCI2H)	32	R/W	0000_0000h

2.1.9 Controlled port of the SecY and RX

The following statistics are kept for the controlled port of the SecY and the RX associated SC (applicable to frames with resolved SCI). The statistics such as ifInDiscards and ifInErrors are not kept since they can be aggregated by software using hardware provided statistics.

Offset	Register	Width (In bits)	Access	Reset value
110h	ifInOctets first half Statistic (MACSEC_IFIO1HS)	32	R/W	0000_0000h
114h	ifInOctets second half Statistic (MACSEC_IFIO2HS)	32	R/W	0000_0000h
118h	ifInUcastPkts Statistic (MACSEC_IFIUPS)	32	R/W	0000_0000h
120h	ifInMulticastPkts Statistic (MACSEC_IFIMPS)	32	R/W	0000_0000h
124h	ifInBroadcastPkts Statistic (MACSEC_IFIBPS)	32	R/W	0000_0000h
128h	RX Secure Channel configuration (MACSEC_RXSCCFG)	32	R/W	0000_0000h
12Ch	replayWindow (MACSEC_RPW)	32	R/W	0000_0000h
140h	InOctetsValidated first half Statistic (MACSEC_INOV1HS)	32	R/W	0000_0000h
144h	InOctetsValidated second half Statistic (MACSEC_INOV2HS)	32	R/W	0000_0000h
148h	InOctetsDecrypted first half Statistic (MACSEC_INOD1HS)	32	R/W	0000_0000h
14Ch	InOctetsDecrypted second half Statistic (MACSEC_INOD2HS)	32	R/W	0000_0000h

2.1.10 Controlled port of the SecY and the associated SC

The following statistics are kept for the controlled port of the SecY and the associated SC. The statistics for the uncontrolled port can be obtained by subtracting the controlled port statistics from the Common Port (MAC) statistics. The statistics such as InPktsUnusedSA and ifOutErrors are not kept since they can be aggregated by software using hardware provided statistics.

Offset	Register	Width (In bits)	Access	Reset value
210h	ifOutOctets first half Statistic (MACSEC_IFOO1HS)	32	R/W	0000_0000h
214h	ifOutOctets second half Statistic (MACSEC_IFOO2HS)	32	R/W	0000_0000h
218h	ifOutUcastPkts Statistic (MACSEC_IFOUPS)	32	R/W	0000_0000h
21Ch	OutPktsUntagged Statistic (MACSEC_OPUS)	32	R/W	0000_0000h
220h	ifOutMulticastPkts Statistic (MACSEC_IFOMPS)	32	R/W	0000_0000h
224h	ifOutBroadcastPkts Statistic (MACSEC_IFOBPS)	32	R/W	0000_0000h
228h	TX Secure Channel configuration (MACSEC_TXSCCFG)	32	R/W	0000_0000h
22Ch	OutPktsTooLong Statistic (MACSEC_OPTLS)	32	R/W	0000_0000h
240h	OutOctetsProtected first half Statistic (MACSEC_OOP1HS)	32	R/W	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
244h	OutOctetsProtected second half Statistic (MACSEC_OOP2HS)	32	R/W	0000_0000h
248h	OutOctetsEncrypted first half Statistic (MACSEC_OOE1HS)	32	R/W	0000_0000h
24Ch	OutOctetsEncrypted second half Statistic (MACSEC_OOE2HS)	32	R/W	0000_0000h
280h	TX Security Association A configuration and status (MACSEC_TXSAACS)	32	W1C	0000_0000h
284h	TX Security Association A nextPN (MACSEC_TXSAANPN)	32	R/W	0000_0000h
288h	TX Security Association Extended A nextPN (MACSEC_TXSAAXNPN)	32	R/W	0000_0000h
28Ch	TX Security Association A OutPktsProtected Statistic (MACSEC_TXSAAOPPS)	32	R/W	0000_0000h
290h	TX Security Association A OutPktsEncrypted Statistic (MACSEC_TXSAAOPES)	32	R/W	0000_0000h
2A0h	TX Security Association A hash 1 of 4 (MACSEC_TXSAHA1Q)	32	R/W	0000_0000h
2A4h	TX Security Association A hash 2 of 4 (MACSEC_TXSAHA2Q)	32	R/W	0000_0000h
2A8h	TX Security Association A hash 3 of 4 (MACSEC_TXSAHA3Q)	32	R/W	0000_0000h
2ACh	TX Security Association A hash 4 of 4 (MACSEC_TXSAHA4Q)	32	R/W	0000_0000h
2C0h	TX Security Association B configuration and status (MACSEC_TXSABCS)	32	W1C	0000_0000h
2C4h	TX Security Association B nextPN (MACSEC_TXSABNPN)	32	R/W	0000_0000h
2C8h	TX Security Association Extended B nextPN (MACSEC_TXSABXNPN)	32	R/W	0000_0000h
2CCh	TX Security Association B OutPktsProtected Statistic (MACSEC_TXSABOPPS)	32	R/W	0000_0000h
2D0h	TX Security Association B OutPktsEncrypted Statistic (MACSEC_TXSABOPES)	32	R/W	0000_0000h
2E0h	TX Security Association B hash 1 of 4 (MACSEC_TXSAHB1Q)	32	R/W	0000_0000h
2E4h	TX Security Association B hash 2 of 4 (MACSEC_TXSAHB2Q)	32	R/W	0000_0000h
2E8h	TX Security Association B hash 3 of 4 (MACSEC_TXSAHB3Q)	32	R/W	0000_0000h
2ECh	TX Security Association B hash 4 of 4 (MACSEC_TXSAHB4Q)	32	R/W	0000_0000h

2.1.11 TX configuration, Status, and Statistic registers

The TX configuration, Status and Statistic registers are used to configure the TX SCI, configure the TX SCs and their associated TX SAs, and access the per TX SA statistics. The accessed TX SC is configured via the TX SC access select (see MACSEC_TXSCA). For storing the TX SAKs and SAHs, the TX SC selection is done via memory map.

Offset	Register	Width (In bits)	Access	Reset value
200h	TX Secure Channel Identifier first half (MACSEC_TXSCI1H)	32	R/W	0000_0000h
204h	TX Secure Channel Identifier second half (MACSEC_TXSCI2H)	32	R/W	0000_0000h

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2.1.12 Configuring the TX SC Selection table

The following registers are used to configure the TX SC Selection table. The number of Records are limited by the number of Secured Channels on TX in the SoC. It supports 1 to 31 Records.

Offset	Register	Width (In bits)	Access	Reset value
800h	TX SC Selection Record 0 part 0 (MACSEC_TXSCSR0P0)	32	R/W	0000_0000h
804h	TX SC Selection Record 0 part 1 (MACSEC_TXSCSR0P1)	32	R/W	0000_0000h
808h	TX SC Selection Record 0 part 2 (MACSEC_TXSCSR0P2)	32	R/W	0000_0000h
80Ch	TX SC Selection Record 0 part 3 (MACSEC_TXSCSR0P3)	32	R/W	0000_0000h
810h	TX SC Selection Record 1 part 0 (MACSEC_TXSCSR1P0)	32	R/W	0000_0000h
814h	TX SC Selection Record 1 part 1 (MACSEC_TXSCSR1P1)	32	R/W	0000_0000h
818h	TX SC Selection Record 1 part 2 (MACSEC_TXSCSR1P2)	32	R/W	0000_0000h
81Ch	TX SC Selection Record 1 part 3 (MACSEC_TXSCSR1P3)	32	R/W	0000_0000h
820h	TX SC Selection Record 2 part 0 (MACSEC_TXSCSR2P0)	32	R/W	0000_0000h
824h	TX SC Selection Record 2 part 1 (MACSEC_TXSCSR2P1)	32	R/W	0000_0000h
828h	TX SC Selection Record 2 part 2 (MACSEC_TXSCSR2P2)	32	R/W	0000_0000h
82Ch	TX SC Selection Record 2 part 3 (MACSEC_TXSCSR2P3)	32	R/W	0000_0000h
830h	TX SC Selection Record 3 part 0 (MACSEC_TXSCSR3P0)	32	R/W	0000_0000h
834h	TX SC Selection Record 3 part 1 (MACSEC_TXSCSR3P1)	32	R/W	0000_0000h
838h	TX SC Selection Record 3 part 2 (MACSEC_TXSCSR3P2)	32	R/W	0000_0000h
83Ch	TX SC Selection Record 3 part 3 (MACSEC_TXSCSR3P3)	32	R/W	0000_0000h

2.1.13 Configuring the Uncontrolled Port Filter table

The following registers are used to configure the Uncontrolled Port Filter table used on RX and TX.

Offset	Register	Width (In bits)	Access	Reset value
A00h	Uncontrolled Port filter Record 0 data part 0 (MACSEC_UPFR0D0)	32	R/W	0000_0000h
A04h	Uncontrolled Port filter Record 0 data part 1 (MACSEC_UPFR0D1)	32	R/W	0000_0000h
A08h	Uncontrolled Port filter Record 0 data part 2 (MACSEC_UPFR0D2)	32	R/W	0000_0000h
A0Ch	Uncontrolled Port filter Record 0 mask part 0 (MACSEC_UPFR0M0)	32	R/W	0000_0000h
A10h	Uncontrolled Port filter Record 0 mask part 1 (MACSEC_UPFR0M1)	32	R/W	0000_0000h
A14h	Uncontrolled Port filter Record 0 mask part 2 (MACSEC_UPFR0M2)	32	R/W	0000_0000h
A18h	Uncontrolled Port filter Record 0 Result (MACSEC_UPFR0R)	32	R/W	0000_0000h
A20h	Uncontrolled Port filter Record 1 data part 0 (MACSEC_UPFR1D0)	32	R/W	0000_0000h
A24h	Uncontrolled Port filter Record 1 data part 1 (MACSEC_UPFR1D1)	32	R/W	0000_0000h
A28h	Uncontrolled Port filter Record 1 data part 2 (MACSEC_UPFR1D2)	32	R/W	0000_0000h
A2Ch	Uncontrolled Port filter Record 1 mask part 0 (MACSEC_UPFR1M0)	32	R/W	0000_0000h
A30h	Uncontrolled Port filter Record 1 mask part 1 (MACSEC_UPFR1M1)	32	R/W	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
A34h	Uncontrolled Port filter Record 1 mask part 2 (MACSEC_UPFR1M2)	32	R/W	0000_0000h
A38h	Uncontrolled Port filter Record 1 Result (MACSEC_UPFR1R)	32	R/W	0000_0000h
A40h	Uncontrolled Port filter Record 2 data part 0 (MACSEC_UPFR2D0)	32	R/W	0000_0000h
A44h	Uncontrolled Port filter Record 2 data part 1 (MACSEC_UPFR2D1)	32	R/W	0000_0000h
A48h	Uncontrolled Port filter Record 2 data part 2 (MACSEC_UPFR2D2)	32	R/W	0000_0000h
A4Ch	Uncontrolled Port filter Record 2 mask part 0 (MACSEC_UPFR2M0)	32	R/W	0000_0000h
A50h	Uncontrolled Port filter Record 2 mask part 1 (MACSEC_UPFR2M1)	32	R/W	0000_0000h
A54h	Uncontrolled Port filter Record 2 mask part 2 (MACSEC_UPFR2M2)	32	R/W	0000_0000h
A58h	Uncontrolled Port filter Record 2 Result (MACSEC_UPFR2R)	32	R/W	0000_0000h
A60h	Uncontrolled Port filter Record 3 data part 0 (MACSEC_UPFR3D0)	32	R/W	0000_0000h
A64h	Uncontrolled Port filter Record 3 data part 1 (MACSEC_UPFR3D1)	32	R/W	0000_0000h
A68h	Uncontrolled Port filter Record 3 data part 2 (MACSEC_UPFR3D2)	32	R/W	0000_0000h
A6Ch	Uncontrolled Port filter Record 3 mask part 0 (MACSEC_UPFR3M0)	32	R/W	0000_0000h
A70h	Uncontrolled Port filter Record 3 mask part 1 (MACSEC_UPFR3M1)	32	R/W	0000_0000h
A74h	Uncontrolled Port filter Record 3 mask part 2 (MACSEC_UPFR3M2)	32	R/W	0000_0000h
A78h	Uncontrolled Port filter Record 3 Result (MACSEC_UPFR3R)	32	R/W	0000_0000h
A80h	Uncontrolled Port filter Record 4 data part 0 (MACSEC_UPFR4D0)	32	R/W	0000_0000h
A84h	Uncontrolled Port filter Record 4 data part 1 (MACSEC_UPFR4D1)	32	R/W	0000_0000h
A88h	Uncontrolled Port filter Record 4 data part 2 (MACSEC_UPFR4D2)	32	R/W	0000_0000h
A8Ch	Uncontrolled Port filter Record 4 mask part 0 (MACSEC_UPFR4M0)	32	R/W	0000_0000h
A90h	Uncontrolled Port filter Record 4 mask part 1 (MACSEC_UPFR4M1)	32	R/W	0000_0000h
A94h	Uncontrolled Port filter Record 4 mask part 2 (MACSEC_UPFR4M2)	32	R/W	0000_0000h
A98h	Uncontrolled Port filter Record 4 Result (MACSEC_UPFR4R)	32	R/W	0000_0000h
AA0h	Uncontrolled Port filter Record 5 data part 0 (MACSEC_UPFR5D0)	32	R/W	0000_0000h
AA4h	Uncontrolled Port filter Record 5 data part 1 (MACSEC_UPFR5D1)	32	R/W	0000_0000h
AA8h	Uncontrolled Port filter Record 5 data part 2 (MACSEC_UPFR5D2)	32	R/W	0000_0000h
AACh	Uncontrolled Port filter Record 5 mask part 0 (MACSEC_UPFR5M0)	32	R/W	0000_0000h
AB0h	Uncontrolled Port filter Record 5 mask part 1 (MACSEC_UPFR5M1)	32	R/W	0000_0000h
AB4h	Uncontrolled Port filter Record 5 mask part 2 (MACSEC_UPFR5M2)	32	R/W	0000_0000h
AB8h	Uncontrolled Port filter Record 5 Result (MACSEC_UPFR5R)	32	R/W	0000_0000h
AC0h	Uncontrolled Port filter Record 6 data part 0 (MACSEC_UPFR6D0)	32	R/W	0000_0000h
AC4h	Uncontrolled Port filter Record 6 data part 1 (MACSEC_UPFR6D1)	32	R/W	0000_0000h
AC8h	Uncontrolled Port filter Record 6 data part 2 (MACSEC_UPFR6D2)	32	R/W	0000_0000h
ACCh	Uncontrolled Port filter Record 6 mask part 0 (MACSEC_UPFR6M0)	32	R/W	0000_0000h
AD0h	Uncontrolled Port filter Record 6 mask part 1 (MACSEC_UPFR6M1)	32	R/W	0000_0000h
AD4h	Uncontrolled Port filter Record 6 mask part 2 (MACSEC_UPFR6M2)	32	R/W	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
AD8h	Uncontrolled Port filter Record 6 Result (MACSEC_UPFR6R)	32	R/W	0000_0000h
AE0h	Uncontrolled Port filter Record 7 data part 0 (MACSEC_UPFR7D0)	32	R/W	0000_0000h
AE4h	Uncontrolled Port filter Record 7 data part 1 (MACSEC_UPFR7D1)	32	R/W	0000_0000h
AE8h	Uncontrolled Port filter Record 7 data part 2 (MACSEC_UPFR7D2)	32	R/W	0000_0000h
AECh	Uncontrolled Port filter Record 7 mask part 0 (MACSEC_UPFR7M0)	32	R/W	0000_0000h
AF0h	Uncontrolled Port filter Record 7 mask part 1 (MACSEC_UPFR7M1)	32	R/W	0000_0000h
AF4h	Uncontrolled Port filter Record 7 mask part 2 (MACSEC_UPFR7M2)	32	R/W	0000_0000h
AF8h	Uncontrolled Port filter Record 7 Result (MACSEC_UPFR7R)	32	R/W	0000_0000h

2.2 MACsec configuration (MACSEC_CFG)

2.2.1 Offset

Register	Offset
MACSEC_CFG	0h

2.2.2 Function

The MACSEC_CFG register is used to configure the global attributes of the MACsec.

2.2.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0				UFDCM	EXTSCS	UFDCD	VCT	0						SLD	SLC
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0		KSSHD	UPSCMT	UECT	ESCBT	USFT		ITT	KFT	UFT		VFN	KSS	BYPN	S0I
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

2.2.4 Fields

Field	Function
31-28 —	Reserved, should be cleared.
27 UFDCM	UFDCM Unmodified Frame Delay Compensation Mode MACSec TX delay compensation mode based on different payload pattern, it can be either based on frame processing cycles or the number of octets accumulated in TX FIFO. This field is used when UFDCD is 1. 0b - TX Frame Processing Cycle Mode. Delay compensation is added based on frame processing cycles. (default) 1b - TX FIFO Depth Mode. Delay compensation is added based on TX FIFO depth.

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Field	Function
26 EXTSCS	EXTSCS External SC Selection This field is valid if MACSEC_IP_REV_2[IP_INT] is set to 2 0b - Uncontrolled portl filter and TX SC selection are used on transmit path for port/SC selection. (default) 1b - SC selection and uncontrolled/controlled selection are determined by transmit input sideband signals.
25 UFDCD	UFDCD Unmodified Frame Delay Compensation Desired A compensation delay is added to the unmodified frames to match the latency of the encrypted frames. This is done to keep frame processing latency constant through MACsec TX path. 0b - Delay compensation not desired -disabled (default) 1b - Delay compensation desired - enabled
24 VCT	VCT Clear Tag support: 0b - not supported (standard behavior) 1b - enabled, see Section 3.5 for details
23-18 —	Reserved, should be cleared.
17 SLD	SLD SL Disable. This field determines how the SL field in the SecTag of transmitted frames is formatted: 0b - Set SL in the SecTAG of transmitted frames based the number of octets between the last octet of the SecTAG and the first octet of the ICV, if that number is less than 48, SL is set to the number of octets. Otherwise, SL is set to zero.(default) 1b - Unconditionally set SL in the SecTAG of transmitted frames to 0. User needs to guarantee that each transmitted frame has no less than 48 octets after MAC SA or clear tags if present excluding FCS. Otherwise, MACSec may send out transmitted frames with SL 0 but the number of octets between the last octet of the SecTAG and the first octet of the ICV is less than 48, this will fail the SL checking on the MACSec RX path.
16 SLC	SLC SL Check. This field determines how the SL field in the SecTag of received frames is checked: 0b - Verify that the two ms bits of the SL octet are b00 and that the value coded in the 6 ls bits of the SL octet is <48. If either test fails the frame is treated according to the ITT field. 1b - Verify that the value coded in the 6 ls bits of the SL octet is <48. If not the frame is treated according to the ITT field. This setting allows the 2 ms bits to be used for other purposes without causing SecTag errors.
15-14 —	Reserved, should be cleared.
13 KSSHD	KSSHD The key store secured hash disable bit is used to disable hash values privacy control. 0b - The hash values aren't readable when MACSEC_CFG[KSS] is 1. 1b - The hash values are readable when MACSEC_CFG[KSS] is 1.
12 UPSCMT	UPSCMT This field determines how frames transmitted with uncontrolled port filter mismatched and SC classification mismatched are treated. 0b - discard on the controlled port. 1b - deliver on the uncontrolled port.
11	UECT

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Field	Function
UECT	This field determines how frames received with the Encryption (E) bit cleared and the Changed Text (C) bit set (reserved and not supported) are treated ***Note:*** *In this context, a discarded frame is sent to the controlled port marked with error unless the frame is sent to the uncontrolled port, the frame is only delivered on the controlled or the uncontrolled port.* 0b - discard on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disable mode of operation 1b - deliver on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disabled mode of operation
10 ESCBT	ESCBT This field determines how frames received with the Secure Channel (SC) bit cleared, the End Station (ES) bit cleared, and the Single Copy Broadcast bit set (SCB) (EPON mode not supported) are treated ***Note:*** *In this context, a discarded frame is sent to the controlled port marked with error unless the frame is sent to the uncontrolled port, the frame is only delivered on the controlled or the uncontrolled port.* 0b - discard on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disable mode of operation 1b - deliver on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disabled mode of operation
9-8 USFT	USFT This field determines how frames received with unknown SCI are treated ***Note:*** *In this context, a discarded frame is sent to the controlled port marked with error unless the frame is sent to the uncontrolled port, the frame is only delivered on the controlled or the uncontrolled port.* unknown_sci_frame_treatment: 00b - discard on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict mode of operation 01b - discard on the uncontrolled port and deliver on the controlled port (with the SecTag and ICV removed) if the C bit in the SecTAG is clear. If the C bit in the SecTAG is set then discard on the controlled port. For the controlled port, this provides Check or Disabled mode of operation. 10b - deliver on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict mode of operation. 11b - deliver on the uncontrolled port if the C bit in the SecTAG is set and discard on the controlled port if the C bit in the SecTAG is set. If the C bit in the SecTAG is clear then deliver on the controlled port (with the SecTag and ICV removed) but discard on the uncontrolled port. For the controlled port, this provides Check or Disabled mode of operation
7 ITT	ITT This field determines how frames received with an invalid SecTAG ***Note:*** *Due to the MACsec block's cut-through nature, the ITT setting does not control treatment of frames with invalid ICV. Invalid ICV is an error condition that is not detectable by MACsec until the frame has already begun forwarding on the chosen port. Thus, frames with Invalid ICV are unconditionally discarded and counted in the InPktsBadTag counter.* are treated ***Note:*** *In this context, a discarded frame is sent to the controlled port marked with error unless the frame is sent to the uncontrolled port, the frame is only delivered on the controlled or the uncontrolled port.* Frame received with invalid tags 0b - discard on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disabled mode of operation 1b - deliver on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disabled mode of operation
6 KFT	KFT This field determines how frames received with the Encryption (E) bit set and the Changed Text (C) bit clear (reserved for use by the KaY) are treated ***Note:*** *In this context, a discarded frame is sent to the controlled port marked with error unless the frame is sent to the uncontrolled port, the frame is only delivered on the controlled or the uncontrolled port.*

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Field	Function
	kay_frame_treatment: 0b - deliver on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disabled mode of operation 1b - discard on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict, Check or Disable mode of operation
5-4 UFT	UFT This field determines how frames received without the MAC security tag (SecTAG) are treated ***Note:*** *In this context, a discarded frame is sent to the controlled port marked with error unless the frame is sent to the uncontrolled port, the frame is only delivered on the controlled or the uncontrolled port.* untag_frame_treatment: 00b - deliver on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict mode of operation 01b - discard on the uncontrolled port and discard on the controlled port. For the controlled port, this provides Strict mode of operation 10b - discard on the uncontrolled port and deliver on the controlled port unmodified (MACsec header not removed). For the controlled port, this provides Check or Disabled mode of operation 11b - reserved
3 VFN	VFN ValidateFrame == Null 802.1AE-2018 10.7.8 This field determines how frames received on the controlled port are treated. The validateFrame is configured on a per RX SC basis for Disabled, Check or Strict. For the Null setting, it is configured on the port. 0b - ValidateFrame follows the value configured in MACSEC_RXSCCFG[VF] 1b - ValidateFrame == Null, received frames are delivered to the controlled port without modification
2 KSS	KSS Key Store Secured The key store secured bit is used to provide privacy and allow debug read access to the RX and TX keys along with their associated hash values stored in the MACsec. After hard reset, the keys and hash values are readable. Once the KSS is set, the keys and hash values are no longer readable. KSS can not be cleared unless the MACsec is hard reset. Since the keys and hash value are cleared on a hard reset, it makes the key storage secured. This field should be written to one in mission mode to make the keys write only. 0b - keys and hash values are readable 1b - keys and hash values are not readable, privacy mode
1 BYPN	BYPN Bypass mode 0b - bypass MACsec function on RX and TX directions, frames are transferred with no modification and none of the statistics are incremented. RX frames are delivered only to the uncontrolled port. 1b - normal mode; MACsec function on RX and TX is applied
0 S0I	S0I RX SCI0 is the implicit SCI for point to point ({ES,SC,SCB}=000) 0b - disable the use of SCI0 for implicit point-to-point SCI 1b - use SCI0 as the implicit SCI for point-to-point scenario

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2.3 MACsec EtherType (MACSEC_ET)

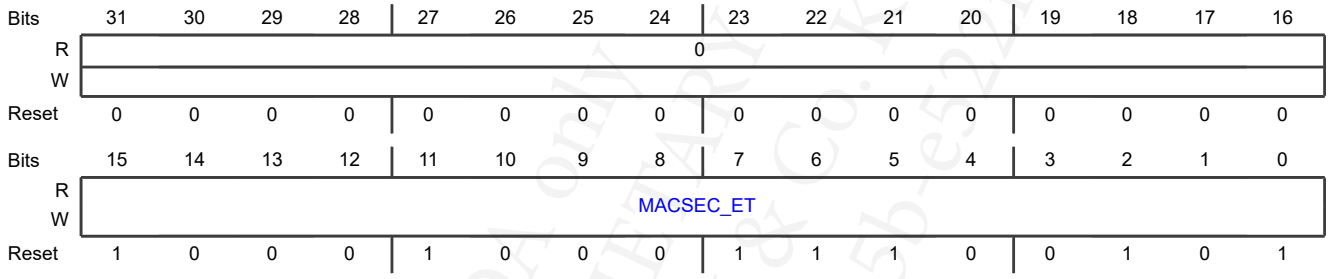
2.3.1 Offset

Register	Offset
MACSEC_ET	4h

2.3.2 Function

The MACSEC_ET register provide the ability to configure the MACsec EtherType. The default EtherType is 0x88E5.

2.3.3 Diagram



2.3.4 Fields

Field	Function
31-16 —	Reserved, should be cleared.
15-0 MACSEC_ET	MACSEC_ET MACsec Ethertype, defaults to 0x88E5 (802.1AE-2006/802.1AEbn-2011 9.4)

2.4 MACsec Configured TPID1 (MACSEC_CTPID1)

2.4.1 Offset

Register	Offset
MACSEC_CTPID1	8h

2.4.2 Function

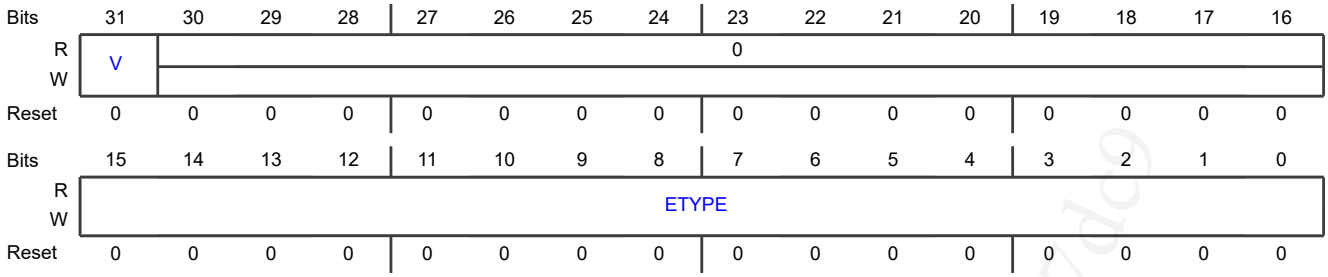
The MACSEC_CTPID1 register configures distinct Ethertype values (or TPID values) to indicate a VLAN tag in addition to the standard VLAN TPID values 0x8100 and 0x88A8. Used to identify VLAN for the Clear Tag feature and used for uncontrolled port filtering, see Section 3.5 for details.

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2.4.3 Diagram



2.4.4 Fields

Field	Function
31 V	V Valid 0b - not valid 1b - valid
30-16 —	Reserved, should be cleared.
15-0 ETYPE	ETYPE Configured Ethertype ***Note:*** *Must be configured to a value greater than 0x5dc but excluding the value of MACSEC_ET*

2.5 MACsec Configured TPID2 (MACSEC_CTPID2)

2.5.1 Offset

Register	Offset
MACSEC_CTPID2	Ch

2.5.2 Function

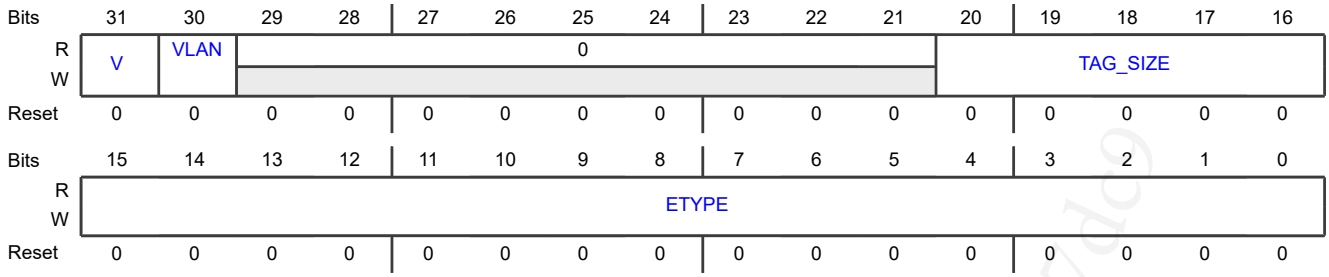
The MACSEC_CTPID2 register configures distinct Ethertype values (or TPID values) to indicate a VLAN tag in addition to the standard TPID values 0x8100 and 0x88A8. Used to identify VLAN tags for the VLAN Clear Tag feature and used for uncontrolled port filtering, see Section 3.5 for details.

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2.5.3 Diagram



2.5.4 Fields

Field	Function
31 V	V Valid 0b - not valid 1b - valid
30 VLAN	VLAN VLAN 0b - CTPID2 is a custom TAG 1b - CTPID2 is a VLAN TAG, can be used to identify a VLAN and used to extract PCP/DEI
29-21 —	Reserved, should be cleared.
20-16 TAG_SIZE	TAG_SIZE Tag Size Size of the custom tag in bytes including the ethertype. Not valid when VLAN=1 (size 4 is used) Valid range is 3 to 16 bytes, values 0-2 will be treated as invalid and not yield a match.
15-0 ETYPE	ETYPE Configured Ethertype ***Note:*** *Must be configured to a value greater than 0x5dc but excluding the value of MACSEC_ET.*

2.6 MACsec VLAN Classification Control (MACSEC_VCLCTR)

2.6.1 Offset

Register	Offset
MACSEC_VCLCTR	10h

2.6.2 Function

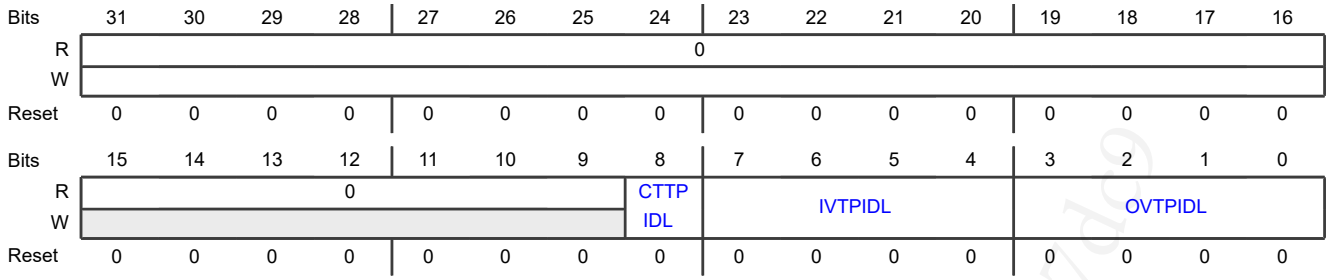
The MACSEC_VCLCTR register configures which of the 4 VLAN / custom TPIDs are supported in a given position. Used to validate the tags for the Clear Tag feature and used for uncontrolled port filtering, see Section 3.5 for details.

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2.6.3 Diagram



2.6.4 Fields

Field	Function
31-9 —	Reserved, should be cleared.
8 CTTPIDL	CTTPIDL Custom Tag TPID list: bitmap identifying which TPIDs are acceptable as custom tag Bit: 8 Custom Tag as defined by MACSEC_CTPID2[ETYPE]
7-4 IVTPIDL	IVTPIDL Inner VLAN TPID List: bitmap identifying which TPIDs are acceptable as Inner VLAN tag Bit: 7 Standard C-VLAN 0x8100 6 Standard S-VLAN 0x88A8 5 Custom VLAN as defined by MACSEC_CTPID1[ETYPE] 4 Custom VLAN as defined by MACSEC_CTPID2[ETYPE]
3-0 OVTPIDL	OVTPIDL Outer VLAN TPID List: bitmap identifying which TPIDs are acceptable as Outer VLAN tag Bit: 3 Standard C-VLAN 0x8100 2 Standard S-VLAN 0x88A8 1 Custom VLAN as defined by MACSEC_CTPID1[ETYPE] 0 Custom VLAN as defined by MACSEC_CTPID2[ETYPE]

2.7 Maximum Frame Length (MACSEC_MFL)

2.7.1 Offset

Register	Offset
MACSEC_MFL	40h

2.7.2 Function

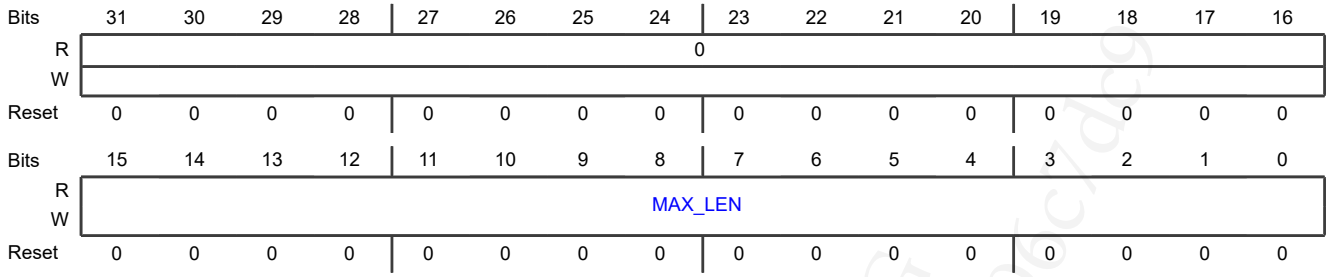
The MACSEC_MFL register provide the ability to configure the maximum frame length value used on TX. The frame length includes the MAC DA/SA, one or two VLAN tags (when VCT=1), one custom tag, the SecTAG, user data and ICV, excluding the padding and FCS inserted by the MAC. The confidentiality and integrity protection can grow the frame size, the MAX_LEN is set to limit the resulting frame size (original frame size plus 8/16B SecTag, plus 16B ICV). If a frame grows to a size larger than the MAX_LEN, it will be truncated

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to MACSEC_MFL and it will be marked as abort for the MAC to invalidate the FCS, the Section 2.64 will be incremented.

2.7.3 Diagram



2.7.4 Fields

Field	Function
31-16 —	Reserved, should be cleared.
15-0 MAX_LEN	MAX_LEN maximum frame length in bytes (802.1AE-2006/802.1AEbn-2011 10.7.18 and figure 10-4) The value zero is used to disable the max length check.

2.8 TX Packet Number exhaustion threshold (MACSEC_TPNET)

2.8.1 Offset

Register	Offset
MACSEC_TPNET	44h

2.8.2 Function

The MACSEC_TPNET register provide the ability to configure a TX Packet Number exhaustion threshold. Once the threshold is reached, an interrupt event is asserted to warn the key management Software that the Section 2.70 or Section 2.79 on TX has reached the exhaustion threshold and that it should re-key before the PN is exhausted.

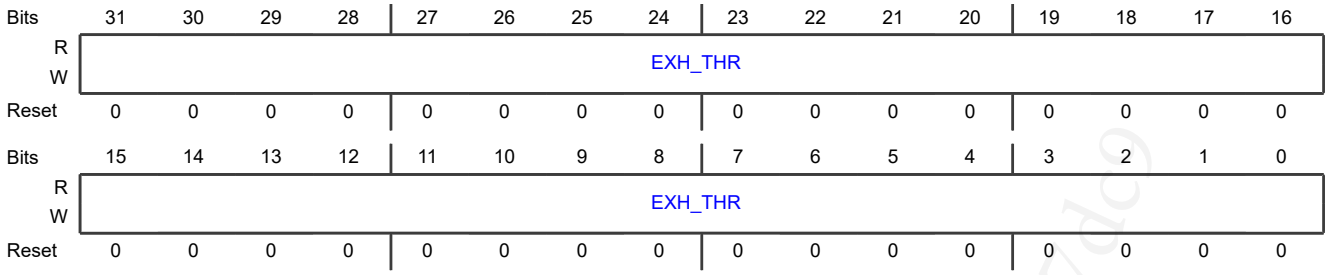
For XPN-capable cipher suite, the PN Exhaustion value will be used for upper 32 bits concatenated with lower 32 bit 0s.

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2.8.3 Diagram



2.8.4 Fields

Field	Function
31-0 EXH_THR	EXH_THR Exhaustion threshold. When NextPN is at or crosses the EXH_THR, an interrupt event is asserted (see Section 2.89)

2.9 RX SC access select (MACSEC_RXSCA)

2.9.1 Offset

Register	Offset
MACSEC_RXSCA	80h

2.9.2 Function

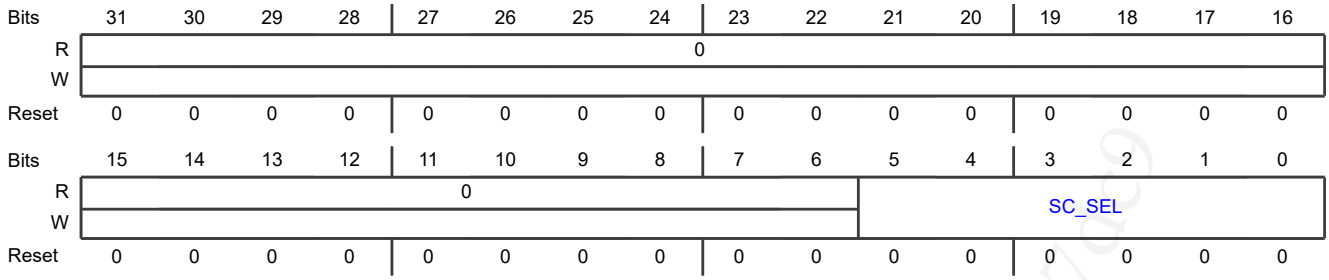
All the context, configuration and statistics associated to an RX SC are grouped into the 0x100 to 0x1FC range in the memory map, except RX SAK and SAH registers. There are a total of 1 to 64 RX SC supported but only one can be accessed at any time. The MACSEC_RXSCA register is used to select the RX Secure Channel to be accessed.

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2.9.3 Diagram



2.9.4 Fields

Field	Function
31-6 —	Reserved, should be cleared.
5-0 SC_SEL	SC_SEL SC select

2.10 RX SC Key access select (MACSEC_RXSCKA)

2.10.1 Offset

Register	Offset
MACSEC_RXSCKA	84h

2.10.2 Function

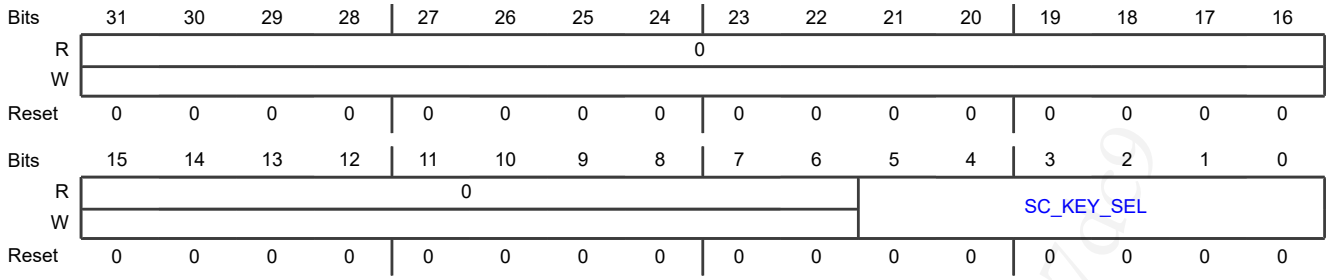
All the keys associated to a RX SC include RX SAK and SAH registers. There are a total of 1 to 64 RX SC supported but only one can be accessed at any time. The MACSEC_RXSCKA register is used to select the RX Secure Channel to be accessed for trust software.

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2.10.3 Diagram



2.10.4 Fields

Field	Function
31-6 —	Reserved, should be cleared.
5-0 SC_KEY_SEL	SC_KEY_SEL SC Key select

2.11 TX SC access select (MACSEC_TXSCA)

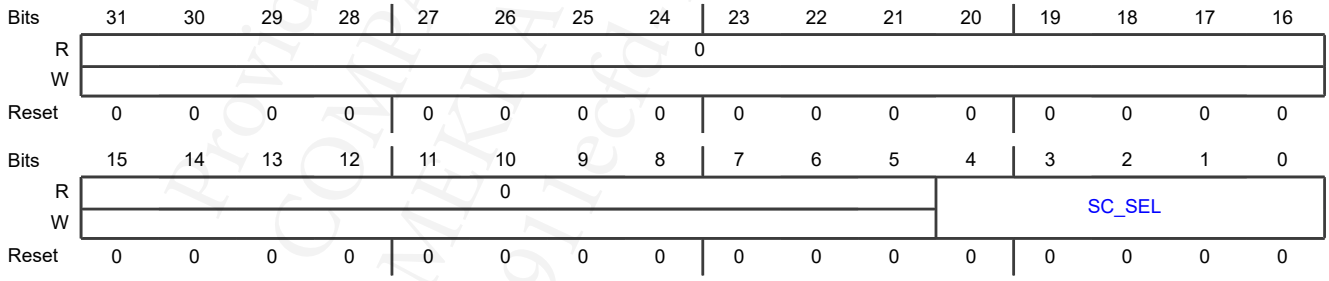
2.11.1 Offset

Register	Offset
MACSEC_TXSCA	C0h

2.11.2 Function

All the context, configuration and statistics associated to a TX SC are grouped into the 0x200 to 0x2FC range in the memory map except TX SAK and SAH registers. There are a total of 1 to 31 TX SC supported but only one can be accessed at any time. The MACSEC_TXSCA register is used to select the TX Secure Channel to be accessed.

2.11.3 Diagram



2.11.4 Fields

Field	Function
31-5 —	Reserved, should be cleared.

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Field	Function
4-0 SC_SEL	SC_SEL SC select

2.12 TX SC access select (MACSEC_TXSCKA)

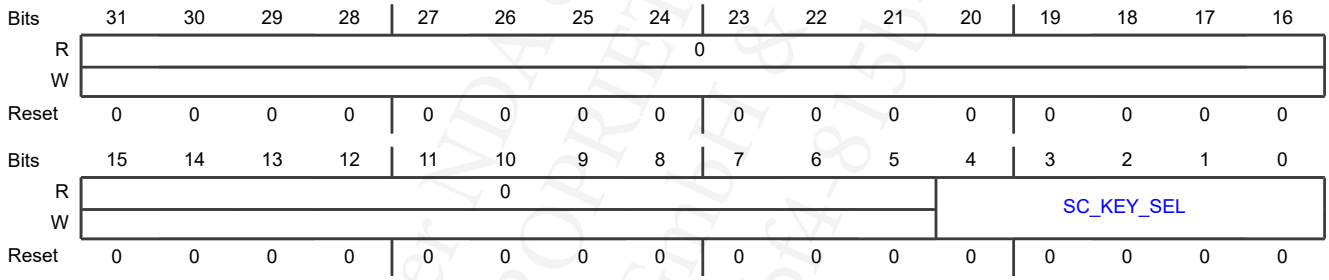
2.12.1 Offset

Register	Offset
MACSEC_TXSCKA	C4h

2.12.2 Function

All the keys associated to a TX SC include TX SAK and SAH registers. There are a total of 1 to 31 TX SC supported but only one can be accessed at any time. The MACSEC_TXSCKA register is used to select the TX Secure Channel to be accessed for trust software.

2.12.3 Diagram



2.12.4 Fields

Field	Function
31-5 —	Reserved, should be cleared.
4-0 SC_KEY_SEL	SC_KEY_SEL SC select

2.13 RX Secure Channel Identifier first half (MACSEC_RXSCI1H)

2.13.1 Offset

Register	Offset
MACSEC_RXSCI1H	100h

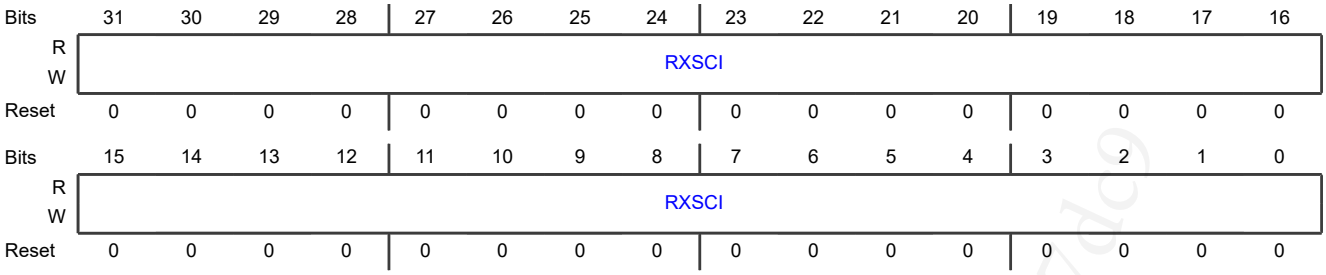
2.13.2 Function

The MACSEC_RXSCI1H register is used to program the first half of the SCI.

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2.13.3 Diagram



2.13.4 Fields

Field	Function
31-0 RXSCI	RXSCI Receive SCI bit 0 to 31 to be used for SC

2.14 RX Secure Channel Identifier second half (MACSEC_RXSCI2H)

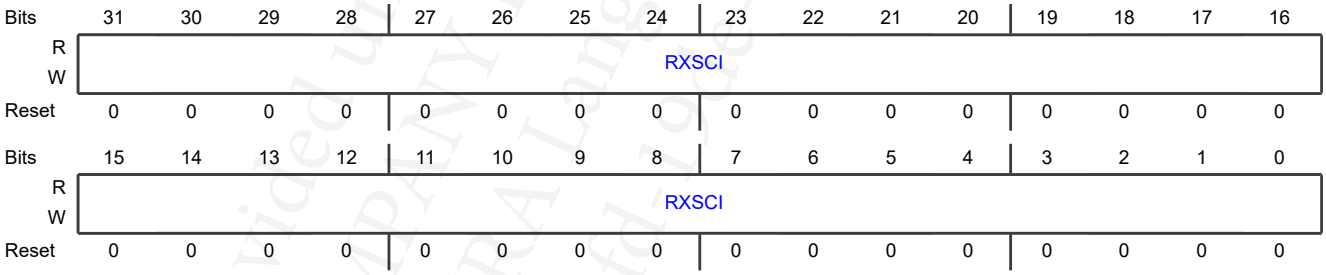
2.14.1 Offset

Register	Offset
MACSEC_RXSCI2H	104h

2.14.2 Function

The MACSEC_RXSCI2H register is used to program the second half of the SCI.

2.14.3 Diagram



2.14.4 Fields

Field	Function
31-0 RXSCI	RXSCI Receive SCI bit 32 to 63 to be used for SC

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2.15 ifInOctets first half Statistic (MACSEC_IFIO1HS)

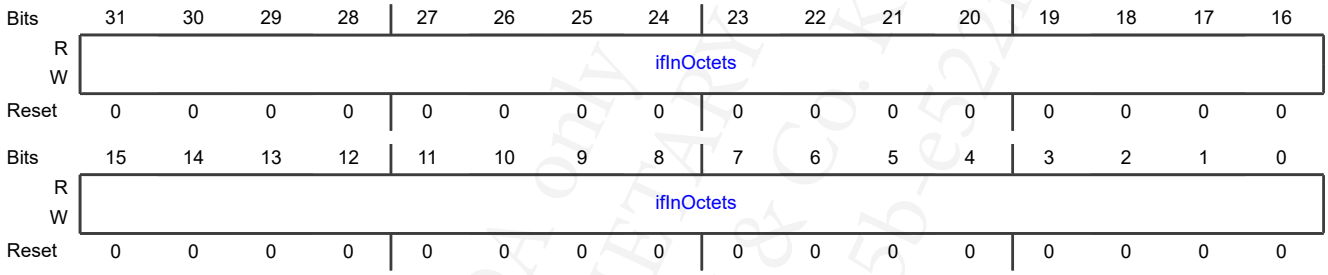
2.15.1 Offset

Register	Offset
MACSEC_IFIO1HS	110h

2.15.2 Function

The MACSEC_IFIO1HS register provides the count of the Controlled Port Interface Input Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

2.15.3 Diagram



2.15.4 Fields

Field	Function
31-0 ifInOctets	ifInOctets Controlled Port Interface Input Octets 0..31 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifInOctets count is the sum of all the octets of the MSDUs delivered to the user of the Controlled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses for frames with resolved SCI.

Section 2

2.16 ifInOctets second half Statistic (MACSEC_IFIO2HS)

2.16.1 Offset

Register	Offset
MACSEC_IFIO2HS	114h

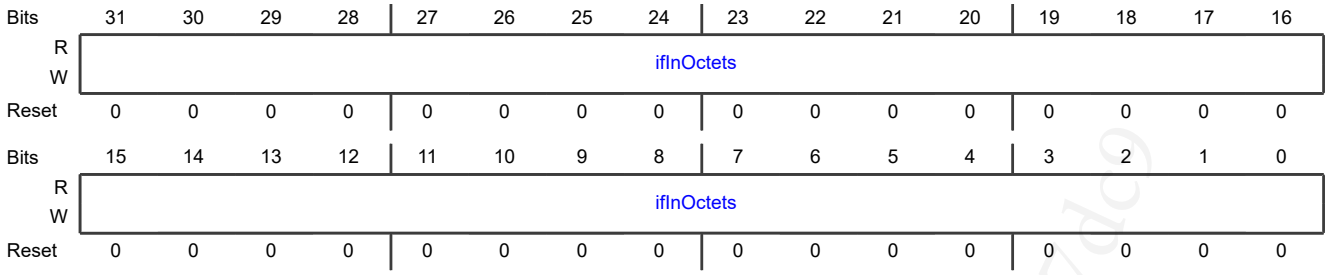
2.16.2 Function

The MACSEC_IFIO2HS register provides the count of the Controlled Port Interface Input Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

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2.16.3 Diagram



2.16.4 Fields

Field	Function
31-0 ifInOctets	ifInOctets Controlled Port Interface Input Octets 32..63 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifInOctets count is the sum of all the octets of the MSDUs delivered to the user of the Controlled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses for frames with resolved SCI.

2.17 ifInUcastPkts Statistic (MACSEC_IFIUPS)

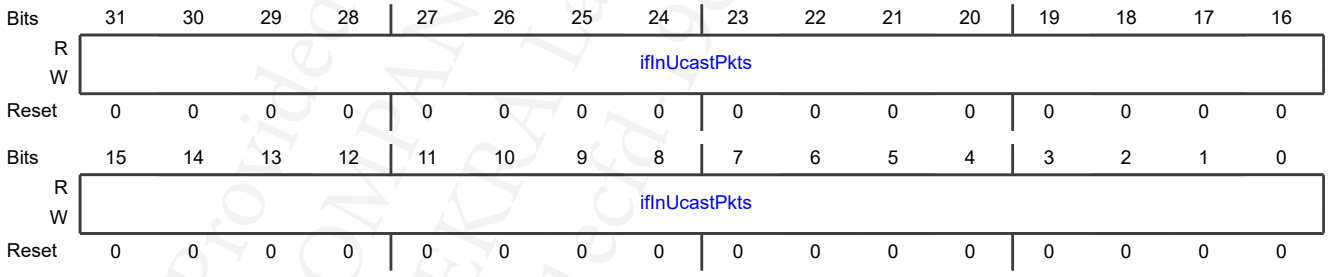
2.17.1 Offset

Register	Offset
MACSEC_IFIUPS	118h

2.17.2 Function

The register MACSEC_IFIUPS provides the count of the Controlled Port Interface Input Unicast Frames.

2.17.3 Diagram



2.17.4 Fields

Field	Function
31-0 ifInUcastPkts	ifInUcastPkts The number of frames with resolved SCI, delivered to the user of a controlled port, which were not addressed to a multicast or broadcast address. (802.1AE-2006/802.1AEbn-2011 10.7.6)

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2.18 ifInMulticastPkts Statistic (MACSEC_IFIMPS)

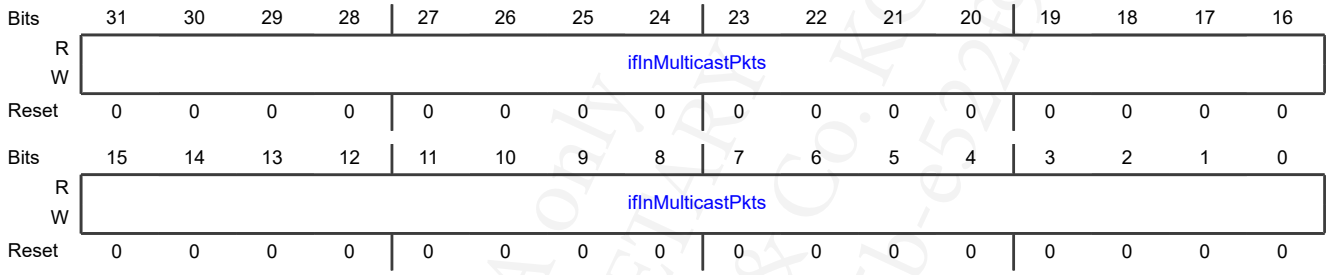
2.18.1 Offset

Register	Offset
MACSEC_IFIMPS	120h

2.18.2 Function

The MACSEC_IFIMPS register provides the count of the Controlled Port interface Input Multicast Frames.

2.18.3 Diagram



2.18.4 Fields

Field	Function
31-0 ifInMulticastPkts	ifInMulticastPkts The number of frames with resolved SCI, delivered to the user of a controlled port, which are addressed to a multicast address. (802.1AE-2006/802.1AEbn-2011 10.7.6)

2.19 ifInBroadcastPkts Statistic (MACSEC_IFIBPS)

2.19.1 Offset

Register	Offset
MACSEC_IFIBPS	124h

2.19.2 Function

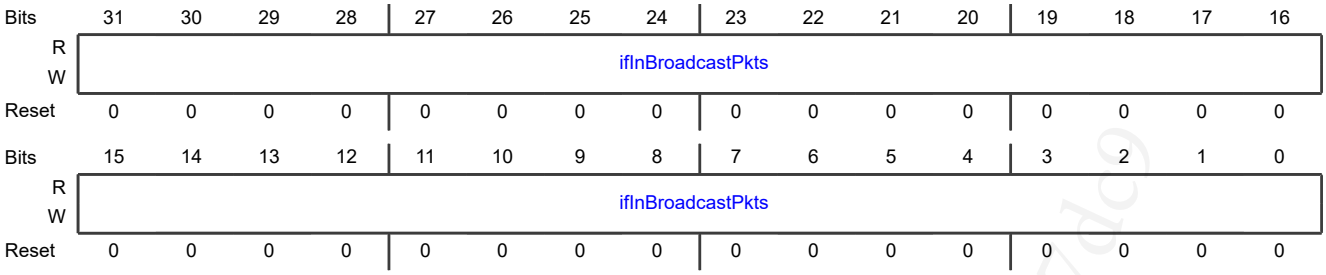
The MACSEC_IFIBPS register provides the count of the Controlled Port interface Input Broadcast Frames.

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2.19.3 Diagram



2.19.4 Fields

Field	Function
31-0 ifInBroadcast Pkts	ifInBroadcastPkts The number of frames with resolved SCI, delivered to the user of a controlled port, which are addressed to a broadcast address. (802.1AE-2006/802.1AEbn-2011 10.7.6)

2.20 RX Secure Channel configuration (MACSEC_RXSCCFG)

2.20.1 Offset

Register	Offset
MACSEC_RXSCCFG	128h

2.20.2 Function

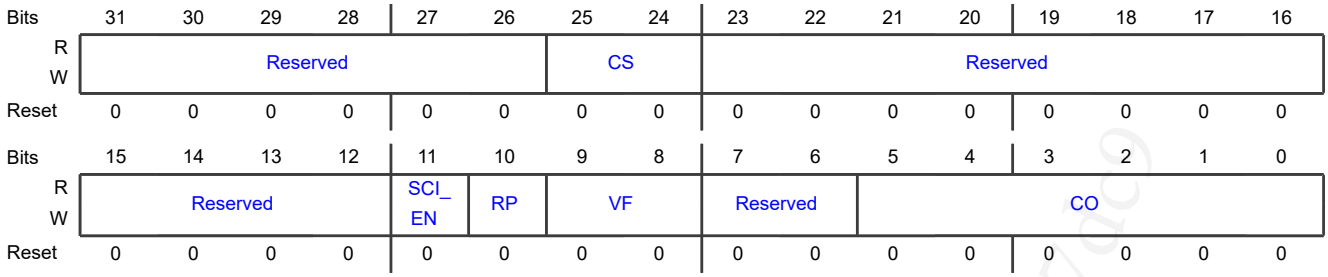
The MACSEC_RXSCCFG register is used to configure the SecY attributes of the SC.

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2.20.3 Diagram



2.20.4 Fields

Field	Function
31-26 —	Reserved, should be cleared.
25-24 CS	CS CipherSuite This field indicates the Cipher Suite to be used for this SC (802.1AE-2006/802.1AEbn-2011/802.1AE-2018 14.4) 00b - GCM-AES-128 01b - GCM-AES-256 10b - GCM-AES-XPN-128 11b - GCM-AES-XPN-256
23-12 —	Reserved, should be cleared.
11 SCI_EN	SCI_EN SCI enable This field indicates whether to include the SCI when performing RX SCI lookup (see Section 3.2.1 for details) 0b - disabled 1b - enabled, SCI is to be included in the SCI search
10 RP	RP replayProtect (802.1AE-2006/802.1AEbn-2011 10.7.8) To enable or disable the replay protection function 0b - disabled 1b - enabled
9-8 VF	VF validateFrames (802.1AE-2006/802.1AEbn-2011 10.7.8) To control the validation function for received frames 00b - Disable to disable the validation function 01b - Check to enable the validation function but only for checking without filtering out invalid frames 10b - Strict to enable the validation function and also strictly filter out those invalid frames 11b - reserved ,
7-6 —	Reserved, should be cleared.
5-0 CO	CO Confidentiality offset (802.1AE-2006/802.1AEbn-2011 10.7.24, 10.7.25) The number of initial octets of each MSDU without confidentiality protection

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Field	Function
	00_0000b - no offset 00_0001b-11_1111b - the first 1 to 63 Byte of data are integrity protected, not confidentiality protected. This configuration is ignored when a XPN-capable Cipher Suites is configured for this secure channel

2.21 replayWindow (MACSEC_RPW)

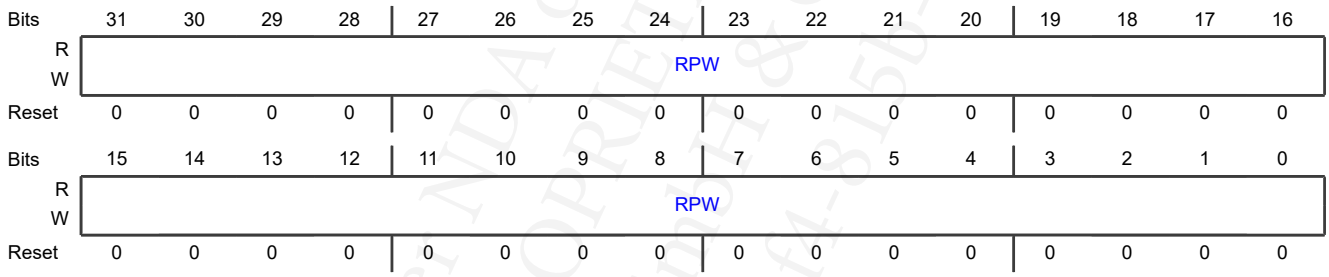
2.21.1 Offset

Register	Offset
MACSEC_RPW	12Ch

2.21.2 Function

The MACSEC_RPW register provide the ability to configure the verification replay window. Should be configured while the SC is disabled.

2.21.3 Diagram



2.21.4 Fields

Field	Function
31-0 RPW	RPW replayWindow (802.1AE-2006/802.1AEbn-2011 10.7.8) If the Current Cipher Suite uses extended packet numbering, the maximum value of replayWindow used in the Secure Frame Verification process (802.1AEbn-2018 10.6) is 2^30 -1. replayWindow greater than the maximum value will cause the incorrect recovered PN value. (802.1AEbn-2018 10.6.8)

2.22 InOctetsValidated first half Statistic (MACSEC_INOV1HS)

2.22.1 Offset

Register	Offset
MACSEC_INOV1HS	140h

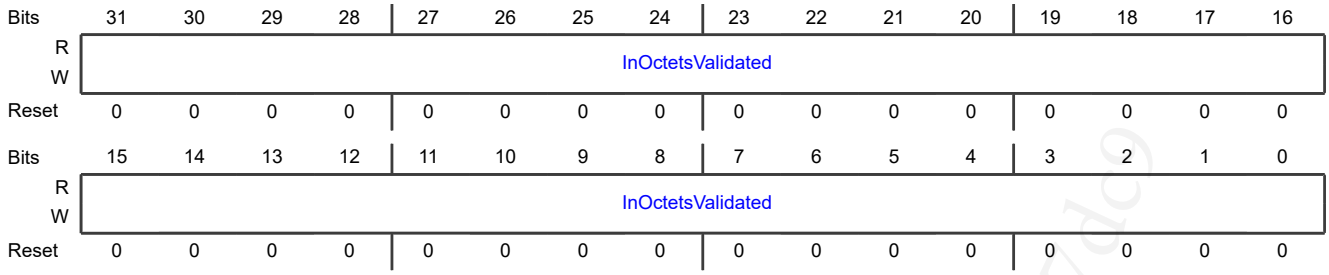
2.22.2 Function

The MACSEC_INOV1HS register provides the count of Controlled Port validated Input Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

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2.22.3 Diagram



2.22.4 Fields

Field	Function
31-0 InOctets Validated	InOctetsValidated The number of octets of plain text recovered from received frames with resolved SCI that were integrity protected but not encrypted 0..31. (802.1AE-2006/802.1AEbn-2011 10.7.10)

2.23 InOctetsValidated second half Statistic (MACSEC_INOV2HS)

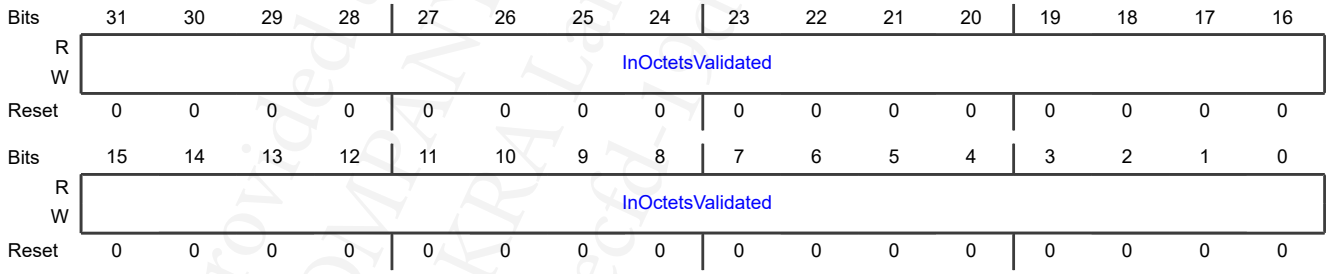
2.23.1 Offset

Register	Offset
MACSEC_INOV2HS	144h

2.23.2 Function

The MACSEC_INOV2HS register provides the count of Controlled Port validated Input Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

2.23.3 Diagram



2.23.4 Fields

Field	Function
31-0 InOctets Validated	InOctetsValidated The number of octets of plain text recovered from received frames with resolved SCI that were integrity protected but not encrypted 32..63. (802.1AE-2006/802.1AEbn-2011 10.7.10)

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2.24 InOctetsDecrypted first half Statistic (MACSEC_INOD1HS)

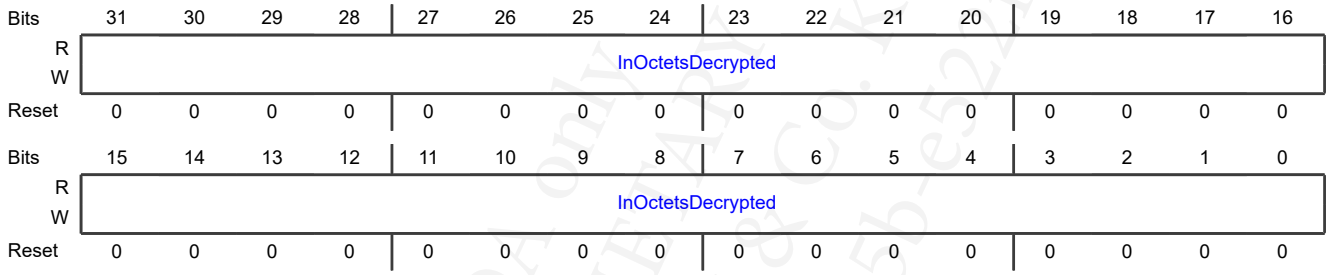
2.24.1 Offset

Register	Offset
MACSEC_INOD1HS	148h

2.24.2 Function

The MACSEC_INOD1HS register provides the count of Controlled Port Input Decrypted Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

2.24.3 Diagram



2.24.4 Fields

Field	Function
31-0 InOctets Decrypted	InOctetsDecrypted The number of octets of plain text recovered from received frames with resolved SCI that were integrity protected and encrypted 0..31. (802.1AE-2006/802.1AEbn-2011 10.7.10)

2.25 InOctetsDecrypted second half Statistic (MACSEC_INOD2HS)

2.25.1 Offset

Register	Offset
MACSEC_INOD2HS	14Ch

2.25.2 Function

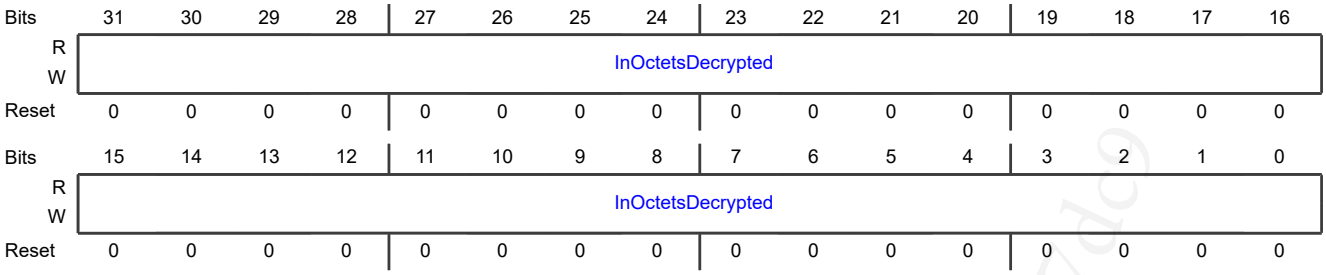
The MACSEC_INOD2HS register provides the count of Controlled Port Input Decrypted Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

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2.25.3 Diagram



2.25.4 Fields

Field	Function
31-0 InOctets Decrypted	InOctetsDecrypted The number of octets of plain text recovered from received frames with resolved SCI that were integrity protected and encrypted 32..63. (802.1AE-2006/802.1AEbn-2011 10.7.10)

2.26 RX Secure Channel InPktsUnchecked Statistic (MACSEC_RXSCIPUS)

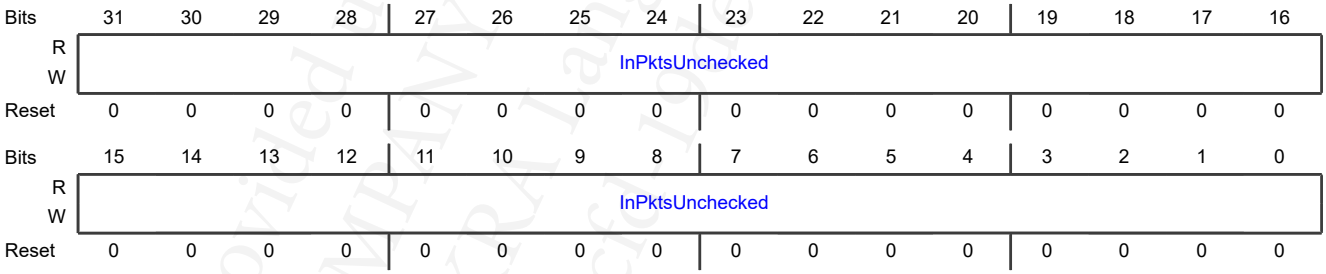
2.26.1 Offset

Register	Offset
MACSEC_RXSCIPUS	150h

2.26.2 Function

The register MACSEC_RXSCIPUS provides the count of Input Frames Unchecked for Secure Channel.

2.26.3 Diagram



2.26.4 Fields

Field	Function
31-0 InPkts Unchecked	InPktsUnchecked The number of frames with resolved SCI, delivered to the user of a controlled port, that are not validated with the validateFrame set to disable. (802.1AE-2006/802.1AEbn-2011 10.7.9)

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2.27 RX Secure Channel InPktsDelayed Statistic (MACSEC_RXSCIPDS)

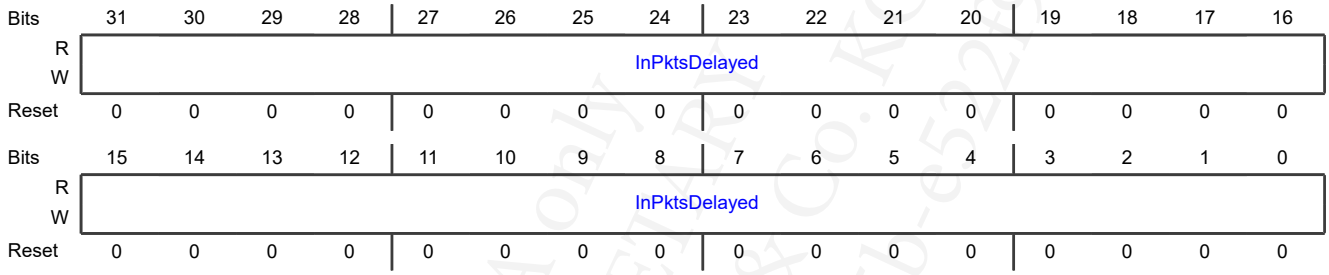
2.27.1 Offset

Register	Offset
MACSEC_RXSCIPDS	154h

2.27.2 Function

The register MACSEC_RXSCIPDS provides the count of Input Frames Delayed for Secure Channel.

2.27.3 Diagram



2.27.4 Fields

Field	Function
31-0 InPktsDelayed	InPktsDelayed The number of frames with resolved SCI, delivered to the user of a controlled port, that have their PN smaller than the lowest_PN with the validateFrame set to disable or replayProtect disabled. (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.28 RX Secure Channel InPktsLate Statistic (MACSEC_RXSCIPLS)

2.28.1 Offset

Register	Offset
MACSEC_RXSCIPLS	158h

2.28.2 Function

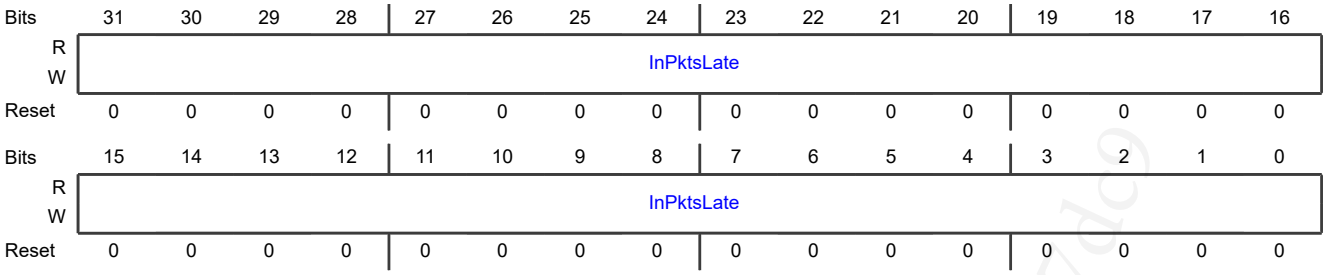
The register MACSEC_RXSCIPLS provides the count of Input Frames Late for Secure Channel.

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2.28.3 Diagram



2.28.4 Fields

Field	Function
31-0 InPktsLate	InPktsLate The number of frames with resolved SCI, discarded on the controlled port, that have their PN smaller than the lowest_PN with the validateFrame set to Check or Strict and replayProtect enabled. (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.29 RX AN a InNotUsingSA Statistic (MACSEC_RXAN0INUSS - MACSEC_RXAN3INUSS)

2.29.1 Offset

Register	Offset
MACSEC_RXAN0INUSS	160h
MACSEC_RXAN1INUSS	164h
MACSEC_RXAN2INUSS	168h
MACSEC_RXAN3INUSS	16Ch

2.29.2 Function

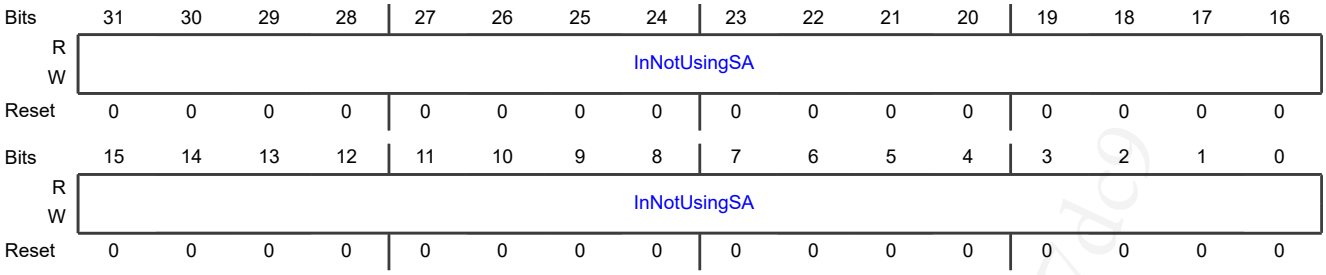
The MACSEC_RXAN*n*INUSS registers counts the Input Not Using SA for the AN *n* of the Secure Channel. Since there are only 2 SA context kept per SC and frames can be received on all 4 AN values, this statistic has to be kept on a per AN basis.

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2.29.3 Diagram



2.29.4 Fields

Field	Function
31-0 InNotUsingSA	InNotUsingSA The number of frames received with resolved SCI and discarded on disabled or not provisioned SA with validateFrame in the strict mode or the C bit is set. (802.1AE-2006/802.1AEbn-2011 10.7.9) ***Note:*** *In 802.1AE-2018 standard, this is now counted per SC in InPktsNoSAError, Software must aggregate this counter at SC level.*

2.30 RX AN a InPktsUnusedSA Statistic (MACSEC_RXAN0IPUSS - MACSEC_RXAN3IPUSS)

2.30.1 Offset

Register	Offset
MACSEC_RXAN0IPUSS	170h
MACSEC_RXAN1IPUSS	174h
MACSEC_RXAN2IPUSS	178h
MACSEC_RXAN3IPUSS	17Ch

2.30.2 Function

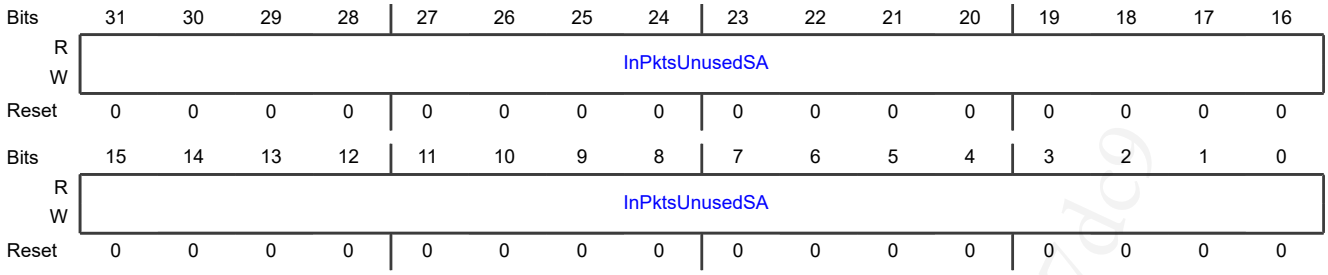
The MACSEC_RXAN*n*IPUSS registers counts the Input Frames Unused SA for the AN *n* of the Secure Channel. Since there are only 2 SA context kept per SC and frames can be received on all 4 AN values, this statistic has to be kept on a per AN basis.

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2.30.3 Diagram



2.30.4 Fields

Field	Function
31-0 InPkts UnusedSA	InPktsUnusedSA The number of frames received with resolved SCI on disabled or not provisioned SA with validateFrame not in the strict mode and the C bit is cleared. (802.1AE-2006/802.1AEbn-2011 10.7.9) ***Note:*** *In 802.1AE-2018 standard, this is now counted per SC in InPktsNoSA, Software must aggregate this counter at SC level.*

2.31 RX Security Association A configuration and status (MACSEC_RXSAACS)

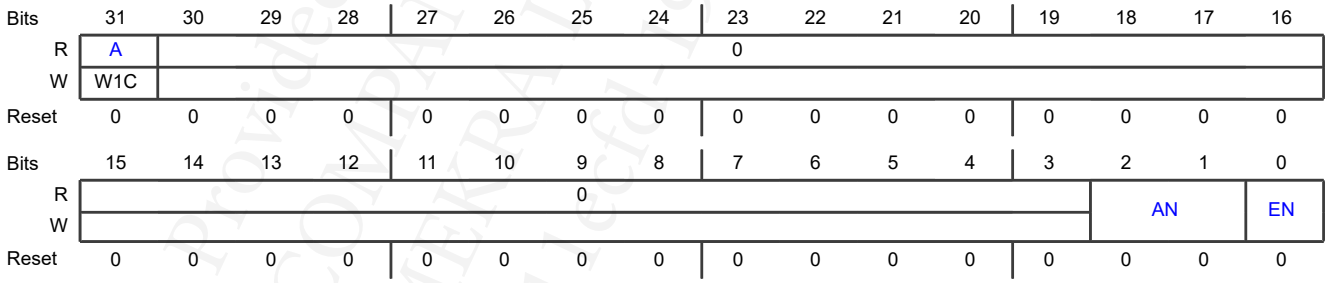
2.31.1 Offset

Register	Offset
MACSEC_RXSAACS	180h

2.31.2 Function

The MACSEC_RXSAACS register is used to configure and read the status of the Security Association A for the Secure Channel.

2.31.3 Diagram



2.31.4 Fields

Field	Function
31 A	A Active (RO/w1c) Must be cleared when the SA is initialized Will be set to 1 once it has been used at least once to receive

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Field	Function
	This bit is also used to track if the key hash has been generated for the SAK. This why it must be cleared every time the SAK is modified.
30-3 —	Reserved, should be cleared
2-1 AN	AN AN association, SA A is associated with the programmed AN
0 EN	EN enableReceive (802.1AE-2006/802.1AEbn-2011 10.7.14) When set, it is an indication that the SA has been configured and can be used. When cleared, frame received on the SA will be marked for discard and counted in Section 2.29 or Section 2.30 depending on validateFrames setting.

2.32 RX Security Association A nextPN (MACSEC_RXSAANPN)

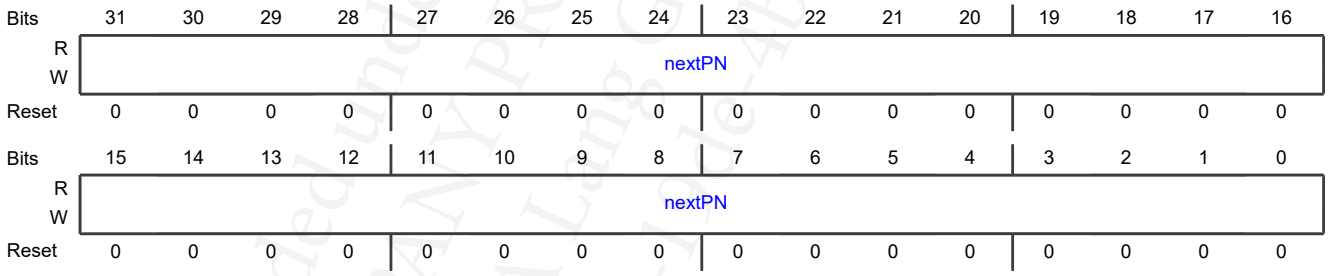
2.32.1 Offset

Register	Offset
MACSEC_RXSAANPN	184h

2.32.2 Function

The MACSEC_RXSAANPN register provide the ability to read and configure the Next Packet Number expected on RX.

2.32.3 Diagram



2.32.4 Fields

Field	Function
31-0 nextPN	nextPN RX Security Association A x nextPN (802.1AE-2006/802.1AEbn-2011 10.6.5) nextPN is used to track the next expected Packet Number for this SA. It holds the value of the highest PN seen on this SA plus one. It should be initialized to one. Management Software can change this value any time. For XPN-capable cipher suite, nextPN is the bit 31-0 of nextXPN.

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2.33 RX Security Association Extended A nextPN (MACSEC_RXSAAXNPN)

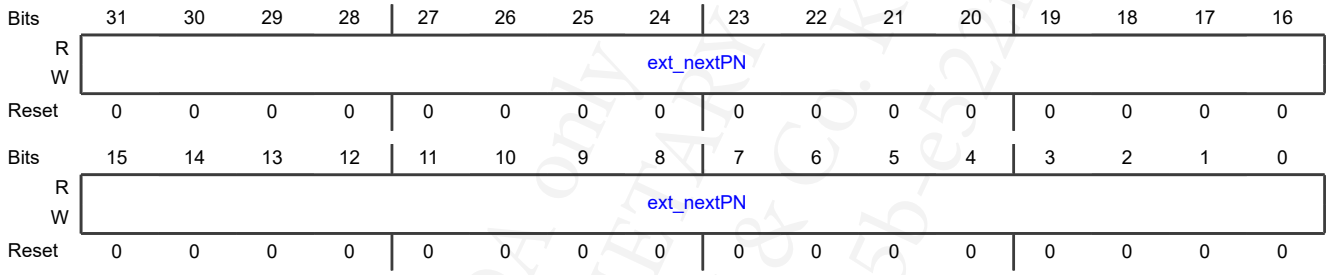
2.33.1 Offset

Register	Offset
MACSEC_RXSAAXNPN	188h

2.33.2 Function

The MACSEC_RXSAAXNPN register provide the ability to read and configure the bit 63-32 of Next Extended Packet Number accepted on RX when XPN capable cipher suite is configured.

2.33.3 Diagram



2.33.4 Fields

Field	Function
31-0 ext_nextPN	ext_nextPN RX Security Association A extended nextPN (802.1AEbw-2013) Extended nextPN is used to support XPN capable cipher suite. extended_nextPN is the bit 63-32 of next XPN.It should be initialized to zero. Management Software can change this value any time. It is not used when non-XPN capable cipher suite is configured for this secure channel.

2.34 RX Security Association A lowestPN (MACSEC_RXSAALPN)

2.34.1 Offset

Register	Offset
MACSEC_RXSAALPN	18Ch

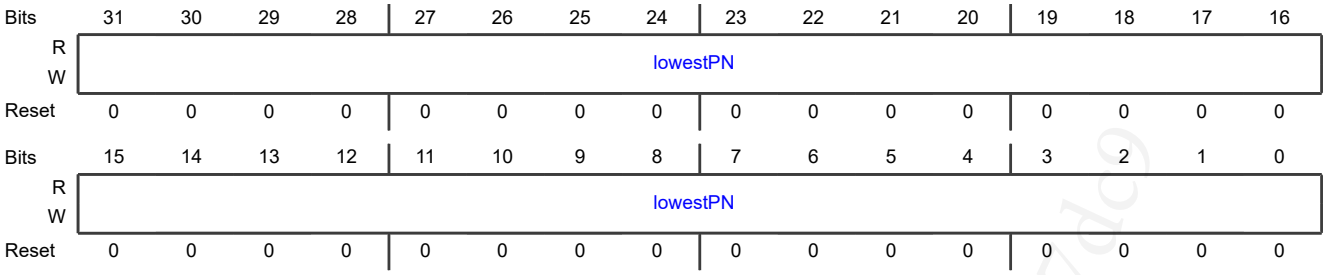
2.34.2 Function

The MACSEC_RXSAALPN register provide the ability to read and configure the Lowest Packet Number accepted on RX.

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2.34.3 Diagram



2.34.4 Fields

Field	Function
31-0 lowestPN	lowestPN RX Security Association A x lowestPN (802.1AE-2006/802.1AEbn-2011 10.6.5) lowestPN is used to track the lowest acceptable PN value used in the anti-replay algorithm. Management Software can change this value to open or close the window. The hardware will update this value to max(current lowestPN, nextPN - replayWindow) on every updates of Section 2.32 For non XPN-capable cipher suite, a 32-bit lowestPN value of zero will be treated as the same as 2^32. For XPN-capable cipher suite, lowestPN is the bit 31-0 of lowestXPN (802.1AEbw-2013), a 64-bit lowest XPN value of zero will be treated as the same as 2^64.

2.35 RX Security Association Extended A lowestPN (MACSEC_RXSAAXLPN)

2.35.1 Offset

Register	Offset
MACSEC_RXSAAXLPN	190h

2.35.2 Function

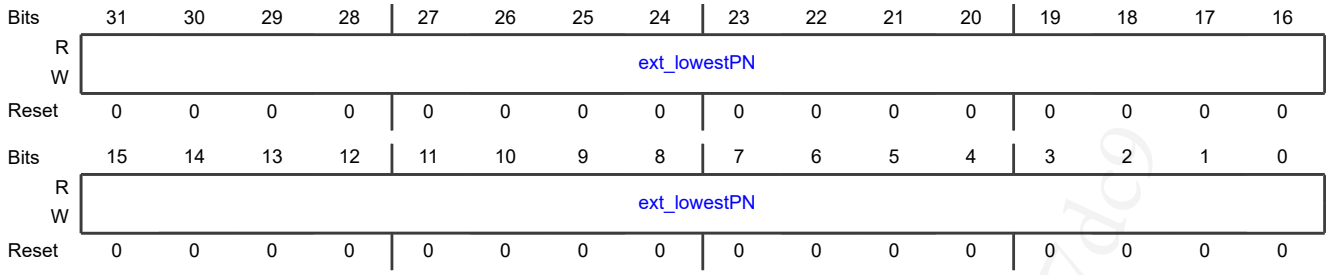
The MACSEC_RXSAAXLPN register provide the ability to read and configure the bit 63-32 of Lowest Extended Packet Number accepted on RX when XPN capable cipher suite is configured.

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2.35.3 Diagram



2.35.4 Fields

Field	Function
31-0 ext_lowestPN	ext_lowestPN RX Security Association A extended lowestPN Extended_lowestPN is used to support XPN capable cipher suite. extended_nextPN is the bit 63-32 of lowestXPN(802.1AEbw-2013). It is not used when non-XPN capable cipher suite is configured for this secure channel.

2.36 RX Security Association A InPktsOK Statistic (MACSEC_RXSAAIPOS)

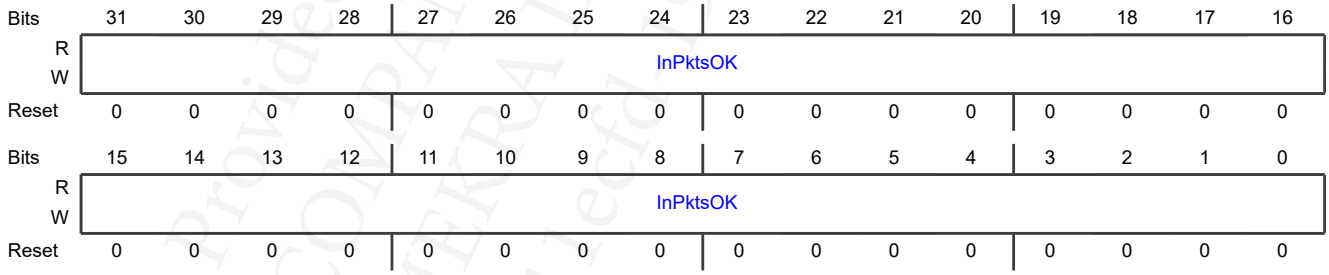
2.36.1 Offset

Register	Offset
MACSEC_RXSAAIPOS	194h

2.36.2 Function

The MACSEC_RXSAAIPOS register counts the Input Frames OK for the Security Association A of the Secure Channel.

2.36.3 Diagram



2.36.4 Fields

Field	Function
31-0 InPktsOK	InPktsOK The number of frames with resolved SCI, delivered to the user of a controlled port, that have passed all frame validation with the validateFrame not set to disable. (802.1AE-2006/802.1AEbn-2011 10.7.9)

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Field	Function
	Note: *In 802.1AE-2018 standard, this is now counted per SC, Software must aggregate this counter at SC level.*

2.37 RX Security Association A hash 1 of 4 (MACSEC_RXSAHA1Q)

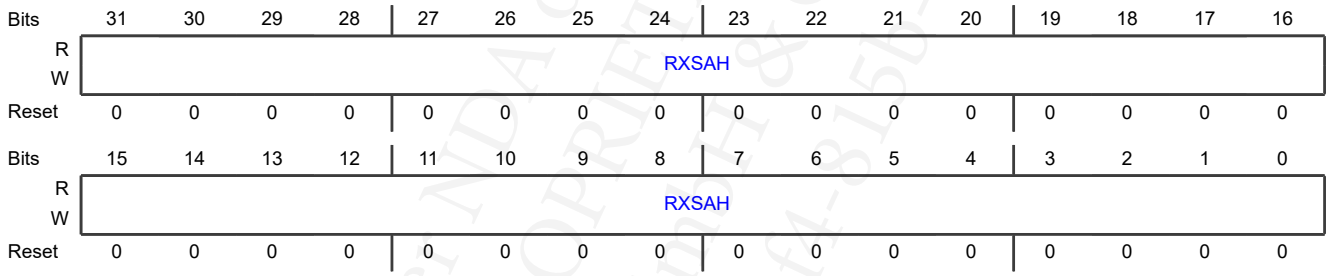
2.37.1 Offset

Register	Offset
MACSEC_RXSAHA1Q	1A0h

2.37.2 Function

The MACSEC_RXSAHA1Q register is used to access the hardware generated Security Association A hash for the Secure Channel. The hash is valid when Section 2.31 A bit is set. The hash is the H=E(K,0) value used by the GHASH algorithm.

2.37.3 Diagram



2.37.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 0-31

2.38 RX Security Association A hash 2 of 4 (MACSEC_RXSAHA2Q)

2.38.1 Offset

Register	Offset
MACSEC_RXSAHA2Q	1A4h

2.38.2 Function

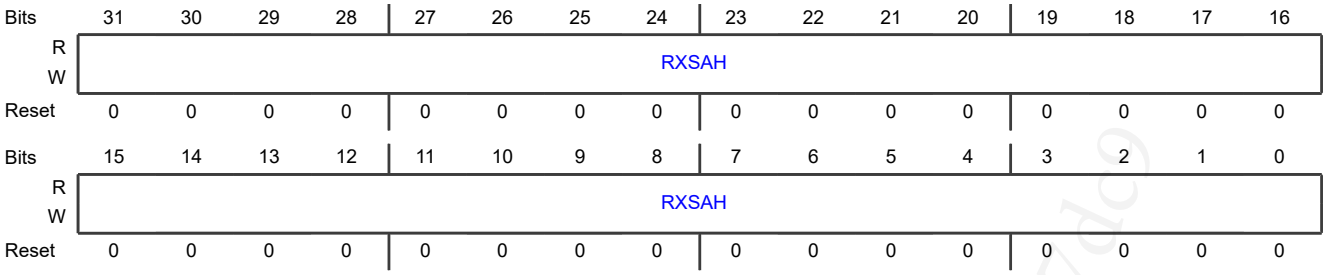
The MACSEC_RXSAHA2Q register is used to access the hardware generated Security Association A hash for the Secure Channel.

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2.38.3 Diagram



2.38.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 32-63

2.39 RX Security Association A hash 3 of 4 (MACSEC_RXSAHA3Q)

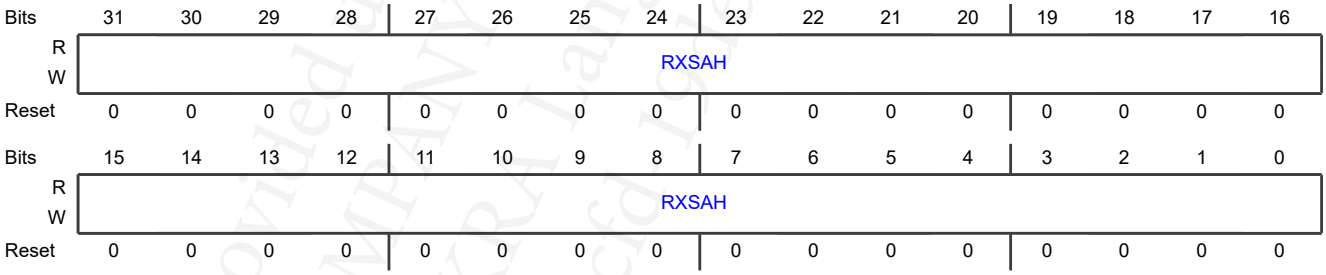
2.39.1 Offset

Register	Offset
MACSEC_RXSAHA3Q	1A8h

2.39.2 Function

The MACSEC_RXSAHA3Q register is used to access the hardware generated Security Association A hash for the Secure Channel.

2.39.3 Diagram



2.39.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 64-95

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2.40 RX Security Association A hash 4 of 4 (MACSEC_RXSAHA4Q)

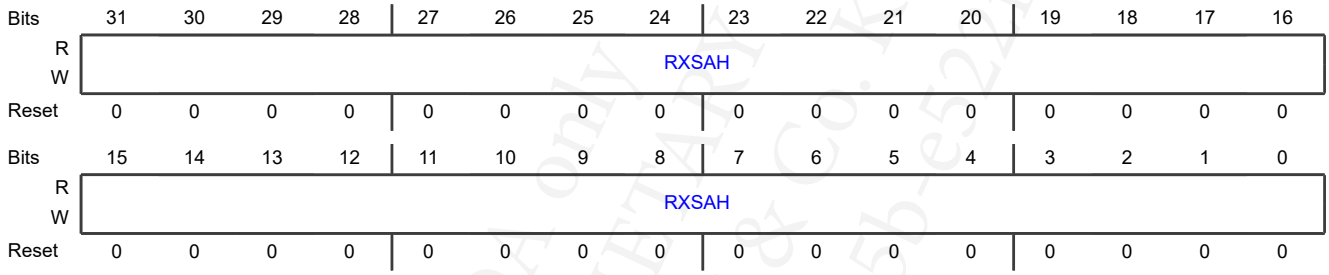
2.40.1 Offset

Register	Offset
MACSEC_RXSAHA4Q	1ACh

2.40.2 Function

The MACSEC_RXSAHA4Q register is used to access the hardware generated Security Association A hash for the Secure Channel.

2.40.3 Diagram



2.40.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 96-127

2.41 RX Security Association A InPktsInvalid Statistic (MACSEC_RXSAAIPIS)

2.41.1 Offset

Register	Offset
MACSEC_RXSAAIPIS	1B0h

2.41.2 Function

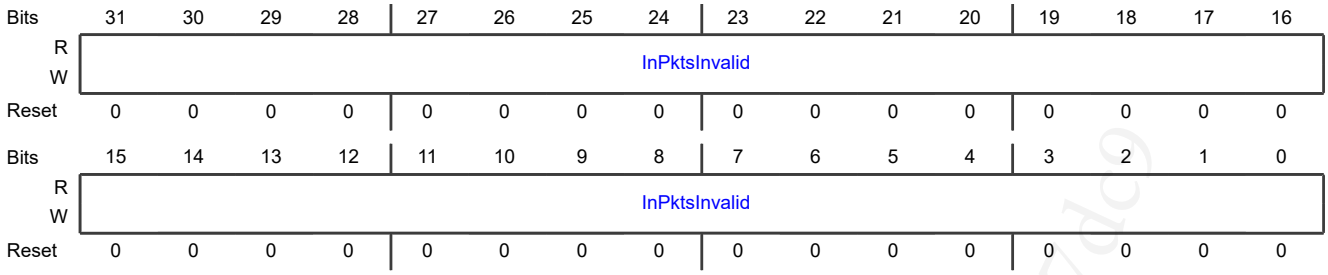
The MACSEC_RXSAAIPIS register counts the Input Frames Invalid for the Security Association A of the Secure Channel.

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2.41.3 Diagram



2.41.4 Fields

Field	Function
31-0 InPktsInvalid	InPktsInvalid The number of frames with resolved SCI, delivered to the user of a controlled port, that have failed frame validation with the validateFrame set to check. (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.42 RX Security Association A InPktsNotValid Statistic (MACSEC_RXSAAIPNVS)

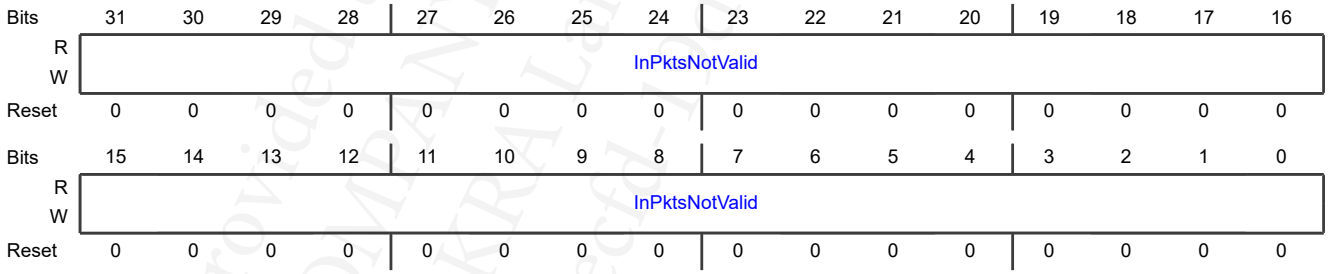
2.42.1 Offset

Register	Offset
MACSEC_RXSAAIPNVS	1B4h

2.42.2 Function

The MACSEC_RXSAAIPNVS register counts the Input Frames Not Valid for the Security Association A of the Secure Channel.

2.42.3 Diagram



2.42.4 Fields

Field	Function
31-0 InPktsNotValid	InPktsNotValid The number of frames with resolved SCI, discarded on the controlled port, that have failed frame validation with the validateFrame set to strict or the c bit is set. (802.1AE-2006/802.1AEbn-2011 10.7.9)

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2.43 RX Security Association B configuration and status (MACSEC_RXSABCS)

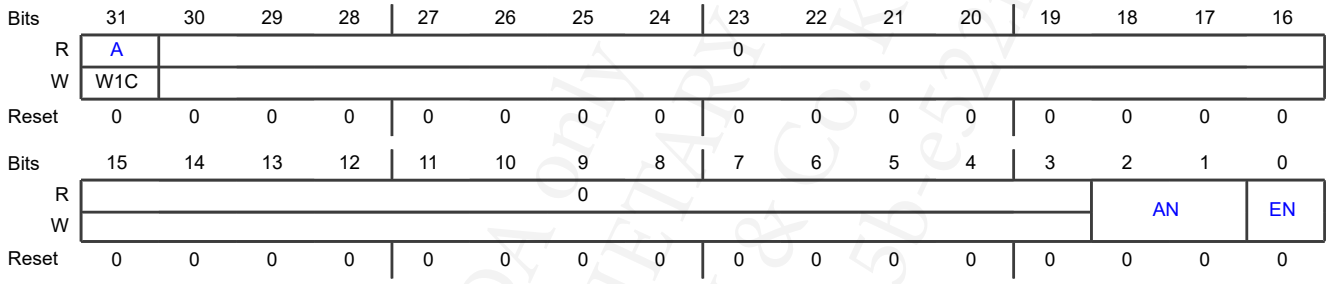
2.43.1 Offset

Register	Offset
MACSEC_RXSABCS	1C0h

2.43.2 Function

The MACSEC_RXSABCS register is used to configure and read the status of the Security Association B for the Secure Channel.

2.43.3 Diagram



2.43.4 Fields

Field	Function
31 A	A Active (RO/w1c) Must be cleared when the SA is initialized Will be set to 1 once it has been used at least once to receive
30-3 —	Reserved, should be cleared
2-1 AN	AN AN association, SA A is associated with the programmed AN
0 EN	EN enableReceive (802.1AE-2006/802.1AEbn-2011 10.7.14)

2.44 RX Security Association B nextPN (MACSEC_RXSABNPN)

2.44.1 Offset

Register	Offset
MACSEC_RXSABNPN	1C4h

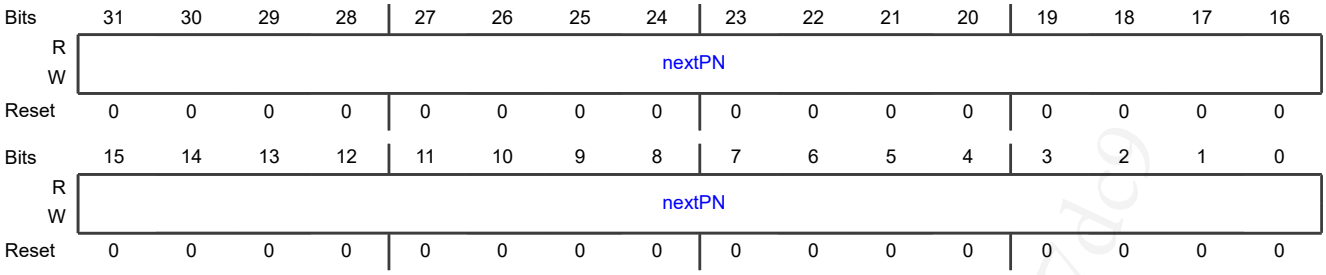
2.44.2 Function

The MACSEC_RXSABNPN register provide the ability to read and configure the Next Packet Number expected on RX.

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2.44.3 Diagram



2.44.4 Fields

Field	Function
31-0 nextPN	nextPN RX Security Association B x nextPN (802.1AE-2006/802.1AEbn-2011 10.6.5) nextPN is used to track the next expected Packet Number for this SA. It holds the value of the highest PN seen on this SA plus one. It should be initialized to one. Management Software can change this value any time. For XPN-capable cipher suite, nextPN is the bit 31-0 of nextXPN.

2.45 RX Security Association Extended B nextPN (MACSEC_RXSABXNPN)

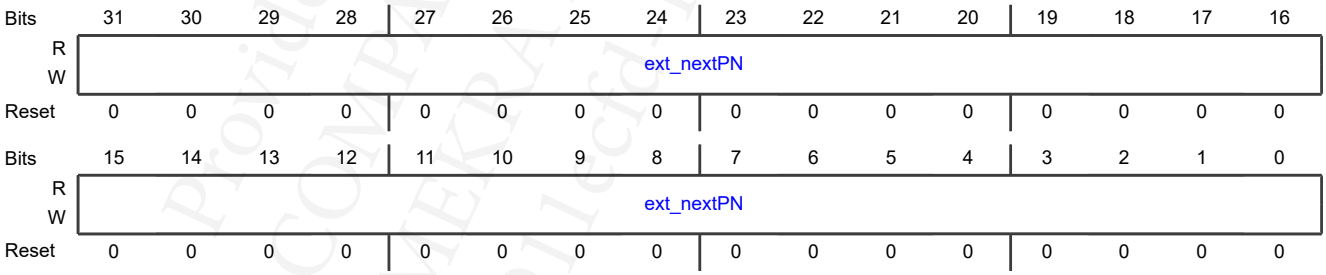
2.45.1 Offset

Register	Offset
MACSEC_RXSABXNPN	1C8h

2.45.2 Function

The MACSEC_RXSABXNPN register provide the ability to read and configure the bit 63-32 of Next Extended Packet Number accepted on RX when XPN capable cipher suite is configured.

2.45.3 Diagram



2.45.4 Fields

Field	Function
31-0 ext_nextPN	ext_nextPN RX Security Association B extended nextPN (802.1AEbw-2013)

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Field	Function
	Extended nextPN is used to support XPN capable cipher suite. extended_nextPN is the bit 63-32 of next XPN.It should be initialized to zero. Management Software can change this value any time. It is not used when non-XPN capable cipher suite is configured for this secure channel.

2.46 RX Security Association B lowestPN (MACSEC_RXSABLPN)

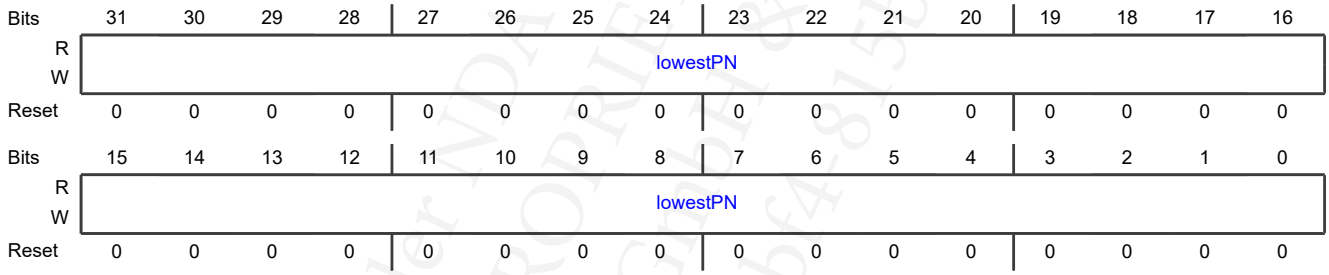
2.46.1 Offset

Register	Offset
MACSEC_RXSABLPN	1CCh

2.46.2 Function

The MACSEC_RXSABLPN register provide the ability to read and configure the Lowest Packet Number accepted on RX.

2.46.3 Diagram



2.46.4 Fields

Field	Function
31-0 lowestPN	lowestPN RX Security Association B x lowestPN (802.1AE-2006/802.1AEbn-2011 10.6.5) lowestPN is used to track the lowest acceptable PN value used in the anti-replay algorithm. Management Software can change this value to open or close the window. The hardware will update this value to max(current lowestPN, nextPN - replayWindow) on every updates of Section 2.44 For non XPN-capable cipher suite, a 32-bit lowestPN value of zero will be treated as the same as 2^32. For XPN-capable cipher suite, lowestPN is the bit 31-0 of lowestXPN (802.1AEbw-2013), a 64-bit lowest XPN value of zero will be treated as the same as 2^64.

2.47 RX Security Association Extended B lowestPN (MACSEC_RXSABXLPN)

2.47.1 Offset

Register	Offset
MACSEC_RXSABXLPN	1D0h

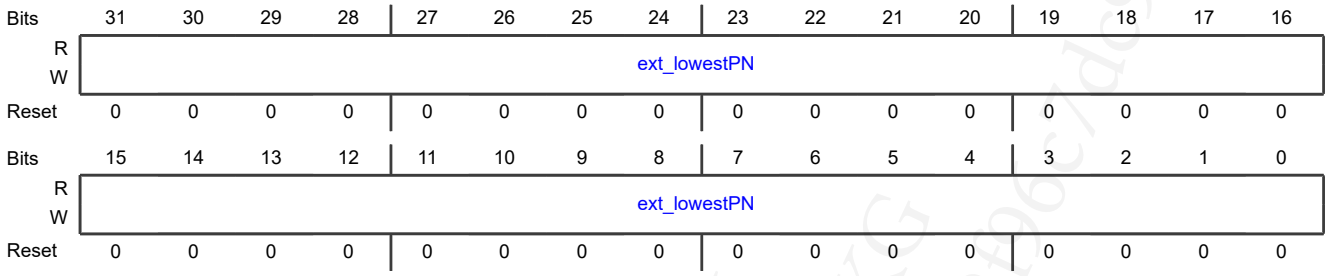
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2.47.2 Function

The MACSEC_RXSABXLPN register provide the ability to read and configure the bit 63-32 of Lowest Extended Packet Number accepted on RX when XPN capable cipher suite is configured.

2.47.3 Diagram



2.47.4 Fields

Field	Function
31-0 ext_lowestPN	ext_lowestPN RX Security Association B extended lowestPN Extended_lowestPN is used to support XPN capable cipher suite. extended_nextPN is the bit 63-32 of lowestXPN(802.1AEbw-2013). It is not used when non-XPN capable cipher suite is configured for this secure channel.

2.48 RX Security Association B InPktsOK Statistic (MACSEC_RXSABIPOS)

2.48.1 Offset

Register	Offset
MACSEC_RXSABIPOS	1D4h

2.48.2 Function

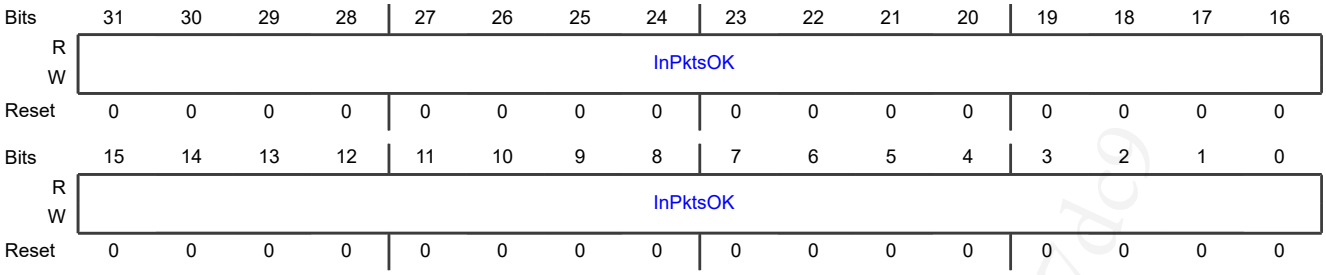
The MACSEC_RXSABIPOS register counts the Input Frames OK for the Security Association B of the Secure Channel.

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2.48.3 Diagram



2.48.4 Fields

Field	Function
31-0 InPktsOK	InPktsOK Input Frames OK (802.1AE-2006/802.1AEbn-2011 10.7.9) The number of frames with resolved SCI, delivered to the user of a controlled port, that have passed all frame validation with the validateFrame not set to disable. ***Note:*** *In 802.1AE-2018 standard, this is now counted per SC, Software must aggregate this counter at SC level.* Note:

2.49 RX Security Association B hash 1 of 4 (MACSEC_RXSAHB1Q)

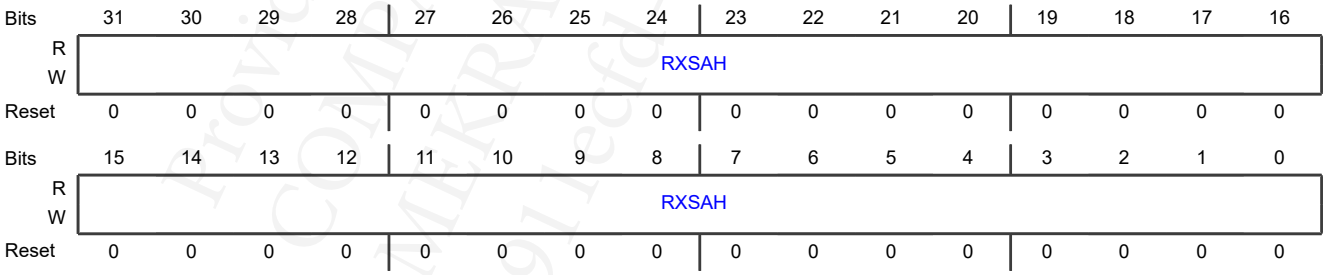
2.49.1 Offset

Register	Offset
MACSEC_RXSAHB1Q	1E0h

2.49.2 Function

The MACSEC_RXSAHB1Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

2.49.3 Diagram



2.49.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 0-31

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2.50 RX Security Association B hash 2 of 4 (MACSEC_RXSAHB2Q)

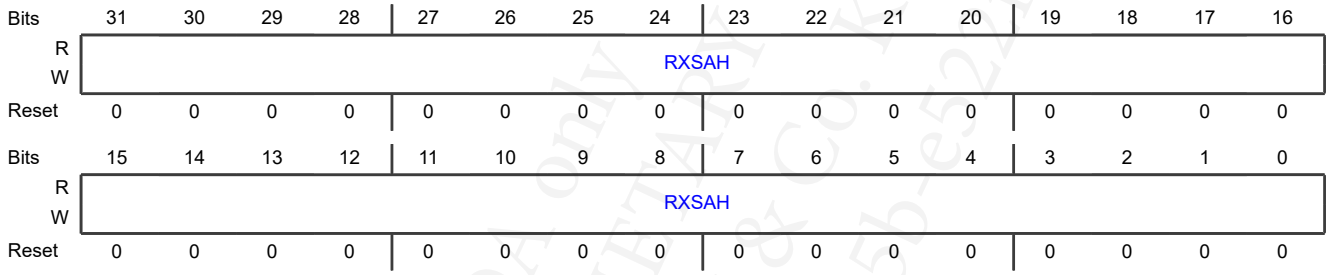
2.50.1 Offset

Register	Offset
MACSEC_RXSAHB2Q	1E4h

2.50.2 Function

The MACSEC_RXSAHB2Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

2.50.3 Diagram



2.50.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 32-63

2.51 RX Security Association B hash 3 of 4 (MACSEC_RXSAHB3Q)

2.51.1 Offset

Register	Offset
MACSEC_RXSAHB3Q	1E8h

2.51.2 Function

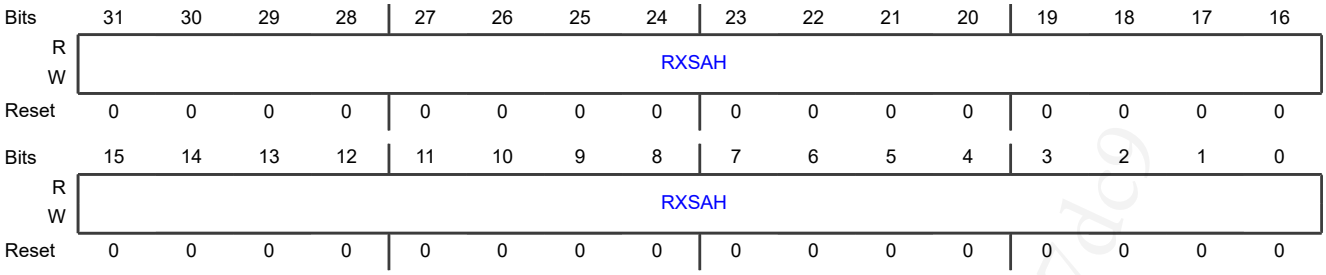
The MACSEC_RXSAHB3Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

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2.51.3 Diagram



2.51.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 64-95

2.52 RX Security Association B hash 4 of 4 (MACSEC_RXSAHB4Q)

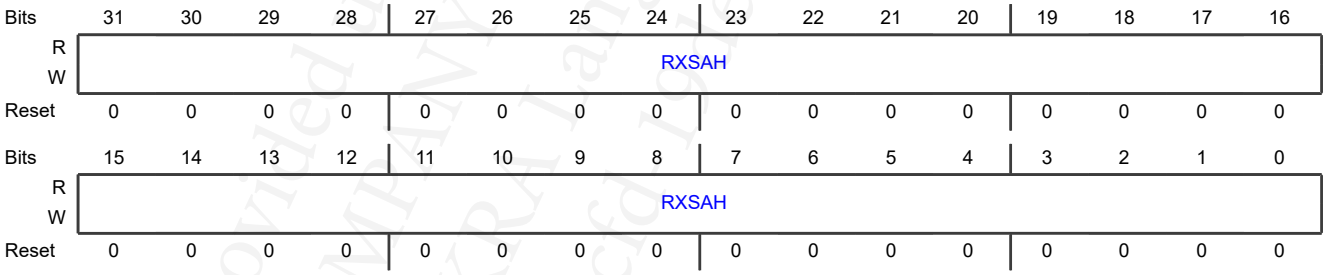
2.52.1 Offset

Register	Offset
MACSEC_RXSAHB4Q	1ECh

2.52.2 Function

The MACSEC_RXSAHB4Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

2.52.3 Diagram



2.52.4 Fields

Field	Function
31-0 RXSAH	RXSAH RX Security Association hash bit 96-127

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2.53 RX Security Association B InPktsInvalid Statistic (MACSEC_RXSABIPIS)

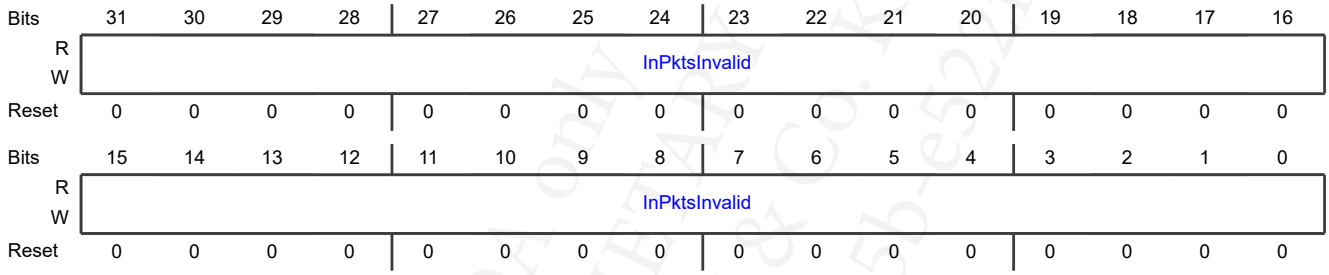
2.53.1 Offset

Register	Offset
MACSEC_RXSABIPIS	1F0h

2.53.2 Function

The MACSEC_RXSAAIPIS register counts the Input Frames Invalid for the Security Association B of the Secure Channel.

2.53.3 Diagram



2.53.4 Fields

Field	Function
31-0 InPktsInvalid	InPktsInvalid The number of frames with resolved SCI, delivered to the user of a controlled port, that have failed frame validation with the validateFrame set to check (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.54 RX Security Association B InPktsNotValid Statistic (MACSEC_RXSABIPNVS)

2.54.1 Offset

Register	Offset
MACSEC_RXSABIPNVS	1F4h

2.54.2 Function

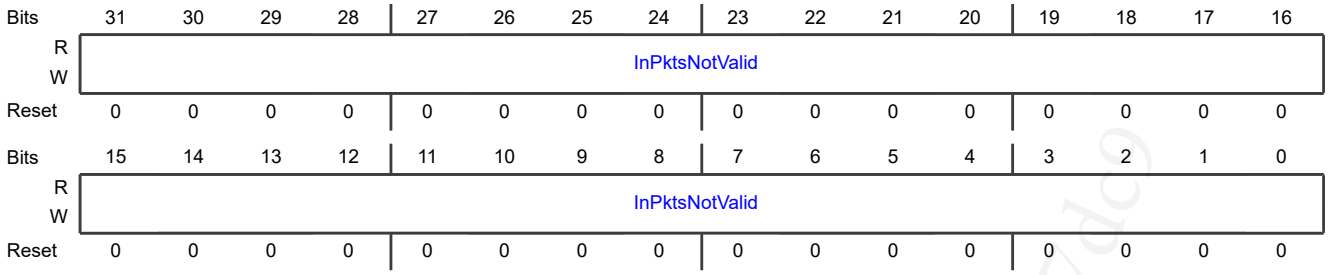
The MACSEC_RXSABIPNVS register counts the Input Frames Not Valid for the Security Association B of the Secure Channel.

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2.54.3 Diagram



2.54.4 Fields

Field	Function
31-0 InPktsNotValid	InPktsNotValid The number of frames with resolved SCI, discarded on the controlled port, that have failed frame validation with the validateFrame set to strict or the c bit is set. (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.55 TX Secure Channel Identifier first half (MACSEC_TXSCI1H)

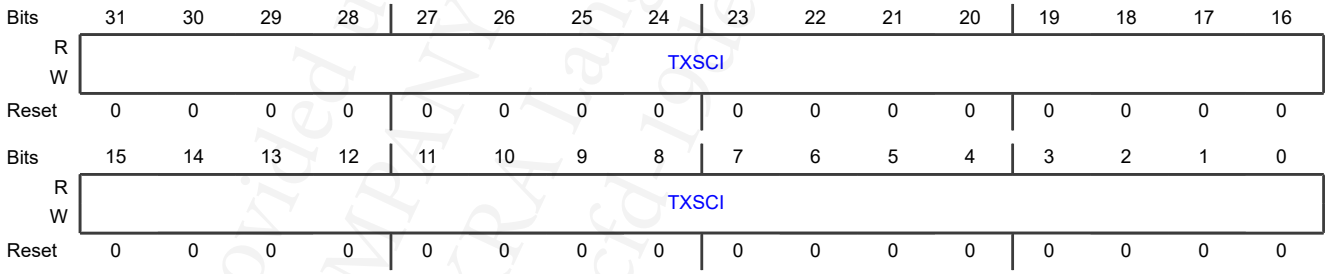
2.55.1 Offset

Register	Offset
MACSEC_TXSCI1H	200h

2.55.2 Function

The MACSEC_TXSCI1H register is used to program the first half of the SCI.

2.55.3 Diagram



2.55.4 Fields

Field	Function
31-0 TXSCI	TXSCI Transmit SCI bit 0 to 31 to be used for SC

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2.56 TX Secure Channel Identifier second half (MACSEC_TXSCI2H)

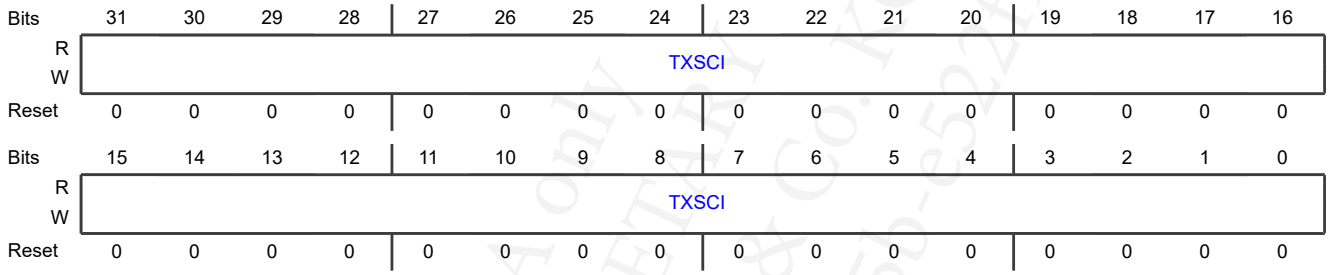
2.56.1 Offset

Register	Offset
MACSEC_TXSCI2H	204h

2.56.2 Function

The MACSEC_TXSCI2H register is used to program the second half of the SCI.

2.56.3 Diagram



2.56.4 Fields

Field	Function
31-0 TXSCI	TXSCI Transmit SCI bit 32 to 63 to be used for SC x

2.57 ifOutOctets first half Statistic (MACSEC_IFOO1HS)

2.57.1 Offset

Register	Offset
MACSEC_IFOO1HS	210h

2.57.2 Function

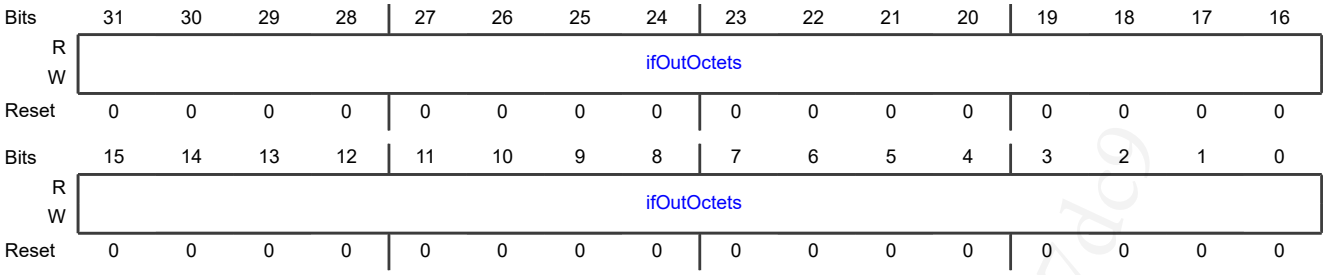
The MACSEC_IFOO1HS register provides the count of the Controlled Port interface Output Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

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2.57.3 Diagram



2.57.4 Fields

Field	Function
31-0 ifOutOctets	ifOutOctets Controlled Port Interface Output Octets 0..31 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifOutOctets count is the sum of the all octets of the MSDUs delivered by the user of the Controlled Port to the Secure Frame Generation process (10.5), plus the octets of the destination and source MAC addresses.

2.58 ifOutOctets second half Statistic (MACSEC_IFOO2HS)

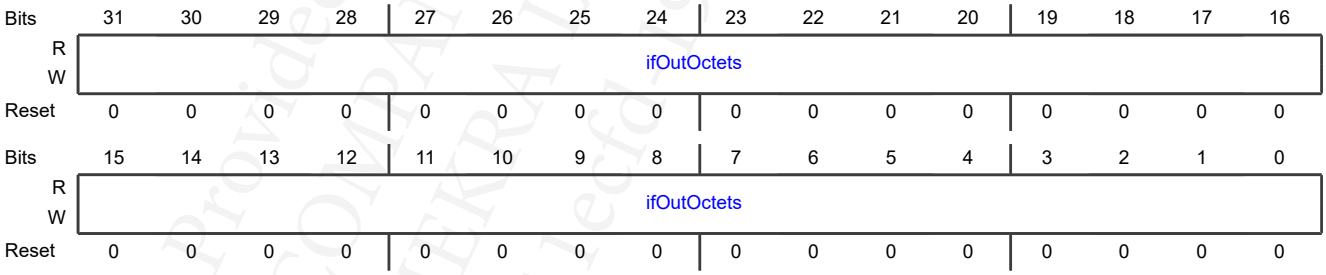
2.58.1 Offset

Register	Offset
MACSEC_IFOO2HS	214h

2.58.2 Function

The MACSEC_IFOO2HS register provides the count of the Controlled Port interface Output Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

2.58.3 Diagram



2.58.4 Fields

Field	Function
31-0 ifOutOctets	ifOutOctets Controlled Port Interface Output Octets 32..63 (802.1AE-2006/802.1AEbn-2011 10.7.6)

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Field	Function
	The ifOutOctets count is the sum of the all octets of the MSDUs delivered by the user of the Controlled Port to the Secure Frame Generation process (10.5), plus the octets of the destination and source MAC addresses.

2.59 ifOutUcastPkts Statistic (MACSEC_IFOUPS)

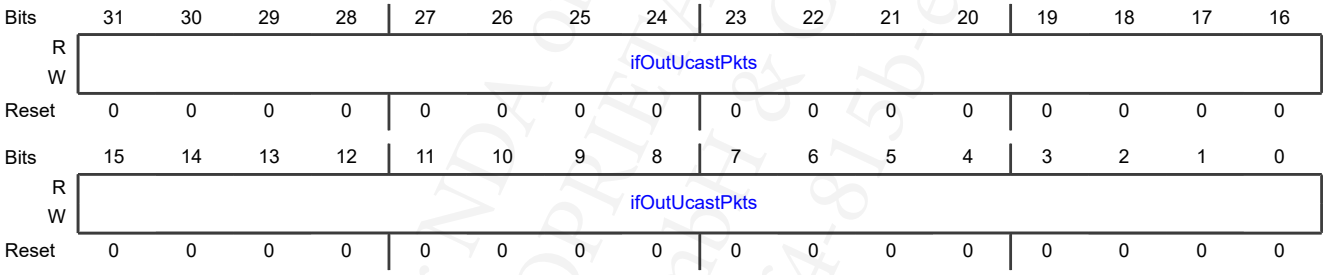
2.59.1 Offset

Register	Offset
MACSEC_IFOUPS	218h

2.59.2 Function

The MACSEC_IFOUPS register provides the count of the Controlled Port interface Output Unicast Frames.

2.59.3 Diagram



2.59.4 Fields

Field	Function
31-0 ifOutUcastPkts	ifOutUcastPkts The number of frames, that the user of the controlled port requested to be transmitted, which were not addressed to a multicast or broadcast address. (802.1AE-2006/802.1AEbn-2011 10.7.6)

2.60 OutPktsUntagged Statistic (MACSEC_OPUS)

2.60.1 Offset

Register	Offset
MACSEC_OPUS	21Ch

2.60.2 Function

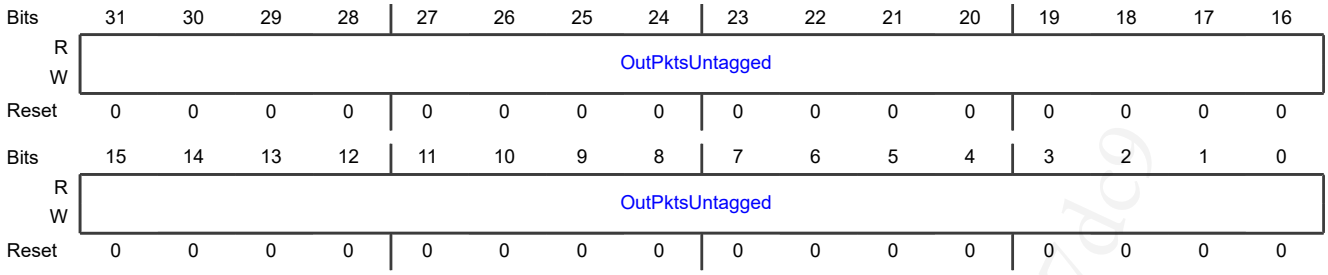
The MACSEC_OPUS register provides the count of the Controlled Port output Frames Untagged.

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2.60.3 Diagram



2.60.4 Fields

Field	Function
31-0 OutPkts Untagged	OutPktsUntagged The number of frames, that the user of the controlled port requested to be transmitted, with protectFrame false. (802.1AE-2006/802.1AEbn-2011 10.7.18)

2.61 ifOutMulticastPkts Statistic (MACSEC_IFOMPS)

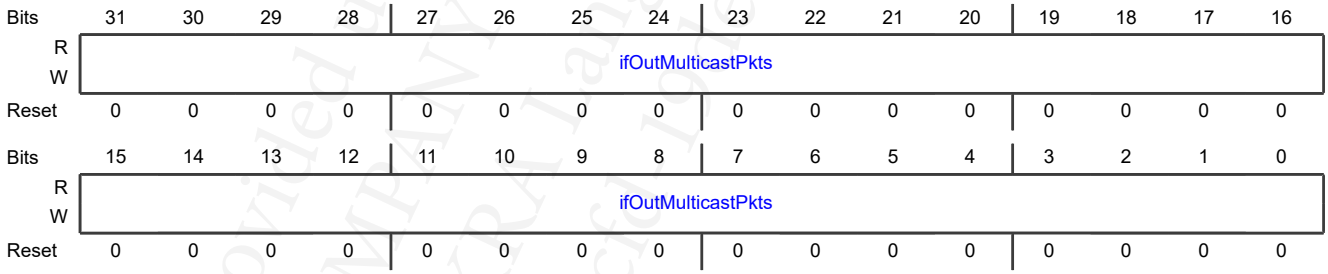
2.61.1 Offset

Register	Offset
MACSEC_IFOMPS	220h

2.61.2 Function

The MACSEC_IFOMPS register provides the count of the Controlled Port interface Output Multicast Frames.

2.61.3 Diagram



2.61.4 Fields

Field	Function
31-0 ifOut MulticastPkts	ifOutMulticastPkts The number of frames, that the user of the controlled port requested to be transmitted, which were addressed to a multicast address. (802.1AE-2006/802.1AEbn-2011 10.7.6)

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2.62 ifOutBroadcastPkts Statistic (MACSEC_IFOBPS)

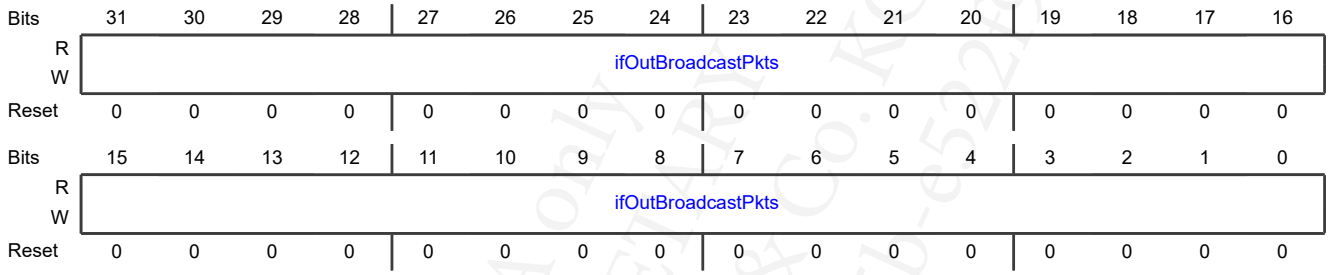
2.62.1 Offset

Register	Offset
MACSEC_IFOBPS	224h

2.62.2 Function

The MACSEC_IFOBPS register provides the count of the Controlled Port interface Output Broadcast Frames.

2.62.3 Diagram



2.62.4 Fields

Field	Function
31-0 ifOut BroadcastPkts	ifOutBroadcastPkts The number of frames, that the user of the controlled port requested to be transmitted, which were addressed to broadcast address (802.1AE-2006/802.1AEbn-2011 10.7.6)

2.63 TX Secure Channel configuration (MACSEC_TXSCCFG)

2.63.1 Offset

Register	Offset
MACSEC_TXSCCFG	228h

2.63.2 Function

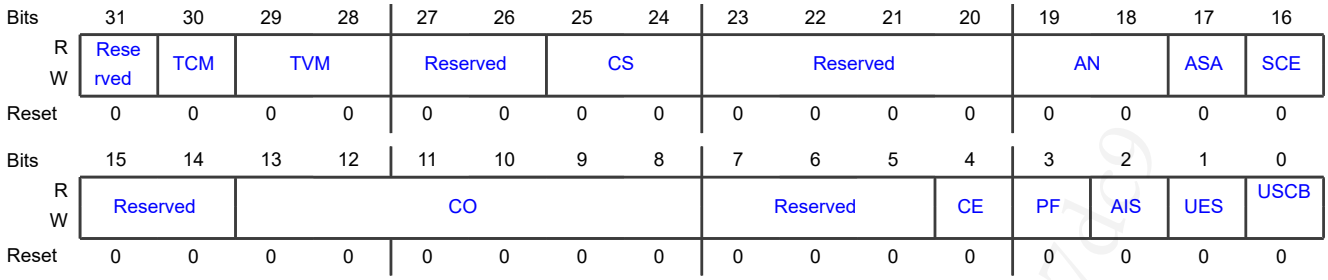
The MACSEC_TXSCCFG register is used to configure the TX SC.

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2.63.3 Diagram



2.63.4 Fields

Field	Function
31 —	Reserved, should be cleared
30 TCM	TCM TX Custom clear tag Mode. This field is used to select how many Custom tag are transferred clear on the TX path for this TX SC. MACSEC_CFG[VCT] must be enabled in order for this field to take effect. 0b - no Custom tags are transferred clear 1b - up to 1 Custom tag is transferred clear if present
29-28 TVM	TVM TX VCT mode. This field is used to select how many VLANs are transferred clear on the TX path for this TX SC. MACSEC_CFG[VCT] must be enabled in order for this field to take effect. 00b - no VLAN tags are transferred clear 01b - up to 1 VLAN tag is transferred clear if present 10b - up to 2 VLAN tags are transferred clear if present 11b - reserved, will behave like 0
27-26 —	Reserved, should be cleared.
25-24 CS	CS CipherSuite This field indicates the Cipher Suite to be used for this SC (802.1AE-2006/802.1AEbn-2011/802.1AE-2018 14.4) 00b - GCM-AES-128 01b - GCM-AES-256 10b - GCM-AES-XPN-128 11b - GCM-AES-XPN-256
23-20 —	Reserved, should be cleared
19-18 AN	AN AN association, AN value to be used by frame generation
17 ASA	ASA active SA to be used on TX for SC 0b - TX SA A 1b - TX SA B
16 SCE	SCE TX SC Enable

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Field	Function
	0b - disable. frame received on disabled SC are marked as error and an error interrupt is set Section 2.91 1b - enable
15-14 —	Reserved, should be cleared
13-8 CO	CO Confidentiality offset (802.1AE-2006/802.1AEbn-2011 10.7.25) 00_0000b - no offset 00_0001b-11_1111b - the first 1 to 63 Byte of data are integrity protected, not confidentiality protected. This configuration is ignored when a XPN-capable Cipher Suites is configured for this secure channel
7-5 —	Reserved, should be cleared
4 CE	CE confidentiality enable 0b - no confidentiality protection, only integrity protection 1b - confidentiality protection and integrity protection
3 PF	PF protectFrames (802.1AE-2006/802.1AEbn-2011 10.5) 0b - False, frames are not protected (security disabled) 1b - True, frames are protected
2 AIS	AIS alwaysIncludeSCI (802.1AE-2006/802.1AEbn-2011 10.5.3) default False 0b - False 1b - True, an explicit SCI is present in the SecTAG
1 UES	UES useES (802.1AE-2006/802.1AEbn-2011 10.5.3) default False 0b - False 1b - True, set the ES bit in the Sec TAG, ignored if alwaysIncludeSCI is set
0 USCB	USCB useSCB (802.1AE-2006/802.1AEbn-2011 10.5.3) default False 0b - False 1b - True, set the SCB bit in the Sec TAG, ignored if alwaysIncludeSCI is set

2.64 OutPktsTooLong Statistic (MACSEC_OPTLS)

2.64.1 Offset

Register	Offset
MACSEC_OPTLS	22Ch

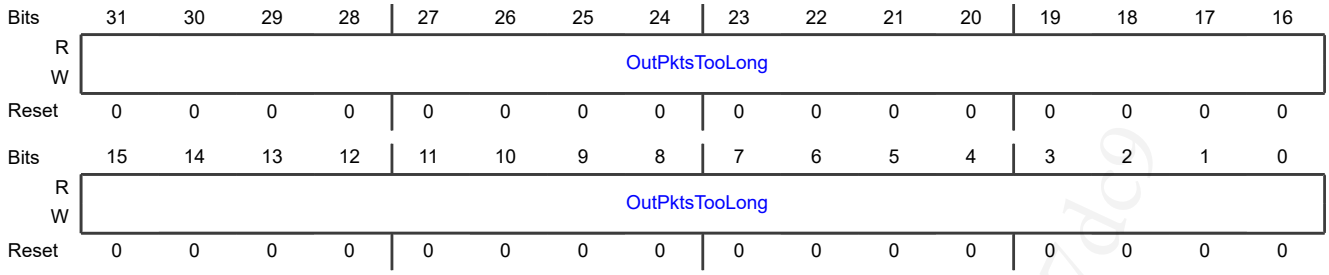
2.64.2 Function

The MACSEC_OPTLS register provides the count of the Controlled Port output Frames Too Long.

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2.64.3 Diagram



2.64.4 Fields

Field	Function
31-0 OutPktsTooLong	OutPktsTooLong The number of frames, that the user of the controlled port requested to be transmitted, marked as error due to length being larger than Section 2.7. (802.1AE-2006/802.1AEbn-2011 10.7.18)

2.65 OutOctetsProtected first half Statistic (MACSEC_OOP1HS)

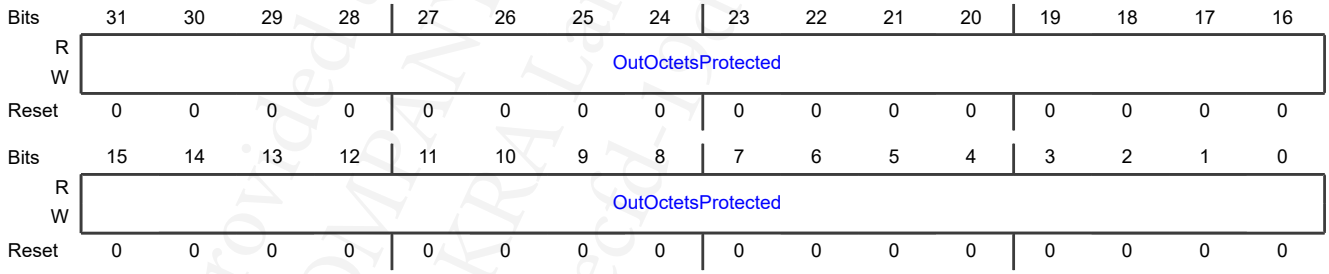
2.65.1 Offset

Register	Offset
MACSEC_OOP1HS	240h

2.65.2 Function

The MACSEC_OOP1HS register provides the count of Controlled Port Output Protected Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

2.65.3 Diagram



2.65.4 Fields

Field	Function
31-0 OutOctets Protected	OutOctetsProtected The number of octets of User Data in transmitted frames that were integrity protected but not encrypted 0..31. (802.1AE-2006/802.1AEbn-2011 10.7.19)

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2.66 OutOctetsProtected second half Statistic (MACSEC_OOP2HS)

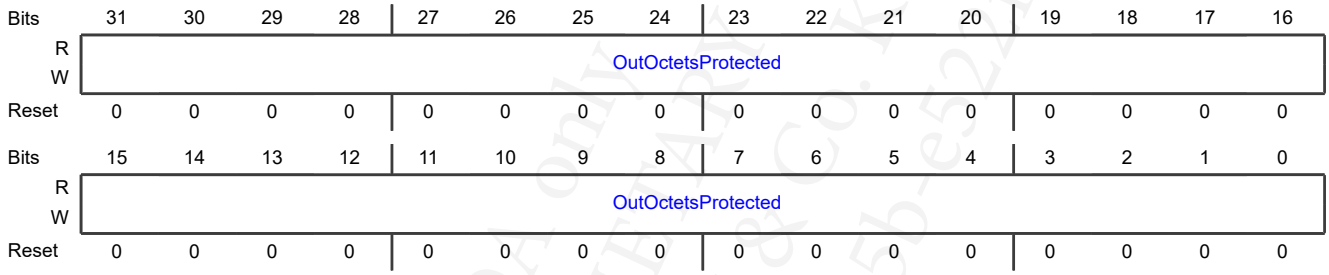
2.66.1 Offset

Register	Offset
MACSEC_OOP2HS	244h

2.66.2 Function

The MACSEC_OOP2HS register provides the count of Controlled Port Output Validated Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

2.66.3 Diagram



2.66.4 Fields

Field	Function
31-0 OutOctets Protected	OutOctetsProtected The number of octets of User Data in transmitted frames that were integrity protected but not encrypted 32..63. (802.1AE-2006/802.1AEbn-2011 10.7.19)

2.67 OutOctetsEncrypted first half Statistic (MACSEC_OOE1HS)

2.67.1 Offset

Register	Offset
MACSEC_OOE1HS	248h

2.67.2 Function

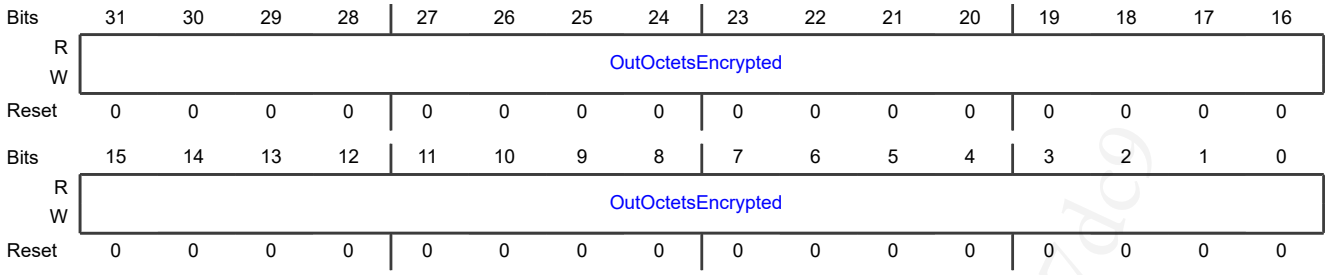
The MACSEC_OOE1HS register provides the count of Controlled Port Output Encrypted Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

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2.67.3 Diagram



2.67.4 Fields

Field	Function
31-0 OutOctets Encrypted	OutOctetsEncrypted The number of octets of User Data in transmitted frames that were both integrity protected and encrypted 0..31. (802.1AE-2006/802.1AEbn-2011 10.7.19)

2.68 OutOctetsEncrypted second half Statistic (MACSEC_OOE2HS)

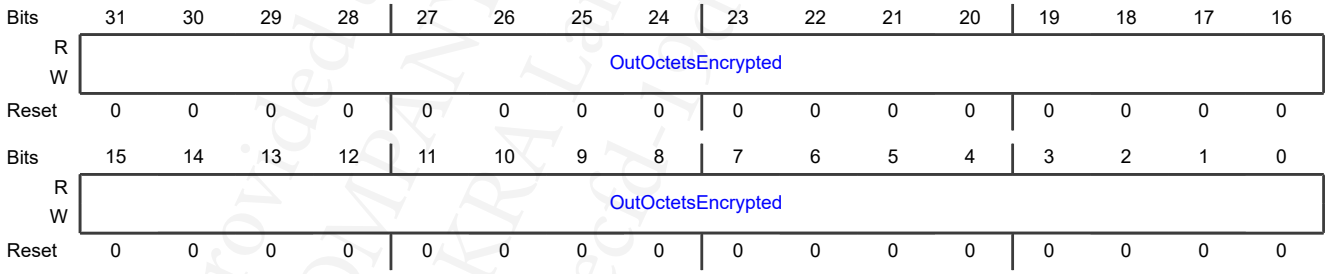
2.68.1 Offset

Register	Offset
MACSEC_OOE2HS	24Ch

2.68.2 Function

The MACSEC_OOE2HS register provides the count of Controlled Port Output Encrypted Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

2.68.3 Diagram



2.68.4 Fields

Field	Function
31-0 OutOctets Encrypted	OutOctetsEncrypted The number of octets of User Data in transmitted frames that were both integrity protected and encrypted 32..63. (802.1AE-2006/802.1AEbn-2011 10.7.19)

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2.69 TX Security Association A configuration and status (MACSEC_TXSAACS)

2.69.1 Offset

Register	Offset
MACSEC_TXSAACS	280h

2.69.2 Function

The MACSEC_TXSAACS register is used to configure the Security Association A for the Secure Channel.

2.69.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	A	0														
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

2.69.4 Fields

Field	Function
31 A	A Active (RO/w1c) Must be cleared when the SA is initialized Will be set to 1 once it has been used at least once This bit is also used to track if the key hash has been generated for the SAK. This why it must be cleared every time the SAK is modified.
30-0 —	Reserved, should be cleared

2.70 TX Security Association A nextPN (MACSEC_TXSAANPN)

2.70.1 Offset

Register	Offset
MACSEC_TXSAANPN	284h

2.70.2 Function

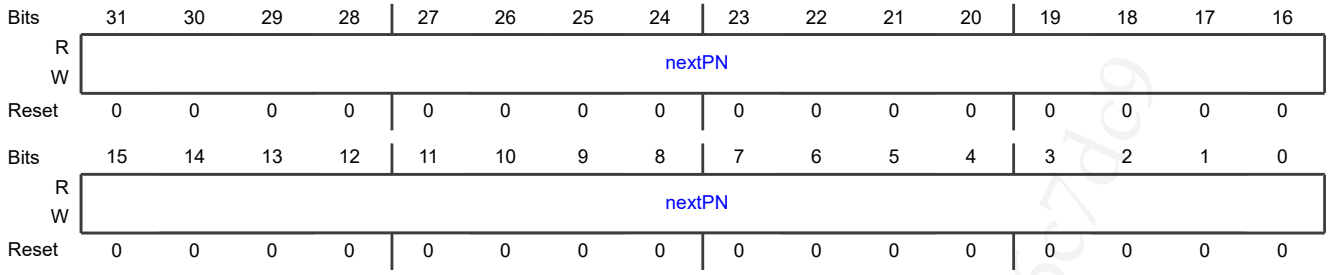
The MACSEC_TXSAANPN register provide the ability to read and configure the Next Packet Number to be used on the TX MACsec for the Security Association A of the Secure Channel. Must be initialized to non zero (default is 0x1) before the TX SC is enabled. A value of zero will cause frames to be marked as abort for MAC to invalidate the FCS and an error interrupt Section 2.91 is asserted if protectFrame control is set. This PN will be incremented by one on every frame transmission for the SA, except for the cases where the frame is marked as abort and counted in Section 2.114.

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For XPN-capable cipher suite, nextPN is the bit 31-0 of nextXPN.

2.70.3 Diagram



2.70.4 Fields

Field	Function
31-0 nextPN	nextPN TX Security Association A nextPN to be used for frame generation (802.1AE-2006/802.1AEbn-2011 10.6.5)

2.71 TX Security Association Extended A nextPN (MACSEC_TXSAAXNPN)

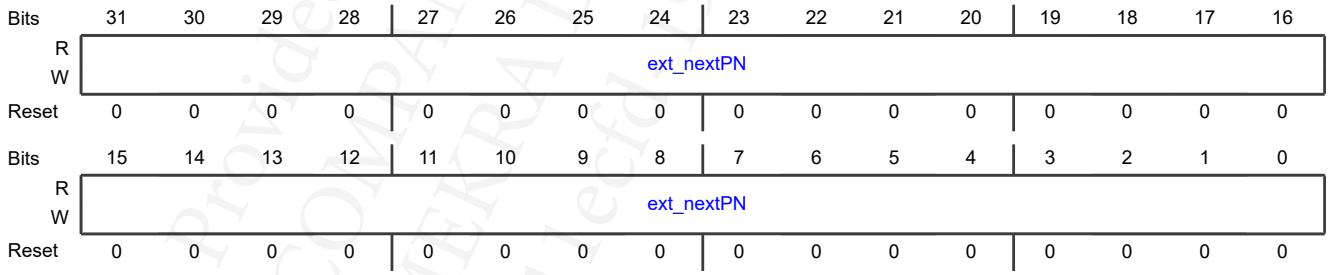
2.71.1 Offset

Register	Offset
MACSEC_TXSAAXNPN	288h

2.71.2 Function

The MACSEC_TXSAAXNPN register provide the ability to read and configure the bit 63-32 of Next Extended Packet Number accepted on TX when XPN capable cipher suite is configured.

2.71.3 Diagram



2.71.4 Fields

Field	Function
31-0 ext_nextPN	ext_nextPN TX Security Association A extended nextPN (802.1AEbw-2013) Extended nextPN is used to support XPN capable cipher suite. extended_nextPN is the bit 63-32 of next XPN.It should be initialized to zero.

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Field	Function
	Management Software can change this value any time. It is not used when non-XPN capable cipher suite is configured for this secure channel.

2.72 TX Security Association A OutPktsProtected Statistic (MACSEC_TXSAAOPPS)

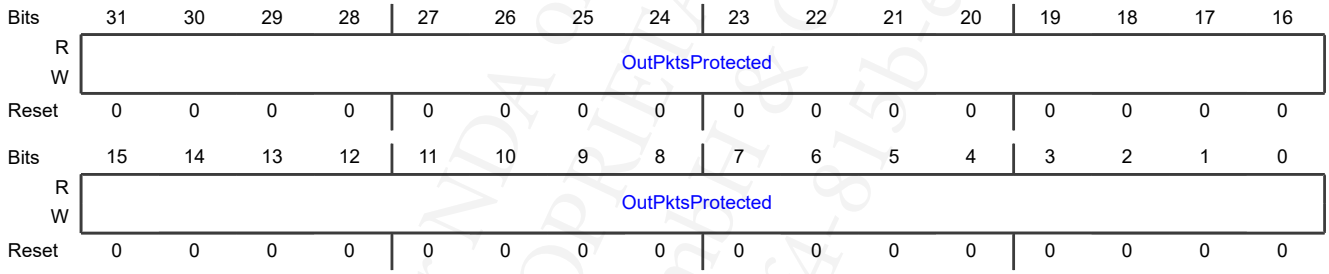
2.72.1 Offset

Register	Offset
MACSEC_TXSAAOPPS	28Ch

2.72.2 Function

The MACSEC_TXSAAOPPS register provides the count of Output Frames Protected for Security Association A of Secure Channel.

2.72.3 Diagram



2.72.4 Fields

Field	Function
31-0 OutPkts Protected	OutPktsProtected The number of frames, that the user of the controlled port requested to be transmitted, which were integrity protected. (802.1AE-2006/802.1AEbn-2011 10.7.18)

2.73 TX Security Association A OutPktsEncrypted Statistic (MACSEC_TXSAAOPES)

2.73.1 Offset

Register	Offset
MACSEC_TXSAAOPES	290h

2.73.2 Function

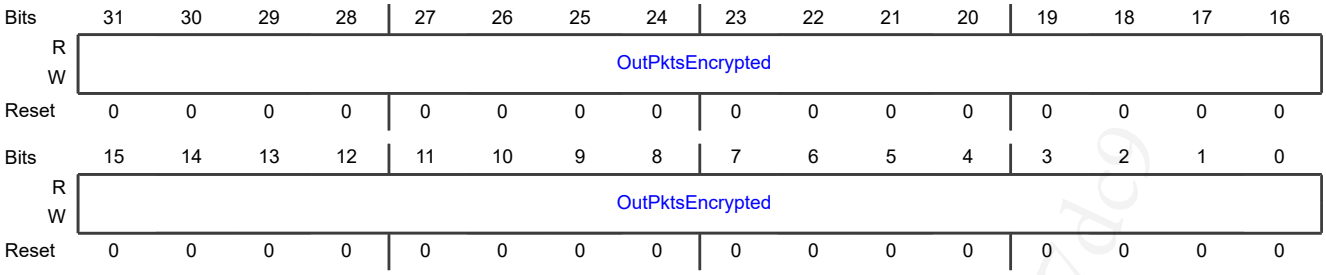
The MACSEC_TXSAAOPES register provides the count of Output Frames Encrypted for Security Association A of Secure Channel.

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2.73.3 Diagram



2.73.4 Fields

Field	Function
31-0 OutPkts Encrypted	OutPktsEncrypted The number of frames, that the user of the controlled port requested to be transmitted, which were confidentiality protected. (802.1AE-2006/802.1AEbn-2011 10.7.18)

2.74 TX Security Association A hash 1 of 4 (MACSEC_TXSAHA1Q)

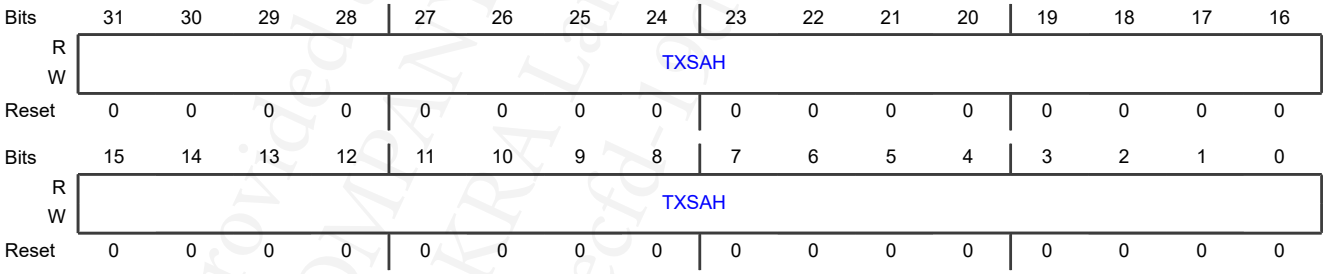
2.74.1 Offset

Register	Offset
MACSEC_TXSAHA1Q	2A0h

2.74.2 Function

The MACSEC_TXSAHA1Q register is used to access the hardware generated Security Association A hash for the Secure Channel.

2.74.3 Diagram



2.74.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 0-31

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2.75 TX Security Association A hash 2 of 4 (MACSEC_TXSAHA2Q)

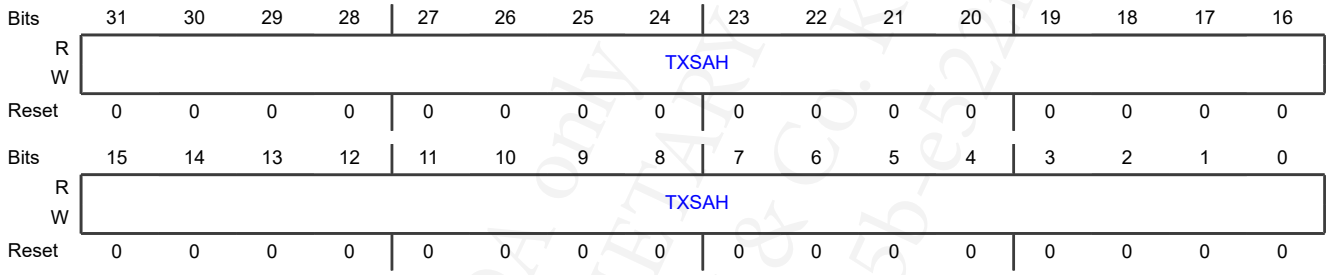
2.75.1 Offset

Register	Offset
MACSEC_TXSAHA2Q	2A4h

2.75.2 Function

The MACSEC_TXSAHA2Q register is used to access the hardware generated Security Association A hash for the Secure Channel.

2.75.3 Diagram



2.75.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 32-63

2.76 TX Security Association A hash 3 of 4 (MACSEC_TXSAHA3Q)

2.76.1 Offset

Register	Offset
MACSEC_TXSAHA3Q	2A8h

2.76.2 Function

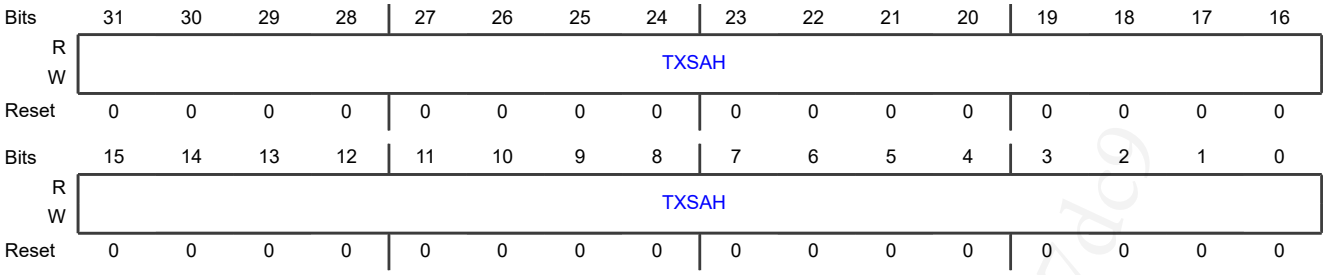
The MACSEC_TXSAHA3Q register is used to access the hardware generated Security Association A key for the Secure Channel.

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2.76.3 Diagram



2.76.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 64-95

2.77 TX Security Association A hash 4 of 4 (MACSEC_TXSAHA4Q)

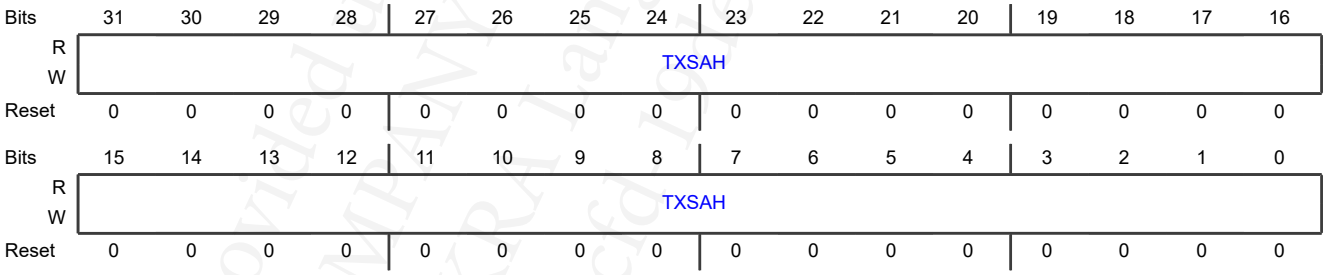
2.77.1 Offset

Register	Offset
MACSEC_TXSAHA4Q	2ACh

2.77.2 Function

The MACSEC_TXSAHA4Q register is used to access the hardware generated Security Association A hash for the Secure Channel.

2.77.3 Diagram



2.77.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 96-127

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2.78 TX Security Association B configuration and status (MACSEC_TXSABCS)

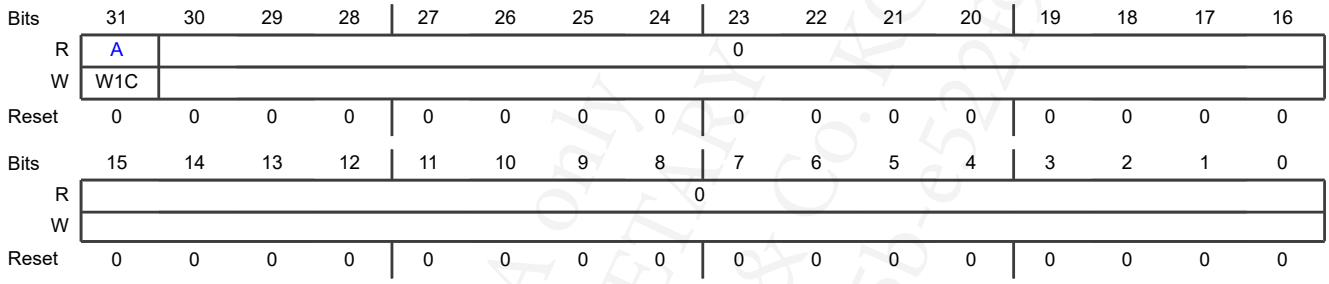
2.78.1 Offset

Register	Offset
MACSEC_TXSABCS	2C0h

2.78.2 Function

The MACSEC_TXSABCS register is used to configure the Security Association B for the Secure Channel.

2.78.3 Diagram



2.78.4 Fields

Field	Function
31 A	A Active (RO/w1c) Must be cleared when the SA is initialized Will be set to 1 once it has been used at least once
30-0 —	Reserved, should be cleared

2.79 TX Security Association B nextPN (MACSEC_TXSABNPN)

2.79.1 Offset

Register	Offset
MACSEC_TXSABNPN	2C4h

2.79.2 Function

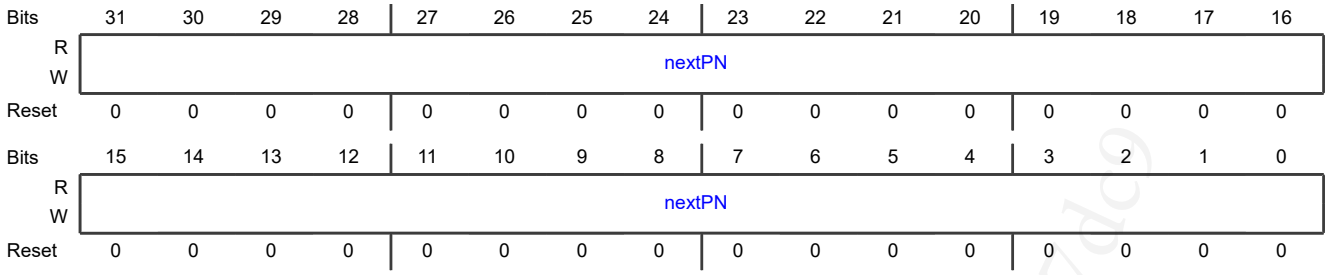
The MACSEC_TXSABNPN register provide the ability to read and configure the Next Packet Number to be used on the TX MACsec for Security Association B of Secure Channel. Must be initialized to non zero (default is 0x1) before the TX SC is enabled. A value of zero will cause frames to be marked as abort for the MAC to invalidate the FCS and an error interrupt MACsec Error Register (MACSEC_ERR) bit x is asserted if protectFrame control is set. This PN will be incremented by one on every frame transmission for the SA, except for the cases where the frame is marked as abort and counted in Section 2.114.

For XPN-capable cipher suite, nextPN is the bit 31-0 of nextXPN.

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2.79.3 Diagram



2.79.4 Fields

Field	Function
31-0 nextPN	nextPN TX Security Association A nextPN to be used for frame generation (802.1AE-2006/802.1AEbn-2011 10.6.5)

2.80 TX Security Association Extended B nextPN (MACSEC_TXSABXNPN)

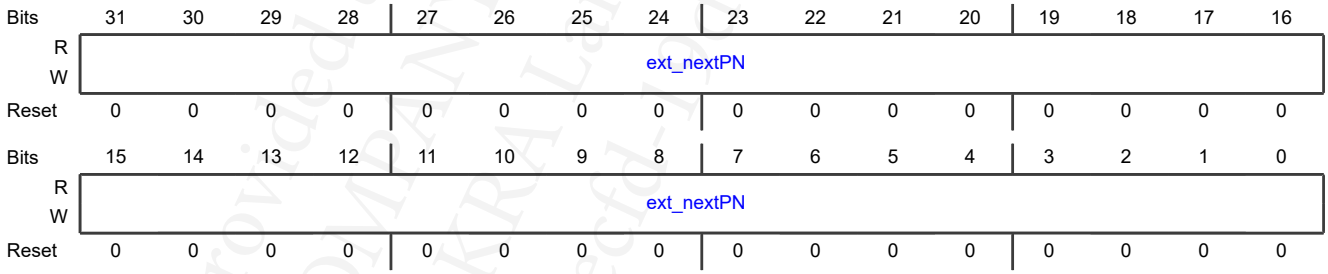
2.80.1 Offset

Register	Offset
MACSEC_TXSABXNPN	2C8h

2.80.2 Function

The MACSEC_TXSABXNPN register provide the ability to read and configure the bit 63-32 of Next Extended Packet Number accepted on TX when XPN capable cipher suite is configured.

2.80.3 Diagram



2.80.4 Fields

Field	Function
31-0 ext_nextPN	ext_nextPN TX Security Association B extended nextPN (802.1AEbw-2013) Extended nextPN is used to support XPN capable cipher suite. extended_nextPN is the bit 63-32 of next XPN.It should be initialized to zero. Management Software can change this value any time. It is not used when non-XPN capable cipher suite is configured for this secure channel.

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2.81 TX Security Association B OutPktsProtected Statistic (MACSEC_TXSABOPPS)

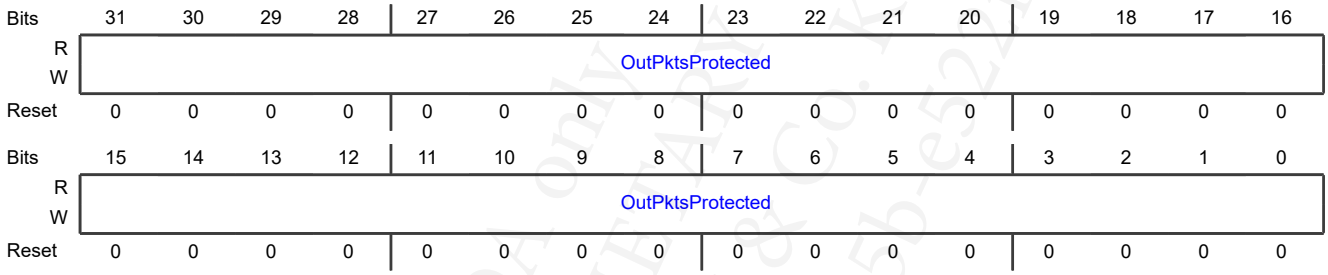
2.81.1 Offset

Register	Offset
MACSEC_TXSABOPPS	2CCh

2.81.2 Function

The MACSEC_TXSABOPPS register provides the count of Output Frames Protected for Security Association B of Secure Channel.

2.81.3 Diagram



2.81.4 Fields

Field	Function
31-0 OutPkts Protected	OutPktsProtected The number of frames, that the user of the controlled port requested to be transmitted, which were integrity protected. (802.1AE-2006/802.1AEbn-2011 10.7.18)

2.82 TX Security Association B OutPktsEncrypted Statistic (MACSEC_TXSABOPES)

2.82.1 Offset

Register	Offset
MACSEC_TXSABOPES	2D0h

2.82.2 Function

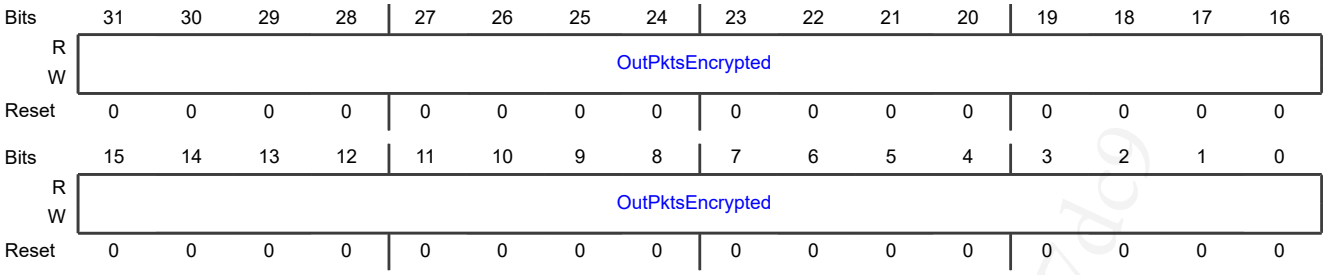
The MACSEC_TXSABOPES register provides the count of Output Frames Encrypted for Security Association B of Secure Channel.

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2.82.3 Diagram



2.82.4 Fields

Field	Function
31-0 OutPkts Encrypted	OutPktsEncrypted The number of frames, that the user of the controlled port requested to be transmitted, which were confidentiality protected. (802.1AE-2006/802.1AEbn-2011 10.7.18)

2.83 TX Security Association B hash 1 of 4 (MACSEC_TXSAHB1Q)

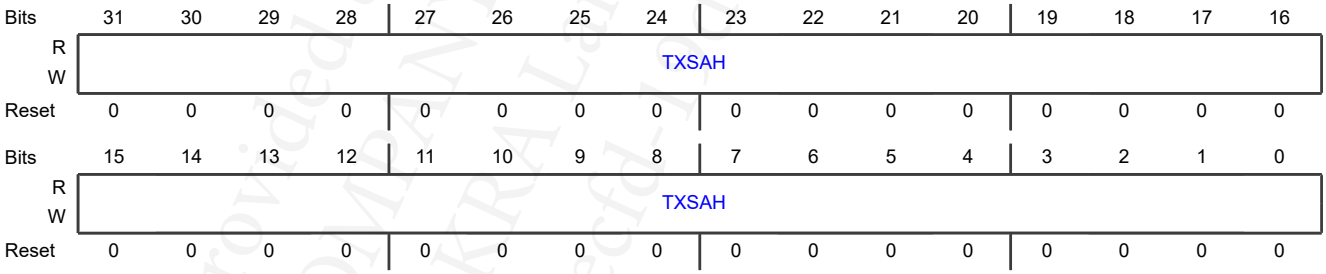
2.83.1 Offset

Register	Offset
MACSEC_TXSAHB1Q	2E0h

2.83.2 Function

The MACSEC_TXSAHB1Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

2.83.3 Diagram



2.83.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 0-31

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2.84 TX Security Association B hash 2 of 4 (MACSEC_TXSAHB2Q)

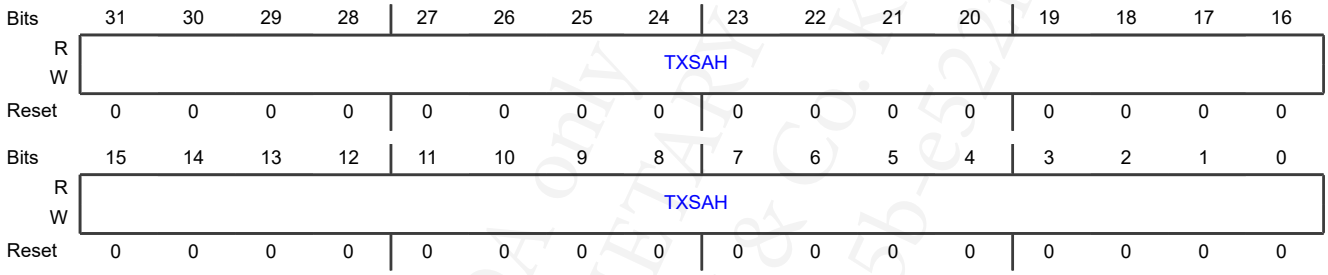
2.84.1 Offset

Register	Offset
MACSEC_TXSAHB2Q	2E4h

2.84.2 Function

The MACSEC_TXSAHB2Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

2.84.3 Diagram



2.84.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 32-63

2.85 TX Security Association B hash 3 of 4 (MACSEC_TXSAHB3Q)

2.85.1 Offset

Register	Offset
MACSEC_TXSAHB3Q	2E8h

2.85.2 Function

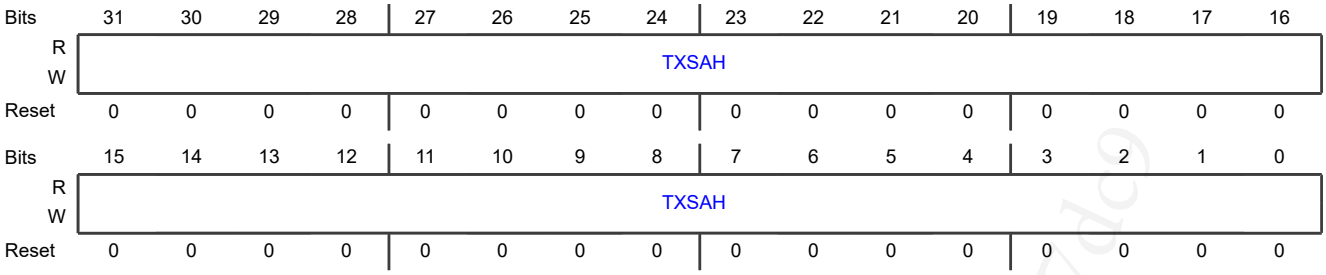
The MACSEC_TXSAHB3Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

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2.85.3 Diagram



2.85.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 64-95

2.86 TX Security Association B hash 4 of 4 (MACSEC_TXSAHB4Q)

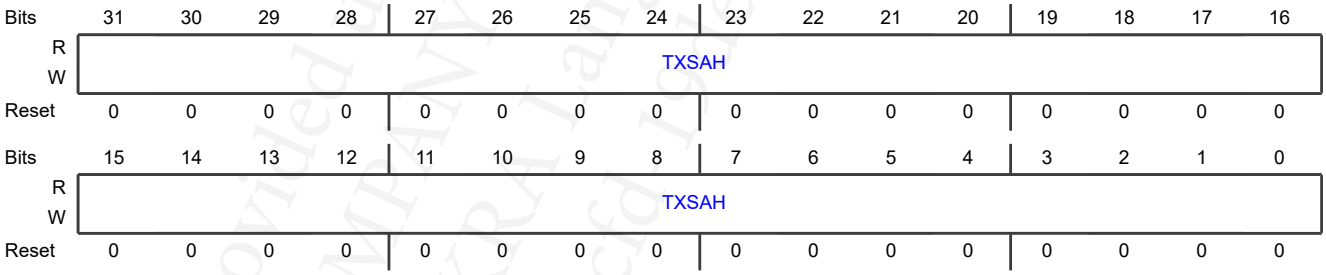
2.86.1 Offset

Register	Offset
MACSEC_TXSAHB4Q	2ECh

2.86.2 Function

The MACSEC_TXSAHB4Q register is used to access the hardware generated Security Association B hash for the Secure Channel.

2.86.3 Diagram



2.86.4 Fields

Field	Function
31-0 TXSAH	TXSAH TX Security Association hash bit 96-127

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2.87 MACsec IP Block Revision 1 register (MACSEC_IP_REV_1)

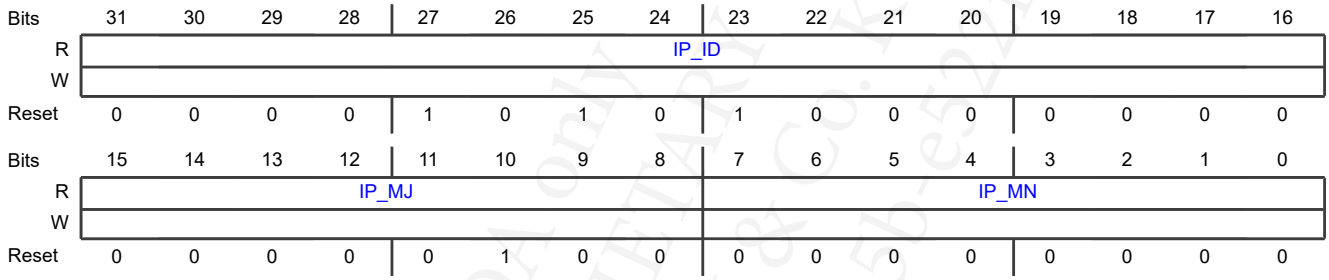
2.87.1 Offset

Register	Offset
MACSEC_IP_REV_1	3F8h

2.87.2 Function

The MACSEC_IP_REV_1 register is a read-only register which provides the unique IP block ID for the MACsec block, as well as major and minor revision numbers.

2.87.3 Diagram



2.87.4 Fields

Field	Function
31-16 IP_ID	IP_ID IP Block ID. For the MACsec, this value is 0x0A80.
15-8 IP_MJ	IP_MJ Major revision. Set to 0x04
7-0 IP_MN	IP_MN Minor revision. This is currently set to 0x00.

2.88 MACsec IP Block Revision 2 register (MACSEC_IP_REV_2)

2.88.1 Offset

Register	Offset
MACSEC_IP_REV_2	3FCh

2.88.2 Function

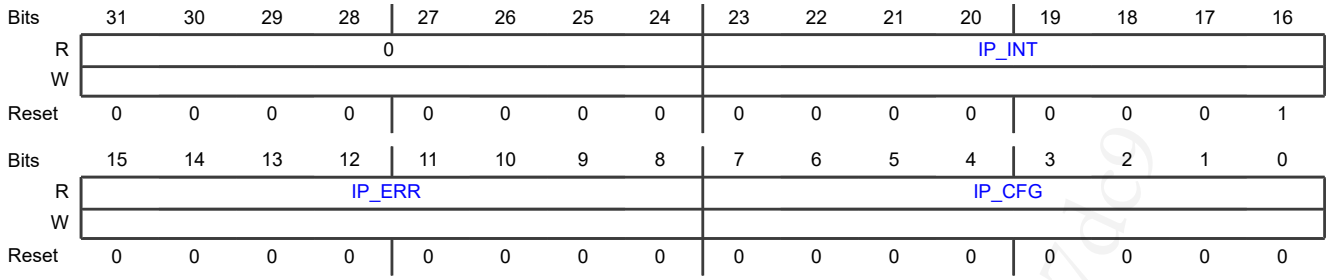
The MACSEC_IP_REV_2 register is a read-only register which provides MACsec block integration and configuration options.

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2.88.3 Diagram



2.88.4 Fields

Field	Function
31-24 —	Reserved, read as 0.
23-16 IP_INT	IP_INT Integration options. 0000_0000b - when TX uncontrolled port and SC selection is done out of band 0000_0001b - when TX uncontrolled port and SC selection is done with lookup table 0000_0010b - TX uncontrolled port and SC selection is done based on MACSEC_CFG[EXTSCS].
15-8 IP_ERR	IP_ERR Errata revision level. Set to 0x00
7-0 IP_CFG	IP_CFG Configuration options. This is currently set to 0x00.

2.89 MACsec Event Register (MACSEC_EVR)

2.89.1 Offset

Register	Offset
MACSEC_EVR	400h

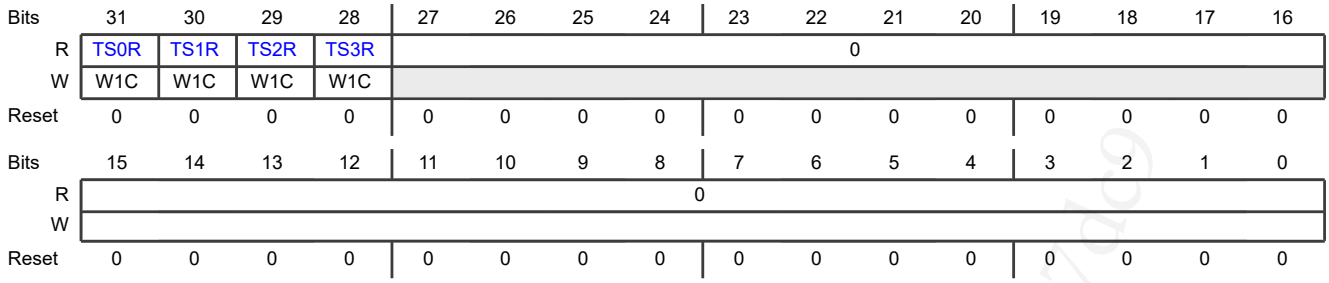
2.89.2 Function

The MACsec contains one dedicated interrupt signal for signaling event conditions to software. The MACSEC_EVR register identifies the source of the interrupt. If an event or condition occurs that sets a bit in the MACSEC_EVR register the MACsec interrupt is generated if the corresponding bit in the MACSEC_EVER register is also set. Once a bit in the MACSEC_EVR register is asserted, it will remain latched asserted until the original event is cleared and the bit in the MACSEC_EVR register is cleared. The bit in the MAC_EVR register is cleared by writing a 1 to that bit position. Clearing the MACSEC_EVR bit clears the MACsec interrupt indication as long as the original event is no longer present (the interrupt source has been cleared) and no other MACSEC_EVR/MACSEC_EVER bit pairs are set. A write of 0 has no effect.

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2.89.3 Diagram



2.89.4 Fields

Field	Function
31 TS0R	TS0R TX SC 0 next_PN exhaustion threshold reached Interrupt Indication that the PN (see Section 2.70) has reached the exhaustion threshold for the current SA of the SC
30 TS1R	TS1R TX SC 1 next_PN exhaustion threshold reached Interrupt Indication that the PN (see Section 2.70) has reached the exhaustion threshold for the current SA of the SC
29 TS2R	TS2R TX SC 2 next_PN exhaustion threshold reached Interrupt Indication that the PN (see Section 2.70) has reached the exhaustion threshold for the current SA of the SC
28 TS3R	TS3R TX SC 3 next_PN exhaustion threshold reached Interrupt Indication that the PN (see Section 2.70) has reached the exhaustion threshold for the current SA of the SC
27-0 —	Reserved, should be cleared

2.90 MACsec Event Enable Register (MACSEC_EVER)

2.90.1 Offset

Register	Offset
MACSEC_EVER	404h

2.90.2 Function

The MACSEC_EVER register provides control over which possible interrupt events are allowed to generate the actual MACsec interrupt. If the corresponding bits in both MACSEC_EVR and MACSEC_EVER registers are set, then the MACsec interrupt is generated. The state of an MACSEC_EVER bit does not affect setting of the corresponding MACSEC_EVR bit, it controls whether or not setting of the MACSEC_EVR bit will cause an interrupt. Setting a MACSEC_EVER bit to 0 will cause the interrupt indication to be cleared if no other MACSEC_EVR/MACSEC_EVER bit pairs are set.

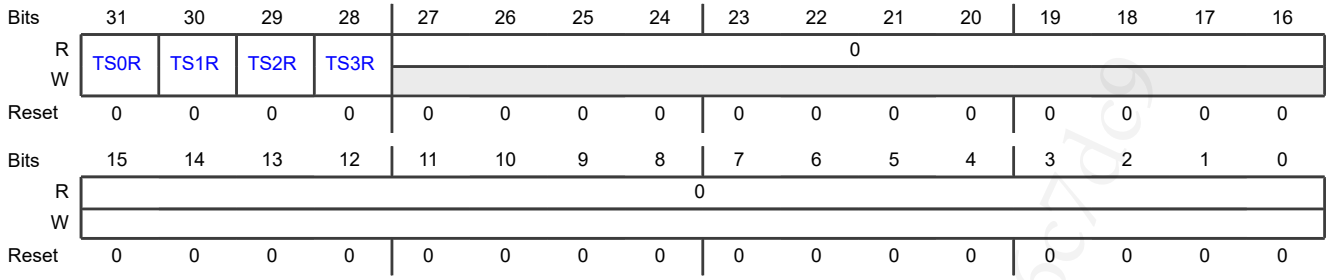
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All implemented bits in this register are R/W. This register is cleared upon hard reset

2.90.3 Diagram



2.90.4 Fields

Field	Function
31 TS0R	TS0R TX SC 0 next_PN exhaustion threshold reached Interrupt Enable. 0 Disable interrupt 1 Enable interrupt
30 TS1R	TS1R TX SC 1 next_PN exhaustion threshold reached Interrupt Enable. 0 Disable interrupt 1 Enable interrupt
29 TS2R	TS2R TX SC 2 next_PN exhaustion threshold reached Interrupt Enable. 0 Disable interrupt 1 Enable interrupt
28 TS3R	TS3R TX SC 3 next_PN exhaustion threshold reached Interrupt Enable. 0 Disable interrupt 1 Enable interrupt
27-0 —	Reserved, should be cleared

2.91 MACsec Error Register (MACSEC_ERR)

2.91.1 Offset

Register	Offset
MACSEC_ERR	40Ch

2.91.2 Function

The MACsec contains one dedicated interrupt indication (that is, MACsec error interrupt) for signaling error conditions to software. The MACSEC_ERR register identifies the source of the interrupt. If an error occurs that sets a bit in the MACSEC_ERR register the MACsec error interrupt is generated if the corresponding bit in the MACSEC_ERER register is also set. Once a bit in the MACSEC_ERR register is asserted, it will remain

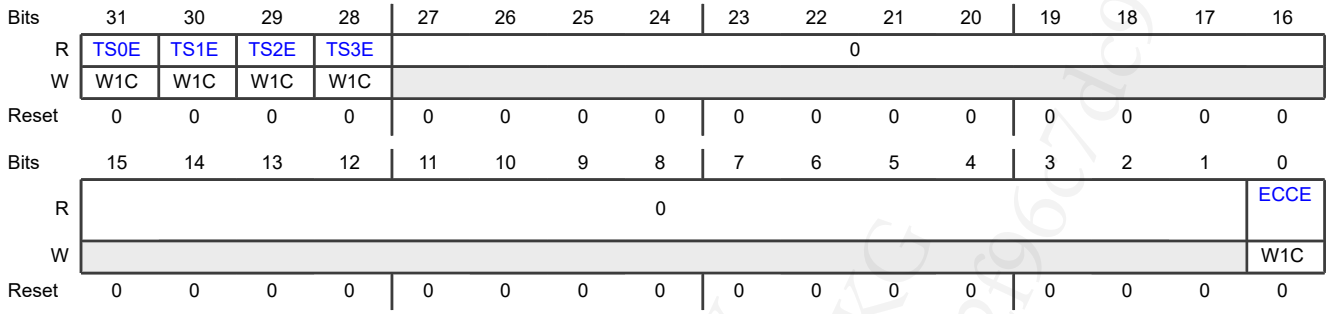
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asserted until the bit in the MACSEC_ERR register is cleared. The bit in the MACSEC_ERR register is cleared by writing a 1 to that bit position. Clearing the MACSEC_ERR bit clears the MACsec error interrupt indication as long as no other MACSEC_ERR/MACSEC_ERER bit pairs are set. A write of 0 has no effect.

2.91.3 Diagram



2.91.4 Fields

Field	Function
31 TS0E	TS0E TX SC 0 error, frame error Interrupt An error occurred on TX SC that cause a frame to be marked as aborted due to: NextPN == 0, SC is disabled Section 2.63 bit SCE is zero, alwaysIncludeSCI = 0, useES = 0, useSCB = 1 unsupported configuration, or a multiple bit ECC error occurred while reading context for the frame. The frame is masked with zeros for security reason and marked as aborted for the MAC to invalidate the FCS. If the frame is shorter than 60 bytes plus 4 bytes FCS if present, the FCS is stripped, the frame is padded with 4 bytes of zeros until the frame length is greater or equal to 60 bytes.
30 TS1E	TS1E TX SC 1 error, frame error Interrupt
29 TS2E	TS2E TX SC 2 error, frame error Interrupt
28 TS3E	TS3E TX SC 3 error, frame error Interrupt
27-1 —	Reserved, should be cleared
0 ECCE	ECCE MACsec memory ECC multiple-bit error Interrupt

2.92 MACsec Error Enable Register (MACSEC_ERER)

2.92.1 Offset

Register	Offset
MACSEC_ERER	410h

2.92.2 Function

The MACSEC_ERER register provides control over which possible interrupt errors are allowed to generate the actual MACsec error interrupt. If the corresponding bits in both MACSEC_ERR and MACSEC_ERER registers

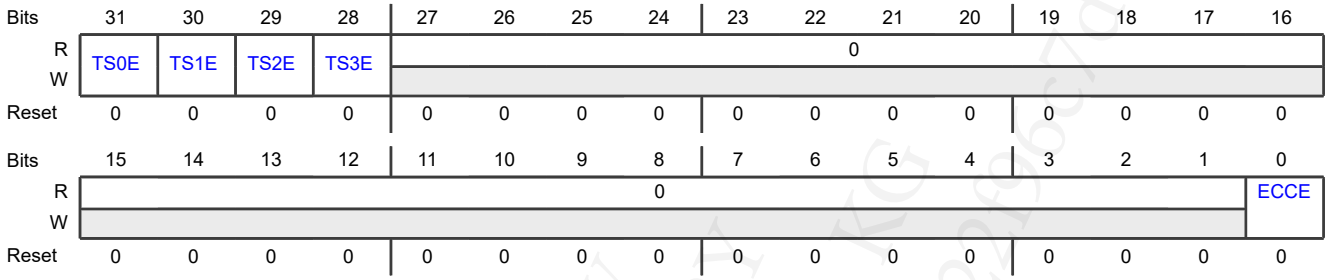
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are set, then the MACsec error interrupt is generated. The state of an MACSEC_ERER bit does not affect setting of the corresponding MACSEC_ERR bit, it controls whether or not setting of the MACSEC_ERR bit will cause an interrupt. Setting a MACSEC_ERER bit to 0 will cause the interrupt indication to be cleared if no other MACSEC_ERR/MACSEC_ERER bit pairs are set.

All implemented bits in this register are R/W. This register is cleared upon hard reset.

2.92.3 Diagram



2.92.4 Fields

Field	Function
31 TS0E	TS0E TX SC 0 error, frame error Interrupt Enable. 0b - Disable interrupt 1b - Enable interrupt
30 TS1E	TS1E TX SC 1 error, frame error Interrupt Enable. 0b - Disable interrupt 1b - Enable interrupt
29 TS2E	TS2E TX SC 2 error, frame error Interrupt Enable. 0b - Disable interrupt 1b - Enable interrupt
28 TS3E	TS3E TX SC 3 error, frame error Interrupt Enable. 0b - Disable interrupt 1b - Enable interrupt
27-1 —	Reserved, should be cleared
0 ECCE	ECCE MACsec memory ECC multiple-bit error Interrupt Enable. 0b - Disable interrupt 1b - Enable interrupt

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2.93 MACsec Memory ECC Error Capture Register (MACSEC_MEEC)

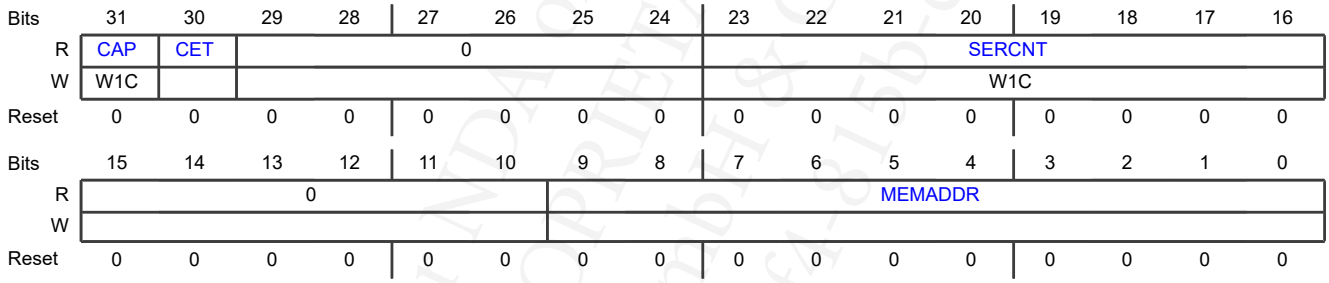
2.93.1 Offset

Register	Offset
MACSEC_MEEC	440h

2.93.2 Function

The MACSEC_MEEC register count single-bit ECC errors on the MACsec internal memory, and capture the address in the MACsec internal memory at which the ECC error (single-bit or multi-bit) occurred. The first ECC error (single-bit or multi-bit) that occurs will lock the captured memory address of the error from any additional single-bit ECC errors. The first multi-bit ECC error that occurs will over-write any captured memory address from a single-bit ECC error and will lock the captured memory address from any additional ECC errors (single-bit or multi-bit). The captured lock can be cleared to enable the capturing of memory address from additional single-bit or multi-bit ECC errors.

2.93.3 Diagram



2.93.4 Fields

Field	Function
31 CAP	CAP Captured Error Indication Any single-bit ECC errors will set the CAP bit, and the erred memory location will be captured in the MEMADDR field and the MEMADDR will be locked from capturing a new single-bit ECC error. For multi-bit ECC errors, the CAP bit will be set, the MEMADDR field will hold the memory address of the erred location and MEMADDR will be locked from re-capturing any other ECC errors (single-bit or multi-bit). 0b - No error captured in MEMADDR field. 1b - ECC error is captured.
30 CET	CET Captured Error Type 0b - Single-Bit ECC Error is captured if CAP=1. No ECC error captured if CAP=0. 1b - Multi-Bit ECC error is captured. CET is never asserted if CAP=0.
29-24 —	Reserved, read as 0.
23-16 SERCNT	SERCNT Soft Error Counter. This counter accumulates single-bit ECC errors that have been detected and corrected by the ECC logic. Multi-bit ECC errors are not counted and will directly set the MACSEC_ERR[ECCE] error status bit. If MACSEC_ERER[ECCE]=1, then a multi-bit ECC error will also assert the MACsec error interrupt.

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Field	Function
	This field is cleared on bitwise w1c. When the count reaches its maximum value (0xFF), it will stick at that value without rollover until this register field is cleared.
15-10 —	Reserved, read as 0.
9-0 MEMADDR	MEMADDR Captured memory address in the MACsec internal memory that caused the ECC error. Valid when CAP is set. Cleared when CAP is written with a one (that is, clears the captured lock).

2.94 MACsec Idle status Register (MACSEC_STATUS)

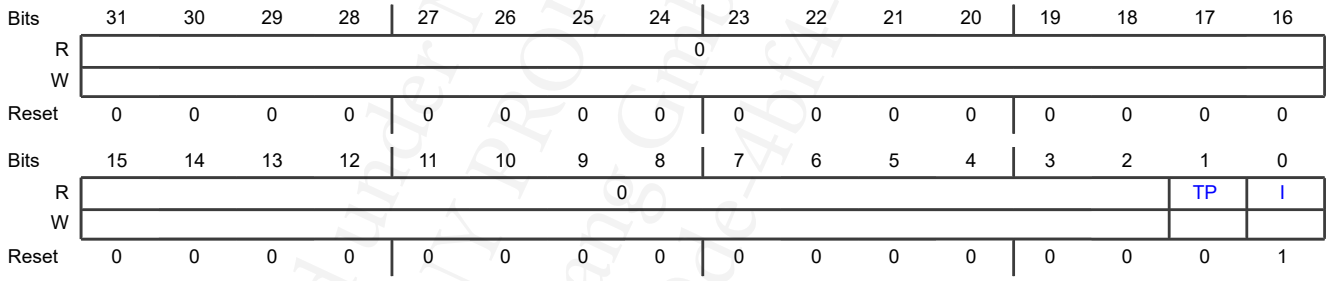
2.94.1 Offset

Register	Offset
MACSEC_STATUS	444h

2.94.2 Function

The MACSEC_STATUS register can be used by software to determine MACSec status include idle and tamper protection, information which may be used to control power management functions and behaviors under secure incidents in the SOC.

2.94.3 Diagram



2.94.4 Fields

Field	Function
31-2 —	Reserved, read as 0.
1 TP	I Tamper Protection This bit is asserted when MACsec received tamper protection signal is 1. 0: MACsec operates as the same as normal. 1: controlled frames are set to discard, uncontrolled frames have no influences.
0 I	I Idle This bit is asserted when MACsec is idle. MACsec is considered idle when the following conditions are met: No frames are being processed on both the RX and TX directions that is, all FIFOs and FSM are empty.

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Field	Function
	No interrupts are pending

2.95 ifInOctetsCp first half Statistic (MACSEC_IFIOCP1HS)

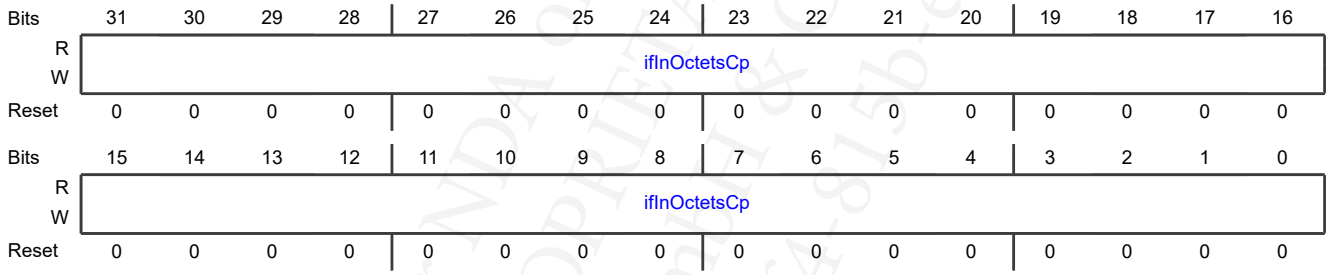
2.95.1 Offset

Register	Offset
MACSEC_IFIOCP1HS	600h

2.95.2 Function

The MACSEC_IFIOCP1HS register provides the global count of the Controlled Port Interface Input Octets for frames with unresolved SCI. The counter is actually 64 bits, this register represents the MSB portion of the counter.

2.95.3 Diagram



2.95.4 Fields

Field	Function
31-0 ifInOctetsCp	ifInOctetsCp Controlled Port Interface Input Octets 0..31 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifInOctetsCp count is the sum of all the octets of the MSDUs delivered to the user of the Controlled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses for frames with unresolved SCI.

2.96 ifInOctetsCp second half Statistic (MACSEC_IFIOCP2HS)

2.96.1 Offset

Register	Offset
MACSEC_IFIOCP2HS	604h

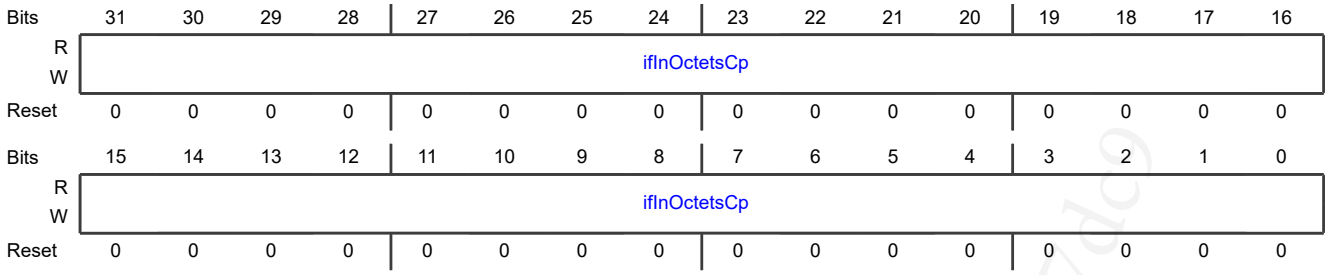
2.96.2 Function

The MACSEC_IFIOCP2HS register provides the count of the Controlled Port Interface Input Octets for frames with unresolved SCI. The counter is actually 64 bits, this register represents the LSB portion of the counter.

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2.96.3 Diagram



2.96.4 Fields

Field	Function
31-0 ifInOctetsCp	ifInOctetsCp Controlled Port Interface Input Octets 32..63 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifInOctetsCp count is the sum of all the octets of the MSDUs delivered to the user of the Controlled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses for frames with unresolved SCI.

2.97 ifInUcastPktsCp Statistic (MACSEC_IFIUPCPS)

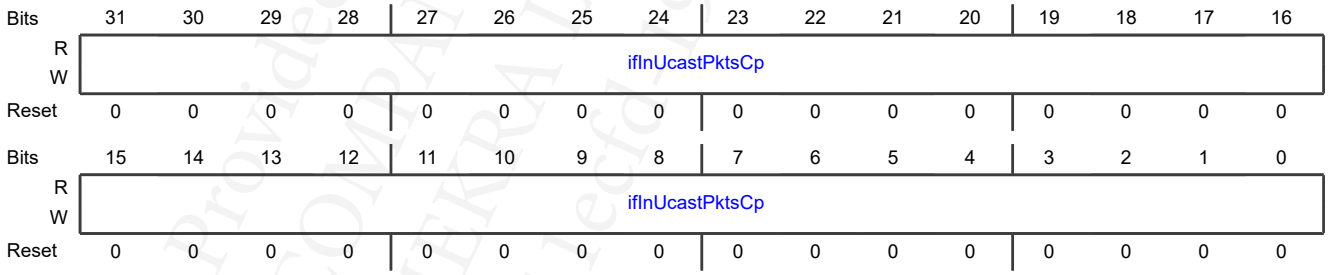
2.97.1 Offset

Register	Offset
MACSEC_IFIUPCPS	608h

2.97.2 Function

The register MACSEC_IFIUPCPS provides the count of the Controlled Port Interface Input Unicast Frames with unresolved SCI.

2.97.3 Diagram



2.97.4 Fields

Field	Function
31-0 ifInUcastPktsCp	ifInUcastPktsCp The number of frames with unresolved SCI, delivered to the user of a controlled port, which were not addressed to a multicast or broadcast address (802.1AE-2006/802.1AEbn-2011 10.7.6).

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2.98 ifInOctetsUp first half Statistic (MACSEC_IFIOUP1HS)

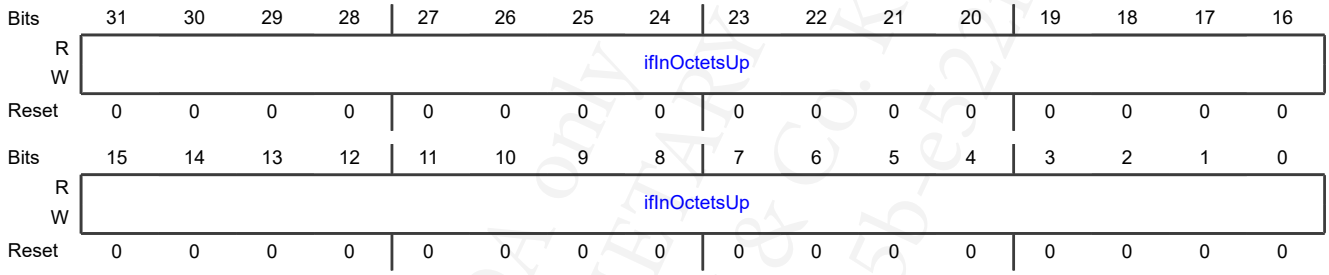
2.98.1 Offset

Register	Offset
MACSEC_IFIOUP1HS	610h

2.98.2 Function

The MACSEC_IFIOUP1HS register provides the count of the Uncontrolled Port Interface Input Octets. The counter is actually 64 bits, this register represents the MSB portion of the counter.

2.98.3 Diagram



2.98.4 Fields

Field	Function
31-0 ifInOctetsUp	ifInOctetsUp Uncontrolled Port Interface Input Octets 0..31 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifInOctetsUp count is the sum of all the octets of the MSDUs delivered to the user of the Uncontrolled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses for frames with unresolved SCI or filtered to the uncontrolled port.

2.99 ifInOctetsUp second half Statistic (MACSEC_IFIOUP2HS)

2.99.1 Offset

Register	Offset
MACSEC_IFIOUP2HS	614h

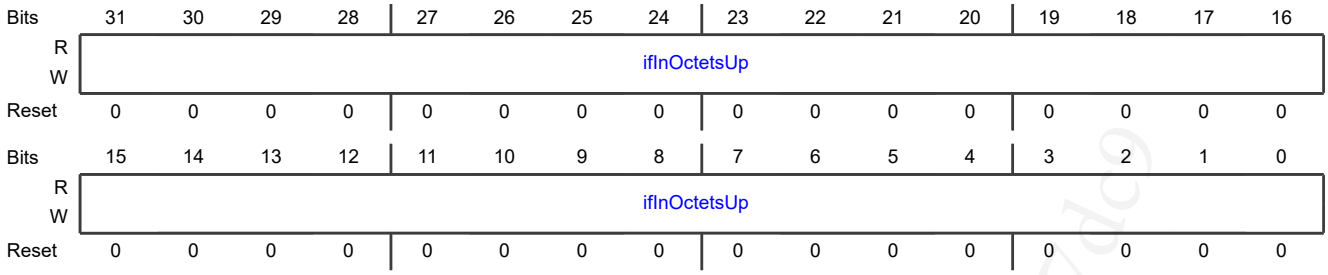
2.99.2 Function

The MACSEC_IFIOUP2HS register provides the count of the Uncontrolled Port Interface Input Octets. The counter is actually 64 bits, this register represents the LSB portion of the counter.

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2.99.3 Diagram



2.99.4 Fields

Field	Function
31-0 ifInOctetsUp	ifInOctetsUp Uncontrolled Port Interface Input Octets 32..63 (802.1AE-2006/802.1AEbn-2011 10.7.6) The ifInOctetsUp count is the sum of all the octets of the MSDUs delivered to the user of the Uncontrolled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses for frames with unresolved SCI or filtered to the uncontrolled port.

2.100 ifInUcastPktsUp Statistic (MACSEC_IFIUPUPS)

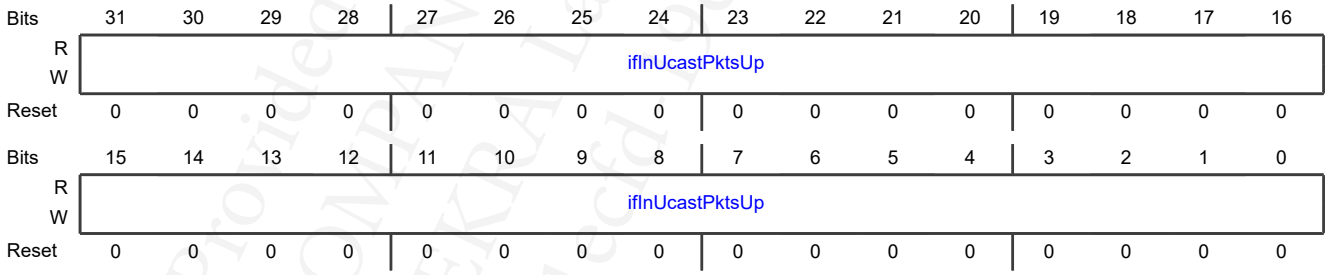
2.100.1 Offset

Register	Offset
MACSEC_IFIUPUPS	618h

2.100.2 Function

The register MACSEC_IFIUPUPS provides the count of the Uncontrolled Port Interface Input Unicast Frames.

2.100.3 Diagram



2.100.4 Fields

Field	Function
31-0 ifInUcastPktsUp	ifInUcastPktsUp The number of frames with unresolved SCI or filtered to the uncontrolled port , delivered to the user of a Uncontrolled port, which were not addressed to a multicast or broadcast address (802.1AE-2006/802.1 AEbn-2011 10.7.6).

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2.101 ifInMulticastPktsCp Statistic (MACSEC_IFIMPCPS)

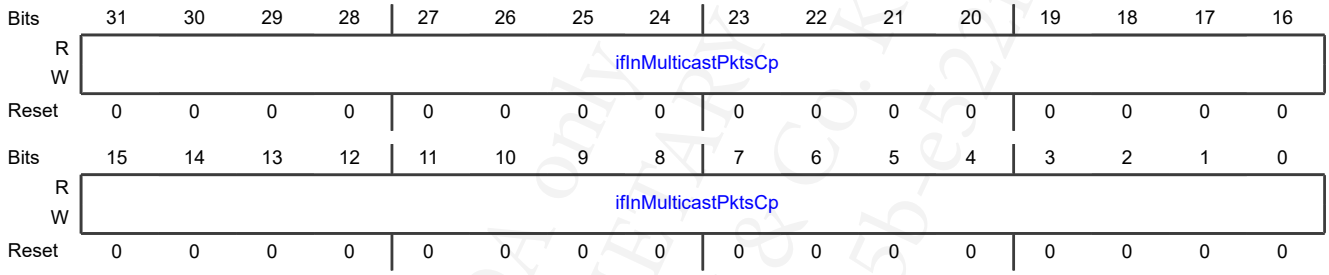
2.101.1 Offset

Register	Offset
MACSEC_IFIMPCPS	620h

2.101.2 Function

The MACSEC_IFIMPCPS register provides the count of the Controlled Port interface Input Multicast Frames with unresolved SCI.

2.101.3 Diagram



2.101.4 Fields

Field	Function
31-0 ifInMulticast PktsCp	ifInMulticastPktsCp The number of frames with unresolved SCI, delivered to the user of a controlled port, which are addressed to a multicast address (802.1AE-2006/802.1AEbn-2011 10.7.6).

2.102 ifInBroadcastPktsCp Statistic (MACSEC_IFIBPCPS)

2.102.1 Offset

Register	Offset
MACSEC_IFIBPCPS	624h

2.102.2 Function

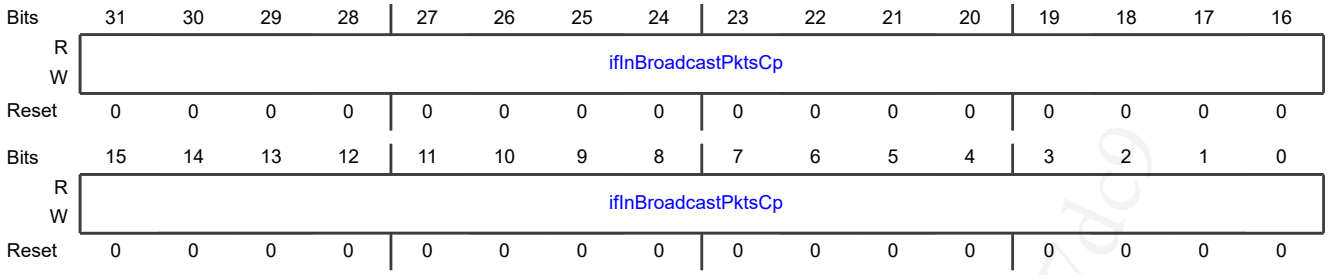
The MACSEC_IFIBPCPS register provides the count of the Controlled Port interface Input Broadcast Frames with unresolved SCI.

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2.102.3 Diagram



2.102.4 Fields

Field	Function
31-0 ifInBroadcast PktsCp	ifInBroadcastPktsCp The number of frames with unresolved SCI, delivered to the user of a Controlled port, which are addressed to a broadcast address (802.1AE-2006/802.1AEbn-2011 10.7.6).

2.103 ifInMulticastPktsUp Statistic (MACSEC_IFIMPUPS)

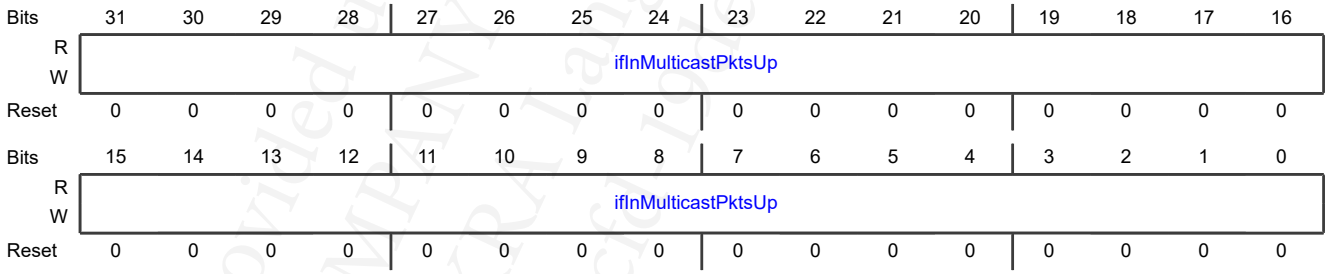
2.103.1 Offset

Register	Offset
MACSEC_IFIMPUPS	628h

2.103.2 Function

The MACSEC_IFIMPUPS register provides the count of the Uncontrolled Port interface Input Multicast Frames.

2.103.3 Diagram



2.103.4 Fields

Field	Function
31-0 ifInMulticast PktsUp	ifInMulticastPktsUp The number of frames with unresolved SCI or filtered to the uncontrolled port , delivered to the user of a Uncontrolled port, which are addressed to a multicast address (802.1AE-2006/802.1AEbn-2011 10.7.6).

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2.104 ifInBroadcastPktsUp Statistic (MACSEC_IFIBPUPS)

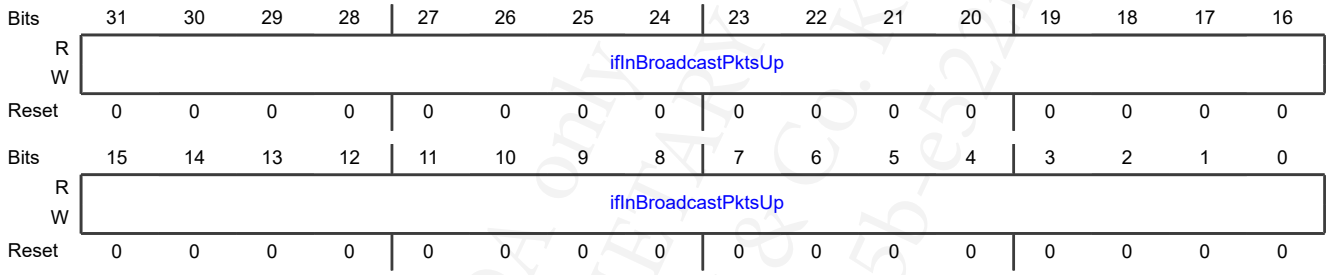
2.104.1 Offset

Register	Offset
MACSEC_IFIBPUPS	62Ch

2.104.2 Function

The MACSEC_IFIBPUPS register provides the count of the Uncontrolled Port interface Input Broadcast Frames.

2.104.3 Diagram



2.104.4 Fields

Field	Function
31-0 ifInBroadcast PktsUp	ifInBroadcastPktsUp The number of frames with unresolved SCI or filtered to the uncontrolled port , delivered to the user of a Uncontrolled port, which are addressed to a broadcast address (802.1AE-2006/802.1AEbn-2011 10.7.6).

2.105 InPktsWithoutTag Statistic (MACSEC_INPWTS)

2.105.1 Offset

Register	Offset
MACSEC_INPWTS	630h

2.105.2 Function

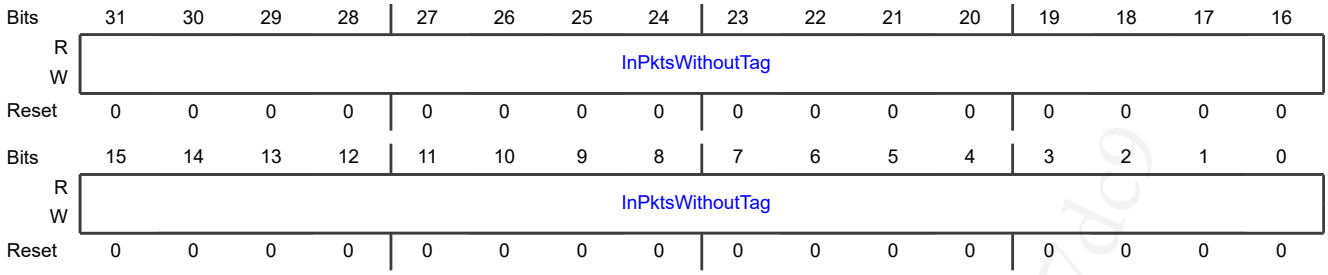
The MACSEC_INPWTS register provides the count of the Controlled Port verified Input Frames Untagged. This global statistic can be use to derive the per SC InPktsUntagged and the InPktsNoTag based on the Software knowledge of the validateFrame setting for each SC.

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2.105.3 Diagram



2.105.4 Fields

Field	Function
31-0 InPkts WithoutTag	InPktsWithoutTag The number of received frames without the MAC security tag (SecTAG). (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.106 InPktsKaY Statistic (MACSEC_INPKAYS)

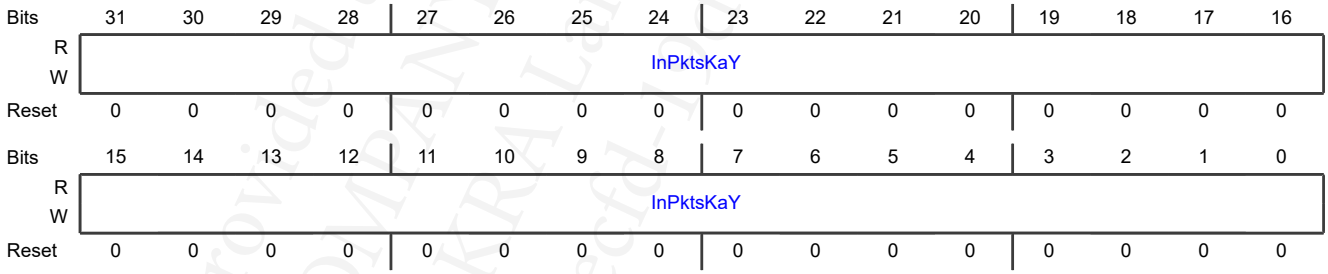
2.106.1 Offset

Register	Offset
MACSEC_INPKAYS	634h

2.106.2 Function

The MACSEC_INPKAYS register provides the count of the Controlled Port verified Input Frames destined to the KaY.

2.106.3 Diagram



2.106.4 Fields

Field	Function
31-0 InPktsKaY	InPktsKaY The number of frames received with the Encryption (E) bit set and the Changed Text (C) bit clear (reserved for use by the KaY)

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2.107 InPktsBadTag Statistic (MACSEC_INPBTS)

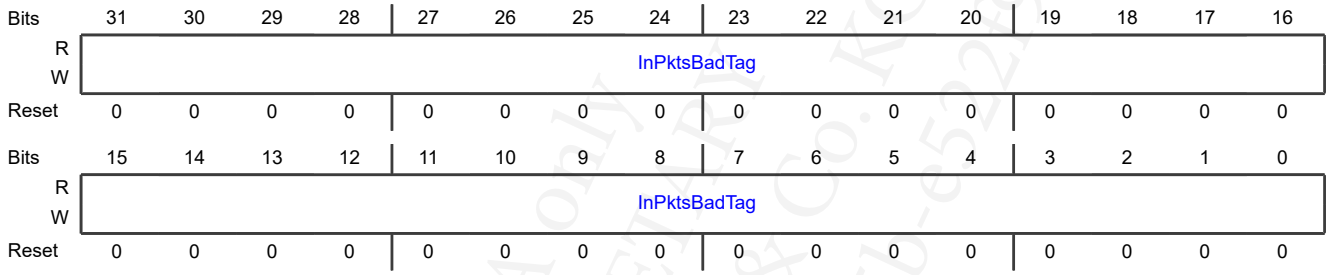
2.107.1 Offset

Register	Offset
MACSEC_INPBTS	638h

2.107.2 Function

The MACSEC_INPBTS register provides the count of the Controlled Port verified Input Frames with Bad Tag.

2.107.3 Diagram



2.107.4 Fields

Field	Function
31-0 InPktsBadTag	InPktsBadTag The number of received frames discarded with an invalid SecTAG or a zero value PN or an invalid ICV. (802.1AE-2006/802.1AEbn-2011 10.7.9)

2.108 InPktsSCINotFound Statistic (MACSEC_IPSNFS)

2.108.1 Offset

Register	Offset
MACSEC_IPSNFS	63Ch

2.108.2 Function

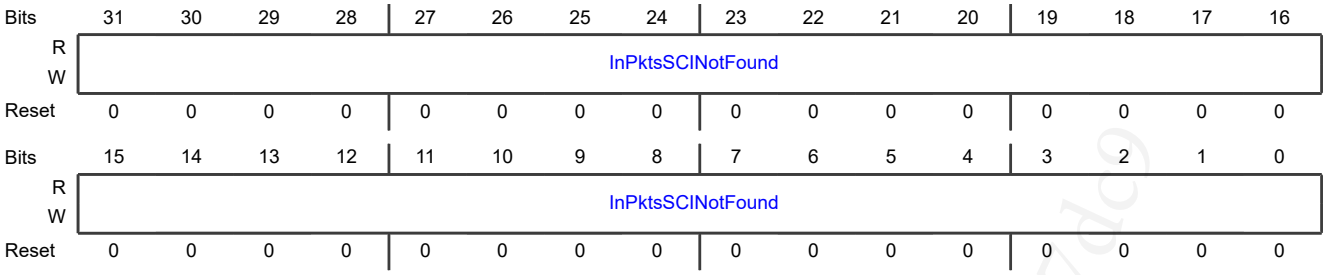
The MACSEC_IPSNFS register provides the count of the Controlled Port verified Input Frames with SCI not found. This global statistic can be use to derive the per SC InPktsUnknownSCI and the InPktsNoSCI based on the Software knowledge of the validateFrame setting for each SC.

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2.108.3 Diagram



2.108.4 Fields

Field	Function
31-0 InPktsSCINot Found	InPktsSCINotFound The number of received frames with SCI not found. (802.1AE-2006/802.1AEbn-2011 10.7.9) ***Note:*** *In 802.1AE-2018 standard, this is now called InPktsNoSA/InPktsNoSAError and must be aggregated by software with MACSEC_RXANnINUSS and MACSEC_RXANnIPUSS on a per SC basis.*

2.109 InPktsUnsupportedEC Statistic (MACSEC_IPUECS)

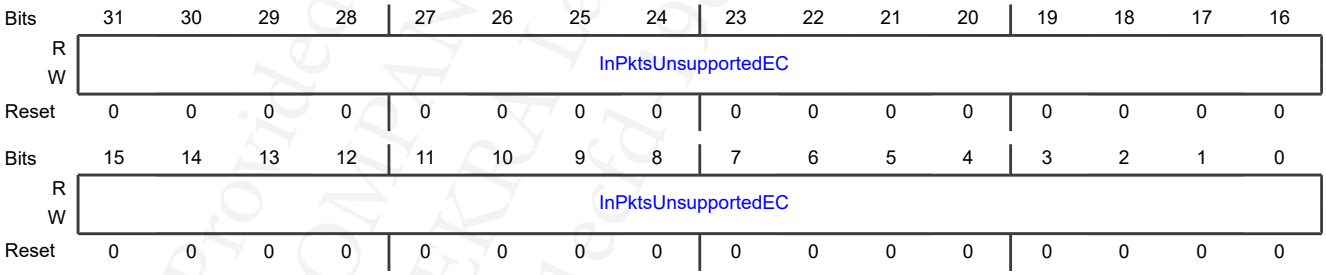
2.109.1 Offset

Register	Offset
MACSEC_IPUECS	640h

2.109.2 Function

The MACSEC_IPUECS register provides the count of the Controlled Port verified Input Frames with Encryption (E) bit cleared and the Changed Text (C) bit set (unsupported).

2.109.3 Diagram



2.109.4 Fields

Field	Function
31-0 InPkts UnsupportedEC	InPktsUnsupportedEC The number of received frames with Encryption (E) bit cleared and the Changed Text (C) bit set.

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2.110 nPktsEponSingleCopyBroadcast Statistic (MACSEC_IPESCBS)

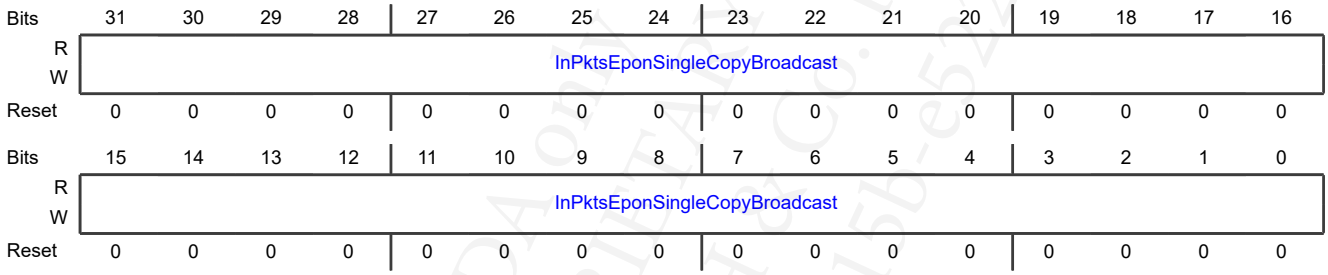
2.110.1 Offset

Register	Offset
MACSEC_IPESCBS	644h

2.110.2 Function

The MACSEC_IPESCBS register provides the count of the Controlled Port verified Input Frames with Secure Channel (SC) bit cleared, the End Station (ES) bit cleared, and the Single Copy Broadcast bit set (SCB) (that is, EPON Single Copy Broadcast type).

2.110.3 Diagram



2.110.4 Fields

Field	Function
31-0 InPktsEpon SingleCopy Broadcast	InPktsEponSingleCopyBroadcast The number of received frames with Secure Channel (SC) bit cleared, the End Station (ES) bit cleared, and the Single Copy Broadcast bit set (SCB) (EPON mode not supported)

2.111 InPktsTooLong Statistic (MACSEC_IPTLS)

2.111.1 Offset

Register	Offset
MACSEC_IPTLS	648h

2.111.2 Function

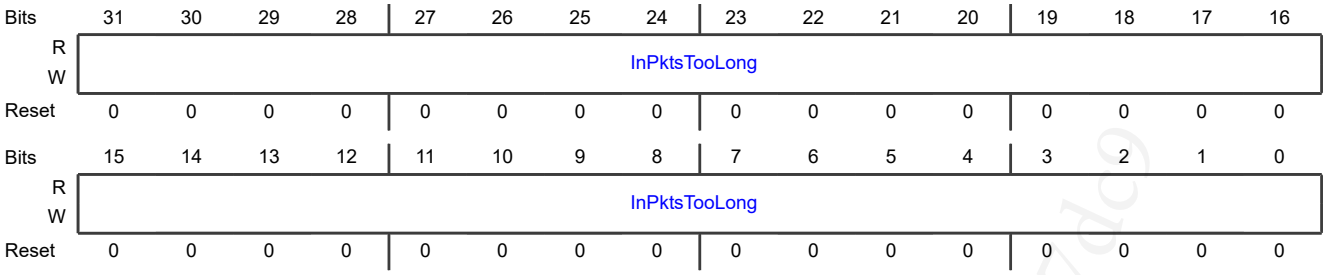
The MACSEC_IPTLS register provides a count of input frames (Controlled or Uncontrolled port) received with more than 65,535 octets of payload. In a properly configured system the MACsec block will not receive frames of this size so this counter should never increment. It is useful as a debug assist only.

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2.111.3 Diagram



2.111.4 Fields

Field	Function
31-0 InPktsTooLong	InPktsTooLong Number of input frames received with more than 65,535 octets of payload.

2.112 InPktsFilteredUp Statistic (MACSEC_IPFUS)

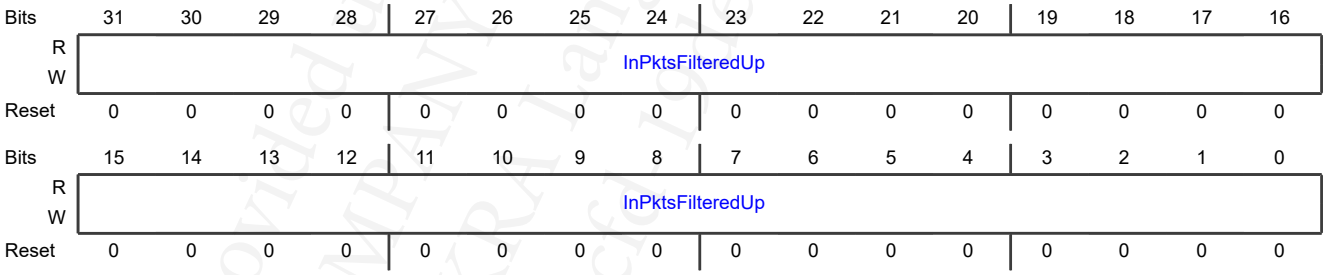
2.112.1 Offset

Register	Offset
MACSEC_IPFUS	64Ch

2.112.2 Function

The MACSEC_IPFUS register provides a count of input frames filtered and not discarded to the uncontrolled port.

2.112.3 Diagram



2.112.4 Fields

Field	Function
31-0 InPktsFilteredUp	InPktsFilteredUp Number of input frames received filtered and not discarded to the uncontrolled port

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2.113 InPktsDiscardedUp Statistic (MACSEC_IPDUS)

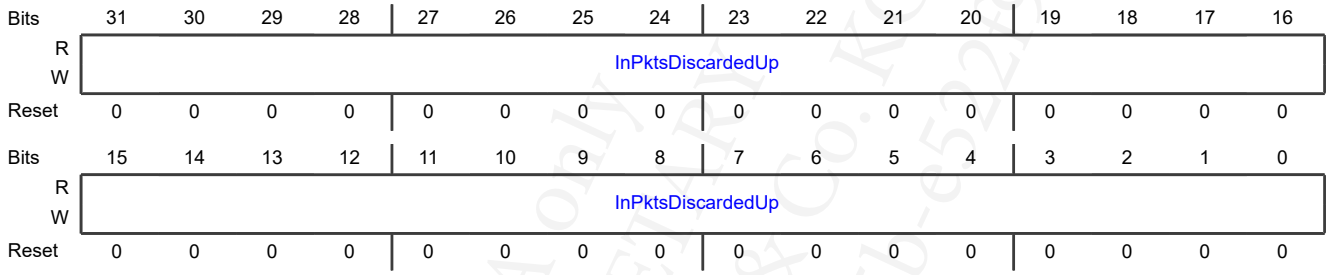
2.113.1 Offset

Register	Offset
MACSEC_IPDUS	650h

2.113.2 Function

The MACSEC_IPDUS register provides a count of input frames filtered and discarded to the uncontrolled port.

2.113.3 Diagram



2.113.4 Fields

Field	Function
31-0 InPkts DiscardedUp	InPktsDiscardedUp Number of input frames received filtered and discarded to the uncontrolled port

2.114 OutPktsDiscarded Statistic (MACSEC_OPDS)

2.114.1 Offset

Register	Offset
MACSEC_OPDS	680h

2.114.2 Function

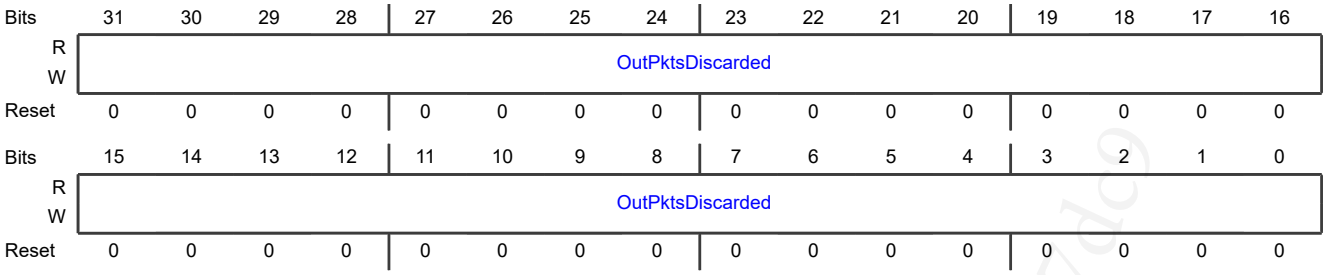
The MACSEC_OPDS register provides the count of frames marked as abort on TX.

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2.114.3 Diagram



2.114.4 Fields

Field	Function
31-0 OutPkts Discarded	OutPktsDiscarded The number of TX frames marked as abort, for the MAC to invalidate the FCS, due to one of the following reason: • A double ECC error on the MACsec memory occurred while the TX data path is reading itâ##s context • TX SC NextPN == 0, illegal configuration • TX SC is disabled • TX SC classification not found • alwaysIncludeSCI = 0, useES = 0, useSCB = 1 unsupported configuration • frame too short: frame received on the controlled port less than 13 bytes plus 4B FCS if present

2.115 RX Security Association A key 1 of 8 (MACSEC_RXSAKA1Q)

2.115.1 Offset

Register	Offset
MACSEC_RXSAKA1Q	700h

2.115.2 Function

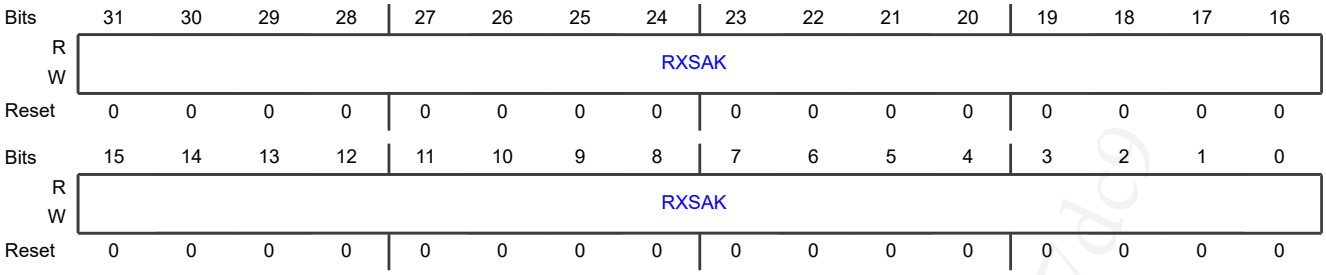
The MACSEC_RXSAKA1Q register is used to configure bits 0-31 of a Security Association A key for the Secure Channel.

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2.115.3 Diagram



2.115.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 0-31

2.116 RX Security Association A key 2 of 8 (MACSEC_RXSAKA2Q)

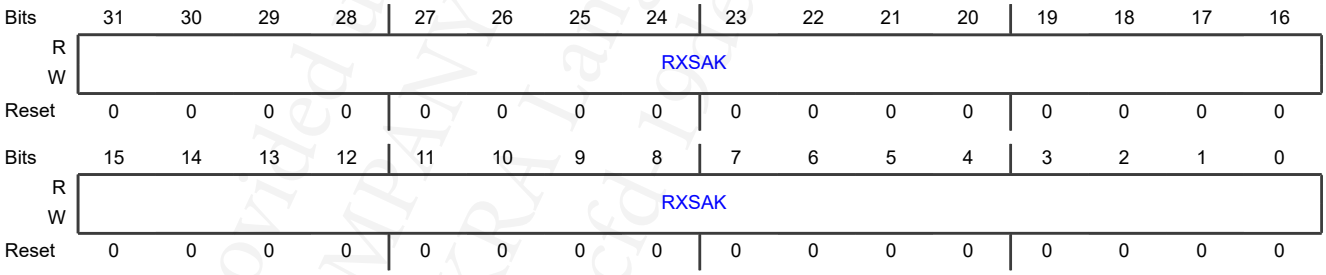
2.116.1 Offset

Register	Offset
MACSEC_RXSAKA2Q	704h

2.116.2 Function

The MACSEC_RXSAKA2Q register is used to configure bits 32-63 of a Security Association A key for the Secure Channel.

2.116.3 Diagram



2.116.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 32-63

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2.117 RX Security Association A key 3 of 8 (MACSEC_RXSAKA3Q)

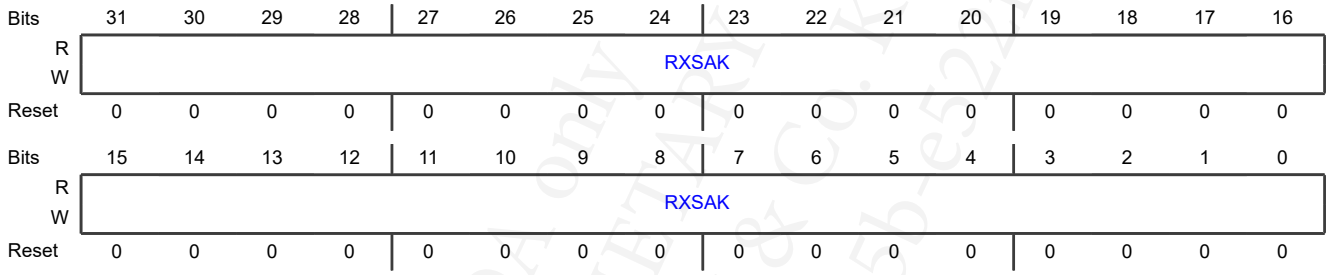
2.117.1 Offset

Register	Offset
MACSEC_RXSAKA3Q	708h

2.117.2 Function

The MACSEC_RXSAKA3Q register is used to configure bits 64-95 of a Security Association A key for the Secure Channel.

2.117.3 Diagram



2.117.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 64-95

2.118 RX Security Association A key 4 of 8 (MACSEC_RXSAKA4Q)

2.118.1 Offset

Register	Offset
MACSEC_RXSAKA4Q	70Ch

2.118.2 Function

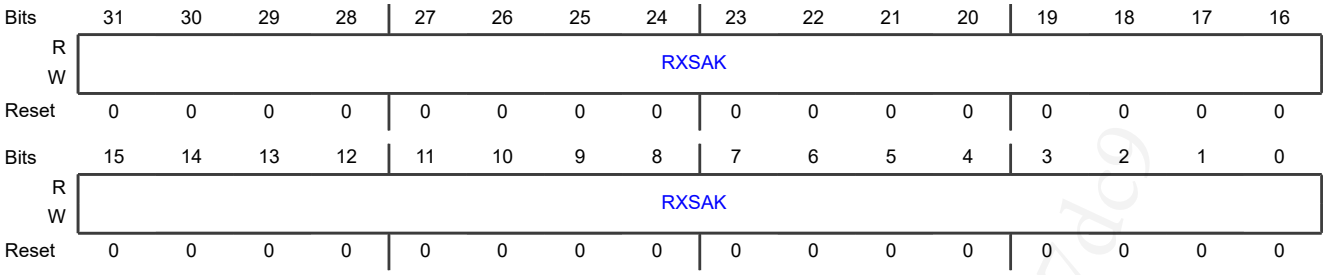
The MACSEC_RXSAKA4Q register is used to configure bits 96-127 of a Security Association A key for the Secure Channel.

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2.118.3 Diagram



2.118.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 96-127

2.119 RX Security Association A key 5 of 8 (MACSEC_RXSAKA5Q)

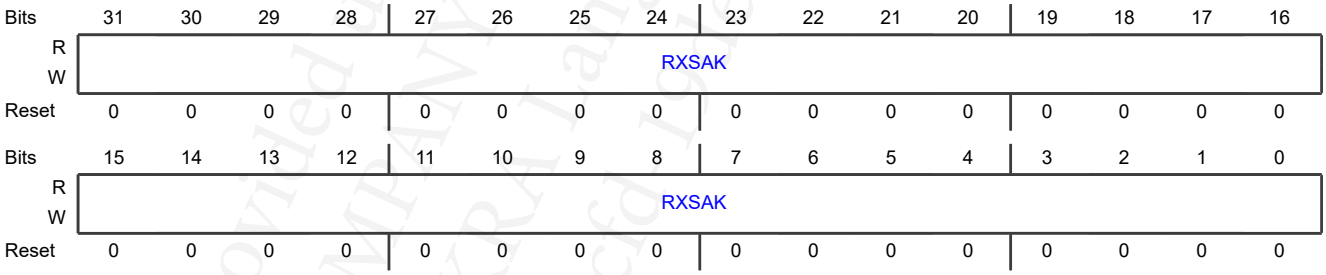
2.119.1 Offset

Register	Offset
MACSEC_RXSAKA5Q	710h

2.119.2 Function

The MACSEC_RXSAKA5Q register is used to configure bits 128-159 of a Security Association A key for the Secure Channel.

2.119.3 Diagram



2.119.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 128-159

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2.120 RX Security Association A key 6 of 8 (MACSEC_RXSAKA6Q)

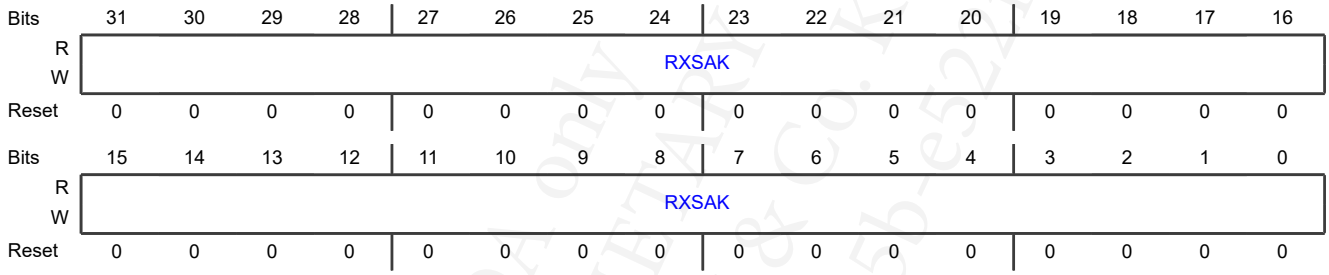
2.120.1 Offset

Register	Offset
MACSEC_RXSAKA6Q	714h

2.120.2 Function

The MACSEC_RXSAKA6Q register is used to configure bits 160-191 of a Security Association A key for the Secure Channel.

2.120.3 Diagram



2.120.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 160-191

2.121 RX Security Association A key 7 of 8 (MACSEC_RXSAKA7Q)

2.121.1 Offset

Register	Offset
MACSEC_RXSAKA7Q	718h

2.121.2 Function

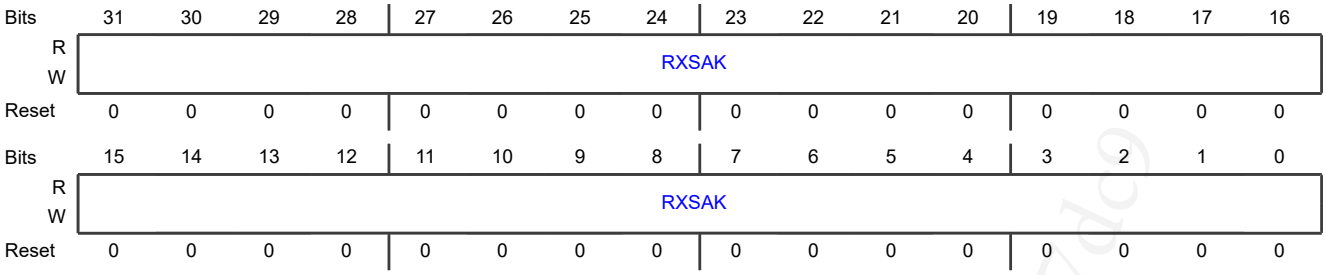
The MACSEC_RXSAKA7Q register is used to configure bits 192-223 of a Security Association A key for the Secure Channel.

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2.121.3 Diagram



2.121.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 192-223

2.122 RX Security Association A key 8 of 8 (MACSEC_RXSAKA8Q)

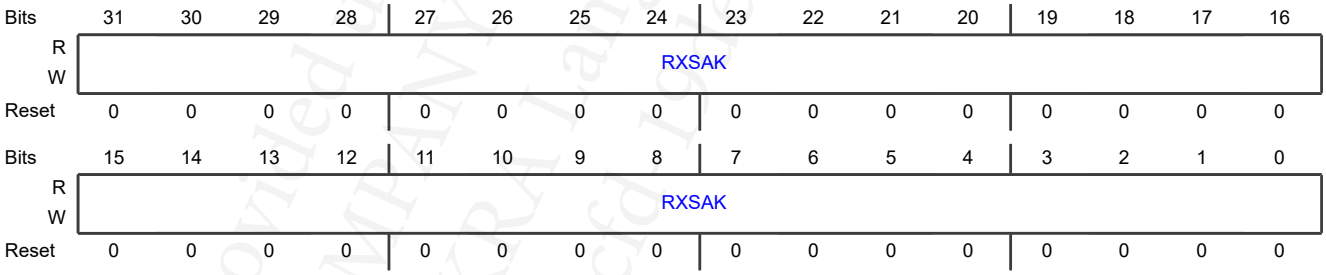
2.122.1 Offset

Register	Offset
MACSEC_RXSAKA8Q	71Ch

2.122.2 Function

The MACSEC_RXSAKA8Q register is used to configure bits 224-255 of a Security Association A key for the Secure Channel.

2.122.3 Diagram



2.122.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 224-255

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2.123 RX Security Association A Short Secure Channel Identifier (MACSEC_RXSAASSCI)

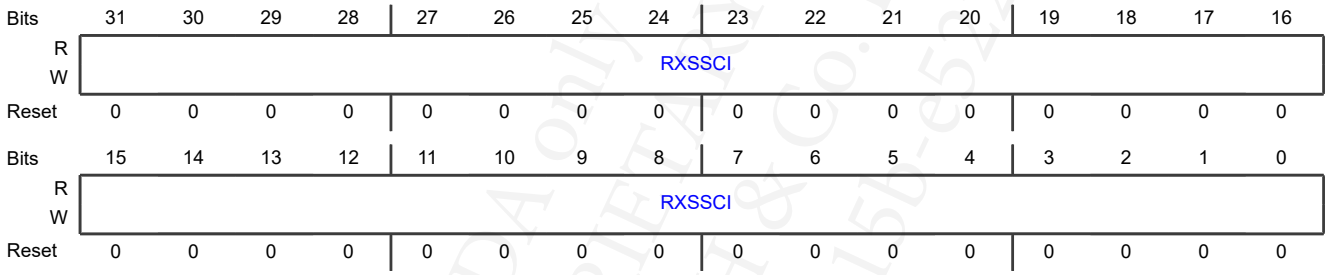
2.123.1 Offset

Register	Offset
MACSEC_RXSAASSCI	720h

2.123.2 Function

The MACSEC_RXSAASSCI register is used to configure SSCI of Security Association B for the Secure Channel, when an XPN Cipher Suite is configured.

2.123.3 Diagram



2.123.4 Fields

Field	Function
31-0 RXSSCI	RXSSCI RX Short Secure Channel Identifier (RXSSCI): A 32-bit value per SA to support XPN Cipher Suite for the secure channel, it is only used when XPN Cipher Suite is configured.

2.124 RX Security Association A SALT 1 of 3 (MACSEC_RXSAASALT1T)

2.124.1 Offset

Register	Offset
MACSEC_RXSAASALT1T	724h

2.124.2 Function

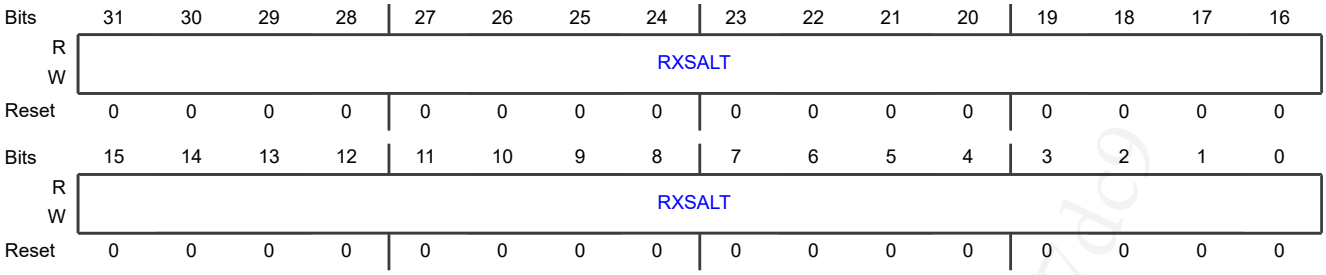
The MACSEC_RXSAASALT1T register is used to configure bits 31-0 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

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2.124.3 Diagram



2.124.4 Fields

Field	Function
31-0 RXSALT	RXSALT RX SALT bits 31-0, it is only used when XPN Cipher Suite is configured.

2.125 RX Security Association A SALT 2 of 3 (MACSEC_RXSAASALT2T)

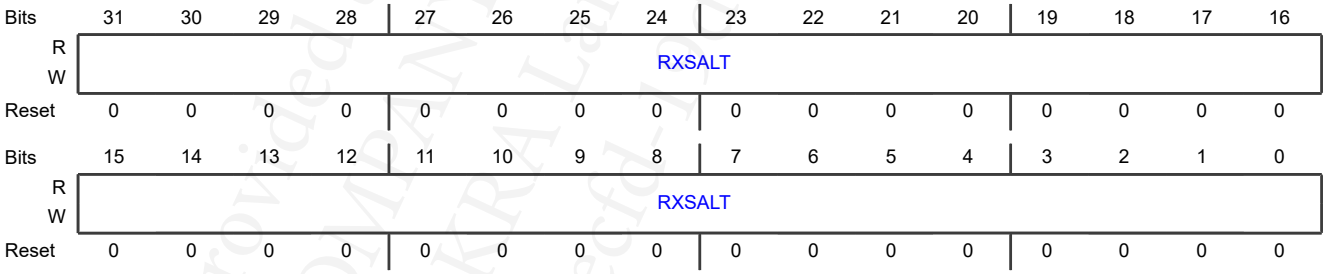
2.125.1 Offset

Register	Offset
MACSEC_RXSAASALT2T	728h

2.125.2 Function

The MACSEC_RXSAASALT2T register is used to configure bits 63-32 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.125.3 Diagram



2.125.4 Fields

Field	Function
31-0 RXSALT	RXSALT RX SALT bits 63-32, it is only used when XPN Cipher Suite is configured.

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2.126 RX Security Association A SALT 3 of 3 (MACSEC_RXSAASALT3T)

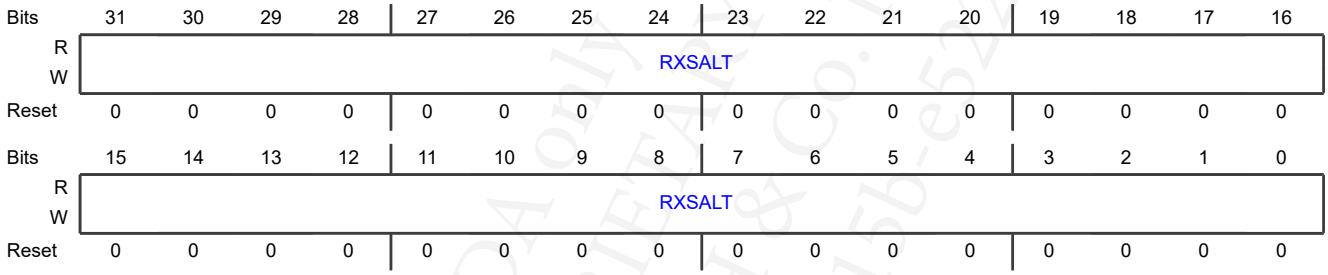
2.126.1 Offset

Register	Offset
MACSEC_RXSAASALT3T	72Ch

2.126.2 Function

The MACSEC_RXSAASALT3T register is used to configure bits 95-64 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.126.3 Diagram



2.126.4 Fields

Field	Function
31-0 RXSALT	RXSALT RX SALT bits 95-64, it is only used when XPN Cipher Suite is configured.

2.127 RX Security Association B key 1 of 8 (MACSEC_RXSAKB1Q)

2.127.1 Offset

Register	Offset
MACSEC_RXSAKB1Q	740h

2.127.2 Function

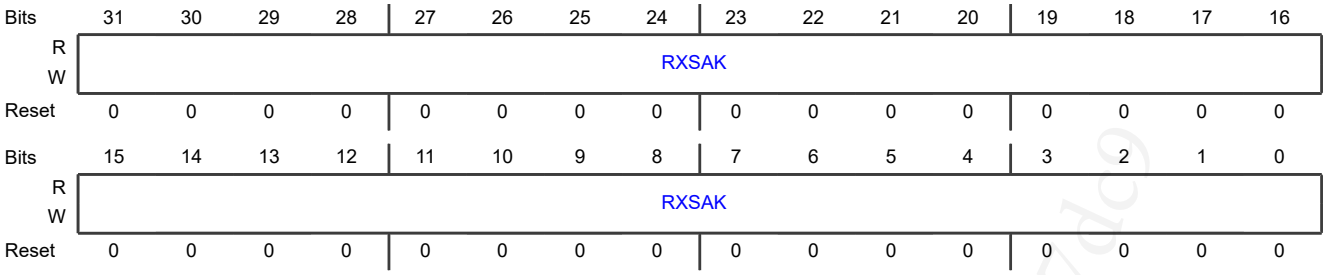
The MACSEC_RXSAKB1Q register is used to configure bits 0-31 of a Security Association B key for the Secure Channel.

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2.127.3 Diagram



2.127.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 0-31

2.128 RX Security Association B key 2 of 8 (MACSEC_RXSAKB2Q)

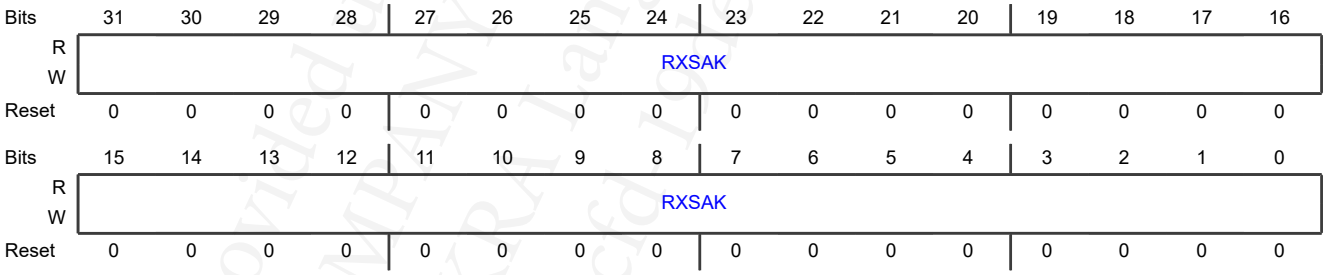
2.128.1 Offset

Register	Offset
MACSEC_RXSAKB2Q	744h

2.128.2 Function

The MACSEC_RXSAKB2Q register is used to configure bits 32-63 of a Security Association B key for the Secure Channel.

2.128.3 Diagram



2.128.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 32-63

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2.129 RX Security Association B key 3 of 8 (MACSEC_RXSAKB3Q)

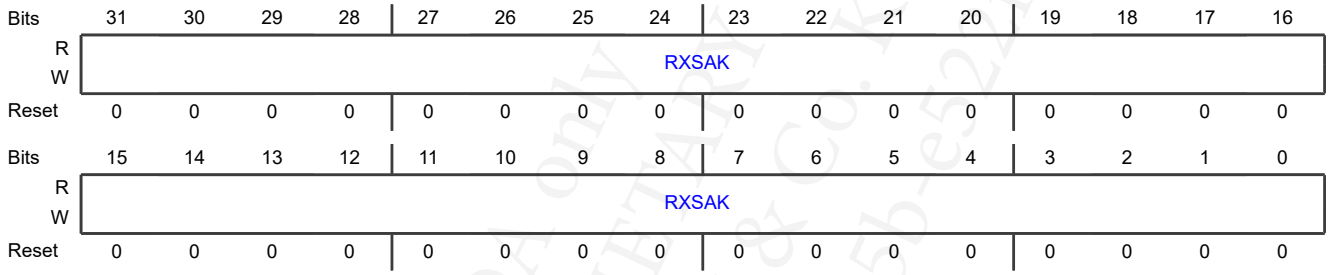
2.129.1 Offset

Register	Offset
MACSEC_RXSAKB3Q	748h

2.129.2 Function

The MACSEC_RXSAKB3Q register is used to configure bits 64-95 of a Security Association B key for the Secure Channel.

2.129.3 Diagram



2.129.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 64-95

2.130 RX Security Association B key 4 of 8 (MACSEC_RXSAKB4Q)

2.130.1 Offset

Register	Offset
MACSEC_RXSAKB4Q	74Ch

2.130.2 Function

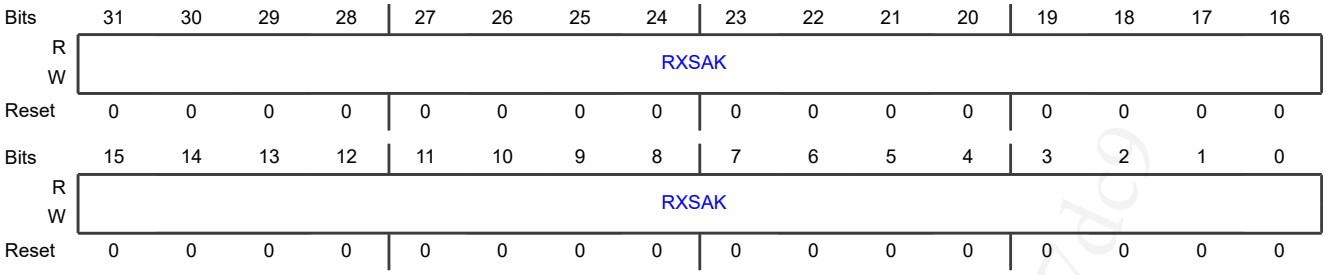
The MACSEC_RXSAKB4Q register is used to configure bits 96-127 of a Security Association B key for the Secure Channel.

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2.130.3 Diagram



2.130.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 96-127

2.131 RX Security Association B key 5 of 8 (MACSEC_RXSAKB5Q)

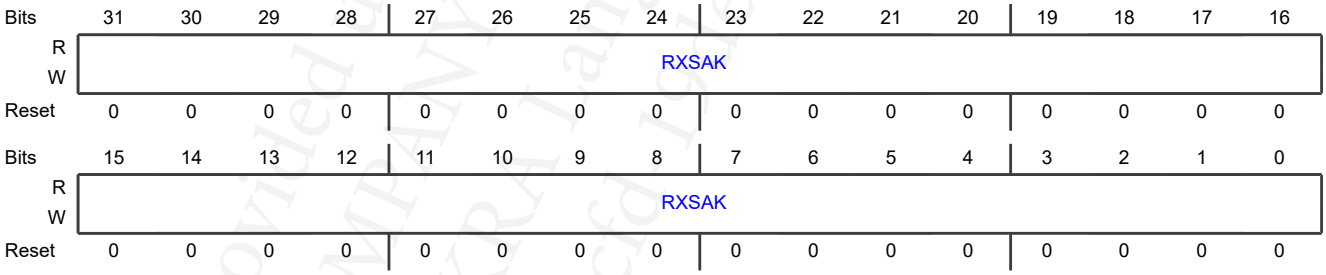
2.131.1 Offset

Register	Offset
MACSEC_RXSAKB5Q	750h

2.131.2 Function

The MACSEC_RXSAKB5Q register is used to configure bits 128-159 of a Security Association B key for the Secure Channel.

2.131.3 Diagram



2.131.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 128-159

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2.132 RX Security Association B key 6 of 8 (MACSEC_RXSAKB6Q)

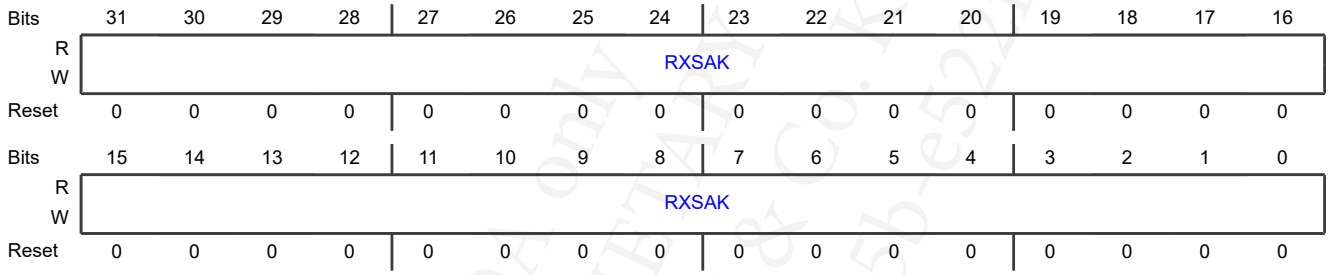
2.132.1 Offset

Register	Offset
MACSEC_RXSAKB6Q	754h

2.132.2 Function

The MACSEC_RXSAKB6Q register is used to configure bits 160-191 of a Security Association B key for the Secure Channel.

2.132.3 Diagram



2.132.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 160-191

2.133 RX Security Association B key 7 of 8 (MACSEC_RXSAKB7Q)

2.133.1 Offset

Register	Offset
MACSEC_RXSAKB7Q	758h

2.133.2 Function

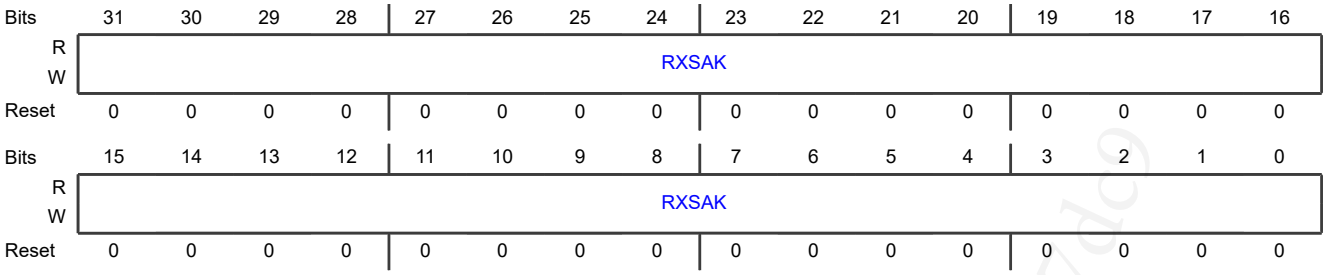
The MACSEC_RXSAKB7Q register is used to configure bits 192-223 of a Security Association B key for the Secure Channel.

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2.133.3 Diagram



2.133.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 192-223

2.134 RX Security Association B key 8 of 8 (MACSEC_RXSAKB8Q)

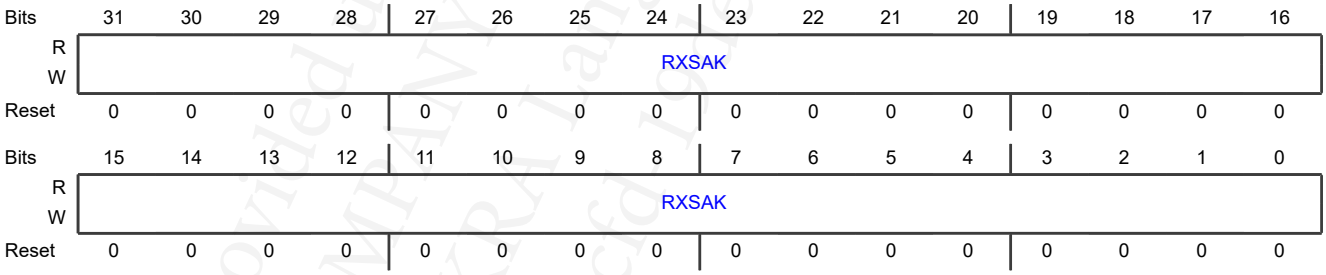
2.134.1 Offset

Register	Offset
MACSEC_RXSAKB8Q	75Ch

2.134.2 Function

The MACSEC_RXSAKB8Q register is used to configure bits 224-255 of a Security Association B key for the Secure Channel.

2.134.3 Diagram



2.134.4 Fields

Field	Function
31-0 RXSAK	RXSAK RX Security Association Key bit 224-255

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2.135 RX Security Association B Short Secure Channel Identifier (MACSEC_RXSABSSCI)

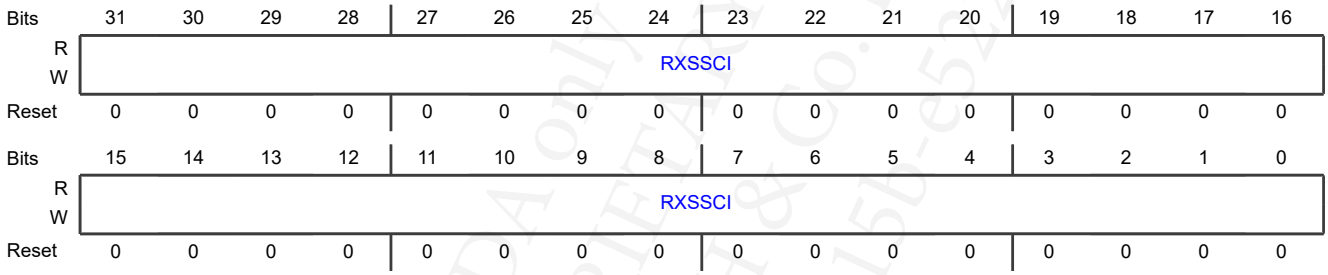
2.135.1 Offset

Register	Offset
MACSEC_RXSABSSCI	760h

2.135.2 Function

The MACSEC_RXSABSSCI register is used to configure SSCI of Security Association B for the Secure Channel, when an XPN Cipher Suite is configured.

2.135.3 Diagram



2.135.4 Fields

Field	Function
31-0 RXSSCI	RXSSCI RX Short Secure Channel Identifier (RXSSCI): A 32-bit value per SA to support XPN Cipher Suite for the secure channel, it is only used when XPN Cipher Suite is configured.

2.136 RX Security Association B SALT 1 of 3 (MACSEC_RXSABSALT1T)

2.136.1 Offset

Register	Offset
MACSEC_RXSABSALT1T	764h

2.136.2 Function

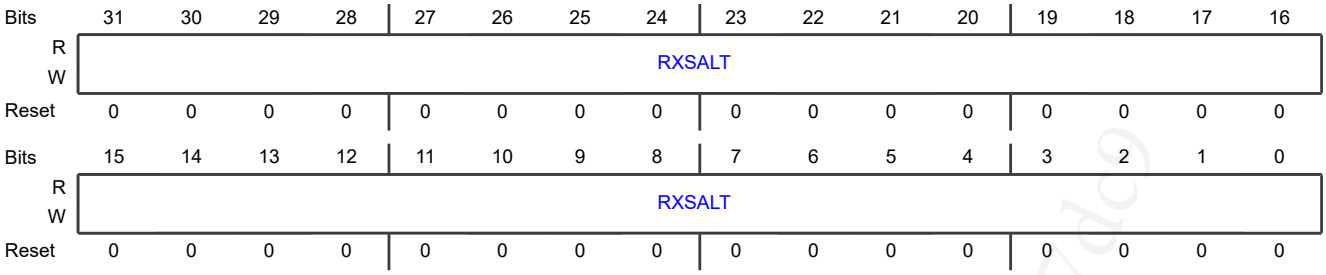
The MACSEC_RXSABSALT1T register is used to configure bits 31-0 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

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2.136.3 Diagram



2.136.4 Fields

Field	Function
31-0 RXSALT	RXSALT RX SALT bits 31-0, it is only used when XPN Cipher Suite is configured.

2.137 RX Security Association B SALT 2 of 3 (MACSEC_RXSABSALT2T)

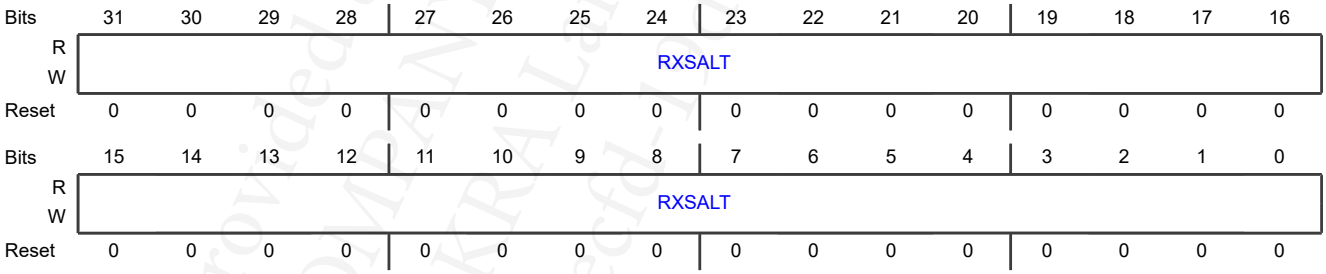
2.137.1 Offset

Register	Offset
MACSEC_RXSABSALT2T	768h

2.137.2 Function

The MACSEC_RXSABSALT2T register is used to configure bits 63-32 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.137.3 Diagram



2.137.4 Fields

Field	Function
31-0 RXSALT	RXSALT RX SALT bits 63-32, it is only used when XPN Cipher Suite is configured.

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2.138 RX Security Association B SALT 3 of 3 (MACSEC_RXSABSALT3T)

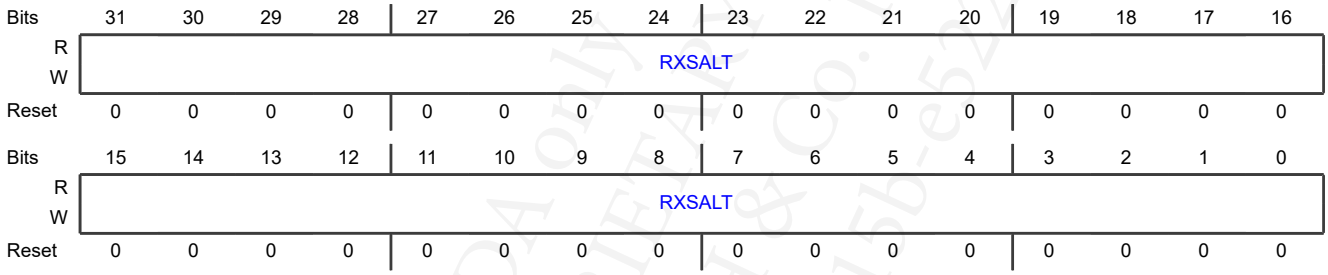
2.138.1 Offset

Register	Offset
MACSEC_RXSABSALT3T	76Ch

2.138.2 Function

The MACSEC_RXSABSALT3T register is used to configure bits 95-64 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.138.3 Diagram



2.138.4 Fields

Field	Function
31-0 RXSALT	RXSALT RX SALT bits 95-64, it is only used when XPN Cipher Suite is configured.

2.139 TX Security Association A key 1 of 8 (MACSEC_TXSAKA1Q)

2.139.1 Offset

Register	Offset
MACSEC_TXSAKA1Q	780h

2.139.2 Function

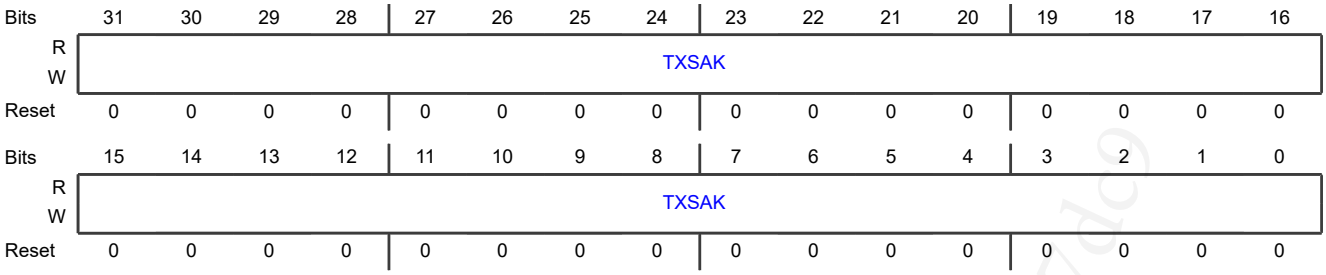
The MACSEC_TXSAKA1Q register is used to configure bits 0-31 of a Security Association A key for the Secure Channel.

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2.139.3 Diagram



2.139.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 0-31

2.140 TX Security Association A key 2 of 8 (MACSEC_TXSAKA2Q)

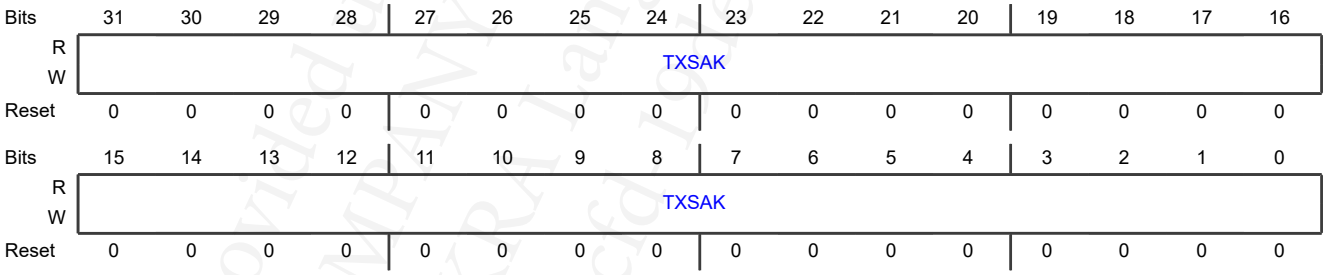
2.140.1 Offset

Register	Offset
MACSEC_TXSAKA2Q	784h

2.140.2 Function

The MACSEC_TXSAKA2Q register is used to configure bits 32-63 of a Security Association A key for the Secure Channel.

2.140.3 Diagram



2.140.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 32-63

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2.141 TX Security Association A key 3 of 8 (MACSEC_TXSAKA3Q)

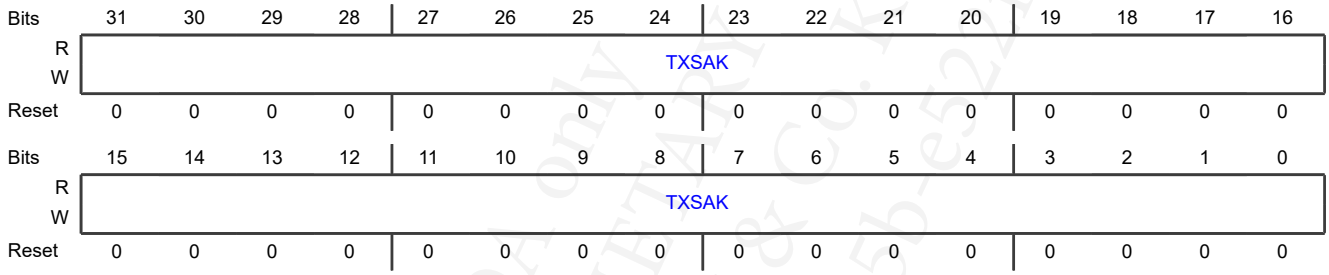
2.141.1 Offset

Register	Offset
MACSEC_TXSAKA3Q	788h

2.141.2 Function

The MACSEC_TXSAKA3Q register is used to configure bits 64-95 of a Security Association A key for the Secure Channel.

2.141.3 Diagram



2.141.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 64-95

2.142 TX Security Association A key 4 of 8 (MACSEC_TXSAKA4Q)

2.142.1 Offset

Register	Offset
MACSEC_TXSAKA4Q	78Ch

2.142.2 Function

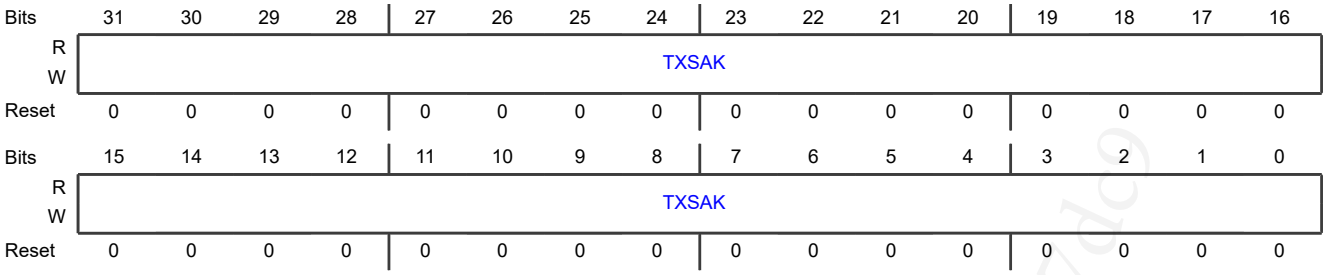
The MACSEC_TXSAKA4Q register is used to configure bits 96-127 of a Security Association A key for the Secure Channel.

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2.142.3 Diagram



2.142.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 96-127

2.143 TX Security Association A key 5 of 8 (MACSEC_TXSAKA5Q)

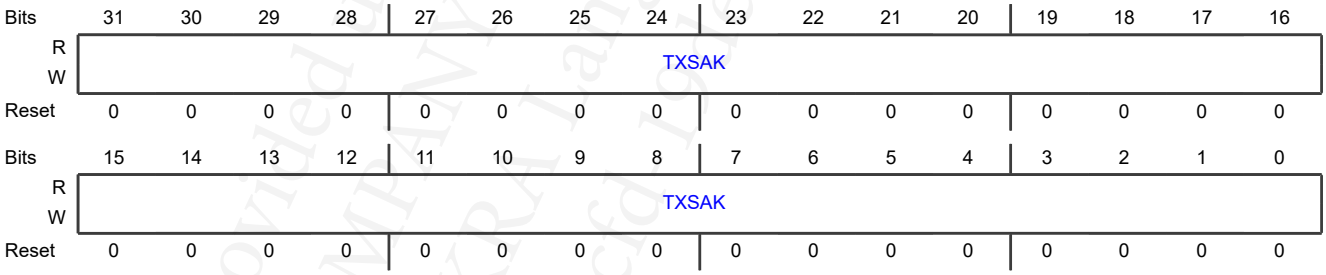
2.143.1 Offset

Register	Offset
MACSEC_TXSAKA5Q	790h

2.143.2 Function

The MACSEC_TXSAKA5Q register is used to configure bits 128-159 of a Security Association A key for the Secure Channel.

2.143.3 Diagram



2.143.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 128-159

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2.144 TX Security Association A key 6 of 8 (MACSEC_TXSAKA6Q)

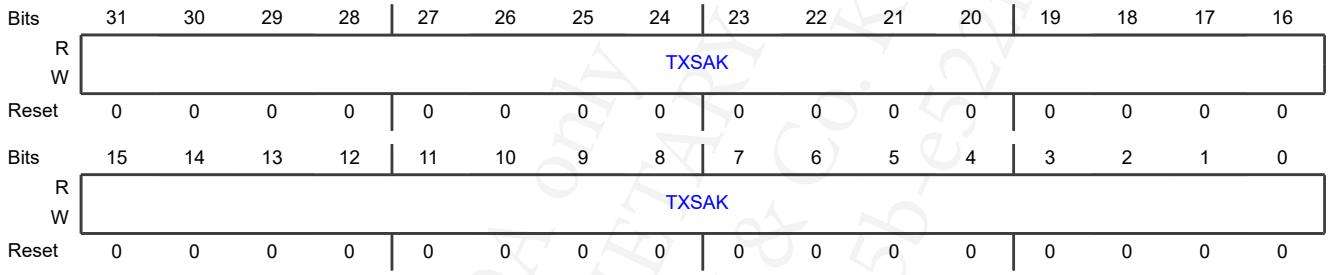
2.144.1 Offset

Register	Offset
MACSEC_TXSAKA6Q	794h

2.144.2 Function

The MACSEC_TXSAKA6Q register is used to configure bits 160-191 of a Security Association A key for the Secure Channel.

2.144.3 Diagram



2.144.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 160-191

2.145 TX Security Association A key 7 of 8 (MACSEC_TXSAKA7Q)

2.145.1 Offset

Register	Offset
MACSEC_TXSAKA7Q	798h

2.145.2 Function

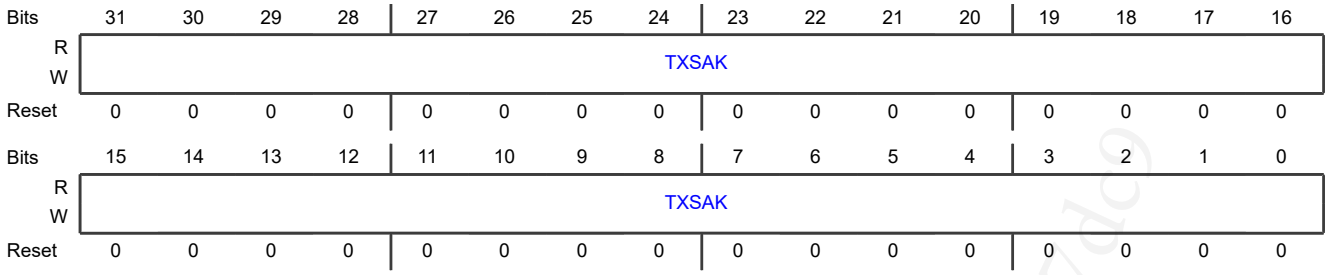
The MACSEC_TXSAKA7Q register is used to configure bits 192-223 of a Security Association A key for the Secure Channel.

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2.145.3 Diagram



2.145.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 192-223

2.146 TX Security Association A key 8 of 8 (MACSEC_TXSAKA8Q)

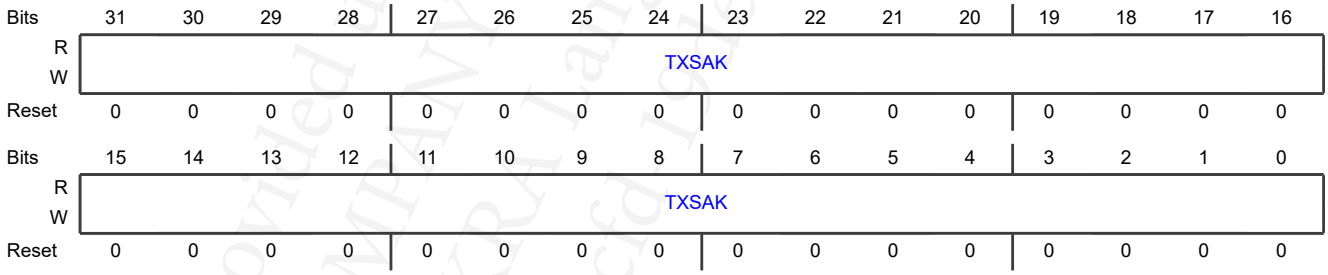
2.146.1 Offset

Register	Offset
MACSEC_TXSAKA8Q	79Ch

2.146.2 Function

The MACSEC_TXSAKA8Q register is used to configure bits 224-255 of a Security Association A key for the Secure Channel.

2.146.3 Diagram



2.146.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 224-255

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2.147 TX Security Association A Short Secure Channel Identifier (MACSEC_TXSAASSCI)

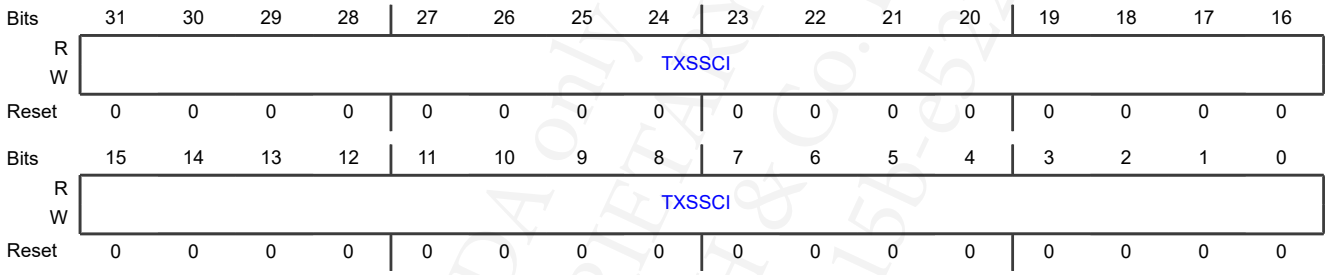
2.147.1 Offset

Register	Offset
MACSEC_TXSAASSCI	7A0h

2.147.2 Function

The MACSEC_TXSAASSCI register is used to configure SSCI of Security Association A for the Secure Channel, when an XPN Cipher Suite is configured.

2.147.3 Diagram



2.147.4 Fields

Field	Function
31-0 TXSSCI	TXSSCI TX Short Secure Channel Identifier (TXSSCI): A 32-bit value per SA to support XPN Cipher Suite for the secure channel, it is only used when XPN Cipher Suite is configured.

2.148 TX Security Association A SALT 1 of 3 (MACSEC_TXSAASALT1T)

2.148.1 Offset

Register	Offset
MACSEC_TXSAASALT1T	7A4h

2.148.2 Function

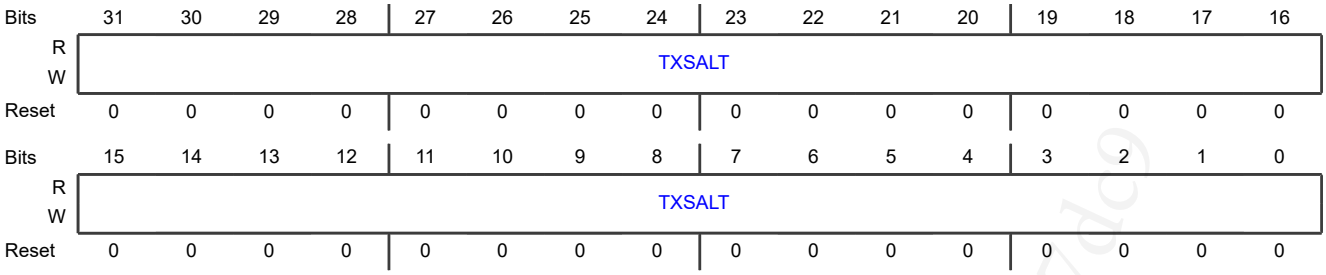
The MACSEC_TXSAASALT1T register is used to configure bits 31-0 of Security Association A SALT for the Secure Channel, when an XPN Cipher Suite is configured.

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2.148.3 Diagram



2.148.4 Fields

Field	Function
31-0 TXSALT	TXSALT TX SALT bits 31-0, it is only used when XPN Cipher Suite is configured.

2.149 TX Security Association A SALT 2 of 3 (MACSEC_TXSAASALT2T)

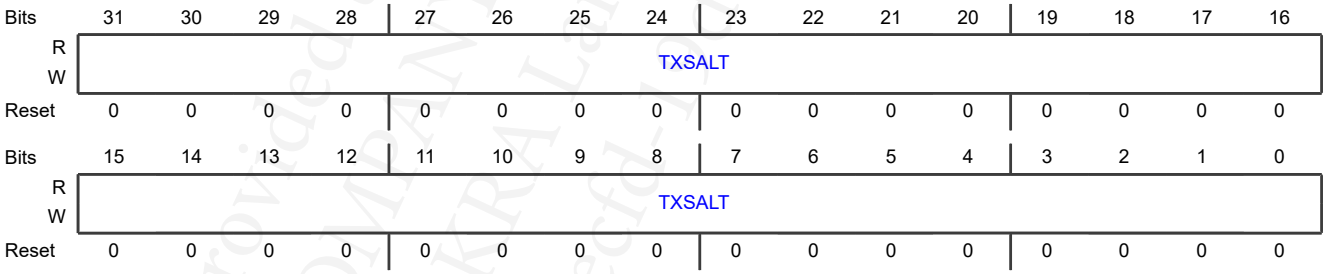
2.149.1 Offset

Register	Offset
MACSEC_TXSAASALT2T	7A8h

2.149.2 Function

The MACSEC_TXSAASALT2T register is used to configure bits 63-32 of Security Association A SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.149.3 Diagram



2.149.4 Fields

Field	Function
31-0 TXSALT	TXSALT TX SALT bits 63-32, it is only used when XPN Cipher Suite is configured.

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2.150 TX Security Association A SALT 3 of 3 (MACSEC_TXSAASALT3T)

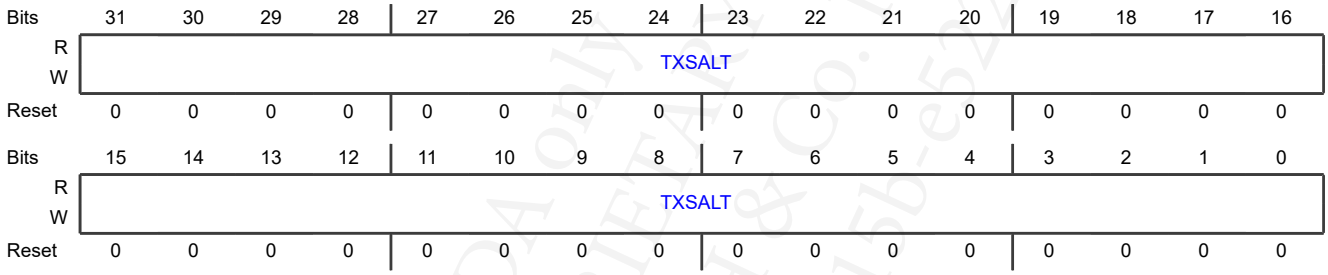
2.150.1 Offset

Register	Offset
MACSEC_TXSAASALT3T	7ACh

2.150.2 Function

The MACSEC_TXSAASALT3T register is used to configure bits 95-64 of Security Association A SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.150.3 Diagram



2.150.4 Fields

Field	Function
31-0 TXSALT	TXSALT TX SALT bits 95-64, it is only used when XPN Cipher Suite is configured.

2.151 TX Security Association A key 1 of 8 (MACSEC_TXSAKB1Q)

2.151.1 Offset

Register	Offset
MACSEC_TXSAKB1Q	7C0h

2.151.2 Function

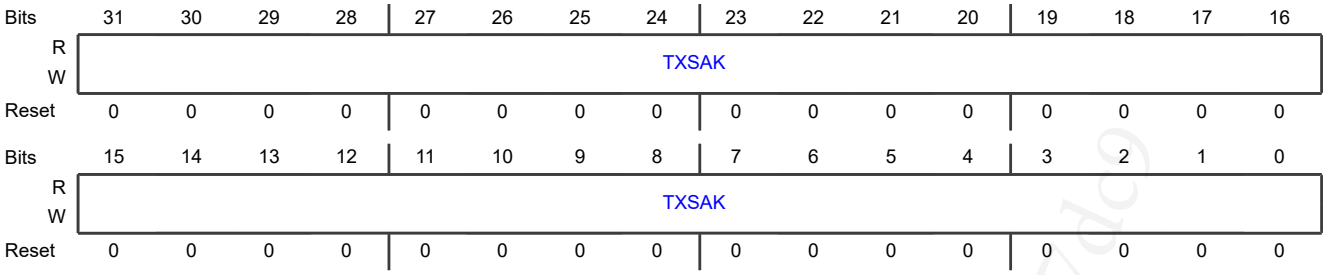
The MACSEC_TXSAKB1Q register is used to configure bits 0-31 of a Security Association B key for the Secure Channel.

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2.151.3 Diagram



2.151.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 0-31

2.152 TX Security Association B key 2 of 8 (MACSEC_TXSAKB2Q)

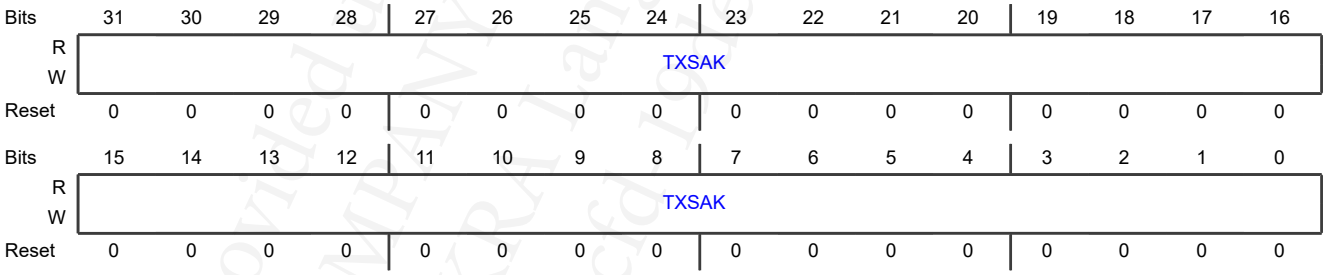
2.152.1 Offset

Register	Offset
MACSEC_TXSAKB2Q	7C4h

2.152.2 Function

The MACSEC_TXSAKB2Q register is used to configure bits 32-63 of a Security Association B key for the Secure Channel.

2.152.3 Diagram



2.152.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 32-63

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2.153 TX Security Association B key 3 of 8 (MACSEC_TXSAKB3Q)

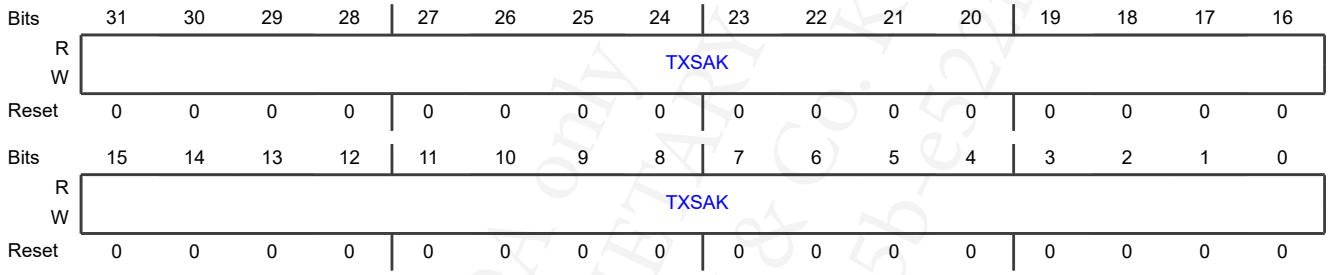
2.153.1 Offset

Register	Offset
MACSEC_TXSAKB3Q	7C8h

2.153.2 Function

The MACSEC_TXSAKB3Q register is used to configure bits 64-95 of a Security Association B key for the Secure Channel.

2.153.3 Diagram



2.153.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 64-95

2.154 TX Security Association B key 4 of 8 (MACSEC_TXSAKB4Q)

2.154.1 Offset

Register	Offset
MACSEC_TXSAKB4Q	7CCh

2.154.2 Function

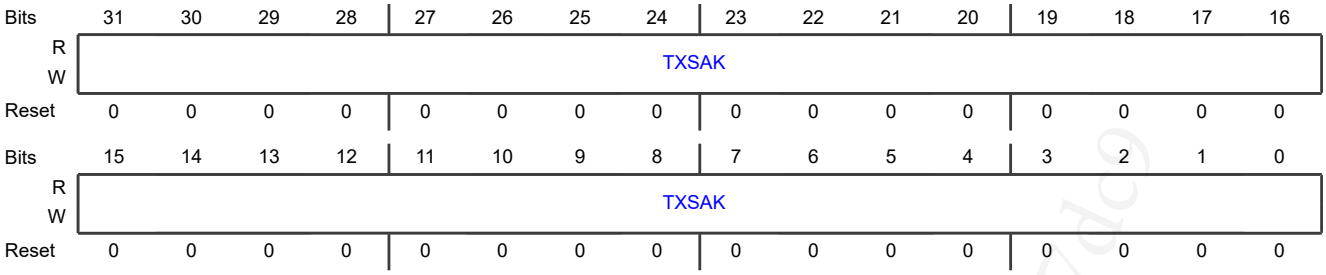
The MACSEC_TXSAKB4Q register is used to configure bits 96-127 of a Security Association B key for the Secure Channel.

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2.154.3 Diagram



2.154.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 96-127

2.155 TX Security Association B key 5 of 8 (MACSEC_TXSAKB5Q)

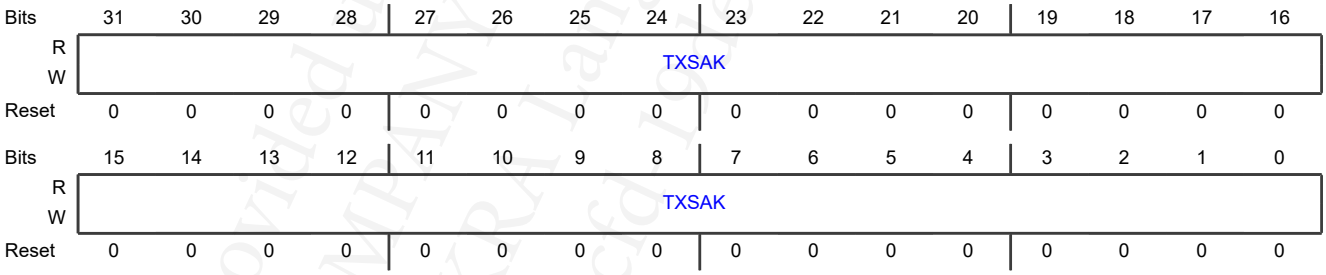
2.155.1 Offset

Register	Offset
MACSEC_TXSAKB5Q	7D0h

2.155.2 Function

The MACSEC_TXSAKB5Q register is used to configure bits 128-159 of a Security Association B key for the Secure Channel.

2.155.3 Diagram



2.155.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 128-159

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2.156 TX Security Association B key 6 of 8 (MACSEC_TXSAKB6Q)

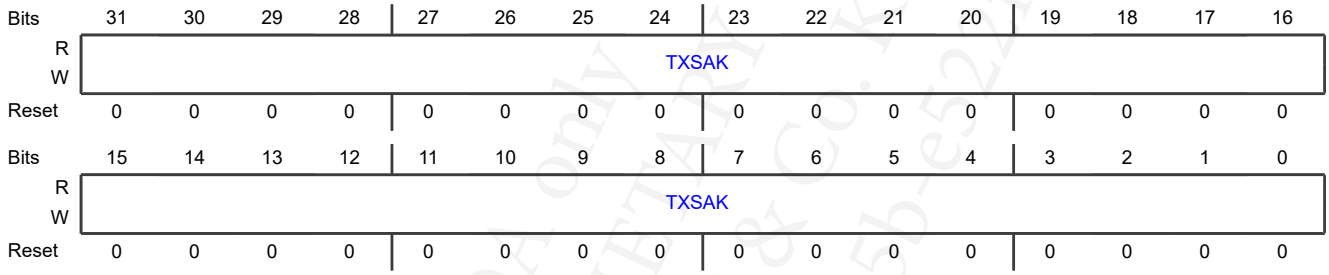
2.156.1 Offset

Register	Offset
MACSEC_TXSAKB6Q	7D4h

2.156.2 Function

The MACSEC_TXSAKB6Q register is used to configure bits 160-191 of a Security Association B key for the Secure Channel.

2.156.3 Diagram



2.156.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 160-191

2.157 TX Security Association B key 7 of 8 (MACSEC_TXSAKB7Q)

2.157.1 Offset

Register	Offset
MACSEC_TXSAKB7Q	7D8h

2.157.2 Function

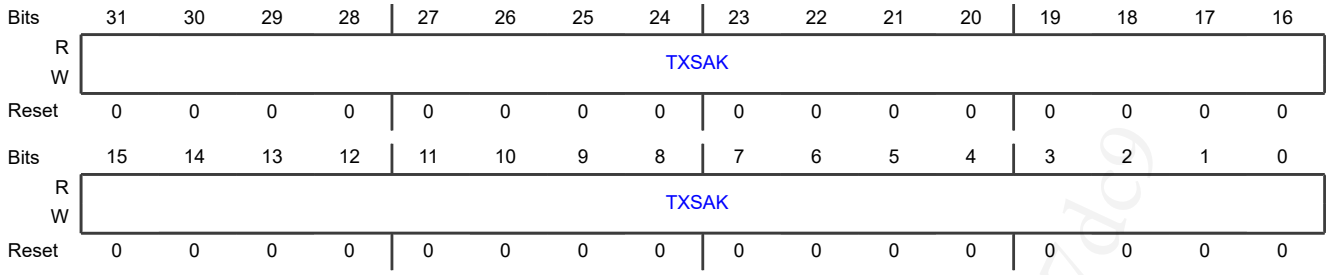
The MACSEC_TXSAKB7Q register is used to configure bits 192-223 of a Security Association B key for the Secure Channel.

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2.157.3 Diagram



2.157.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 192-223

2.158 TX Security Association B key 8 of 8 (MACSEC_TXSAKB8Q)

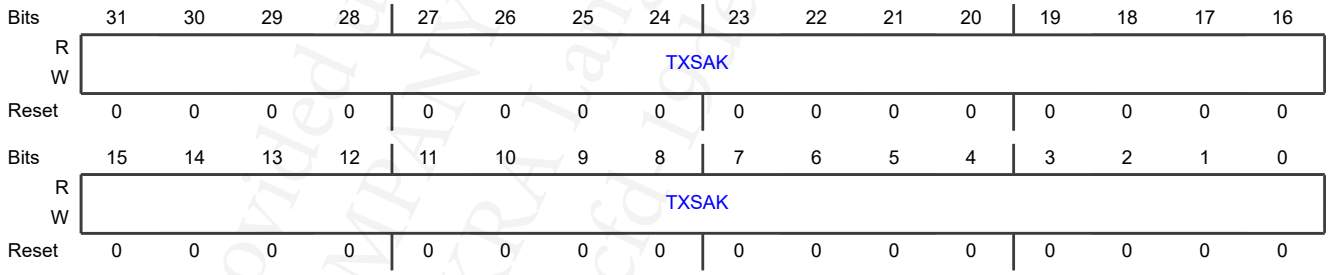
2.158.1 Offset

Register	Offset
MACSEC_TXSAKB8Q	7DCh

2.158.2 Function

The MACSEC_TXSAKB8Q register is used to configure bits 224-255 of a Security Association B key for the Secure Channel.

2.158.3 Diagram



2.158.4 Fields

Field	Function
31-0 TXSAK	TXSAK TX Security Association Key bit 224-255

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2.159 TX Security Association B Short Secure Channel Identifier (MACSEC_TXSABSSCI)

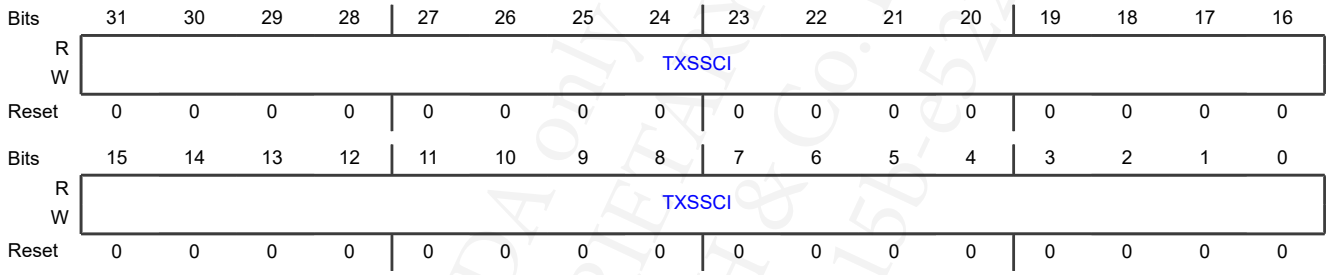
2.159.1 Offset

Register	Offset
MACSEC_TXSABSSCI	7E0h

2.159.2 Function

The MACSEC_TXSABSSCI register is used to configure SSCI of Security Association B for the Secure Channel, when an XPN Cipher Suite is configured.

2.159.3 Diagram



2.159.4 Fields

Field	Function
31-0 TXSSCI	TXSSCI TX Short Secure Channel Identifier (TXSSCI): A 32-bit value per SA to support XPN Cipher Suite for the secure channel, it is only used when XPN Cipher Suite is configured.

2.160 TX Security Association B SALT 1 of 3 (MACSEC_TXSABSALT1T)

2.160.1 Offset

Register	Offset
MACSEC_TXSABSALT1T	7E4h

2.160.2 Function

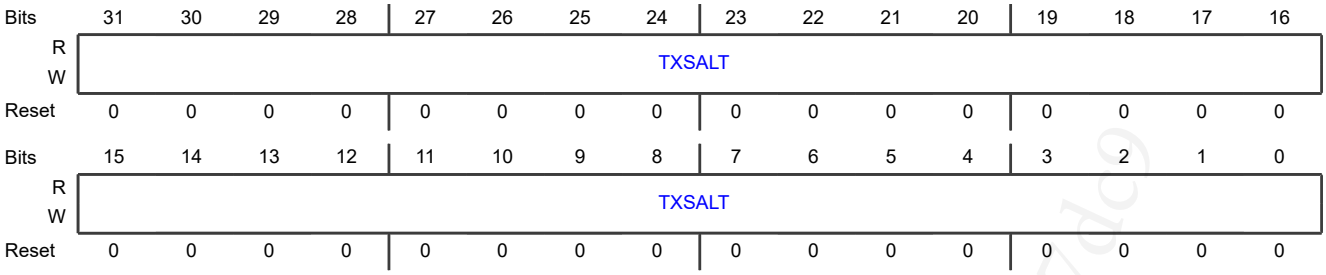
The MACSEC_TXSABSALT1T register is used to configure bits 31-0 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

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2.160.3 Diagram



2.160.4 Fields

Field	Function
31-0 TXSALT	TXSALT TX SALT bits 31-0, it is only used when XPN Cipher Suite is configured.

2.161 TX Security Association B SALT 2 of 3 (MACSEC_TXSABSALT2T)

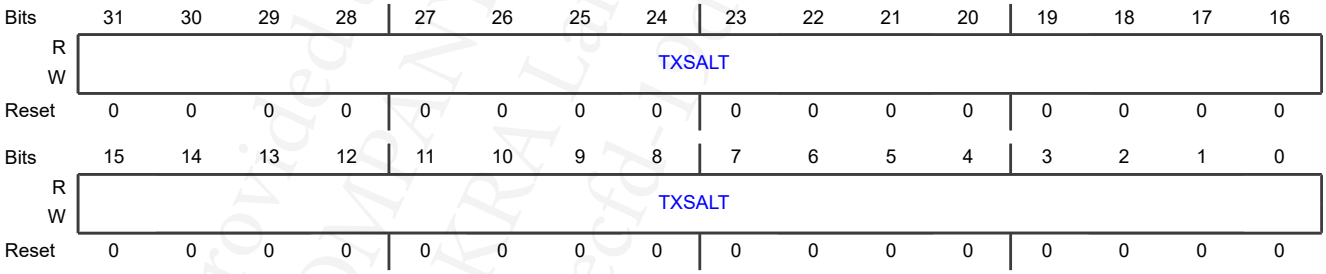
2.161.1 Offset

Register	Offset
MACSEC_TXSABSALT2T	7E8h

2.161.2 Function

The MACSEC_TXSABSALT2T register is used to configure bits 63-32 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.161.3 Diagram



2.161.4 Fields

Field	Function
31-0 TXSALT	TXSALT TX SALT bits 63-32, it is only used when XPN Cipher Suite is configured.

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2.162 TX Security Association B SALT 3 of 3 (MACSEC_TXSABSALT3T)

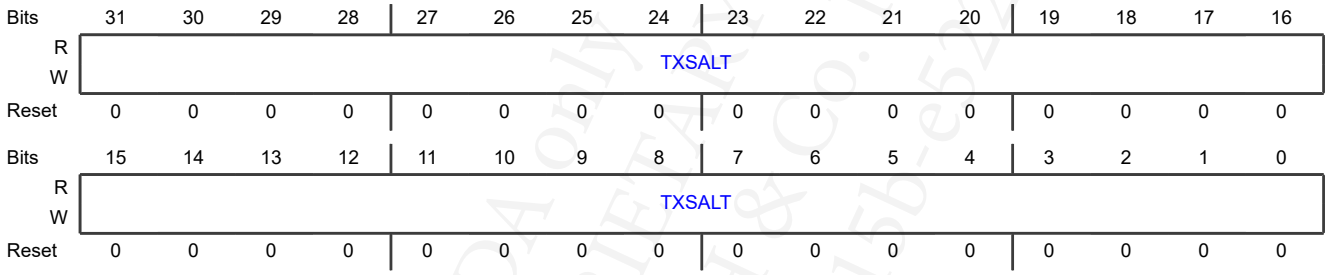
2.162.1 Offset

Register	Offset
MACSEC_TXSABSALT3T	7ECh

2.162.2 Function

The MACSEC_TXSABSALT3T register is used to configure bits 95-64 of Security Association B SALT for the Secure Channel, when an XPN Cipher Suite is configured.

2.162.3 Diagram



2.162.4 Fields

Field	Function
31-0 TXSALT	TXSALT TX SALT bits 95-64, it is only used when XPN Cipher Suite is configured.

2.163 TX SC Selection Record 0 part 0 (MACSEC_TXSCSR0P0)

2.163.1 Offset

Register	Offset
MACSEC_TXSCSR0P0	800h

2.163.2 Function

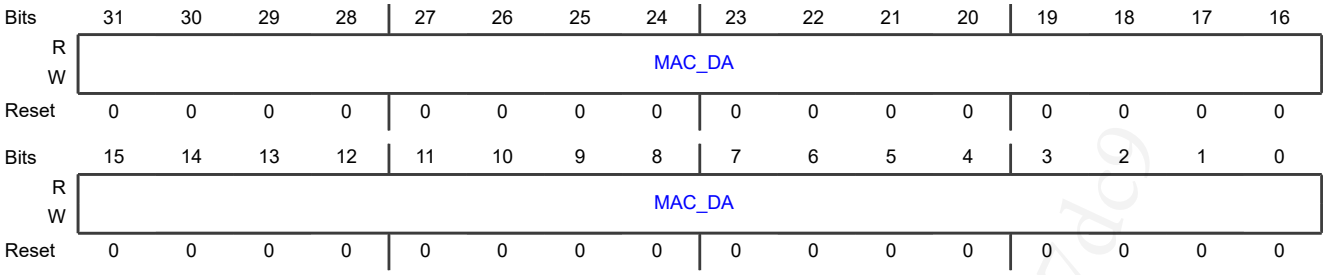
The MACSEC_TXSCSR*0*P0 register is used to configure the record *0*.

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2.163.3 Diagram



2.163.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.164 TX SC Selection Record 0 part 1 (MACSEC_TXSCSR0P1)

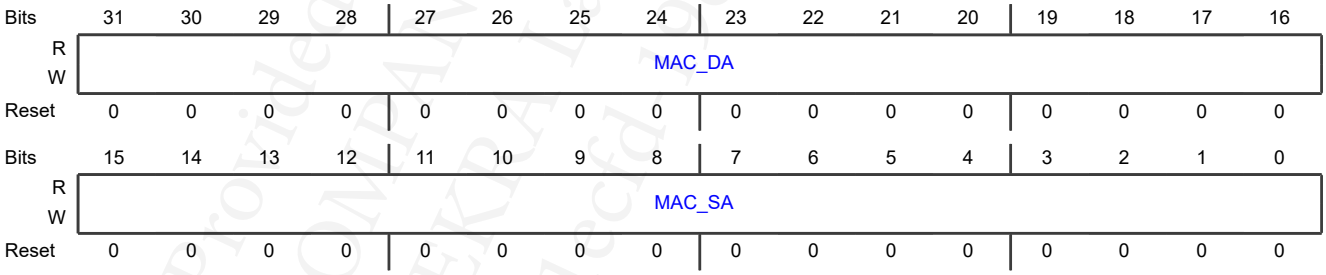
2.164.1 Offset

Register	Offset
MACSEC_TXSCSR0P1	804h

2.164.2 Function

The MACSEC_TXSCSR*0*P1 register is used to configure the record *0*.

2.164.3 Diagram



2.164.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.

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Field	Function
15-0 MAC_SA	MAC_SA MAC SA bits 0-15 (bytes 1,2) This field contains the 2 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[15:8] equals 11h and MAC_SA[7:0] equals 12h

2.165 TX SC Selection Record 0 part 2 (MACSEC_TXSCSR0P2)

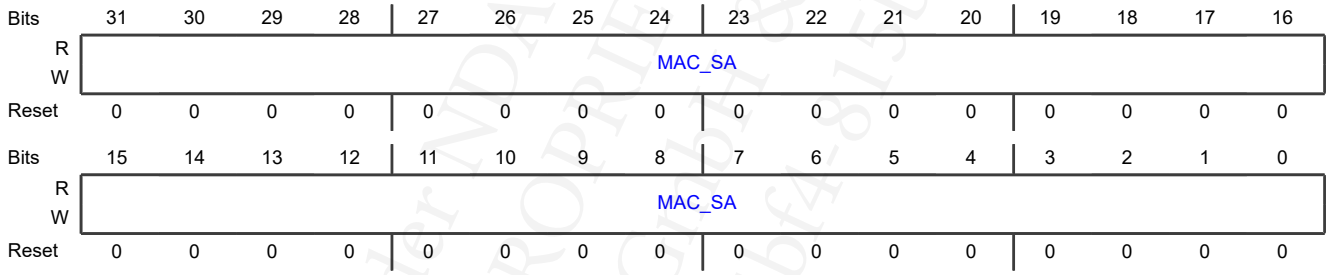
2.165.1 Offset

Register	Offset
MACSEC_TXSCSR0P2	808h

2.165.2 Function

The MACSEC_TXSCSR*0*P2 register is used to configure the record *0*.

2.165.3 Diagram



2.165.4 Fields

Field	Function
31-0 MAC_SA	MAC_SA MAC SA bit 16-47 (bytes 3,4,5,6) This field contains the 4 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[31:24] equals 13h and MAC_SA[7:0] equals 16h.

2.166 TX SC Selection Record 0 part 3 (MACSEC_TXSCSR0P3)

2.166.1 Offset

Register	Offset
MACSEC_TXSCSR0P3	80Ch

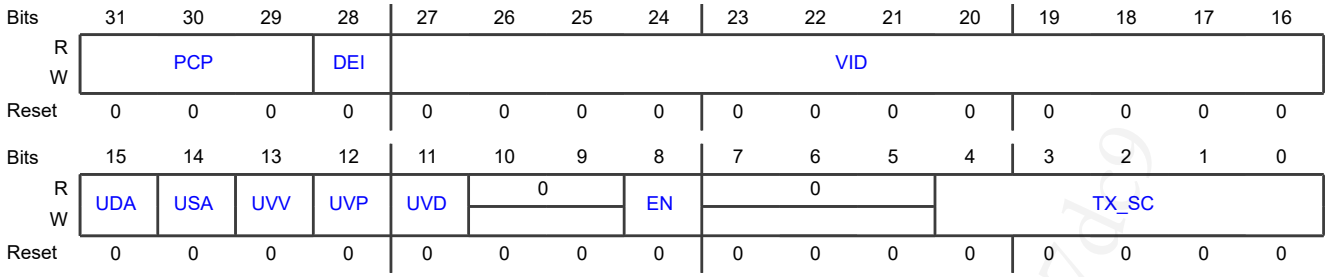
2.166.2 Function

The MACSEC_TXSCSR*0*P3 register is used to configure the record *0*.

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2.166.3 Diagram



2.166.4 Fields

Field	Function
31-29 PCP	PCP Outer VLAN PCP bits 0-2
28 DEI	DEI Outer VLAN DEI
27-16 VID	VID Outer VLAN VID bits 0-11
15 UDA	UDA Use MAC DA as part of the match 0b - MAC DA not used 1b - MAC DA specified in MACSEC_TXSCSRnP0/1 is used to match the incoming frame MAC DA
14 USA	USA Use MAC SA as part of the match compare 0b - MAC SA not used 1b - MAC SA specified in MACSEC_TXSCSRnP1/2 is used to match the incoming frame MAC SA
13 UVV	UVV Use outer VLAN VID as part of the match compare 0b - VLAN VID not used 1b - VLAN VID specified in MACSEC_TXSCSRnP3 is used to match the incoming frame outer VLAN VID. If the outer VLAN is not present, this table record will not match
12 UVP	UVP Use outer VLAN PCP as part of the match compare 0b - VLAN PCP not used 1b - VLAN PCP specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN PCP. If the outer VLAN is not present, this table record will not match
11 UVD	UVD Use outer VLAN DEI as part of the match compare 0b - VLAN DEI not used 1b - VLAN DEI specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN DEI. If the outer VLAN is not present, this table record will not match
10-9 —	Reserved, should be cleared
8 EN	EN Enable 0b - record is not enabled, will not match any MAC DA / MAC SA / VLAN

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Field	Function
	1b - record is enabled, will attempt to match MAC DA / MAC SA / VLAN as specified in MACSEC_TXSCSRn P0/1/2/3
7-5 —	Reserved, should be cleared
4-0 TX_SC	TX_SC TX SC ID If a frame matches this entry, it will be transmitted using the configuration of the SC record specified in this field.

2.167 TX SC Selection Record 1 part 0 (MACSEC_TXSCSR1P0)

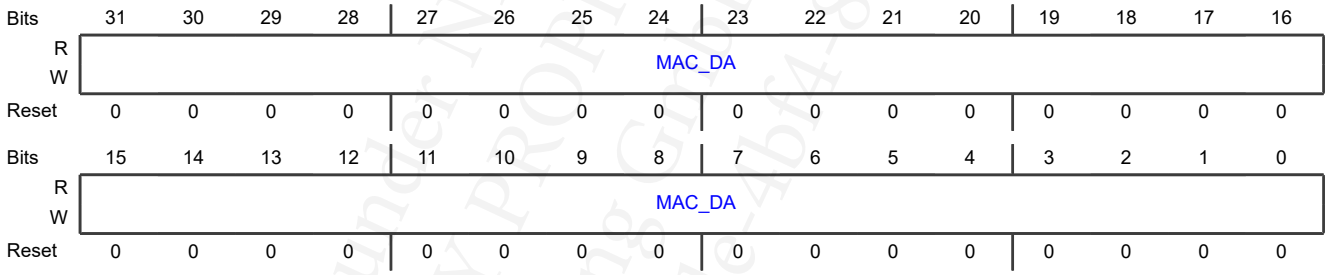
2.167.1 Offset

Register	Offset
MACSEC_TXSCSR1P0	810h

2.167.2 Function

The MACSEC_TXSCSR*1*P0 register is used to configure the record *1*.

2.167.3 Diagram



2.167.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.168 TX SC Selection Record 1 part 1 (MACSEC_TXSCSR1P1)

2.168.1 Offset

Register	Offset
MACSEC_TXSCSR1P1	814h

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391 [illegible] 19dc24b4-815b-e52f96c7dc9

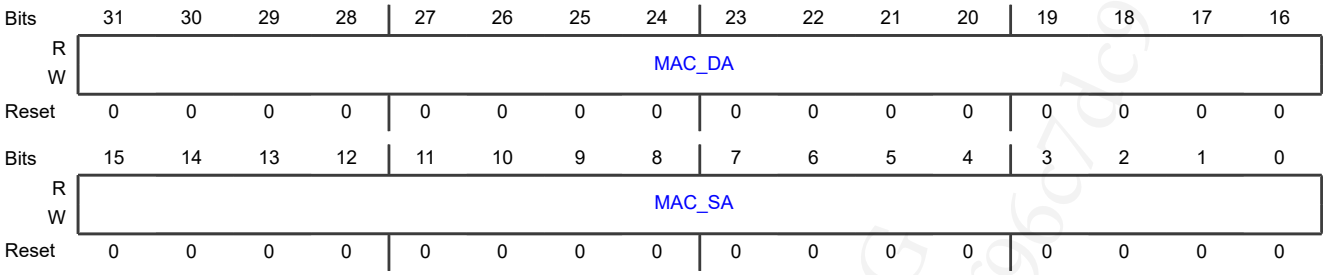
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2.168.2 Function

The MACSEC_TXSCSR*1*P1 register is used to configure the record *1*.

2.168.3 Diagram



2.168.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-0 MAC_SA	MAC_SA MAC SA bits 0-15 (bytes 1,2) This field contains the 2 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[15:8] equals 11h and MAC_SA[7:0] equals 12h

2.169 TX SC Selection Record 1 part 2 (MACSEC_TXSCSR1P2)

2.169.1 Offset

Register	Offset
MACSEC_TXSCSR1P2	818h

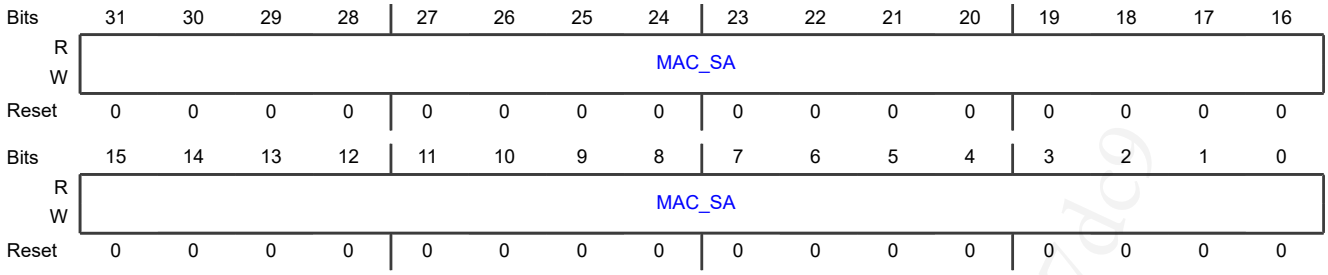
2.169.2 Function

The MACSEC_TXSCSR*1*P2 register is used to configure the record *1*.

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2.169.3 Diagram



2.169.4 Fields

Field	Function
31-0 MAC_SA	MAC_SA MAC SA bit 16-47 (bytes 3,4,5,6) This field contains the 4 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[31:24] equals 13h and MAC_SA[7:0] equals 16h.

2.170 TX SC Selection Record 1 part 3 (MACSEC_TXSCSR1P3)

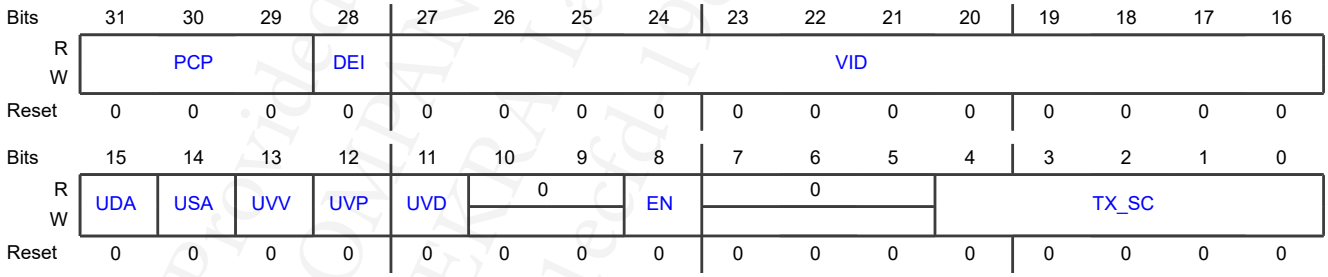
2.170.1 Offset

Register	Offset
MACSEC_TXSCSR1P3	81Ch

2.170.2 Function

The MACSEC_TXSCSR*1*P3 register is used to configure the record *1*.

2.170.3 Diagram



2.170.4 Fields

Field	Function
31-29 PCP	PCP Outer VLAN PCP bits 0-2
28 DEI	DEI Outer VLAN DEI

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Field	Function
27-16 VID	VID Outer VLAN VID bits 0-11
15 UDA	UDA Use MAC DA as part of the match 0b - MAC DA not used 1b - MAC DA specified in MACSEC_TXSCSRnP0/1 is used to match the incoming frame MAC DA
14 USA	USA Use MAC SA as part of the match compare 0b - MAC SA not used 1b - MAC SA specified in MACSEC_TXSCSRnP1/2 is used to match the incoming frame MAC SA
13 UVV	UVV Use outer VLAN VID as part of the match compare 0b - VLAN VID not used 1b - VLAN VID specified in MACSEC_TXSCSRnP3 is used to match the incoming frame outer VLAN VID. If the outer VLAN is not present, this table record will not match
12 UVP	UVP Use outer VLAN PCP as part of the match compare 0b - VLAN PCP not used 1b - VLAN PCP specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN PCP. If the outer VLAN is not present, this table record will not match
11 UVD	UVD Use outer VLAN DEI as part of the match compare 0b - VLAN DEI not used 1b - VLAN DEI specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN DEI. If the outer VLAN is not present, this table record will not match
10-9 —	Reserved, should be cleared
8 EN	EN Enable 0b - record is not enabled, will not match any MAC DA / MAC SA / VLAN 1b - record is enabled, will attempt to match MAC DA / MAC SA / VLAN as specified in MACSEC_TXSCSRn P0/1/2/3
7-5 —	Reserved, should be cleared
4-0 TX_SC	TX_SC TX SC ID If a frame matches this entry, it will be transmitted using the configuration of the SC record specified in this field.

2.171 TX SC Selection Record 2 part 0 (MACSEC_TXSCSR2P0)

2.171.1 Offset

Register	Offset
MACSEC_TXSCSR2P0	820h

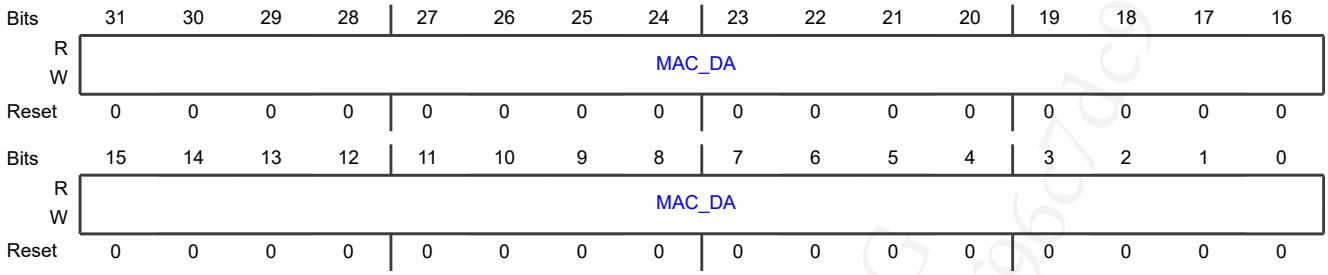
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2.171.2 Function

The MACSEC_TXSCSR2P0 register is used to configure the record *2*.

2.171.3 Diagram



2.171.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.172 TX SC Selection Record 2 part 1 (MACSEC_TXSCSR2P1)

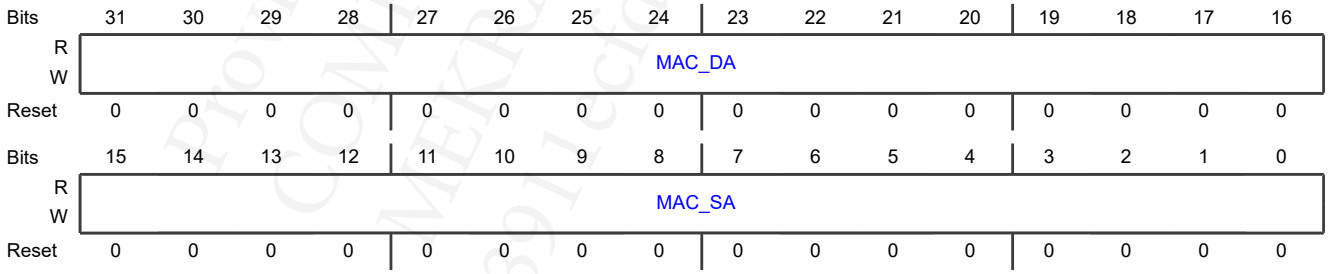
2.172.1 Offset

Register	Offset
MACSEC_TXSCSR2P1	824h

2.172.2 Function

The MACSEC_TXSCSR2P1 register is used to configure the record *2*.

2.172.3 Diagram



2.172.4 Fields

Field	Function
31-16	MAC_DA

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391ccd-1de-4bf4-8152f96d7dc9

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Field	Function
MAC_DA	MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-0 MAC_SA	MAC_SA MAC SA bits 0-15 (bytes 1,2) This field contains the 2 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[15:8] equals 11h and MAC_SA[7:0] equals 12h

2.173 TX SC Selection Record 2 part 2 (MACSEC_TXSCSR2P2)

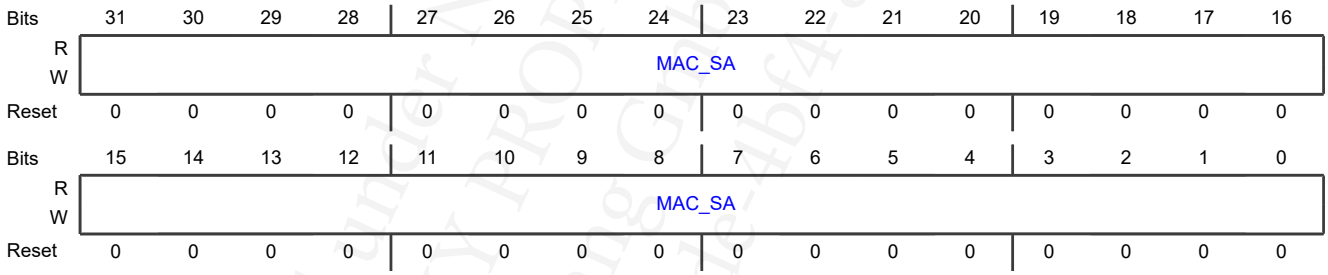
2.173.1 Offset

Register	Offset
MACSEC_TXSCSR2P2	828h

2.173.2 Function

The MACSEC_TXSCSR2P2 register is used to configure the record *2*.

2.173.3 Diagram



2.173.4 Fields

Field	Function
31-0 MAC_SA	MAC_SA MAC SA bit 16-47 (bytes 3,4,5,6) This field contains the 4 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[31:24] equals 13h and MAC_SA[7:0] equals 16h.

2.174 TX SC Selection Record 2 part 3 (MACSEC_TXSCSR2P3)

2.174.1 Offset

Register	Offset
MACSEC_TXSCSR2P3	82Ch

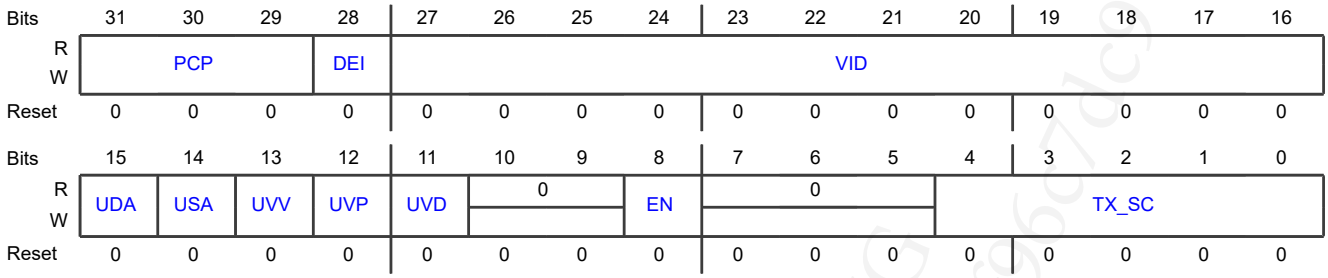
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2.174.2 Function

The MACSEC_TXSCSR2P3 register is used to configure the record *2*.

2.174.3 Diagram



2.174.4 Fields

Field	Function
31-29 PCP	PCP Outer VLAN PCP bits 0-2
28 DEI	DEI Outer VLAN DEI
27-16 VID	VID Outer VLAN VID bits 0-11
15 UDA	UDA Use MAC DA as part of the match 0b - MAC DA not used 1b - MAC DA specified in MACSEC_TXSCSRnP0/1 is used to match the incoming frame MAC DA
14 USA	USA Use MAC SA as part of the match compare 0b - MAC SA not used 1b - MAC SA specified in MACSEC_TXSCSRnP1/2 is used to match the incoming frame MAC SA
13 UVV	UVV Use outer VLAN VID as part of the match compare 0b - VLAN VID not used 1b - VLAN VID specified in MACSEC_TXSCSRnP3 is used to match the incoming frame outer VLAN VID. If the outer VLAN is not present, this table record will not match
12 UVP	UVP Use outer VLAN PCP as part of the match compare 0b - VLAN PCP not used 1b - VLAN PCP specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN PCP. If the outer VLAN is not present, this table record will not match
11 UVD	UVD Use outer VLAN DEI as part of the match compare 0b - VLAN DEI not used 1b - VLAN DEI specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN DEI. If the outer VLAN is not present, this table record will not match
10-9	Reserved, should be cleared

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Field	Function
—	
8 EN	EN Enable 0b - record is not enabled, will not match any MAC DA / MAC SA / VLAN 1b - record is enabled, will attempt to match MAC DA / MAC SA / VLAN as specified in MACSEC_TXSCSRn P0/1/2/3
7-5 —	Reserved, should be cleared
4-0 TX_SC	TX_SC TX SC ID If a frame matches this entry, it will be transmitted using the configuration of the SC record specified in this field.

2.175 TX SC Selection Record 3 part 0 (MACSEC_TXSCSR3P0)

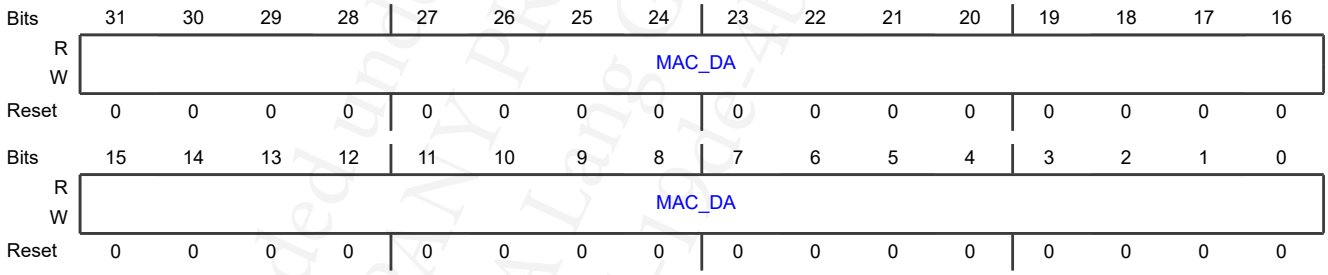
2.175.1 Offset

Register	Offset
MACSEC_TXSCSR3P0	830h

2.175.2 Function

The MACSEC_TXSCSR3P0 register is used to configure the record *3*.

2.175.3 Diagram



2.175.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

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2.176 TX SC Selection Record 3 part 1 (MACSEC_TXSCSR3P1)

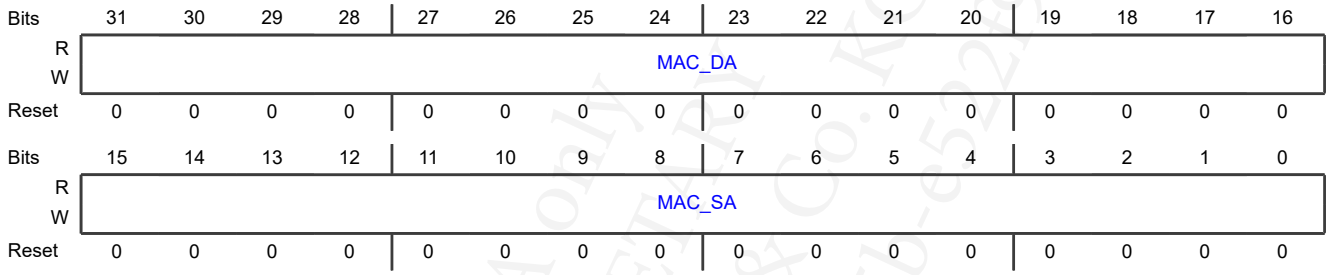
2.176.1 Offset

Register	Offset
MACSEC_TXSCSR3P1	834h

2.176.2 Function

The MACSEC_TXSCSR3P1 register is used to configure the record *3*.

2.176.3 Diagram



2.176.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-0 MAC_SA	MAC_SA MAC SA bits 0-15 (bytes 1,2) This field contains the 2 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[15:8] equals 11h and MAC_SA[7:0] equals 12h

2.177 TX SC Selection Record 3 part 2 (MACSEC_TXSCSR3P2)

2.177.1 Offset

Register	Offset
MACSEC_TXSCSR3P2	838h

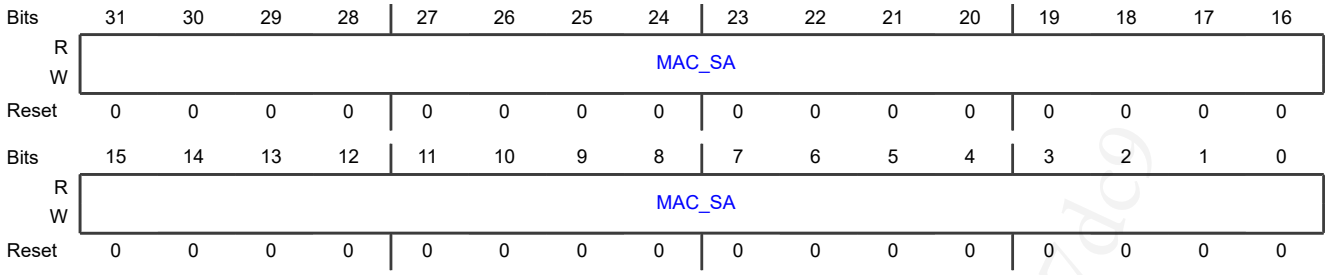
2.177.2 Function

The MACSEC_TXSCSR3P2 register is used to configure the record *3*.

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2.177.3 Diagram



2.177.4 Fields

Field	Function
31-0 MAC_SA	MAC_SA MAC SA bit 16-47 (bytes 3,4,5,6) This field contains the 4 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_SA[31:24] equals 13h and MAC_SA[7:0] equals 16h.

2.178 TX SC Selection Record 3 part 3 (MACSEC_TXSCSR3P3)

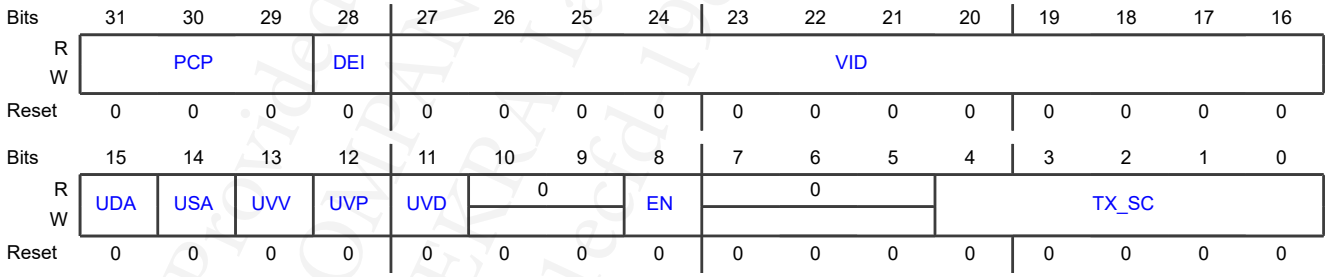
2.178.1 Offset

Register	Offset
MACSEC_TXSCSR3P3	83Ch

2.178.2 Function

The MACSEC_TXSCSR3P3 register is used to configure the record *3*.

2.178.3 Diagram



2.178.4 Fields

Field	Function
31-29 PCP	PCP Outer VLAN PCP bits 0-2
28 DEI	DEI Outer VLAN DEI

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Field	Function
27-16 VID	VID Outer VLAN VID bits 0-11
15 UDA	UDA Use MAC DA as part of the match 0b - MAC DA not used 1b - MAC DA specified in MACSEC_TXSCSRnP0/1 is used to match the incoming frame MAC DA
14 USA	USA Use MAC SA as part of the match compare 0b - MAC SA not used 1b - MAC SA specified in MACSEC_TXSCSRnP1/2 is used to match the incoming frame MAC SA
13 UVV	UVV Use outer VLAN VID as part of the match compare 0b - VLAN VID not used 1b - VLAN VID specified in MACSEC_TXSCSRnP3 is used to match the incoming frame outer VLAN VID. If the outer VLAN is not present, this table record will not match
12 UVP	UVP Use outer VLAN PCP as part of the match compare 0b - VLAN PCP not used 1b - VLAN PCP specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN PCP. If the outer VLAN is not present, this table record will not match
11 UVD	UVD Use outer VLAN DEI as part of the match compare 0b - VLAN DEI not used 1b - VLAN DEI specified in MACSEC_TXSCSRn P3 is used to match the incoming frame outer VLAN DEI. If the outer VLAN is not present, this table record will not match
10-9 —	Reserved, should be cleared
8 EN	EN Enable 0b - record is not enabled, will not match any MAC DA / MAC SA / VLAN 1b - record is enabled, will attempt to match MAC DA / MAC SA / VLAN as specified in MACSEC_TXSCSRn P0/1/2/3
7-5 —	Reserved, should be cleared
4-0 TX_SC	TX_SC TX SC ID If a frame matches this entry, it will be transmitted using the configuration of the SC record specified in this field.

2.179 Uncontrolled Port filter Record 0 data part 0 (MACSEC_UPFR0D0)

2.179.1 Offset

Register	Offset
MACSEC_UPFR0D0	A00h

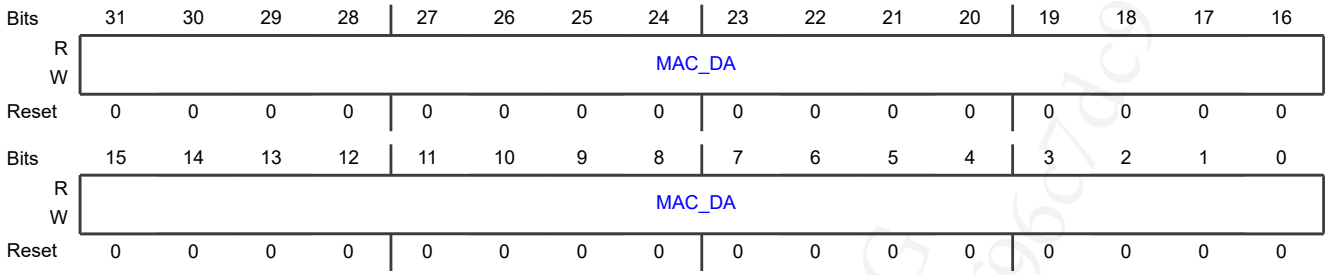
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2.179.2 Function

The MACSEC_UPFR*0*D0 register is used to configure the record *0* data.

2.179.3 Diagram



2.179.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.180 Uncontrolled Port filter Record 0 data part 1 (MACSEC_UPFR0D1)

2.180.1 Offset

Register	Offset
MACSEC_UPFR0D1	A04h

2.180.2 Function

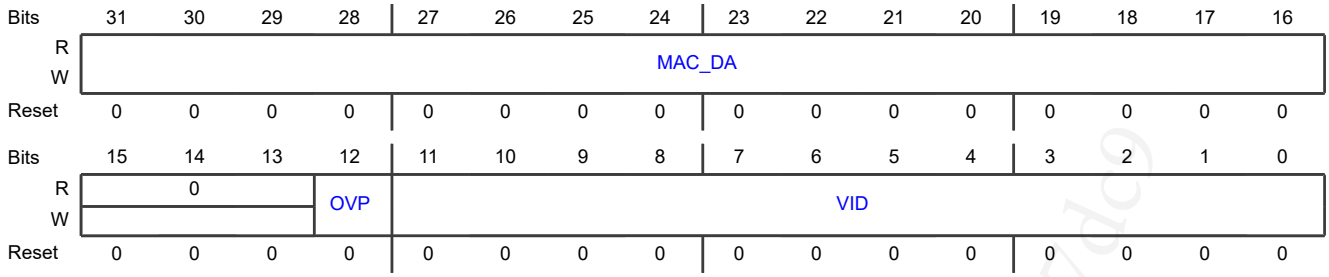
The MACSEC_UPFR*0*D1 register is used to configure the record *0*data.

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2.180.3 Diagram



2.180.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.181 Uncontrolled Port filter Record 0 data part 2 (MACSEC_UPFR0D2)

2.181.1 Offset

Register	Offset
MACSEC_UPFR0D2	A08h

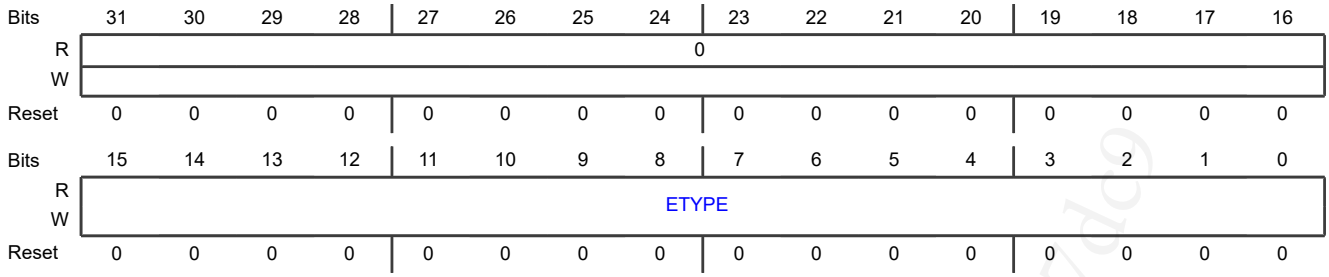
2.181.2 Function

The MACSEC_UPFR*0*D2 register is used to configure the record *0* data.

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2.181.3 Diagram



2.181.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.182 Uncontrolled Port filter Record 0 mask part 0 (MACSEC_UPFR0M0)

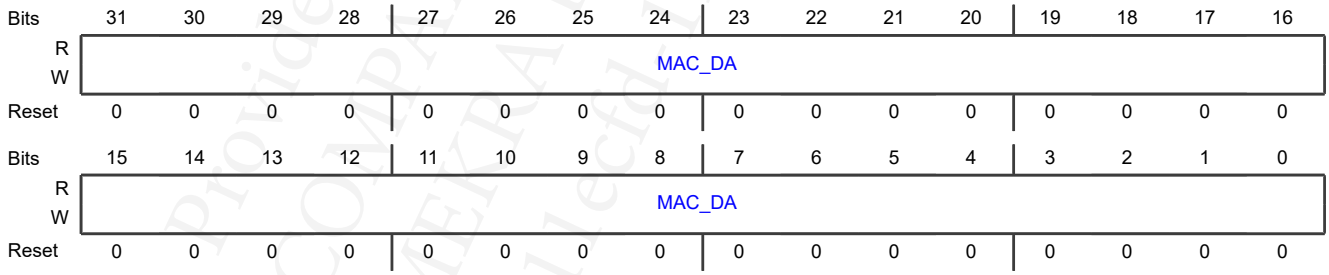
2.182.1 Offset

Register	Offset
MACSEC_UPFR0M0	A0Ch

2.182.2 Function

The MACSEC_UPFR*0*M0 register is used to configure the record *0* mask.

2.182.3 Diagram



2.182.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

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2.183 Uncontrolled Port filter Record 0 mask part 1 (MACSEC_UPFR0M1)

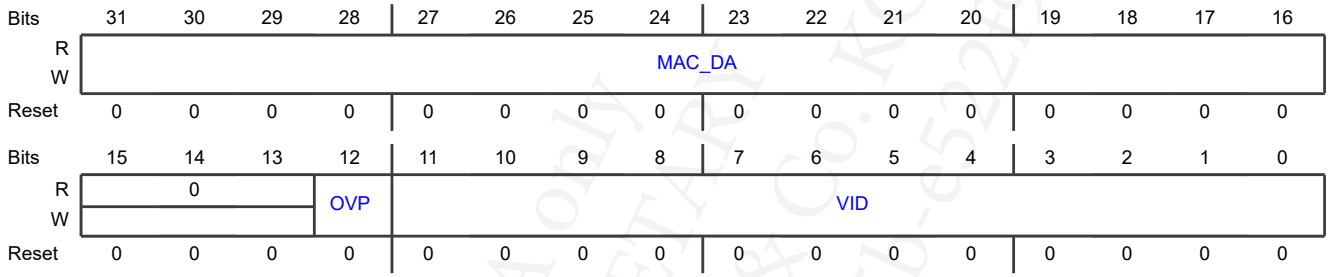
2.183.1 Offset

Register	Offset
MACSEC_UPFR0M1	A10h

2.183.2 Function

The MACSEC_UPFR*0*M1 register is used to configure the record *0* mask.

2.183.3 Diagram



2.183.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.184 Uncontrolled Port filter Record 0 mask part 2 (MACSEC_UPFR0M2)

2.184.1 Offset

Register	Offset
MACSEC_UPFR0M2	A14h

2.184.2 Function

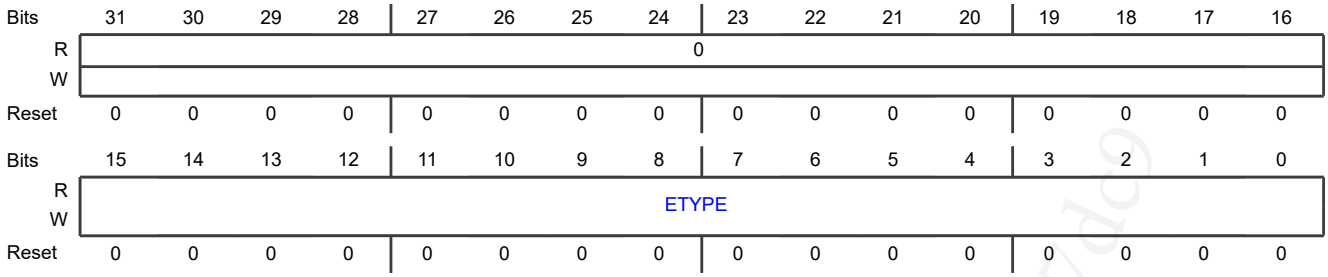
The MACSEC_UPFR*0*M2 register is used to configure the record *0* mask.

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2.184.3 Diagram



2.184.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.185 Uncontrolled Port filter Record 0 Result (MACSEC_UPFR0R)

2.185.1 Offset

Register	Offset
MACSEC_UPFR0R	A18h

2.185.2 Function

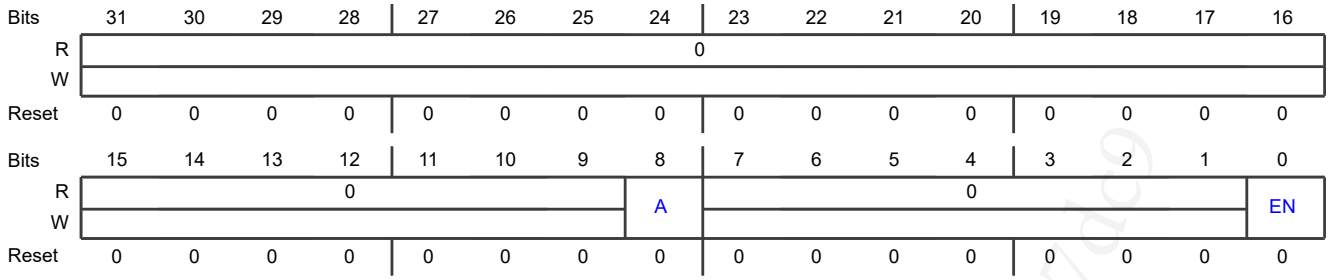
The MACSEC_UPFR0R register is used to configure the record 0 result.

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2.185.3 Diagram



2.185.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

2.186 Uncontrolled Port filter Record 1 data part 0 (MACSEC_UPFR1D0)

2.186.1 Offset

Register	Offset
MACSEC_UPFR1D0	A20h

2.186.2 Function

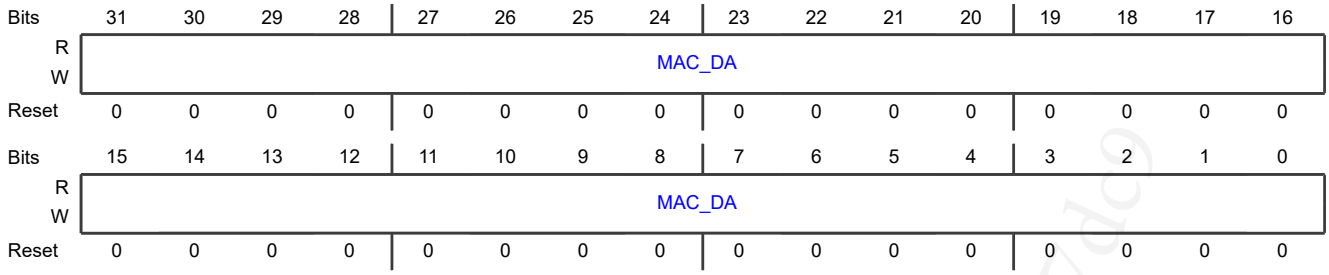
The MACSEC_UPFR*1*D0 register is used to configure the record *1* data.

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2.186.3 Diagram



2.186.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.187 Uncontrolled Port filter Record 1 data part 1 (MACSEC_UPFR1D1)

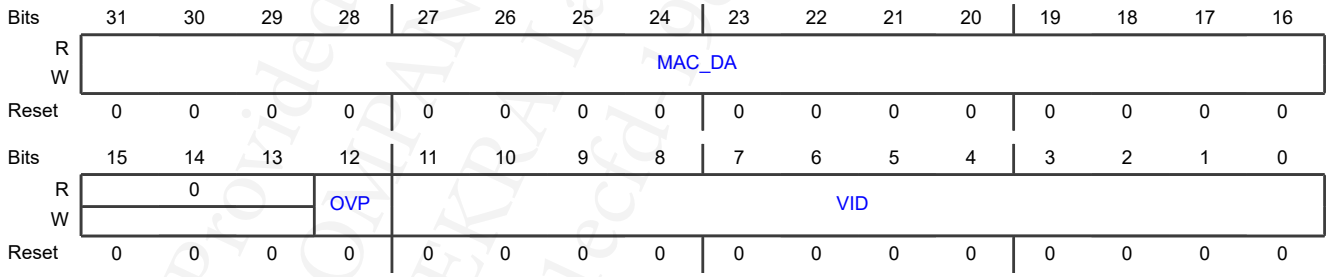
2.187.1 Offset

Register	Offset
MACSEC_UPFR1D1	A24h

2.187.2 Function

The MACSEC_UPFR*1*D1 register is used to configure the record *1*data.

2.187.3 Diagram



2.187.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.188 Uncontrolled Port filter Record 1 data part 2 (MACSEC_UPFR1D2)

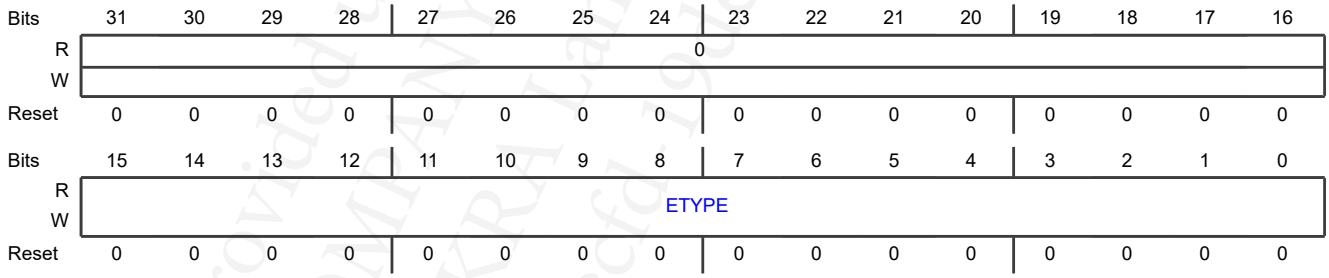
2.188.1 Offset

Register	Offset
MACSEC_UPFR1D2	A28h

2.188.2 Function

The MACSEC_UPFR*1*D2 register is used to configure the record *1* data.

2.188.3 Diagram



2.188.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.189 Uncontrolled Port filter Record 1 mask part 0 (MACSEC_UPFR1M0)

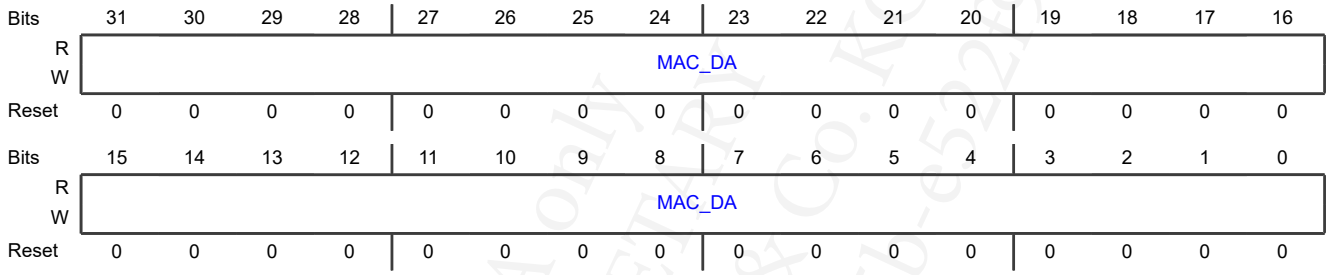
2.189.1 Offset

Register	Offset
MACSEC_UPFR1M0	A2Ch

2.189.2 Function

The MACSEC_UPFR*1*M0 register is used to configure the record *1* mask.

2.189.3 Diagram



2.189.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.190 Uncontrolled Port filter Record 1 mask part 1 (MACSEC_UPFR1M1)

2.190.1 Offset

Register	Offset
MACSEC_UPFR1M1	A30h

2.190.2 Function

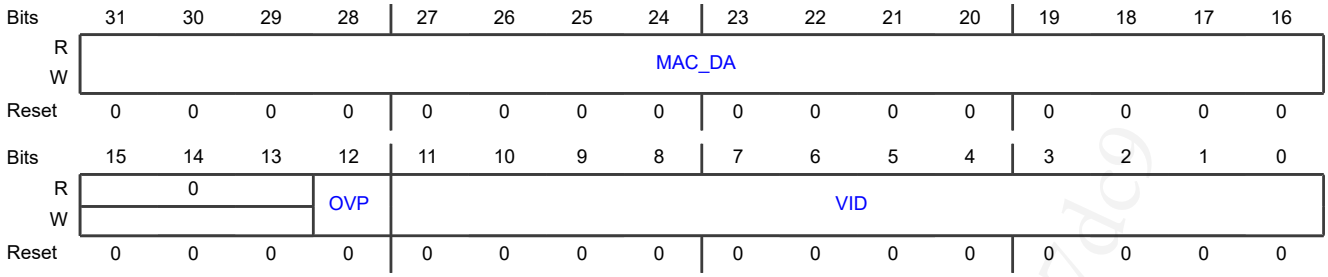
The MACSEC_UPFR*1*M1 register is used to configure the record *1* mask.

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2.190.3 Diagram



2.190.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.191 Uncontrolled Port filter Record 1 mask part 2 (MACSEC_UPFR1M2)

2.191.1 Offset

Register	Offset
MACSEC_UPFR1M2	A34h

2.191.2 Function

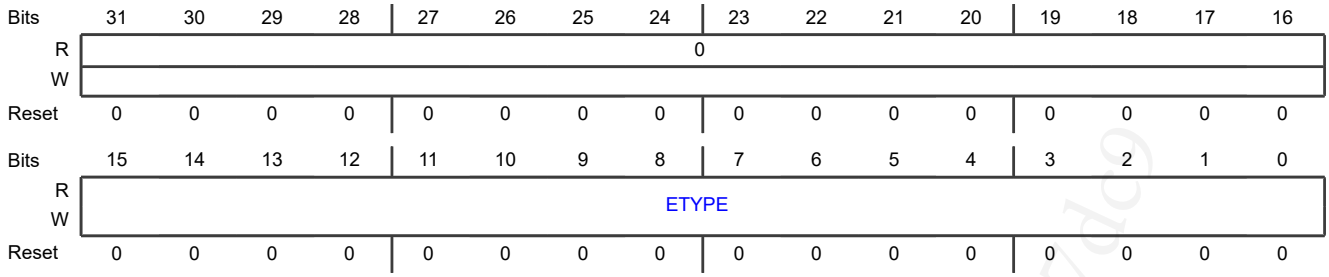
The MACSEC_UPFR*1*M2 register is used to configure the record *1* mask.

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2.191.3 Diagram



2.191.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.192 Uncontrolled Port filter Record 1 Result (MACSEC_UPFR1R)

2.192.1 Offset

Register	Offset
MACSEC_UPFR1R	A38h

2.192.2 Function

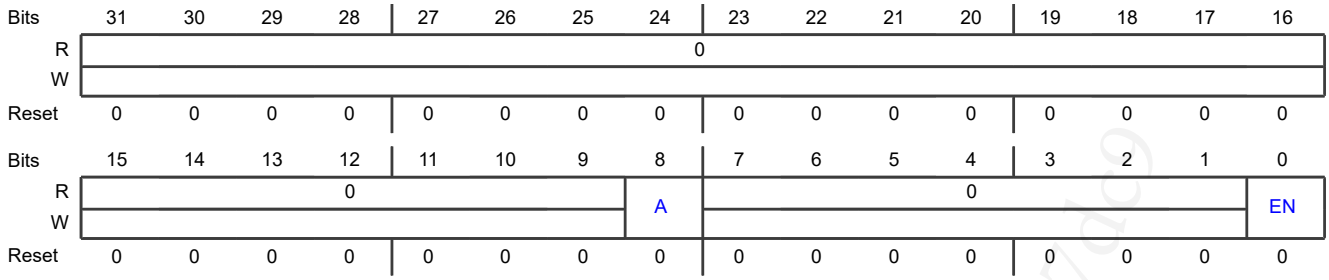
The MACSEC_UPFR1R register is used to configure the record 1 result.

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2.192.3 Diagram



2.192.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

2.193 Uncontrolled Port filter Record 2 data part 0 (MACSEC_UPFR2D0)

2.193.1 Offset

Register	Offset
MACSEC_UPFR2D0	A40h

2.193.2 Function

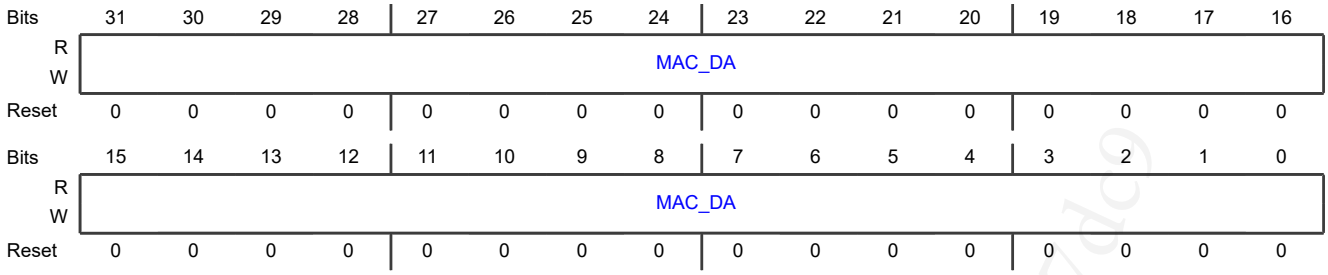
The MACSEC_UPFR2D0 register is used to configure the record *2* data.

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2.193.3 Diagram



2.193.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.194 Uncontrolled Port filter Record 2 data part 1 (MACSEC_UPFR2D1)

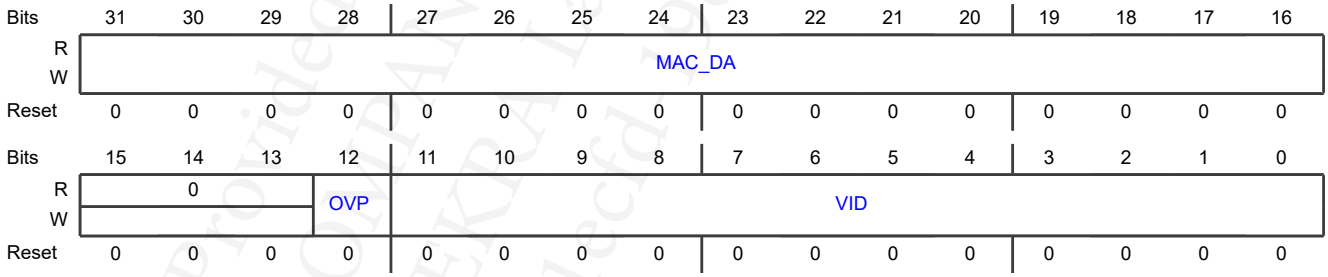
2.194.1 Offset

Register	Offset
MACSEC_UPFR2D1	A44h

2.194.2 Function

The MACSEC_UPFR2D1 register is used to configure the record 2data.

2.194.3 Diagram



2.194.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.195 Uncontrolled Port filter Record 2 data part 2 (MACSEC_UPFR2D2)

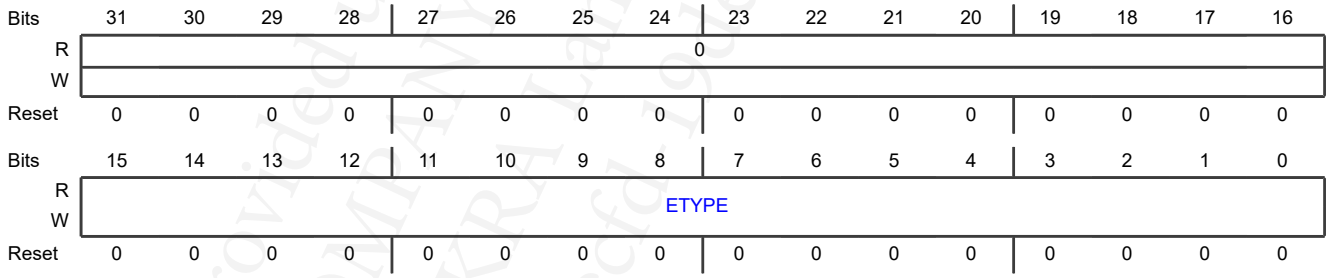
2.195.1 Offset

Register	Offset
MACSEC_UPFR2D2	A48h

2.195.2 Function

The MACSEC_UPFR2D2 register is used to configure the record *2* data.

2.195.3 Diagram



2.195.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.196 Uncontrolled Port filter Record 2 mask part 0 (MACSEC_UPFR2M0)

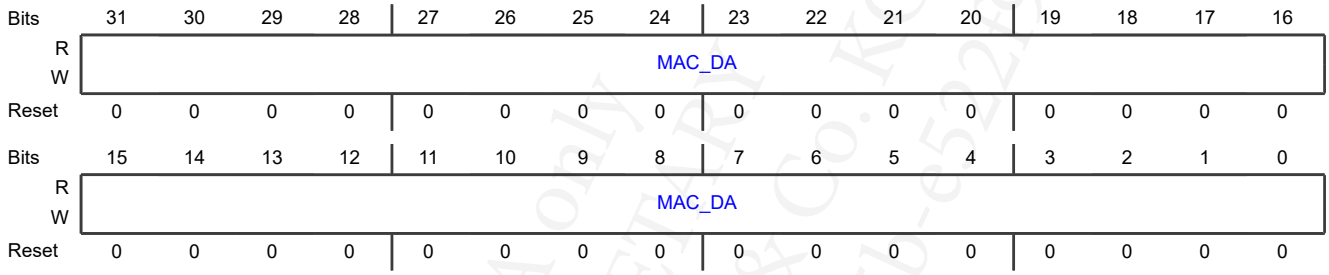
2.196.1 Offset

Register	Offset
MACSEC_UPFR2M0	A4Ch

2.196.2 Function

The MACSEC_UPFR2M0 register is used to configure the record *2* mask.

2.196.3 Diagram



2.196.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.197 Uncontrolled Port filter Record 2 mask part 1 (MACSEC_UPFR2M1)

2.197.1 Offset

Register	Offset
MACSEC_UPFR2M1	A50h

2.197.2 Function

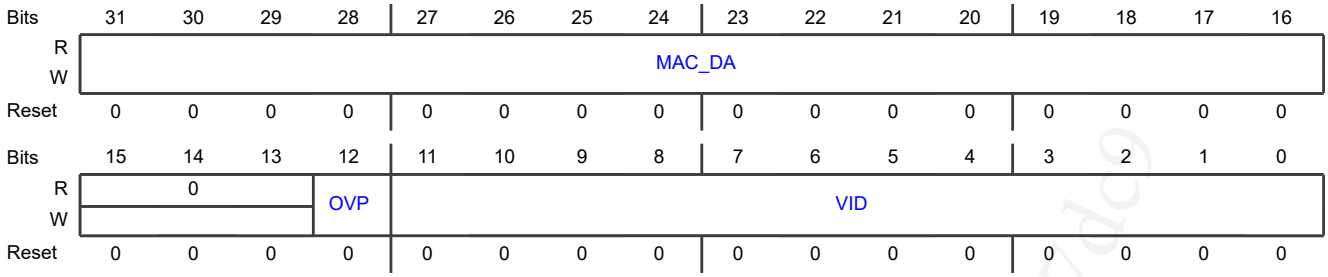
The MACSEC_UPFR2M1 register is used to configure the record *2* mask.

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2.197.3 Diagram



2.197.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.198 Uncontrolled Port filter Record 2 mask part 2 (MACSEC_UPFR2M2)

2.198.1 Offset

Register	Offset
MACSEC_UPFR2M2	A54h

2.198.2 Function

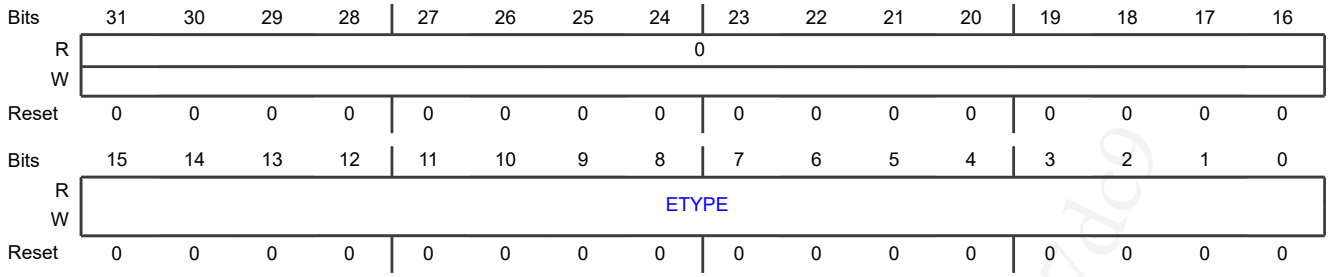
The MACSEC_UPFR2M2 register is used to configure the record *2* mask.

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2.198.3 Diagram



2.198.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.199 Uncontrolled Port filter Record 2 Result (MACSEC_UPFR2R)

2.199.1 Offset

Register	Offset
MACSEC_UPFR2R	A58h

2.199.2 Function

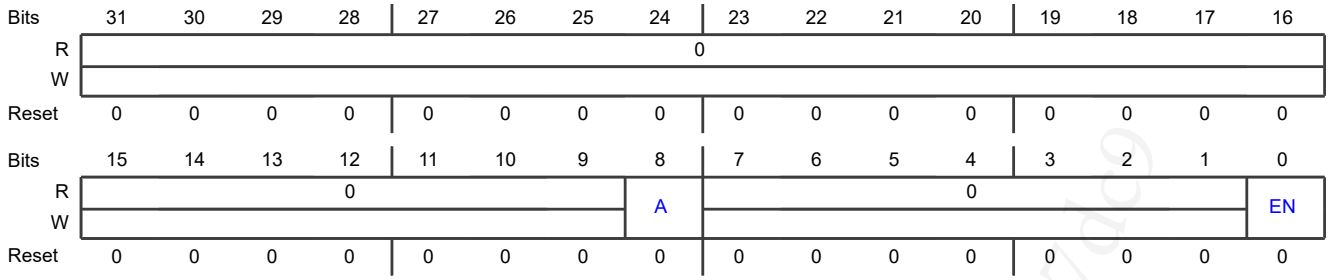
The MACSEC_UPFR2R register is used to configure the record 2 result.

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2.199.3 Diagram



2.199.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

2.200 Uncontrolled Port filter Record 3 data part 0 (MACSEC_UPFR3D0)

2.200.1 Offset

Register	Offset
MACSEC_UPFR3D0	A60h

2.200.2 Function

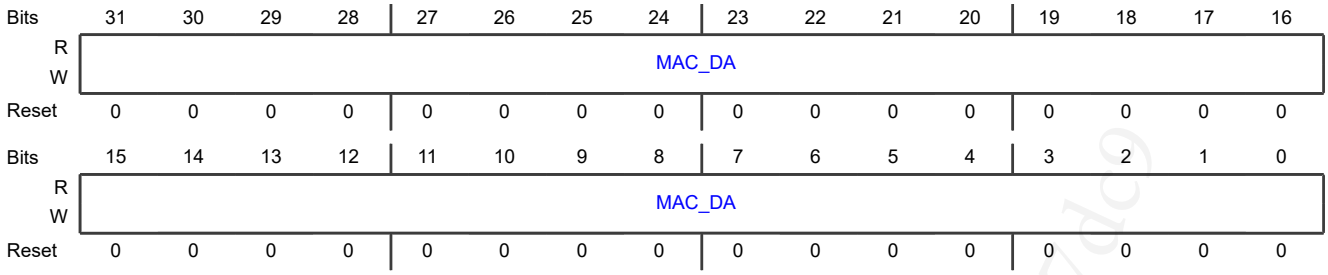
The MACSEC_UPFR3D0 register is used to configure the record *3* data.

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2.200.3 Diagram



2.200.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.201 Uncontrolled Port filter Record 3 data part 1 (MACSEC_UPFR3D1)

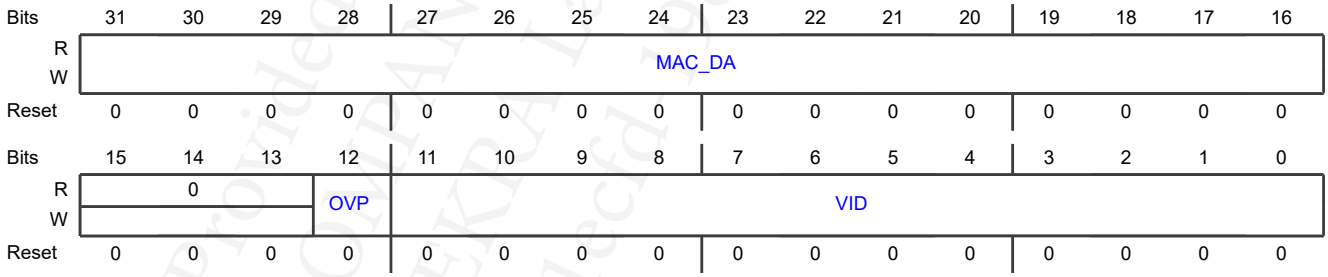
2.201.1 Offset

Register	Offset
MACSEC_UPFR3D1	A64h

2.201.2 Function

The MACSEC_UPFR*3*D1 register is used to configure the record *3*data.

2.201.3 Diagram



2.201.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.202 Uncontrolled Port filter Record 3 data part 2 (MACSEC_UPFR3D2)

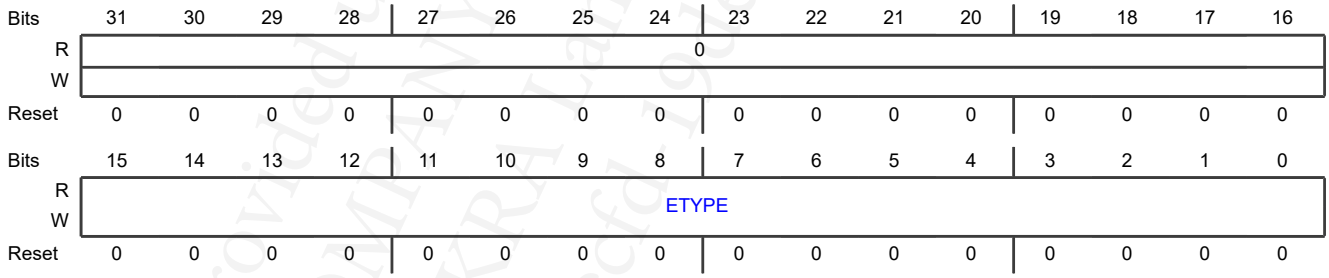
2.202.1 Offset

Register	Offset
MACSEC_UPFR3D2	A68h

2.202.2 Function

The MACSEC_UPFR3D2 register is used to configure the record *3* data.

2.202.3 Diagram



2.202.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.203 Uncontrolled Port filter Record 3 mask part 0 (MACSEC_UPFR3M0)

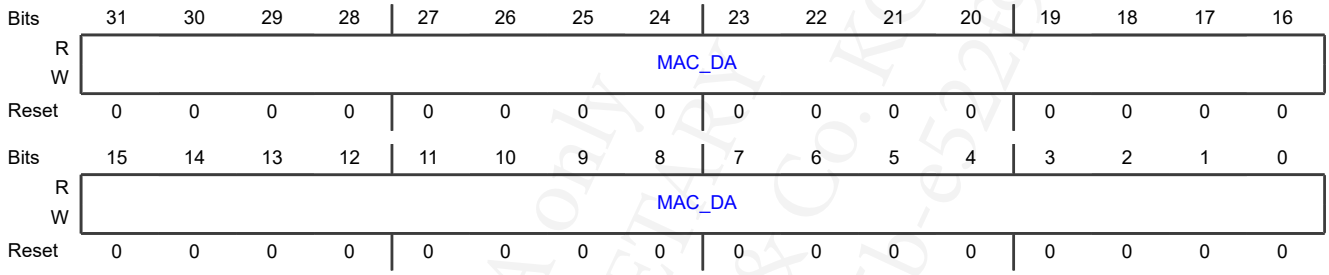
2.203.1 Offset

Register	Offset
MACSEC_UPFR3M0	A6Ch

2.203.2 Function

The MACSEC_UPFR*3*M0 register is used to configure the record *3* mask.

2.203.3 Diagram



2.203.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.204 Uncontrolled Port filter Record 3 mask part 1 (MACSEC_UPFR3M1)

2.204.1 Offset

Register	Offset
MACSEC_UPFR3M1	A70h

2.204.2 Function

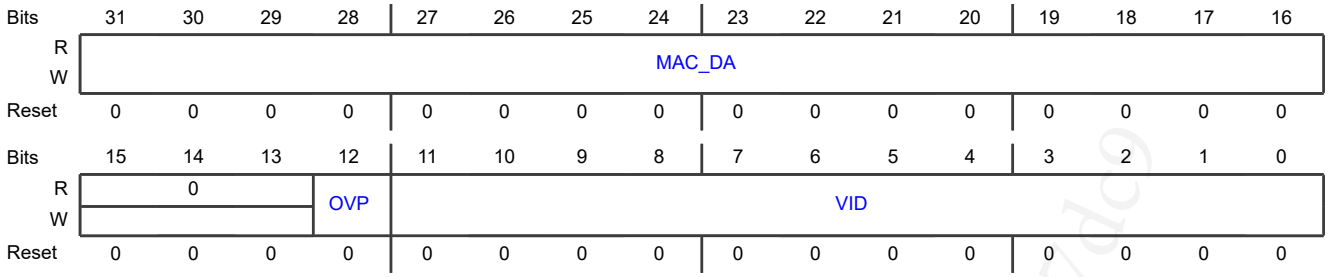
The MACSEC_UPFR*3*M1 register is used to configure the record *3* mask.

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2.204.3 Diagram



2.204.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.205 Uncontrolled Port filter Record 3 mask part 2 (MACSEC_UPFR3M2)

2.205.1 Offset

Register	Offset
MACSEC_UPFR3M2	A74h

2.205.2 Function

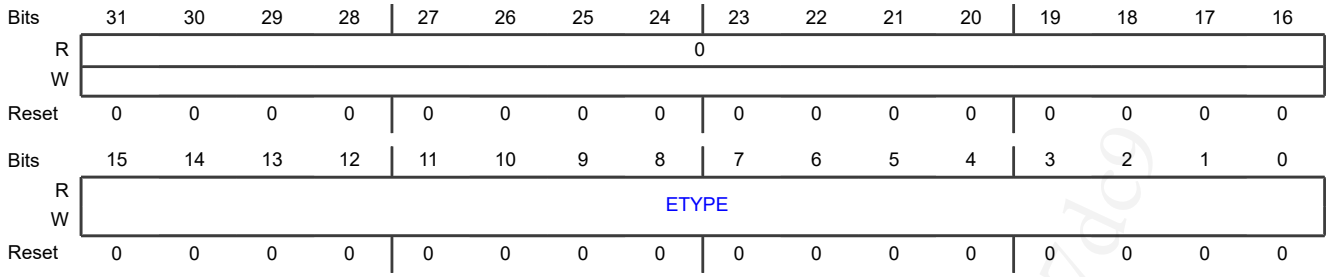
The MACSEC_UPFR3M2 register is used to configure the record *3* mask.

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2.205.3 Diagram



2.205.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.206 Uncontrolled Port filter Record 3 Result (MACSEC_UPFR3R)

2.206.1 Offset

Register	Offset
MACSEC_UPFR3R	A78h

2.206.2 Function

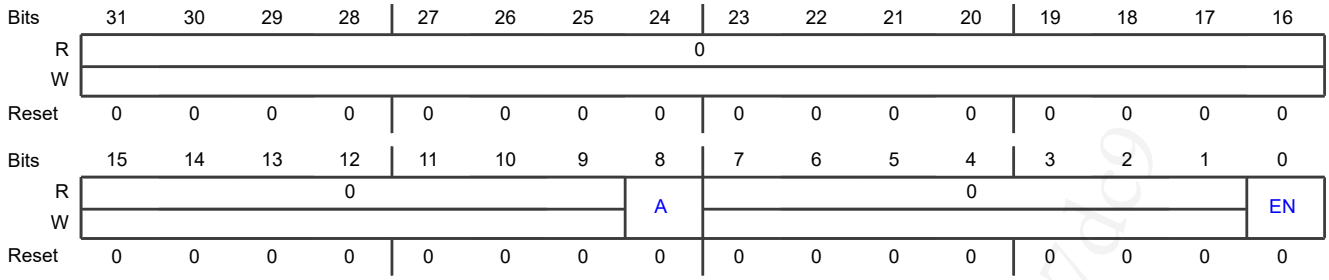
The MACSEC_UPFR3R register is used to configure the record 3 result.

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2.206.3 Diagram



2.206.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

2.207 Uncontrolled Port filter Record 4 data part 0 (MACSEC_UPFR4D0)

2.207.1 Offset

Register	Offset
MACSEC_UPFR4D0	A80h

2.207.2 Function

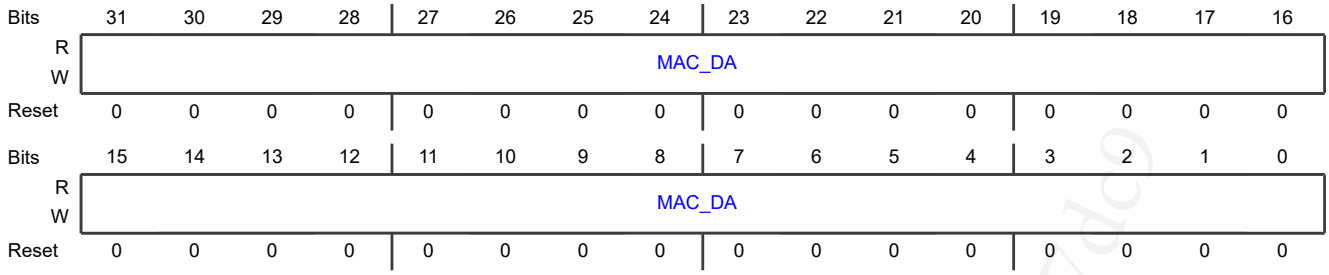
The MACSEC_UPFR*4*D0 register is used to configure the record *4* data.

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2.207.3 Diagram



2.207.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.208 Uncontrolled Port filter Record 4 data part 1 (MACSEC_UPFR4D1)

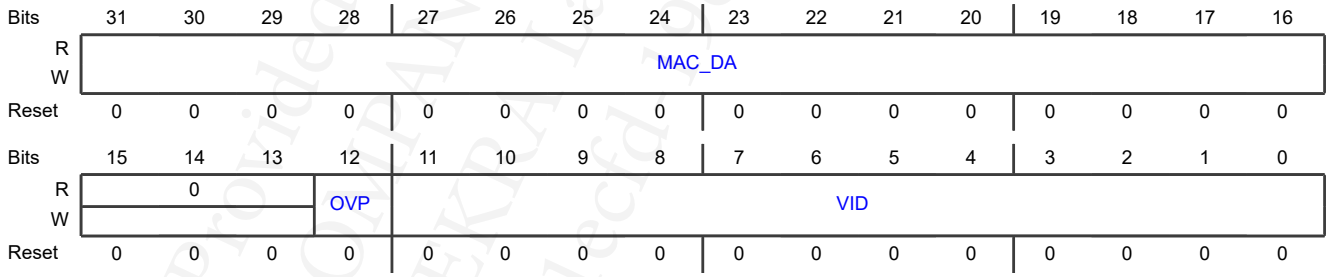
2.208.1 Offset

Register	Offset
MACSEC_UPFR4D1	A84h

2.208.2 Function

The MACSEC_UPFR*4*D1 register is used to configure the record *4*data.

2.208.3 Diagram



2.208.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.209 Uncontrolled Port filter Record 4 data part 2 (MACSEC_UPFR4D2)

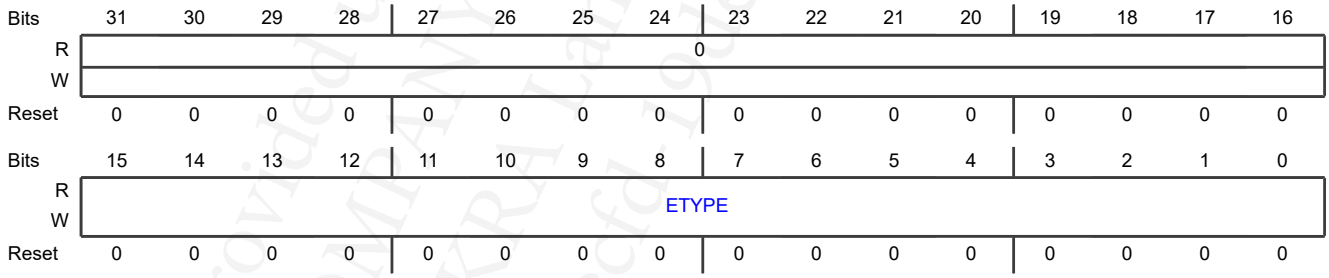
2.209.1 Offset

Register	Offset
MACSEC_UPFR4D2	A88h

2.209.2 Function

The MACSEC_UPFR*4*D2 register is used to configure the record *4* data.

2.209.3 Diagram



2.209.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.210 Uncontrolled Port filter Record 4 mask part 0 (MACSEC_UPFR4M0)

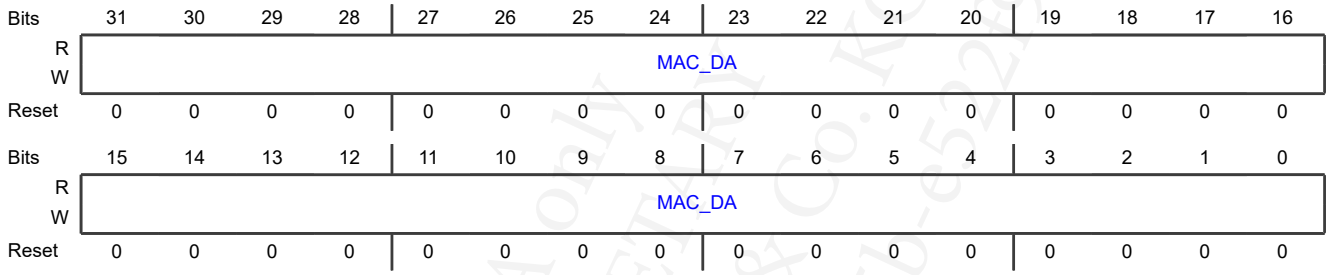
2.210.1 Offset

Register	Offset
MACSEC_UPFR4M0	A8Ch

2.210.2 Function

The MACSEC_UPFR*4*M0 register is used to configure the record *4* mask.

2.210.3 Diagram



2.210.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.211 Uncontrolled Port filter Record 4 mask part 1 (MACSEC_UPFR4M1)

2.211.1 Offset

Register	Offset
MACSEC_UPFR4M1	A90h

2.211.2 Function

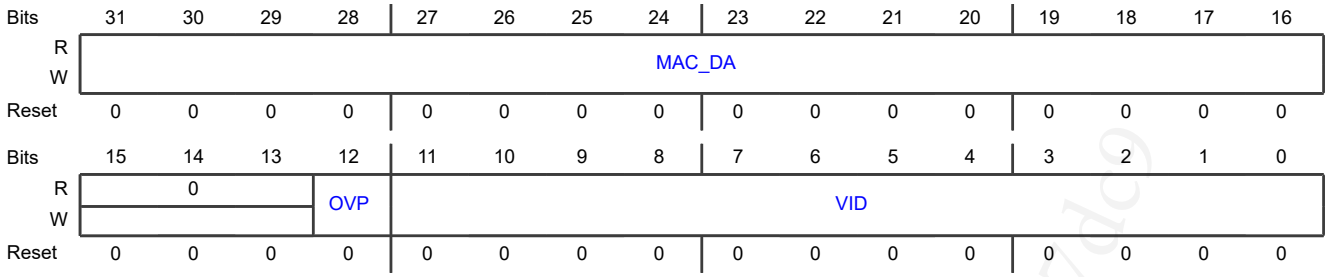
The MACSEC_UPFR*4*M1 register is used to configure the record *4* mask.

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2.211.3 Diagram



2.211.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.212 Uncontrolled Port filter Record 4 mask part 2 (MACSEC_UPFR4M2)

2.212.1 Offset

Register	Offset
MACSEC_UPFR4M2	A94h

2.212.2 Function

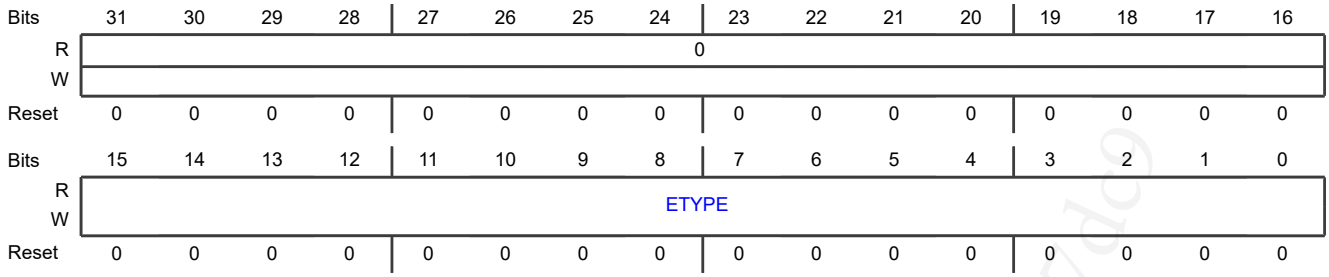
The MACSEC_UPFR*4*M2 register is used to configure the record *4* mask.

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2.212.3 Diagram



2.212.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.213 Uncontrolled Port filter Record 4 Result (MACSEC_UPFR4R)

2.213.1 Offset

Register	Offset
MACSEC_UPFR4R	A98h

2.213.2 Function

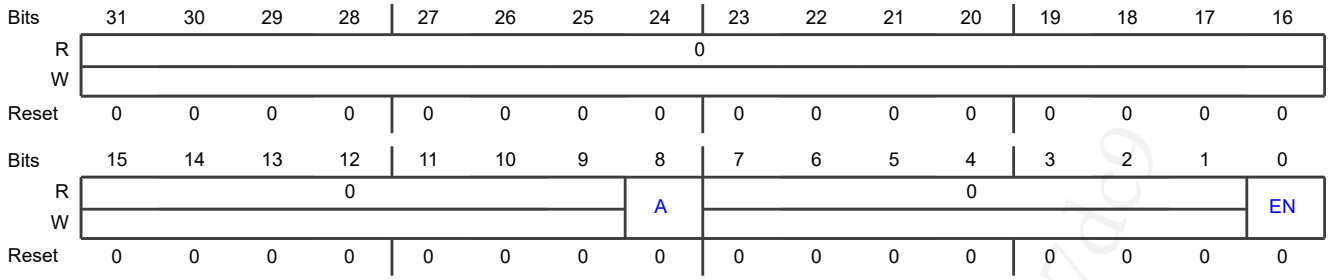
The MACSEC_UPFR4R register is used to configure the record 4 result.

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2.213.3 Diagram



2.213.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

2.214 Uncontrolled Port filter Record 5 data part 0 (MACSEC_UPFR5D0)

2.214.1 Offset

Register	Offset
MACSEC_UPFR5D0	AA0h

2.214.2 Function

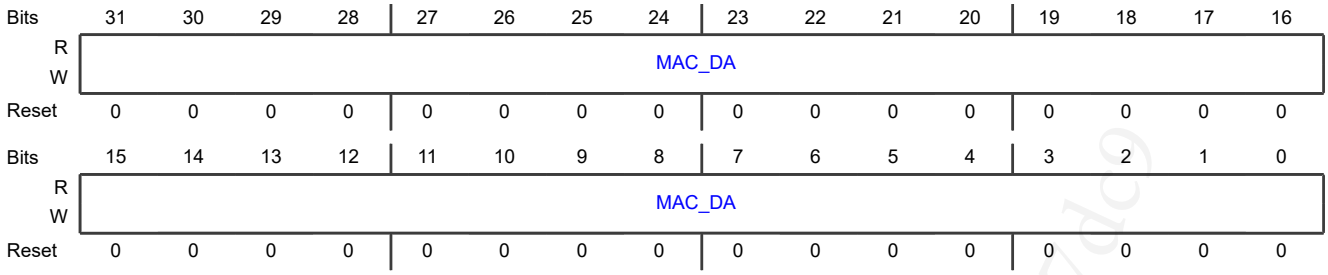
The MACSEC_UPFR*5*D0 register is used to configure the record *5* data.

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2.214.3 Diagram



2.214.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.215 Uncontrolled Port filter Record 5 data part 1 (MACSEC_UPFR5D1)

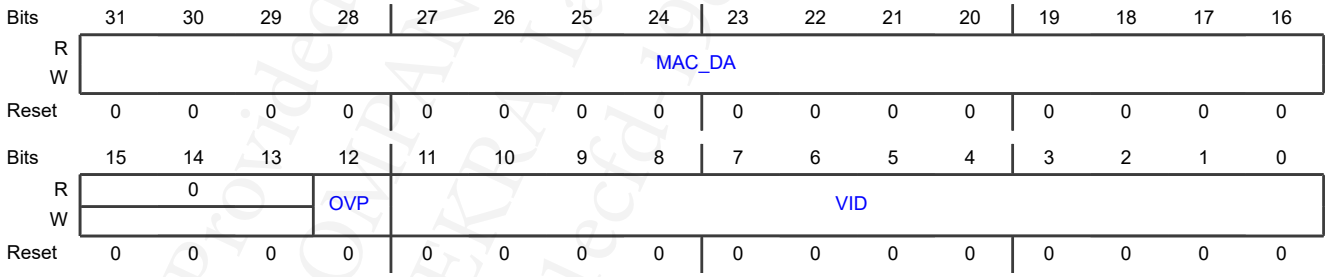
2.215.1 Offset

Register	Offset
MACSEC_UPFR5D1	AA4h

2.215.2 Function

The MACSEC_UPFR*5*D1 register is used to configure the record *5*data.

2.215.3 Diagram



2.215.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.216 Uncontrolled Port filter Record 5 data part 2 (MACSEC_UPFR5D2)

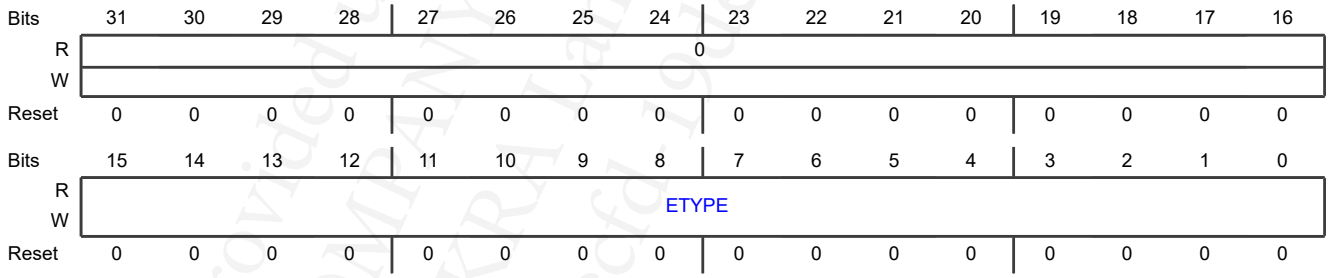
2.216.1 Offset

Register	Offset
MACSEC_UPFR5D2	AA8h

2.216.2 Function

The MACSEC_UPFR*5*D2 register is used to configure the record *5* data.

2.216.3 Diagram



2.216.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.217 Uncontrolled Port filter Record 5 mask part 0 (MACSEC_UPFR5M0)

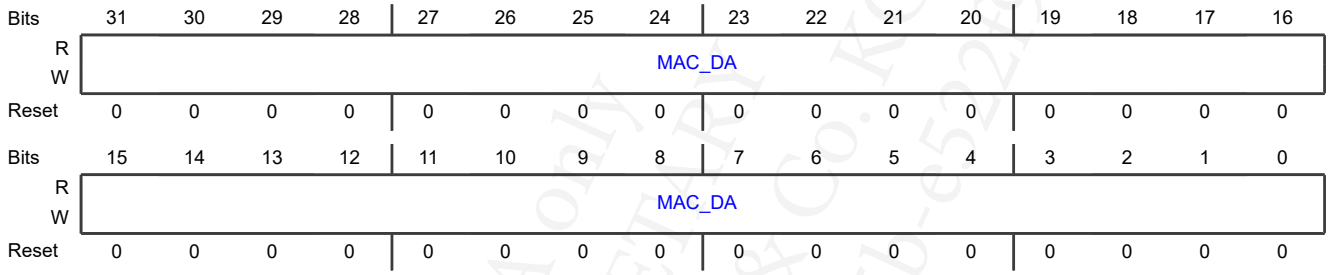
2.217.1 Offset

Register	Offset
MACSEC_UPFR5M0	AACh

2.217.2 Function

The MACSEC_UPFR*5*M0 register is used to configure the record *5* mask.

2.217.3 Diagram



2.217.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.218 Uncontrolled Port filter Record 5 mask part 1 (MACSEC_UPFR5M1)

2.218.1 Offset

Register	Offset
MACSEC_UPFR5M1	AB0h

2.218.2 Function

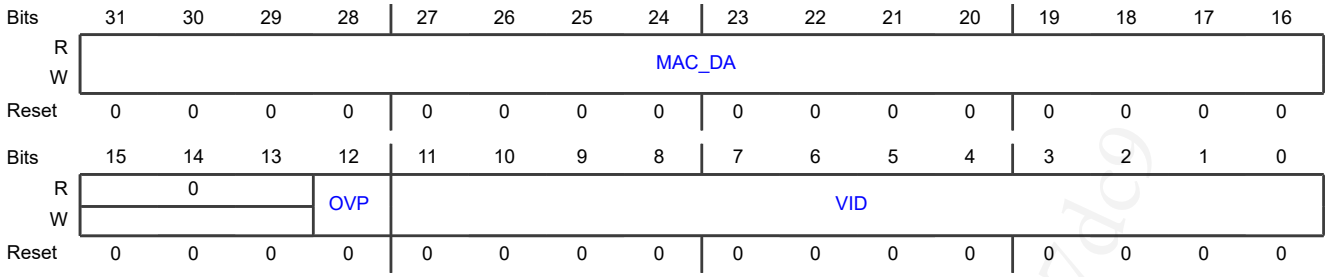
The MACSEC_UPFR*5*M1 register is used to configure the record *5* mask.

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2.218.3 Diagram



2.218.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.219 Uncontrolled Port filter Record 5 mask part 2 (MACSEC_UPFR5M2)

2.219.1 Offset

Register	Offset
MACSEC_UPFR5M2	AB4h

2.219.2 Function

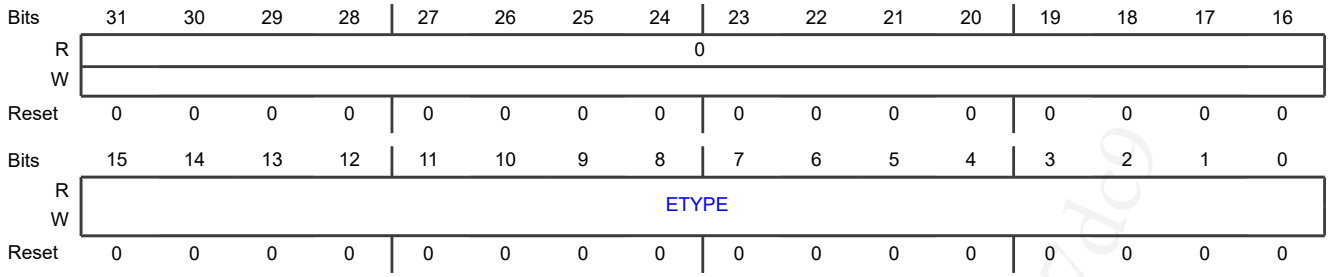
The MACSEC_UPFR*5*M2 register is used to configure the record *5* mask.

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2.219.3 Diagram



2.219.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.220 Uncontrolled Port filter Record 5 Result (MACSEC_UPFR5R)

2.220.1 Offset

Register	Offset
MACSEC_UPFR5R	AB8h

2.220.2 Function

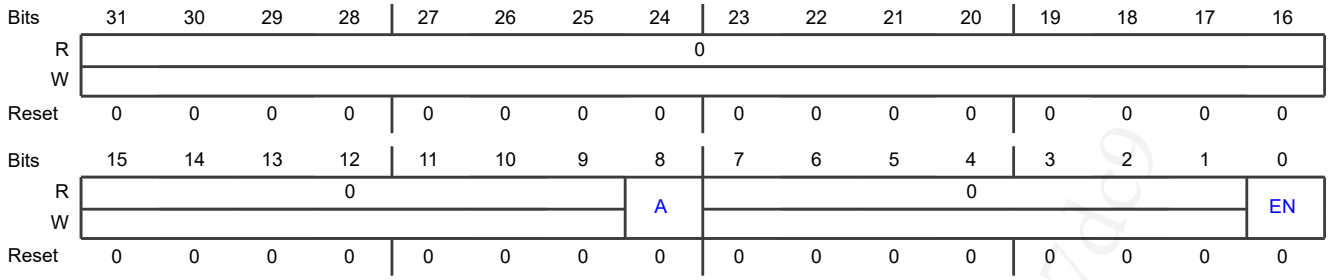
The MACSEC_UPFR5R register is used to configure the record 5 result.

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2.220.3 Diagram



2.220.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

2.221 Uncontrolled Port filter Record 6 data part 0 (MACSEC_UPFR6D0)

2.221.1 Offset

Register	Offset
MACSEC_UPFR6D0	AC0h

2.221.2 Function

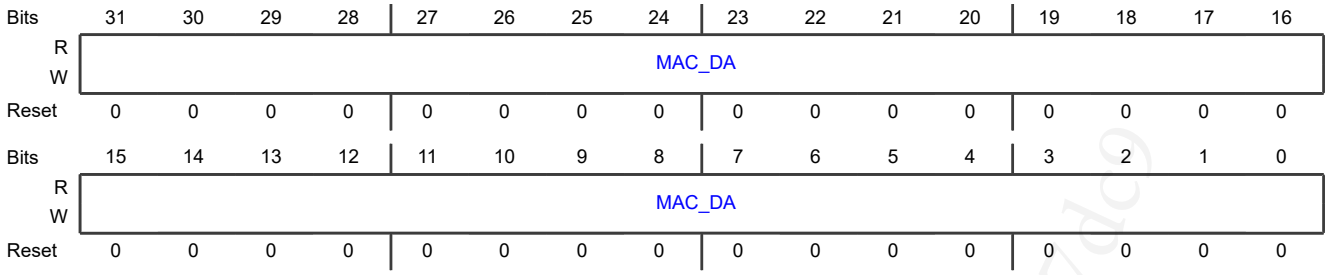
The MACSEC_UPFR*6*D0 register is used to configure the record *6* data.

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2.221.3 Diagram



2.221.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.222 Uncontrolled Port filter Record 6 data part 1 (MACSEC_UPFR6D1)

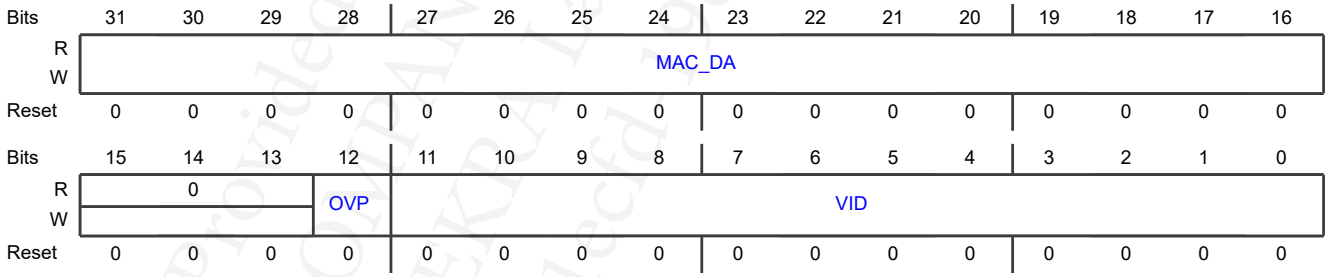
2.222.1 Offset

Register	Offset
MACSEC_UPFR6D1	AC4h

2.222.2 Function

The MACSEC_UPFR*6*D1 register is used to configure the record *6*data.

2.222.3 Diagram



2.222.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.223 Uncontrolled Port filter Record 6 data part 2 (MACSEC_UPFR6D2)

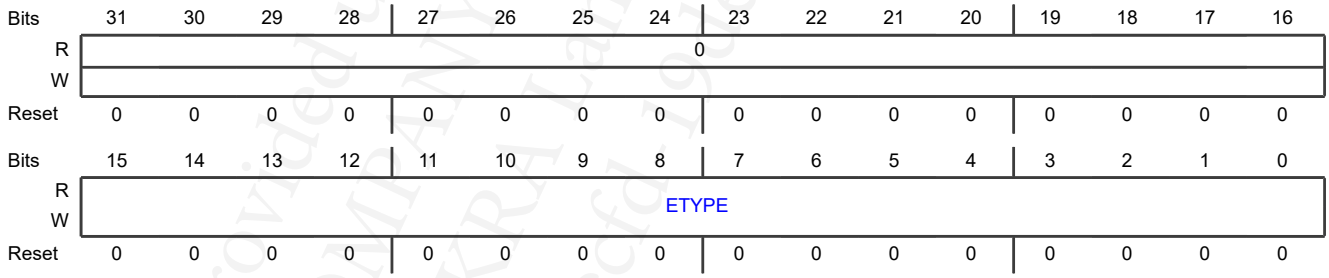
2.223.1 Offset

Register	Offset
MACSEC_UPFR6D2	AC8h

2.223.2 Function

The MACSEC_UPFR*6*D2 register is used to configure the record *6* data.

2.223.3 Diagram



2.223.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.224 Uncontrolled Port filter Record 6 mask part 0 (MACSEC_UPFR6M0)

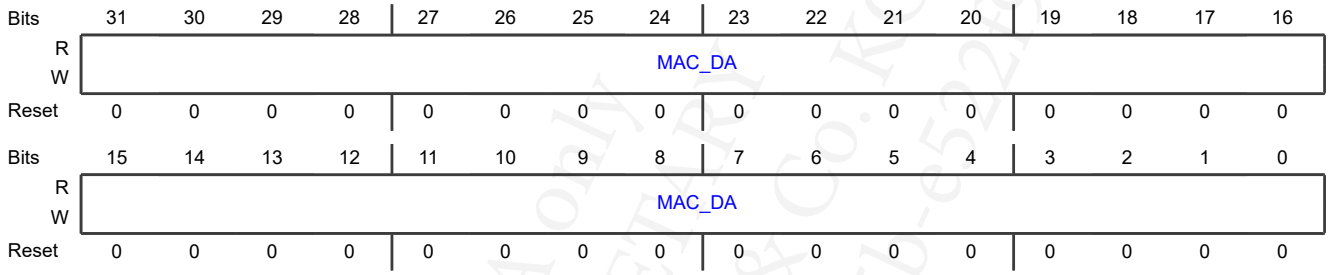
2.224.1 Offset

Register	Offset
MACSEC_UPFR6M0	ACCh

2.224.2 Function

The MACSEC_UPFR*6*M0 register is used to configure the record *6* mask.

2.224.3 Diagram



2.224.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.225 Uncontrolled Port filter Record 6 mask part 1 (MACSEC_UPFR6M1)

2.225.1 Offset

Register	Offset
MACSEC_UPFR6M1	AD0h

2.225.2 Function

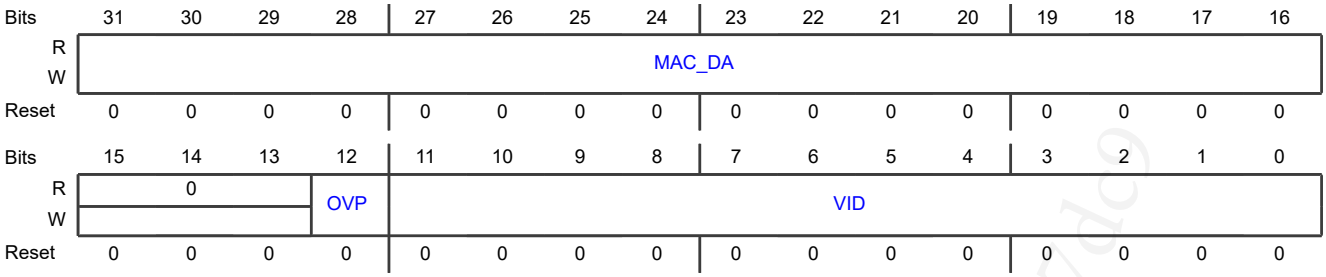
The MACSEC_UPFR*6*M1 register is used to configure the record *6* mask.

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2.225.3 Diagram



2.225.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.226 Uncontrolled Port filter Record 6 mask part 2 (MACSEC_UPFR6M2)

2.226.1 Offset

Register	Offset
MACSEC_UPFR6M2	AD4h

2.226.2 Function

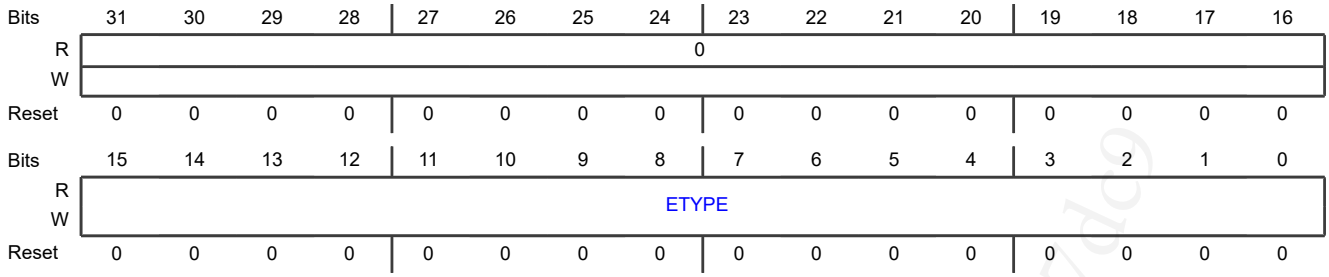
The MACSEC_UPFR*6*M2 register is used to configure the record *6* mask.

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2.226.3 Diagram



2.226.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.227 Uncontrolled Port filter Record 6 Result (MACSEC_UPFR6R)

2.227.1 Offset

Register	Offset
MACSEC_UPFR6R	AD8h

2.227.2 Function

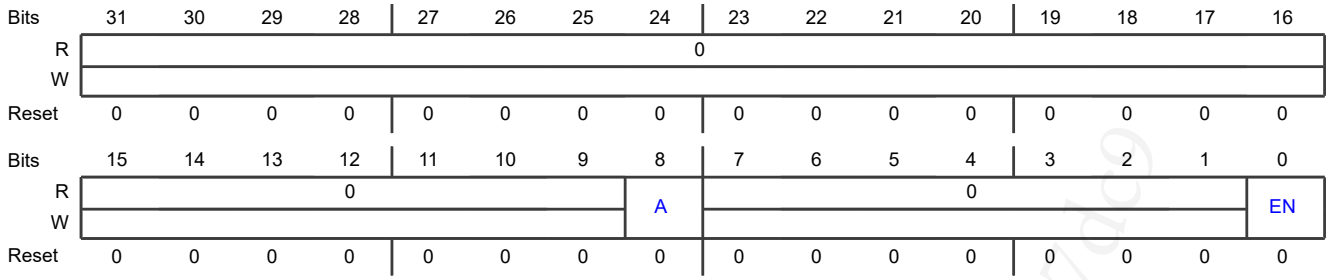
The MACSEC_UPFR6R register is used to configure the record 6 result.

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2.227.3 Diagram



2.227.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRnD0/1/2 and MACSEC_UPFRnM0/1/2

2.228 Uncontrolled Port filter Record 7 data part 0 (MACSEC_UPFR7D0)

2.228.1 Offset

Register	Offset
MACSEC_UPFR7D0	AE0h

2.228.2 Function

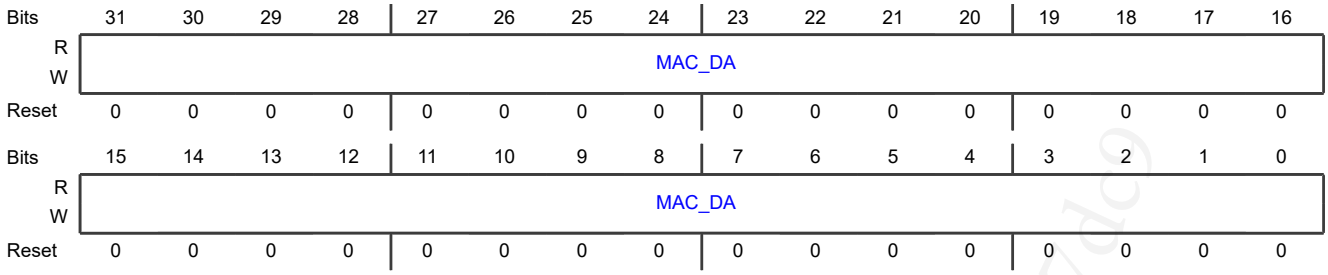
The MACSEC_UPFR7D0 register is used to configure the record 7 data.

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2.228.3 Diagram



2.228.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA data bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire) This field contains the 4 most significant bytes of the MAC address. For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 11h and MAC_DA[7:0] equals 14h.

2.229 Uncontrolled Port filter Record 7 data part 1 (MACSEC_UPFR7D1)

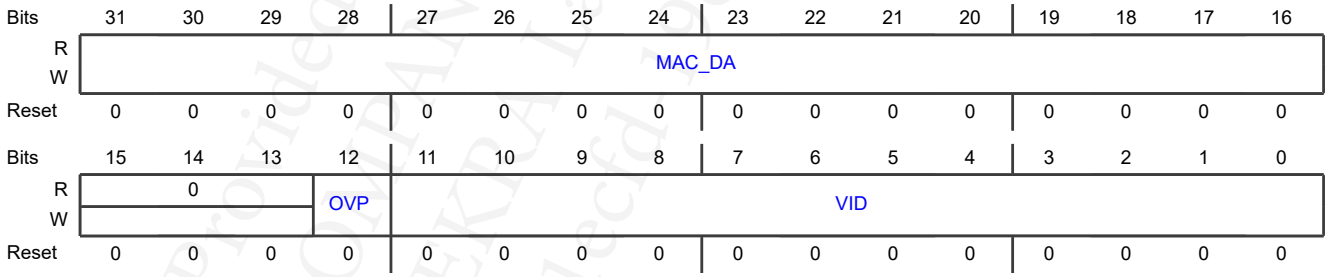
2.229.1 Offset

Register	Offset
MACSEC_UPFR7D1	AE4h

2.229.2 Function

The MACSEC_UPFR7D1 register is used to configure the record 7data.

2.229.3 Diagram



2.229.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA data bit 32-47 (bytes 5,6) This field contains the 2 least significant bytes of the MAC address.

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Field	Function
	For example if MAC address equals 11:12:13:14:15:16 (11 being the most significant byte), then MAC_DA[31:24] equals 15h and MAC_DA[23:16] equals 16h.
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present data bit 0b - Untagged: OVP data is 0, OVP mask is 1, this uncontrolled port filter record is for ETYPE, only the frame with the same ETYPE matches with this record. OVP data is 0, OVP mask is 0, both VID and ETYPE are treated as don't care. 1b - OuterVLANPresent: OVP data is 1, OVP mask is 1, this uncontrolled port filter record is for VID, only the frame with the same outer Vlan ID matches with this record. OVP data is 1, OVP mask is 0, both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID data bits 0-1 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.230 Uncontrolled Port filter Record 7 data part 2 (MACSEC_UPFR7D2)

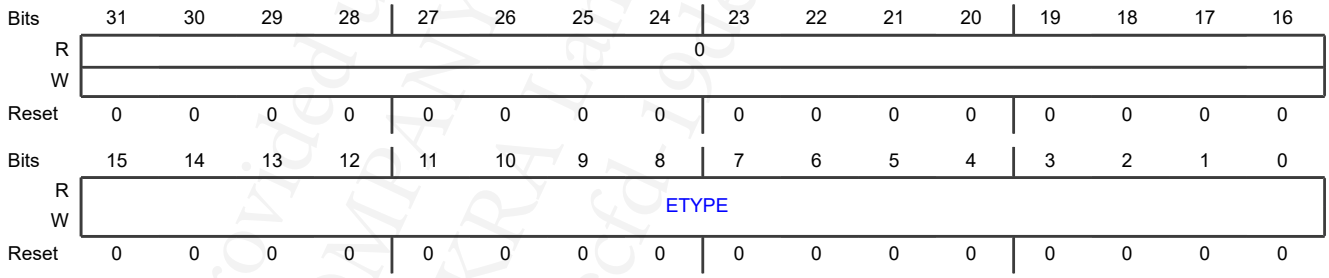
2.230.1 Offset

Register	Offset
MACSEC_UPFR7D2	AE8h

2.230.2 Function

The MACSEC_UPFR7D2 register is used to configure the record 7 data.

2.230.3 Diagram



2.230.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	EtherType EtherType data bit The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

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2.231 Uncontrolled Port filter Record 7 mask part 0 (MACSEC_UPFR7M0)

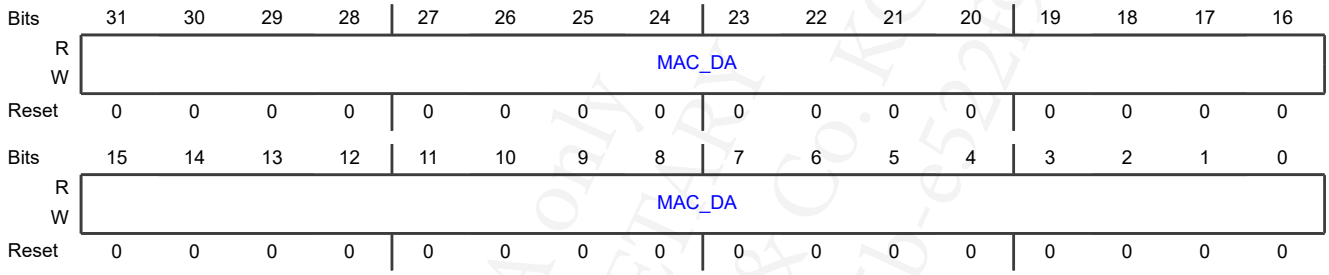
2.231.1 Offset

Register	Offset
MACSEC_UPFR7M0	AECh

2.231.2 Function

The MACSEC_UPFR7M0 register is used to configure the record 7 mask.

2.231.3 Diagram



2.231.4 Fields

Field	Function
31-0 MAC_DA	MAC_DA MAC DA mask bit 0-31 (bytes 1,2,3,4 where byte 1 is most significant byte, first byte on the wire)

2.232 Uncontrolled Port filter Record 7 mask part 1 (MACSEC_UPFR7M1)

2.232.1 Offset

Register	Offset
MACSEC_UPFR7M1	AF0h

2.232.2 Function

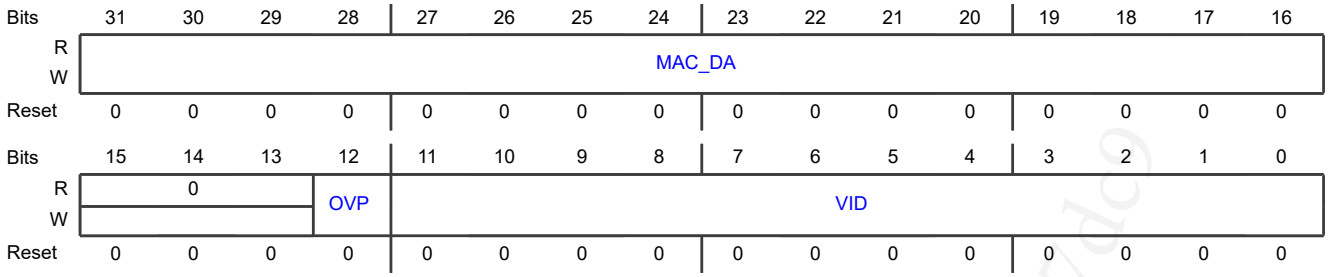
The MACSEC_UPFR7M1 register is used to configure the record 7 mask.

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2.232.3 Diagram



2.232.4 Fields

Field	Function
31-16 MAC_DA	MAC_DA MAC DA mask bit 32-47 (bytes 5,6)
15-13 —	Reserved, should be cleared
12 OVP	OVP Outer VLAN Present mask bit If the OVP mask bit is 0, then both VID and ETYPE are treated as don't care.
11-0 VID	VID Outer VLAN VID mask bits 0-11 The VID is compared with the outer VLAN VID of the frame if deemed present when OVP data is 1 and OVP mask bit is 1.

2.233 Uncontrolled Port filter Record 7 mask part 2 (MACSEC_UPFR7M2)

2.233.1 Offset

Register	Offset
MACSEC_UPFR7M2	AF4h

2.233.2 Function

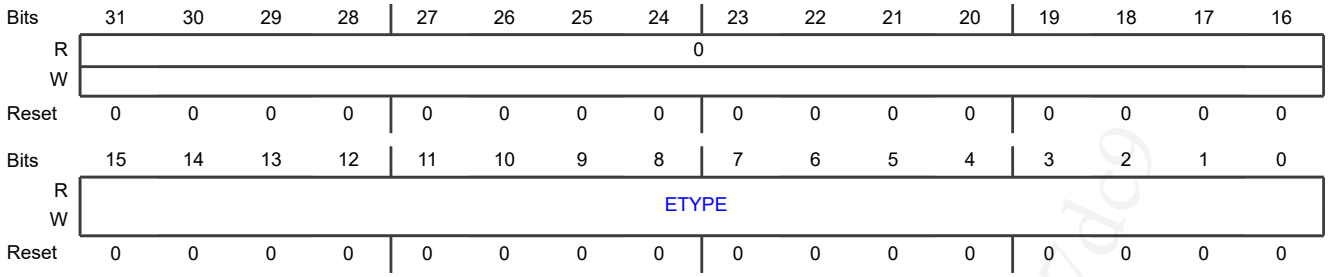
The MACSEC_UPFR7M2 register is used to configure the record 7 mask.

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2.233.3 Diagram



2.233.4 Fields

Field	Function
31-16 —	Reserved, should be cleared
15-0 ETYPE	ETYPE EtherType mask bits The EtherType is compared with 2 bytes after MAC SA of the frame when OVP data is 0 and OVP mask bit is 1.

2.234 Uncontrolled Port filter Record 7 Result (MACSEC_UPFR7R)

2.234.1 Offset

Register	Offset
MACSEC_UPFR7R	AF8h

2.234.2 Function

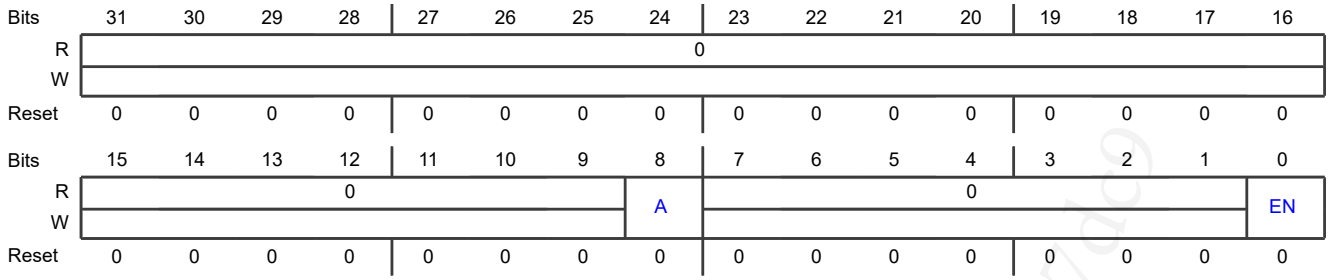
The MACSEC_UPFR7R register is used to configure the record 7 result.

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2.234.3 Diagram



2.234.4 Fields

Field	Function
31-9 —	Reserved, should be cleared
8 A	A Action If a frame matches this entry, it will be process according to the following action. 0b - forward on the uncontrolled port 1b - discard on the uncontrolled port
7-1 —	Reserved, should be cleared
0 EN	EN Enable 0b - record is not enabled, will not match any MAC DA/VLAN/ETYPE 1b - record is enabled, will attempt to match MAC DA/VLAN/ETYPE as specified in MACSEC_UPFRn D0/1/2 and MACSEC_UPFRnM0/1/2

3 MACsec - Functional Description

The security functions provided by the MACsec enable the integrity protection and optionally the confidentiality protection of Ethernet frames using the MAC-level cryptographic algorithms defined in IEEE 802.1AE.

The MACsec implements up to 64 instances of a SecY providing 1 to 64 Secured Channels (SC) on RX and 1 to 31 Secured Channels on TX, each supporting 2 concurrent Security Associations (SA) (referred to as A and B) with selectable Association Number (AN) assignment. The required statistics and management access are also provided.

The MACsec function assumes that Ethernet frames on RX have an Ethernet FCS; that is, FCS may be validated but is never removed at the MAC on RX. The MACsec block does not perform validation of the received FCS. In cases where RX frames are modified by the MACsec function (for example, confidentiality and/or integrity processing), the received FCS is replaced by the MACsec block with a generated FCS. Otherwise the received FCS is forwarded with the unmodified frame.

The Ethernet frames on TX that input to MACsec function have an optional interface signal to specify if FCS is present. The MACsec block does not perform validation of the transmitted FCS. In cases where TX frames are modified by the MACsec function (for example, add SecTag), the transmitted FCS is stripped by the MACsec block, otherwise the transmitted FCS is forwarded with the unmodified frame. It is expected that the MAC computes and appends FCS to the frame on TX based on Ethernet FCS present indication.

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3.1 MACsec frame format

The MACsec frame format, as defined in 802.1AE-2006/802.1AEbn-2011 section 9, is summarized in Figure 1.

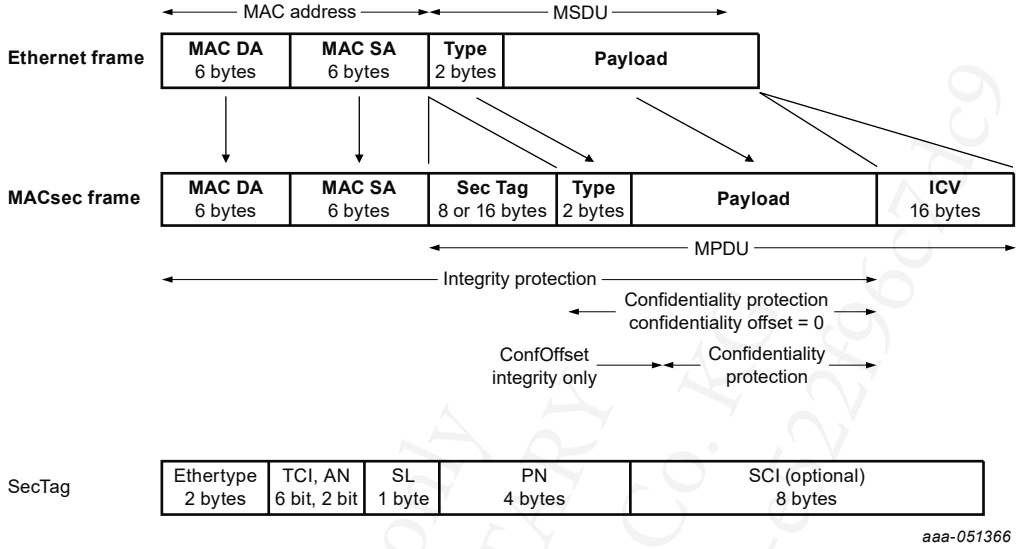


Figure 1. 802.1AE MACsec Frame Format

Table 1. SecTag TCI and AN

Bit(s)		Description
0	V	**Version number.** Only version 0 is supported. Frames with another version value are marked for discard and counted in InPktsBadTag
1	ES	**End Station.** When set, the SCI is derived from the MAC SA and the SC bit must be cleared.
2	SC	**Secure Channel.** When set, the SCI field is present in the SecTag and the ES field must be cleared.
3	SCB	**Single copy Broadcast.** Not supported. Only used when ES is set to determine the end station port number of the SCI.
4	E	**Encryption.** See Table 3 for details.
5	C	**Changed text.** See Table 3 for details.
6 to 7	AN	**Association number.** A 2-bit value used to identify up to four different SAs within the context of an SC.

3.2 RX Frame Processing

Frames received on RX with the MACsec Ethertype of 0x88E5 are first assigned to an scI. Then the frames are validated, decrypted and data integrity validated (see 802.1AE-2006/802.1AEbn-2011 10.6). The SecTAG and ICV are removed before the frame is delivered to the controlled port. Other frames are forwarded with no modification.

Support MACsec frame payload size up to 64K-1B on the controlled port, no size limit on the uncontrolled port.

The MACsec does not discard frames since frames are not buffered (cut through operation). Instead, it will mark the frames as erred on the RX output interface for the red side to discard them.

Figure 2illustrates the MACSec receive frame processing flowchart.

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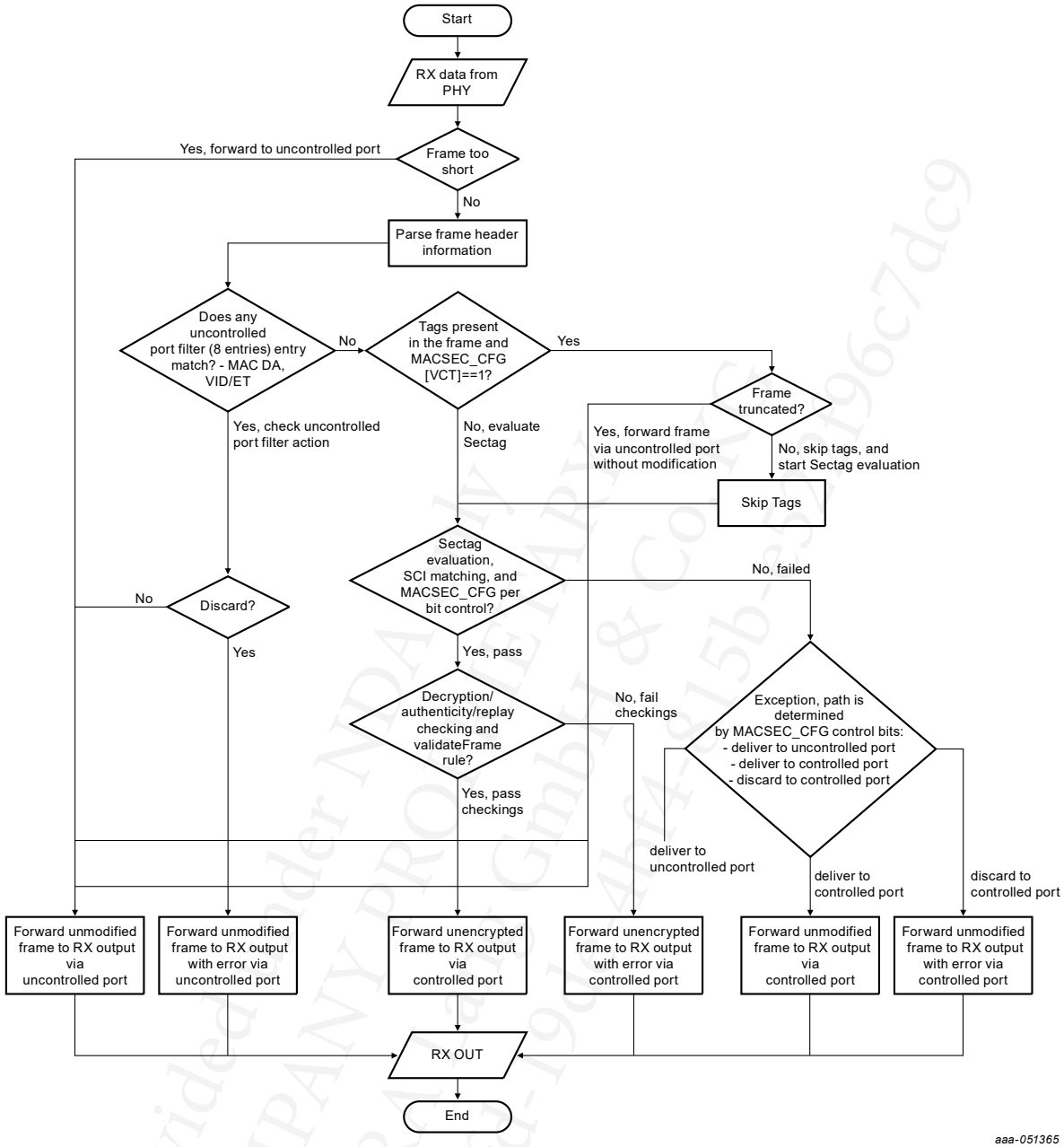


Figure 2. MACsec RX Flowchart

3.2.1 SCI resolution / search

There are 3 ways to resolve the SCI of a frame:

1. Explicit SCI in the SecTag
2. Implicit point-to-point SCI (pre-shared)
3. End station SCI using the MAC SA of the frame

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Table 2. SCI Resolution

SecTag SC	SecTag ES	SecTag SCB	SCI
1	0	0	SCI is explicit in the SecTag
1	X	X	Invalid flag combination, discard frame and count InPktsBadTag
0	0	0	No SCI in the SecTag, implicit point-to-point SCI found in RX SCI 0
0	1	0	No SCI in the SecTag, SCI is {MAC SA, 0x0001}
		1	No SCI in the SecTag, SCI is {MAC SA, 0x0000}
0	0	1	No SCI in the SecTag, EPON Single Copy Broadcast (not supported), frame is marked as error to be discarded by red side or sent to the uncontrolled port.

The RX SCI table is composed of the SCI values programmed in Section 2.13 and Section 2.14 on a per SC basis. The SCI entry is valid only if Section 2.20 SCI_EN bit is set. SC 0 can be reserved via Section 2.2 S0I bit to be an implicit SCI for point-to-point applications, in that case it would be not be considered in the SCI table search.

Once an scI is resolved, a lookup is done in the RX SCI table to associate the SCI with a Secure Channel (SC) number. If the SCI is found in the table, the table index where the entry was found becomes the SC number (0..63). The Association Number (AN) is then used to associate the A or B Security Association (SA) configuration, context, and Security Association Key (SAK) for that SC number.

3.2.2 RX Secure Association

There are two RX SA contexts per SC (A and B). Each SA is associated with a programmable AN value. Up to two SAs can be used concurrently during an AN transient window. Once the transient is over, the unused SA can be disabled and re-configure for Rekeying purposed and be associated with another AN.

At any given time, a frame can be received with an AN not associated with any SA context. In that case the frame is treated like it was received on a disabled SA and counted in RX AN <n> InNotUsingSA Statistic (MACSEC_RXAN<n>INUSS) or RX AN <n> InPktsUnusedSA Statistic (MACSEC_RXAN<n>IPUSS) depending on validateFrame setting.

3.2.3 Frame PDU Validation

The frames are validated according to 802.1AE-2006/802.1AEbn-2011 9.12. The result of validation updates the SecY/SC/SA statistics and state.

Frames with a validation error are marked and passed to the red side for discarding.

Table 3. SecTag E and C bit validation

SecTag C	SecTag E	Outcome
0	0	The frame is only integrity-protected; integrity validation is performed. Frames with an integrity error are discarded if validateFrames is set to strict.
1	1	The frame is confidentiality- and integrity-protected; frame payload is decrypted and integrity validation is performed. Frames with an integrity error are discarded if validateFrames is set to strict.
0	1	The frame is not processed but is reserved for the use of the KaY; the frame is forwarded unmodified.
1	0	The frame is reserved and not supported; the frame is marked as error and is discarded by the red side or sent to the uncontrolled port.

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3.2.4 Frame data integrity

RX frames are verified for integrity. The GCM algorithm is used for the data integrity validation.

The location of the 16-byte Integrity Check Value (ICV) may not be at the end of the frame for a short frame with padding. The SecTag SL field is used to determine the location of the ICV.

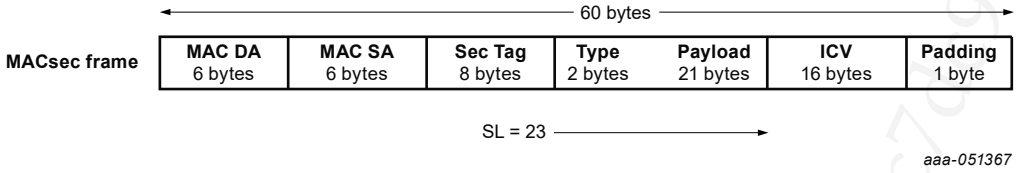


Figure 3. 802.1AE MACsec Frame example with padding

Frames with integrity error are counted in Section 2.41 or Section 2.42, depending on the validateFrame setting. The frames are then marked as erred and passed to the red side for discarding.

3.2.5 Frame data confidentiality

If the SecTag bits E and C are set, the frame payload will be decrypted starting at the optional confidentiality offset and up to the ICV.

The XPN-capable Cipher Suites do not provide Confidentiality Protection with a confidentiality offset.

The AES (128) or AES (256) algorithm is used for data decryption, depending on the setting in the Section 2.20 register.

3.2.6 Replay check

If replayProtect control is enabled in Section 2.20 and the PN of the received frame is less than the lowest acceptable packet number (see Section 2.34) for the SA, the frame is discarded and the Section 2.28 or Section 2.27 counter incremented, depending on validateFrame setting.

If the current cipher suite supports extended packet numbering, PN recovery (IEEE 802.1AE 10.6.2) is implemented before Replay Check. All PN values in the range lowest acceptable PN to lowest acceptable PN plus 2^{31} will be recovered correctly. If the recovered PN value is incorrect, the Cipher Suite validation operation will not return VALID and the frame will be discarded if validateFrames is Strict (IEEE 802.1AE 10.6.5, 10.7.8). A recovered PN value is used to update the lowest acceptable PN only if the validation operation with that PN value returns VALID.

3.2.7 RX Controlled port vs. uncontrolled port

The MACsec has a single physical interface to the red side, but it has the ability to assign a logical port to each frame being received. The two logical ports supported are the controlled port and the uncontrolled port.

This MACsec implementation does not physically replicate all frames to both the controlled port and uncontrolled port. Instead, it provides the ability to filter some frames to the uncontrolled port, filter some frames for discard, and forward frames that are discarded by the controlled port of the SecY (untagged, Kay, bad tag, and SCI not found) to the uncontrolled port. This flexibility point allows for those frames to be processed and counted by all the RX SecY, while having the option (see Section 2.2 USFT, ITT, KFT, and UFT bits for details) to mark the frame for discard or forward the frame to the uncontrolled port.

Frames can be filtered to the uncontrolled port. Filtering is done on MAC DA, either Outer VLAN VID (if present) or EtherType for untagged frames. For example, could be used to filter BPDUs, MKA, EAP, gPTP.

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A ternary match table is used to implement the Uncontrolled Port Filter of eight entries. If multiple entries in the table match, the entry with the lowest numerical values index is chosen as the matching entry. The same table is used on the TX path.

An entry matched when a frame with MAC DA / outer VLAN VID / EtherType for untagged frames matches one of the records defined in MACSEC_UPFRnD0/1/2 and MACSEC_UPFRnM0/1/2. If a filter matches, the frame is sent unmodified over the uncontrolled port and counted in MACSEC_IFIOUP1/2HS, Section 2.100, Section 2.103 , Section 2.104 and Section 2.112 , or is discarded and counted in Section 2.113. Unfiltered frames are sent to the controlled port.

All entry-matching fields are composed of pairs of data and mask values, to support a per-bit ternary match capability. For each {data,mask} bit pair, the following combinations are supported:

- 00 = always match (X)
- 01 = match on 0
- 11 = match on 1
- 10 = always match (X)

In other words, to look for a match on a particular bit, set its mask bit to 1, and its corresponding data bit to the desired matching value, 0 or 1. A mask value of all 1s for a particular field means that field is required. To make a bit don't care, set its mask bit to 0. For all don't care bits, that is, for all mask bits that are 0.

The outer VLAN frame header presence fields can be used to distinguish between frames that do not have the header field at all. OVP data/mask determine either ETYPE or VID is used for the ternary match in an entry. For each {OVP data, OVP mask} pair, the following combinations are supported:

- 00 = Both VID and ETYPE are treated as don't care.
- 01 = ETYPE is used for the ternary match, VID is treated as don't care.
- 11 = VID is used for the ternary match, ETYPE is treated as don't care.
- 10 = Both VID and ETYPE are treated as don't care.

The supported VLAN types are the 2 standard ones: 0x8100 and 0x88A8 plus two customer configured types in the Section 2.4 and Section 2.5 registers. In order to be recognized and skipped, the Ethertype of the VLAN must be one of the above 4 types and the type must be enabled in the MACsec VLAN / Section 2.6 OVTPIDL.

For those frames with unresolved SCI (untagged, Kay, bad tag, SCI not found, unsupported E/C bit combination, and unsupported EPON Single Copy Broadcast), a set of global statistics is provided (see MACsec RX Global Statistics). Since those frames are considered to be received by all the RX SecYs (a virtual replication), the global statistics can be reported on all the SecYs/SCs.

The untagged, Kay, bad tag, SCI not found, unsupported E/C bit combination, and unsupported EPON Single Copy Broadcast statistics are kept in 6 global counters. Those statistics can be reported on a per SC basis by Software with the knowledge of the validateFrame setting for each SC.

For those frames with unresolved SCI will also update a set of global Interface MIB based on the treatment selected in Section 2.2. If the frame is delivered on the controlled port, then the controlled port Interface MIB will be updated. If the frame is delivered on the uncontrolled port, then the uncontrolled port Interface MIB will be updated.

Frames with resolved SCI update the per SC statistics.

3.3 TX Frame Processing

Frames received on TX are assigned an SC from an out-of-band bus, valid on the start of frame on the TX input interface, and are directed to the controlled port or uncontrolled port by the red side (from same out-of-band bus). Frames on the uncontrolled port are transmitted with no modifications.

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The TX controlled vs. uncontrolled port selection and the TX SC determination can optionally be done with an uncontrolled port filter table and a TX SC classification table when the out-of-band information is not available.

Frames on the controlled port:

- Have a SecTag
- Payload integrity is computed and added in the frame trailer (ICV)
- Optionally, the payload is encrypted for confidentiality

The support frame size is 64 K to 1 B on the controlled port, no size limit on the uncontrolled port.

In order to process a frame on the controlled port, the frame must be at least 13 bytes (MAC DS/SA plus 1 byte) plus FCS, if present. Frames delivered to the controlled port smaller than 13 B plus FCS, if present, are marked as aborted for the MAC to invalidate the FCS. In addition, the Section 2.114 is incremented. The aborted frame is padded with 4 bytes of zeros until the resulting frame size is greater or equal to 60 bytes.

Figure 4 illustrates the MACSec transmit frame processing flowchart.

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3911ecfd-19de-4bf4-815b-e522f9

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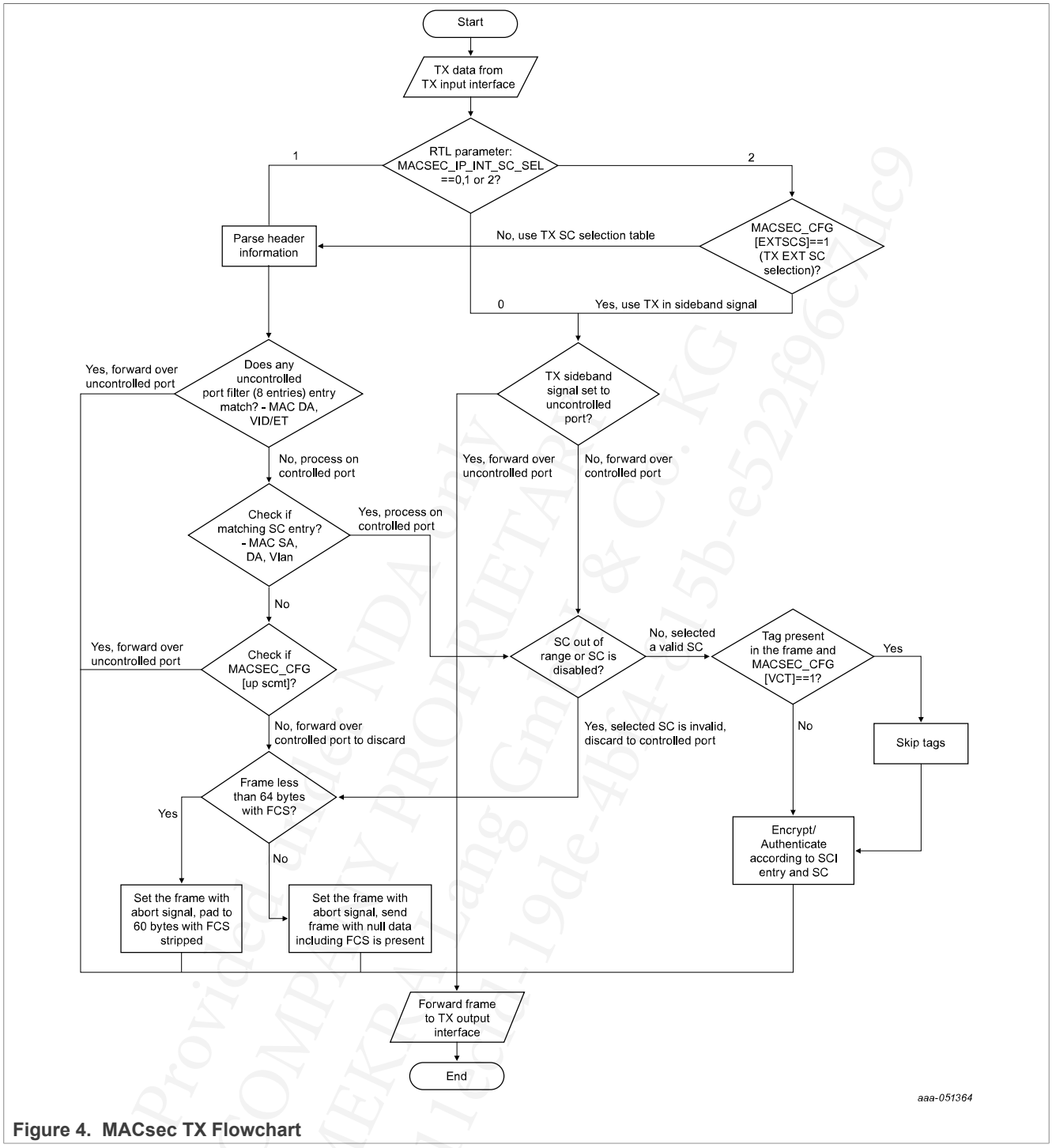


Figure 4. MACsec TX Flowchart

3.3.1 TX Controlled port vs. uncontrolled port

Frames received from the red side can be filtered to the uncontrolled port. Filtering is done on MAC DA, either Outer VLAN VID (if present) or EtherType for untagged frames.

A ternary match table is used to implement the Uncontrolled Port Filter of eight entries. If multiple entries in the table match, the entry with the lowest numerical values index is chosen as the matching entry. The same table is used with the RX path.

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An entry matched when a frame with MAC DA/Outer VLAN VID/ EtherType for untagged frames matches one of the record defined in MACSEC_UPFRnD0/1/2and MACSEC_UPFRnM0/1/2 ignoring the Action field. If a filter matches, the frame is sent over the uncontrolled port, unmodified. Unfiltered frames are sent to the controlled port TX SC selection.

Entry matching is the same as on the RX path, see RX Controlled port vs. uncontrolled port for details.

3.3.2 TX SC selection

Controlled port frames are classified to a secure channel (SC). Classification is done on MAC DA / MAC SA / outer VLAN (PCP/DEI/VID).

An exact match table is used to implement the TX SC Selection table of up to 31 entries limited by the number of Secured Channels on TX in the SoC. If multiple entries in the table match, the entry with the lowest numerical values index is chosen as the matching entry.

An entry matched when a frame with MAC DA / MAC SA / VLAN matches one of the record defined in MACSEC_TXSCSRnP0/1/2/3.

All entry matching fields are composed of MAC DA, MAC SA, outer VLAN VID, outer VLAN PCP, outer VLAN DEI. Each table record can be enabled or disabled, a disabled record will not match. Each field of the table record can be included or ignored for the exact match. A field not used for the exact match is simply ignored. A record is matched when all the enabled fields match the corresponding fields of the frame.

Having the VLAN PCP/DEI bits allows for TX SC selection on priority for a given MAC/VLAN pair. Multiple entries can result to the same TX SC to allow point to multipoint.

Frames not matching a filter and not matching an sc classification entry are treated based on MACSEC_CFG[UPSCMT] Section 2.2, if frames are discarded, the stats are counted in Section 2.114.

3.3.3 SCI Insertion

The SCI is read from the TX SCI context based on an sc (0..30) index provided with the frame.

The three SCI modes supported on TX. The mode is selected using the generation configuration bits found in the Section 2.63 register.

1. Explicit SCI in the SecTag (alwaysIncludeSCI = 1, useES and useSCB are ignored)
   - the SCI in the SecTag is the SCI from the TX SCI table
   - the SCI used as IV for the AES engine is the SCI from the TX SCI table
2. Implicit point-to-point SCI (pre-shared) (alwaysIncludeSCI = 0, useES = 0, useSCB = 0)
   - no SCI in the SecTag
   - the SCI used as IV for the AES engine is the SCI from the TX SCI table entry (note: only one of the TX SC should be using that mode so the far end can differentiate the connections)
3. End station SCI using the MAC SA of the frame. In this mode (alwaysIncludeSCI = 0, useES = 1, useSCB = 0/1)
   - no SCI in the SecTag
   - the SCI used as IV for the AES engine is the SCI from the TX SCI table.
     Note: the port identifier bytes in the TX SCI table must match the behavior of the useSCB bit.
   - the MAC SA is overwritten with the first 6 bytes of the SCI from the TX SCI table

The alwaysIncludeSCI = 0, useES = 0, useSCB = 1 combination is not supported, the frames will be marked as error and counted in the Section 2.114.

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3.3.4 SecTag generation

The SecTag is built according to the configuration bits found in the Section 2.63 register. The SecTag SL field is derived from the frame length to be transmitted.

3.3.5 TX Secure Association

There are two TX SA contexts per SC (A and B). The SC selects the SA in use. The other one is not in use and can be reconfigured to accommodate the next AN and SAK for rekeying.

3.3.6 Frame data integrity

TX frames are integrity protected by the 16-byte Integrity Check Value (ICV) added to the end of the frame.

3.3.7 Frame data confidentiality

TX frames can be encrypted for confidentiality. An optional confidentiality offset may be applied to keep a portion of the Ethernet frame in clear text.

The XPN-capable Cipher Suites do not provide Confidentiality Protection with a confidentiality offset.

The AES (128) or AES (256) algorithm is used for encryption, depending on the setting in the Section 2.63 register.

3.3.8 TX processing latency

Encrypted frames have a TX processing latency of approximately 16 clocks. Unprocessed frames have a processing latency of approximately 7 clocks. The Section 2.2 UFDCD can be used to match those latencies. This is beneficial in systems where constant/predictable processing latency is desirable (i.e., TSN). Note, the very first frame of a Security Association (SA) incurs an additional delay of 9 to 13 clocks due to the one-time Hash calculation of 128 bit and 256 bit keys.

There are two modes supported in MACSec to compensate the latency between unprocessed frame and encrypted frames. When Section 2.2 UFDCM is 0, the latency compensation is done by adding a fixed number of frame-processing clock cycles. When Section 2.2 UFDCM is 1, the delay compensation is added based on TX FIFO depth, this is beneficial in systems with delays on TX Data received by MACSec.

3.4 Key and hash value privacy

The RX/TX keys and the associated hash values E(K,0) are read/writable when Section 2.2 KSS bit is cleared. These keys becomes write only, for privacy reasons, when Section 2.2 KSS bit is set. Once KSS is set, it can no longer be cleared. Section 2.2 KSSHD can disable hash value privacy control. When Section 2.2KSS bit is 1, if Section 2.2 KSSHD is 0, the hash values are not readable, otherwise the hash values are readable.

3.5 Clear Tag support

The support of clear VLAN/custom tag is desirable in networks where the point-to-point or point-to-multipoint MACsec frames are switched between the end points. This feature supports the presence of up to two VLAN tags and a custom tag before the MACsec SecTag.

On RX:

1. The macsec frames are received from the network, if first ethertype is VLAN, then see if MACsec ethertype follows. Support up to two VLANs and a custom tag. If more than two VLANs and a custom tag are present, the frame is treated as an untagged frame, according to Section 2.2 UFT.

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2. If MACsec is found, the frame is decrypted and the VLAN(s)/custom tag(s) are not included in the integrity check (ICV), because the VLAN/custom tag(s) may have changed in the switched network.
3. MACsec and other frames (uncontrolled port) are forwarded with the VLAN/custom tag(s); the VLAN tags are not removed by MACsec.
4. The skipped VLAN/custom tag(s) octets are not counted in ifInOctets, InOctetsValidated / InOctetcDecrypted, ifInOctetsCp or ifInOctetsUp.

On TX:

1. Frames to be transmitted on the control port (to be encrypted) may or may not have VLAN / custom tags(s).
2. If the frame to be encrypted has VLAN/custom tag(s), the tags are preserved and skipped, MACsec ethertype is inserted after the last tag(s). Up to two VLANs and a custom tag are supported. If more than two VLANs and a custom tag are present, the first three tags are skipped, any other tags become part of the payload. The Section 2.63 TVM can be configured to allow zero, one or two VLANs and Section 2.63 TCM to allow zero or one custom tag to be transferred clear.
3. While encrypting MACsec frames, the skipped tag(s) are not included in integrity protection (ICV), because the VLAN/custom tag(s) could change in the switched network.
4. The skipped VLAN/custom tag(s) octets are not counted in if OutOctets or OutOctetsProtected/OutOctetc Encrypted.

The feature can be enabled in the Section 2.2 register. The supported tag types are the two standard VLAN types: 0x8100 and 0x88A8 plus two customer-configured types in the Section 2.4 and Section 2.5 registers. In order to be recognized and skipped, the Ethertype of the tag must be one of the above four types and the type must be enabled in the Section 2.6 IVTPIDL/OVTPIDL/CTTPIDL.

The custom tag is defined by an ethertype and a length. A custom tag is valid when:

- The first two bytes match the MACSEC_CTPID2
- The MACSEC_CTPID is not configured to be VLAN
- There are enough bytes in the frame to cover the length of the tag
- The tag is expected/supported in the position defined in MACSEC_VCLCTR[CTTPIDL]

The custom tag is always after VLAN(s) (if VLANs are present).

A maximum of 3 tags or 16 bytes can be transferred in the clear.

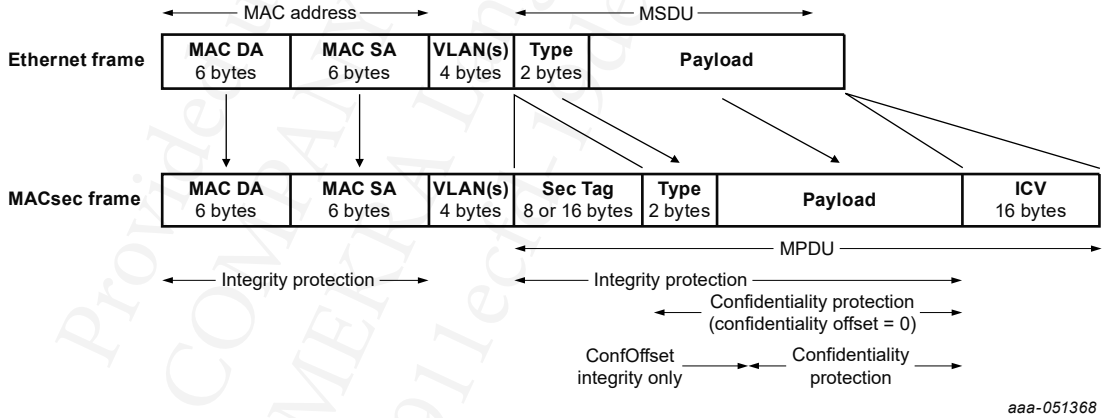


Figure 5. VLAN Clear Tag MACsec Frame Format

3.6 MACsec Bypass

The MACsec can be put in bypass such that it becomes transparent on the data path.

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The Section 2.2 BYPN bit is configured to be in bypass after hard reset. The bit must be set to enable the RX and TX MACsec function. Whether the MACsec is configured in bypass or not, all the Skyblue registers are accessible.

For SOC where the MACsec function is to be removed, the MACsec uses a configuration input signal that can be statically configured (fused). When the macsec_bypass signal is set, the MACsec function becomes bypassed. Furthermore, the Skyblue register access is disabled—writes are ignored and reads return zeros.

While in bypass, all RX frames are delivered to the red side on the uncontrolled port.

3.7 Double ECC error handling

In the event where a double ECC error on the MACsec memory occurs while the RX data path is reading some context, the frame associated with that read is marked as erred. If a double ECC error on the MACsec memory occurs while the TX data path is reading some context, the frame associated with that read is marked as abort and counted in Section 2.114. In both cases, the error is captured in the Section 2.93.

3.8 Tamper Protection

The MACSec supports tamper protection (MACSec controlled port transfer disable) based on an input signal. The input signal is captured in Section 2.94 TP bit to indicate the status of input MACSec-controlled port transfer disable. Tamper protection has no impact on uncontrolled port traffic.

When an RX-controlled frame is impacted by tamper protection, the frame is delivered to a controlled port without frame modification. The frame is then tagged to be discarded. The stats are counted in Section 2.108.

When a TX controlled frame is impacted by tamper protection, the frame is delivered to a controlled port with all data set to 0. The frame is then tagged to be discarded. The stats are counted in Section 2.114.

While MACSec is under tamper protection, all SAK and SAH registers are erased.

3.9 Functional Safety

MACSec functional safety feature meets the ASIL-B out of context, defined by the ISO 26262 functional safety standard.

MACSec has the following functional safety requirements:

- **ASIL-B:** MACSec must not forward a frame with corrupted integrity. In other words, the next hop device must be able to trust the packet received if the FCS is deemed valid.
- **ASIL-B:** MACSec must not forward a frame to an incorrect Secure Channel. Given that software relies on the filtering functions inside MACSec, software must be able to trust that the frame is decrypted and forwarded to a Secure Channel with correct FCS.

MACSec functional safety feature includes the following mechanisms in order to implement the functional safety requirements.

- Detection of a fault that violates the aforementioned functional safety requirements
- Reaction when a fault is detected in order to enter a safe state, and/or report the fault
- Recovery after a fault has occurred in order to return into an operational state; note that recovery is supported to improved MACSec availability in the system but is not needed from a functional safety standpoint
- Injection of faults to facilitate testing and for diagnostic coverage of the MACSec safety mechanisms

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3.9.1 MACSec Fault Detection

MACSec implements a number of fault detection mechanisms in order to detect transient and permanent faults in the IP block. It is intended to detect any single fault that violates the safety requirements. Some of the mechanisms can also correct single-bit errors, and detect multiple-bits errors.

Note: *While the term fault is used here exclusively in the context of functional safety, it also includes ECC errors that are present whether functional safety is supported or not.*

MACSec includes the following fault detection mechanisms:

- **Error Correcting Code (ECC)**: All data stored in SRAMs internally to MACSec are protected with an ECC. This method allows correction of single event upset (single bit error), and detection of multiple event updates (multiple bit errors). ECCs are calculated over address and data fields, stored in the SRAM, then checked when read out.
- **Parity**: All internal data structures that are critical to the safety requirements are protected with single odd parity bit per byte of data. In general, this involves processing Finite State Machine (FSM) context used to update the processing metadata, internal flops that hold the Secure Channel Selection result, or FIFOs for frame data processing.

In addition to the above, configuration registers that are safety critical should be read every Fault Tolerant Time Interval (FTTI) by software to ensure configuration has not changed.

3.9.2 MACSec Fault Reaction

Upon detection of a safety fault, and depending on the severity, MACSec may enter a safe state and report the fault via fault detection interface. For external faults, MACSec discards the impacted frame if parity error is observed with a frame.

3.9.3 MACSec Fault Recovery

Once notified of a fault, software can invoke the necessary diagnostics, and then proceed with the recovery process for availability.

- For correctable faults, or faults local to a frame context, no recovery is required. Processing continues normally.
- For function specific faults, software must issue a Function Level Reset (FLR) for the impacted function, reconfigure the function, and resume processing. Processing continues normally on all other functions.

3.9.4 MACSec Fault Injection

MACSec provides the means to inject faults in most of the safety mechanisms implemented, as listed below. All safety mechanisms are re-enabled after function level reset (flr) or soft reset is issued. MACSec fault injection testing should only be done when the system is out of service, on shutdown phase.

ECC - In all the locations where the ECC is checked (on all memory read interfaces), MACSec is capable of flipping one bit to cause a single-bit error and flipping two bits to cause a multibit error.

Parity - In all the locations where parity is checked, MACSec is capable of injecting a parity error, causing the check to fail.

Fault Injection Interface must be armed with the selected safety detection mechanism, and the next transaction coming through this safety detection mechanism will trigger the fault detection and reporting. It is up to software to target a specific function, or a specific component (using component disable) if a specific error detection and report is desired.

Only one fault injection can be configured at any given time.

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In order to achieve diagnostic coverage, software needs to develop functional vectors in order trigger all fault injection options above.

4 MACsec - Initialization Information

4.1 Startup or re-start

After a hard reset on the MACsec, the MACsec is in bypass mode. All frames are delivered unmodified to the uncontrolled port.

The MACsec internal memory is initialized to all zeros after a hard async reset. If an access to the MACsec is attempted while the initialization is taking place, the access does not complete until the memory initialization is completed. The initialization time is 448 MACsec clock ticks.

Once the MACsec is switched to mission mode (bypass is disabled), RX non-MACsec frames and KaY frames are delivered on the uncontrolled port, other MACsec frames are discarded until one or multiple SCs are configured. Frames on TX can only be sent on the uncontrolled port.

4.1.1 RX SC and SA configuration

To configure an rx SC for the first time, the following must be done:

- Configure RX SC selector via Section 2.9
- Configure RX SC key selector via Section 2.10
- Write a key into the SA A or B context via:
  - Section 2.115
  - Section 2.116
  - and so on
- Initialize SALT into the SA A or B context if the Current Cipher Suite supports extended packet numbering.
- Initialize SSCI into the SA A or B context if the Current Cipher Suite supports extended packet numbering.
- Initialize the SA A or B stats to zero, initialize the SC stats.
- Initialize the Section 2.32 to 1, Section 2.33 to 0, Section 2.34 to 1, and Section 2.35 to 0.
- Configure Section 2.31 by clearing the A bit, enabling the SA (EN) and assigning an AN value to the SA.
- Insert the associated SCI using Section 2.13
- Configure the SC Section 2.21
- Configure the SC Section 2.20 and enable it (SCI_EN)
- The SC is now ready to receive frames on the controlled port

4.1.2 TX SC and SA configuration

To configure a TX SC for the first time, the following must be done:

1. Configure TX SC selector via Section 2.11
2. Configure TX SC key selector via Section 2.12
3. Write a key in the SA A or B via:
   - Section 2.139
   - Section 2.140
   - and so on
4. Initialize SALT into the SA A or B context if the Current Cipher Suite supports extended packet numbering.
5. Initialize SSCI into the SA A or B context if the Current Cipher Suite supports extended packet numbering.
6. Initialize the SA A or B stats to zero, initialize the SC stats.
7. Initialize the Section 2.70 to 1, Section 2.71 to 0.

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8. Configure Section 2.69 by clearing the A bit.
9. Insert the associated SCI using Section 2.55
10. Configure the SC Section 2.63 and enable it (SCE)

The SC is now ready to transmit frames on the controlled port.

4.2 Run time configuration change

The dynamic aspect of MACsec requires run time changes to SCs. For a given SC, it will cycle through keys on a regular basis. The following section describes the procedure to change a SA key for an sc without interrupting the data flow.

4.2.1 Rekey on RX

To rekey on RX, the following must be done:

- Configure RX SC selector via Section 2.9
- Configure RX SC key selector via Section 2.10
- If the SA to be used for the new key is already in service, it must be disabled (assuming it is no longer being used and the other SA is the active one) by clearing the Section 2.31 EN bit
- All the SA stats should be read, stored in the appropriate Software structure and cleared.
- Write a new key into the SA A or B context via:
  - Section 2.115
  - Section 2.116
  - and so on
- Initialize SALT into the SA A or B context, if the Current Cipher Suite supports extended packet numbering.
- Initialize SSCI into the SA A or B context, if the Current Cipher Suite supports extended packet numbering.
- Initialize the Section 2.32 to 1, Section 2.33 to 0, Section 2.34 to 1, and Section 2.35 to 0.
- Configure Section 2.31 by clearing the A bit, enabling the SA (EN) and assigning an AN value to the SA.

Rekey is done and the new SA is ready to be used.

4.2.2 Rekey on TX

To rekey on TX, the following must be done:

- Because only one SA can be active at a time on TX, select the inactive SA
- Configure TX SC selector via Section 2.11
- Configure TX SC key selector via Section 2.12
- Write the new key in the SA A or B via
  - Section 2.139
  - Section 2.140
  - and so on
- Initialize SALT into the SA A or B context, if the Current Cipher Suite supports extended packet numbering.
- Initialize SSCI into the SA A or B context, if the Current Cipher Suite supports extended packet numbering.
- Initialize the SA A or B stats to zero
- Initialize the Section 2.70 to 1, Section 2.71 to 0.
- Configure Section 2.69 by clearing the A bit.
- When ready to switch over, configure the SC Section 2.63 to used the other SA and the appropriate AN
- Capture all the SA stats of the newly retired SA

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4.2.3 Retire or re-configure an sc

To retire or reconfigure an sc, the following must be done:

- Disable the RX and TX SC using Section 2.20 and Section 2.63 SCI_EN bit
- make modification and re-enable

5 MACsec conformance to 802.1AE standard

The following tables are from Annex A of the IEEE 802.1AE standard. An additional column has been added to show which MACsec design feature confirms the item in question.

See Annex A of the IEEE 802.1AE for explanation of all symbols in the table. Note that the section numbers given in the References column are for the standard and not this block guide.

In the Support column, SW refers to support provided by software and HW refers to hardware.

Table 4. Major capabilities

Item	Feature	Status	References	Support	Features
SAP	Does the implementation of each MAC Security Entity support the Controlled and Uncontrolled Ports, and use a Common Port as specified in Clause 10?	M	5.3(a), 10, A.6	YES	Section 3.2.7
STAT	Does the implementation support the MAC status and point-to-point parameters for the Controlled and Uncontrolled Ports as specified in 6.4, 6.5, and 10.7?	M	5.3(b), 6.4, 6.5, 10.7, A.7	SW	N.A.
GEN	Does the implementation process transmit requests from the Controlled Port as required by the specification of Secure Frame Generation (10.5)?	M	5.3(c), 10.5, A.8	YES	Section 3.3
VER	Does the implementation process receive indications from the Common Port as required by the specification of Secure Frame Verification (10.6), prior to causing receive indications at the Controlled Port?	M	5.3(d), 10.6, A.9	YES	Section 3.2
FMT	Does the implementation encode and decode MACsec PDUs as specified in Clause 9?	M	5.3(e), 9, A.10	YES	Section 3.1, Section 2.63
SCI	Does the implementation use a globally unique 48-bit MAC Address and a 16-bit Port Identifier unique within the scope of that address assignment to identify the transmit SCI, as specified in 8.2.1?	M	5.3(f), 8.2.1	YES	Section 2.55
PERF	Does the implementation satisfy the performance requirements specified in Table 10-1 and 8.2.2?	M	5.3(g), 10.1, Table 10-1, 8.2.2	YES	by design
FCS	Does the implementation introduce an undetected frame error rate greater than that achievable by preserving the original FCS, as required by 10.4?	X	5.3(n), 10.4, 6.10	NO	by design
KAY	Does the implementation support the LMI operations required by the Key Agreement Entity as specified in Clause 10?	M	5.3(h), 10, A.11	YES	Table 10
MGT	Does the implementation provide the management functionality specified in 10.7?	M	5.3(i), 10.7, A.11.1	YES	Table 11

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Table 4. Major capabilities...*continued*

Item	Feature	Status	References	Support	Features
MIB	Does the implementation support network management using the MIB specified in Clause 13?	O	5.3(a), 13	SW + HW	Section 2.15 and many more
SNMP	Does the implementation support access to the MIB specified in Clause 13 using SNMP v3 or higher?	O	5.4(b), 13	SW	N.A.
SNMX	Does the implementation support access to MACsec parameters using any version of SNMP prior to v3?	X	5.3(p)	SW	N.A.
MSC	Does the implementation support more than one receive SC?	O	5.4(c)	YES	Section 3.2.1
MSAK	Does the implementation support more than two receive SAKs?	O	5.4(d)	NO	N.A.
CS	Does the implementation protect and validate MACsec PDUs by using implemented Cipher Suites as specified in 14.1?	M	5.3(j), 14.1	YES	Section 3.2.4, Section 3.2.5
CSI	Does the implementation support Integrity Protection using the Default Cipher Suite specified in Clause 14?	M	5.3(k), 14, 14.5	YES	Section 3.2.4
CSC	Does the implementation support Confidentiality Protection using the Default Cipher Suite without a Confidentiality Offset as specified in Clause 14?	¬CSO:O CSO:M	5.4(e), 14, 14.5	YES	Section 3.2.5
CSO	Does the implementation support Confidentiality Protection using the Default Cipher Suite with a Confidentiality Offset as specified in Clause 14?	O	5.4(f), 14, 14.5	YES	Section 2.20, Section 2.63
CSA	Does the implementation include Cipher Suites that are specified in Clause 14 in addition to the Default Cipher Suite? (This PICS requires the completion of a copy of Table A.12 for each such Cipher Suite implemented.)	O	5.4(g), A.12	NO	N.A.
CSX	Does the implementation include any Cipher Suite that is additional to those specified in Clause 14 and does not meet all the criteria specified in 14.2, 14.3, 14.4.1?	X	5.3(o), 14.2, 14.3, 14.4.1	NO	N.A.
CSV	Does the implementation include Cipher Suites other than those specified in Clause 14, but meeting the criteria specified in 14.2, 14.3, 14.4.1? (This PICS requires the completion of a copy of Table A.13 for each such Cipher Suite implemented.)	O	5.4(h), A.13	NO	N.A.
CSR	Does the implementation support a minimum of one receive SC and two receive SAKs, and one of the two receive SAKs at a time for transmission as specified in 5.3(l), for each Cipher Suite implemented?	M	5.3(l), 14	YES	Section 3
CSS	Does this completed PICS specify the maximum number of receive SCs and SAKs for each Cipher Suite implemented?	M	5.3(m), A.12, A.13	YES	See below.
CSRC	What is the maximum number of receive SCs supported by the Default Cipher Suite implementation?		5.3(i)	16	Section 3

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Table 4. Major capabilities...*continued*

Item	Feature	Status	References	Support	Features
CSRK	What is the maximum number of receive SAKs supported by the Default Cipher Suite implementation?		5.3(i)	2 x16	Section 3
FULL	Is a claim for full conformance being made for the implementation?	CSV:X¬CSV:O	5.3	NO	By design we don't conform to: SAP-2
VAR	Is a claim for conformance with cipher suite variance being made for the implementation?		5.3	NO	N.A.

Table 5. Support and use of Service Access Points

Item	Feature	Status	References	Support	Features
SAP-1	Does each transmit request from the Uncontrolled Port result in a single request to the Common Port with the same parameters?	M	10.4	YES	Section 3.3
SAP-2	Does each receive indication from the Common Port result in a single indication to the Uncontrolled Port with the same parameters if any of the users of the Common Port wishes to receive the indication?	M	10.4	NO	N.A. Replication to the uncontrolled port is not supported.
SAP-3	Does each transmit request from the Controlled Port result in at most one request to the Common Port?	M	10.4	YES	Section 3.3
SAP-4	Does each receive indication from the Common Port result in at most one indication to the Controlled Port?	M	10.4	YES	Section 3.3
SAP-5	Are any transmit requests made to the Common Port that do not correspond to requests made at the Uncontrolled or Controlled Port?	X	10.4	NO	
SAP-6	Are any receive indications caused at the Uncontrolled or Controlled Port that do not correspond to indications from the Common Port?	X	10.4	NO	
SAP-7	Is the order of requests made at the Common Port unchanged from the order of corresponding requests from the Uncontrolled Port?	M	10.4	YES	by design
SAP-8	Is the order of requests made at the Common Port unchanged from the order of corresponding requests from the Controlled Port?	M	10.4	YES	by design
SAP-9	Is the order of receive indications caused at the Uncontrolled Port the same as the order of reception from the Common Port?	M	10.4	YES *	by design
SAP-10	Is each transmit request from the Controlled Port processed in accordance with the specification of the Secure Frame Generation process, prior to discarding the request or making a corresponding request to the Common Port?	M	10.4, 10.5	YES	by design.
SAP-11	Is each receive indication from the Common Port processed in accordance with the specification of the Secure Frame Verification process prior to causing a possible	M	10.4, 10.5	YES	By design.

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Table 5. Support and use of Service Access Points...*continued*

Item	Feature	Status	References	Support	Features
	corresponding indication at the Controlled Port?				

Table 6. MAC status and point-to-point parameters

Item	Feature	Status	References	Support	Features
STAT-1	Are the values for MAC_Operational and oper PointToPointMAC for the Uncontrolled Port identical to those for the Common Port?	M	6.4, 10.7.2	SW	N.A.
STAT-2	Is MAC_Operational False for the Controlled Port, and frames neither accepted or delivered on the port, if the SA identified by the encoding SA is not available for use and protectFrames is set?	M	6.4, 10.5.1, 7.1	SW	N.A.
STAT-3	Is MAC_Operational False for the Controlled Port and frames neither accepted nor delivered, if the nextPN for the encodingSA is zero or 2^{32}?	M	6.4, 10.5.2	SW	N.A.
STAT-4	Is MAC_Operational True only if MAC_Enabled is True and MAC_Operational for the Common Port is True?	M	6.4, 10.7.4	SW	N.A.
STAT-5	Is the value of operPointToPointMAC for the Controlled Port always as specified in 10.7.4.	M	6.5, 10.7.4	SW	N.A.

Table 7. Secure Frame Generation

Item	Feature	Status	References	Support	Features
GEN-1	Does each transmit request from the Controlled Port result in an identical transmit request at the Common Port if the management control protectFrames is False?	M	10.5	YES	Section 2.63 and Section 3.3
GEN-2	Does each transmit request at the Common Port resulting from a request at the Common Port convey request parameters, that is, a frame, protected in accordance with Clause 10.5 if the management control protectFrames is True?	M	10.5	YES	Section 2.63 and Section 3.3
GEN-3	Is each protected frame assigned to the SA with AN corresponding to the current value of encodingSA as specified by the KaY?	M	10.5.1	YES	Section 2.63
GEN-4	Are frames to be protected discarded if the assigned SA cannot be used?	M	10.5.1	YES	Section 2.63 and Section 2.114
GEN-5	Is the PN value of zero used?	X	10.5.2	NO	
GEN-6	Following assignment of a PN to a protected frame, is the next frame to be protected for the same SA assigned the next higher value of PN?	M	10.5.2	YES	Section 2.70
GEN-7	Is the SecTAG encoded as specified in Clause 9?	M	10.5.3, 9	YES	Section 3.3.4
GEN-8	Is the ES bit set or clear as required by the management controls useES and always IncludeSCI?	M	10.5.3	YES	Section 2.63

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Table 7. Secure Frame Generation*...continued*

Item	Feature	Status	References	Support	Features
GEN-9	Is the SC bit set or clear and the SCI explicitly encoded or not as required by the management controls useES, use SCB, always IncludeSCI, and by the number of receive SCs?	M	10.5.3	YES	Section 2.63
GEN-10	Is the SCB bit set or clear as required by the management controls useSCB and always IncludeSCI?	M	10.5.3	YES	Section 2.63
GEN-11	Is the E bit set if the frame is confidentiality protected, and clear otherwise?	M	9.5	YES	Section 2.63
GEN-12	Is the C bit set if the octets of the Secure Data differ from those of the User Data or the ICV is not 16 octets, and clear otherwise?	M	9.5	YES	Section 2.63
GEN-13	Is each frame transmitted from the Controlled Port protected using a Cipher Suite as specified in Clause 14 if protectFrames is set?	M	10.5	YES	Section 3.3.7
GEN-14	Is OutOctetsEncrypted incremented by the number of octets in the User Data if confidentiality protections is provided, and Out OctetsProtected incremented otherwise?	M	10.5.4	YES	Section 2.67, Section 2.67
GEN-15	Is the protected frame transmitted if the MACsec PDU (SecTAG, Secure Data, and ICV) does not exceed the maximum data unit size supported by the Common Port and discarded otherwise?	M	10.5.5	YES	Section 2.7

Table 8. Secure Frame Verification

Item	Feature	Status	References	Support	Features
VER-1	For each receive indication, does the Secure Frame Verification process examine the user data for a SecTAG and validate frames with a SecTAG as specified in 9.12, extracting and decoding the SecTAG as specified in 9.3 through 9.9, and extracting the User Data and ICV as specified in 9.10 and 9.11?	M	10.6, 9.3 through 9.9, 9.10, 9.11, 9.12	YES	Section 3.2
VER-2	Is a received frame without a SecTAG delivered to the Controlled Port if validate Frames is not Strict, and discarded otherwise?	M	10.6	YES	Section 2.2
VER-3	Is a received frame with the SecTAG E bit set and C bit clear discarded and not delivered to the Controlled Port?	M	10.6	YES	Section 2.2
VER-4	Is the received frame discarded if the SC is unknown and validateFrames is Strict or the C bit is set, and delivered to the Controlled Port otherwise?	M	10.6.1	YES	Section 2.2
VER-5	Is the received frame discarded if the SA is unused and validateFrames is Strict or the C bit is set, and delivered to the Controlled Port otherwise?	M	10.6.1	YES	Section 2.2
VER-6	Is the received frame discarded if the PN is less than the lowest acceptable packet number for the SA and replayProtect is enabled?	M	10.6.2, 10.6.4	YES	Section 3.2.6

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Table 8. Secure Frame Verification...*continued*

Item	Feature	Status	References	Support	Features
VER-7	Is the InPktsOverrun counter incremented if a received frame is discarded for reasons not attributed to the data conveyed?	M	10.6.3	SW	N.A. We run at line rate.
VER-8	If validateFrames is Disabled, is Cipher Suite validation omitted and a received frame delivered to the Controlled Port if the C bit is not set?	M	10.6.3, 10.6.5	YES	Section 2.20
VER-9	If validateFrames is not Disabled is the Cipher Suite used to validated the received frame?	M	10.6.3	YES	Section 3.2.4
VER-10	Are frames that are not successfully validated discarded if validateFrames is Strict or the C bit is set?	M	10.6.5	YES	Section 2.42
VER-11	Are the values for the next expected and lowest acceptable PN updated as specified in 10.6.5 following receipt of a MACsec PDU successfully validate by the Cipher Suite, and not modified by received frames otherwise?	M	10.6.5	YES	Section 2.44, Section 2.46
VER-12	Are received frames not discarded by Secure Frame Verification delivered to the Controlled Port after removal of a SecTAG and ICV?	M	10.6	YES	Section 3.2
VER-13	Are all received frames delivered to Controlled Port unmodified if validateFrames is Null?	M	10.6	YES	Section 2.2
VER-14	Is protectFrames set False if validateFrames is set to Null?	M	10.6	SW	Section 2.63

Table 9. MACsec PDU encoding and decoding

Item	Feature	Status	References	Support	Features
FMT-1	Does each MACsec PDU transmitted contain an integral number of octets?	M	9.1	YES	By design we only support integral octets.
FMT-2	Does each MACsec PDU transmitted comprise a SecTAG, formatted as specified in Clause 9, one or more octets of Secure Data, and an ICV of the length specified by the Cipher Suite in use?	M	9.1, 9.2, 9.3, Figure 9-1, 10.5.3	YES	Section 3.3
FMT-3	Is the EtherType encoded in the SecTAG the value specified in Table 9-1?	M	9.3, 9.4	YES	Section 2.3
FMT-4	Is the version number in the SecTAG encoded as zero?	M	9.5	YES	Section 3.1
FMT-5	Is the SC bit clear and the SCI not explicitly encoded if the ES bit is set?	M	9.5	YES	Section 2.63
FMT-6	Is the SC bit set if an SCI is explicitly encoded and clear otherwise?	M	9.5	YES	Section 2.63
FMT-7	Is the SC bit clear if the SCB bit is set?	M	9.5	YES	Section 2.63
FMT-8	Are bits 7 and 8 of octet 4 of the SecTAG zero?	M	9.7	YES	Section 3.1
FMT-9	Is each received MACsec PDU validated as specified in 9.12.	M	9.5	YES	Section 3.2.3

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Table 10. Key Agreement Entity LMI

Item	Feature	Status	References	Support	Features
KAY-1	Does the implementation allow the KaY to read the values of the MAC_Enabled, MAC_ Operational, and operPointToPointMAC parameters?	M	10.7.2	SW	N.A.
KAY-2	Does the implementation allow the KaY to set and clear the ControlledPortEnabled parameter, acting on the parameter as specified?	M	10.7.4, 10.7.5	SW	N.A.
KAY-3	Does the implementation allow the KaY to discover which Cipher Suites are implemented and how many receive SCs each can support?	M	10.2, 10.7.7, 10.7.16, 10.7.24	SW	N.A.
KAY-4	Does the implementation allow the KaY to create a receive SC?	M	10.6.1, 10.7.11	SW	N.A.
KAY-5	Does the implementation allow the KaY to create receive SAs as specified in 10.7.13?	M	10.7.13	SW	N.A.
KAY-6	Does the implementation allow the KaY to control the use of each receive SA and to update the values of the next expected PN and lowest acceptable PN as specified in 10.7.15?	M	10.7.15	SW	N.A.
KAY-7	Does the implementation allow the KaY to create transmit SAs as specified in 10.7.21?	M	10.7.21, 10.5.2	SW	N.A.
KAY-8	Does the implementation allow the KaY to control the use of each transmit SA as specified in 10.7.23?	M	10.7.23, 10.5.1, 10.5.2	SW	N.A.
KAY-9	Does the implementation allow the KaY to monitor the nextPN associated with each transmit SA in order to create a new SA with a fresh SAK prior to PN exhaustion?	M	10.7.2	SW	N.A.
KAY-10	Does the implementation allow the KaY to select the Current Cipher Suite as specified in 10.7.25?	M	10.7.25	SW	N.A.
KAY-11	Does the implementation allow the KaY to create and control an SAK as specified in 10.7.26 and 10.7.28?	M	10.7.26, 10.7.28	SW	N.A.

Table 11. Management - control and status information

Item	Feature	Status	References	Support	Features
Can each of the following parameter values be read by management?					
MGT1-1	The SCI for the SecY	M	10.7.1	YES	Section 2.13
MGT1-2	MAC_Enabled, MAC_Operational, and oper Point-ToPointMAC for the Uncontrolled Port	M	10.7.2	SW	N.A.
MGT1-3	MAC_Enabled, MAC_Operational, and oper PointToPointMAC for the Controlled Port	M	10.7.4	SW	N.A.
MGT1-4	The maximum number of receive SCs and SAKs that can be in simultaneous use	M	10.7.7	SW	N.A.
MGT1-5	validateFrames, replayProtect, and replay Window	M	10.7.8	YES	Section 2.20
MGT1-6	The SCI, receiving, createdTime, startedTime, and stoppedTime for each receive SC	M	10.7.12	SW	N.A.

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Table 11. Management - control and status information *...continued*

Item	Feature	Status	References	Support	Features
MGT1-7	inUse, nextPN, lowestPN, createdTime, started Time, stoppedTime, and Key Identifier for each receive SA	M	10.7.14	SW	N.A.
MGT1-8	The maximum number of SAKs that can be in simultaneous use for transmission	M	10.7.16	SW	N.A.
MGT1-9	protectFrames, useES, useSCB, and always IncludeSCI	M	10.7.17	YES	Section 2.63
MGT1-10	transmitting, createdTime, startedTime, and stoppedTime for the transmit SC	M	10.7.20	SW	N.A.
MGT1-11	inUse, nextPN, lowestPN, createdTime, started Time, stoppedTime, and Key Identifier for each transmit SA	M	10.7.22	SW	N.A.
MGT1-12	The currentCipherSuite identifier and the confidentialityOffset for frames with confidentiality protection	M	10.7.25	SW	N.A.
MGT1-13	transmits, receives, and createdTime for each SAK	M	10.7.27	SW	N.A.
MGT1-14	Can the management information for each implemented Cipher Suite be read?	M	10.7.24	SW	N.A.

Table 12. Management - basic controls

Item	Feature	Status	References	Support	Features
Can the following parameters be written by management?					
MGT2-1	validateFrames	O	10.7.8, 10.6	YES	Section 2.20
MGT2-2	replayProtect	O	10.7.8, 10.6.2, 10.6.4	YES	Section 2.20
MGT2-3	replayWindow	O	10.7.8, 10.6.5	YES	Section 2.21
MGT2-4	protectFrames	O	10.7.17, 10.5	YES	Section 2.63
MGT2-5	useES	O	10.7.17, 10.5.3	YES	Section 2.63
MGT2-6	useSCB	O	10.7.17, 10.5.3	YES	Section 2.63
MGT2-7	alwaysIncludeSCI	O	10.7.17, 10.5.3	YES	Section 2.63
Can write access by management to each of the following parameters be disabled individually?					
MGT2-8	validateFrames	MGT2-1:M	10.7.8	SW	N.A.
MGT2-9	replayProtect	MGT2-2:M	10.7.8	SW	N.A.
MGT2-10	replayWindow	MGT2-3:M	10.7.8	SW	N.A.
MGT2-11	protectFrames	MGT2-4:M	10.7.17	SW	N.A.
MGT2-12	useES	MGT2-5:M	10.7.17	SW	N.A.
MGT2-13	useSCB	MGT2-6:M	10.7.17	SW	N.A.
MGT2-14	alwaysIncludeSCI	MGT2-7:M	10.7.17	SW	N.A.
Can the following parameters be written by management, independently for each Controlled Port for each CipherSuite implemented?					
MGT2-15	enableUse	O	10.7.26	SW	N.A.
MGT2-16	requireConfidentiality	O	10.7.26	SW	N.A.
Can write access by management to each of the following CipherSuite use parameters be disabled individually for each Controlled Port?					
MGT2-17	enableUse	MGT2-15:M	10.7.26	SW	N.A.

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Table 12. Management - basic controls...*continued*

Item	Feature	Status	References	Support	Features
MGT2-18	requireConfidentiality	MGT2-16:M	10.7.26	SW	N.A.

Table 13. Management - control over secure communication

Item	Feature	Status	References	Support	Features
Can the following be created, controlled, or selected by management?p					
MGT3-1	Receive SCs and SAs	O	10.7.11, 10.7.13, 10.7.15	SW	N.A.
MGT3-2	Transmit SAs	O	10.7.21, 10.7.23	SW	N.A.
MGT3-3	The current CipherSuite	O	10.7.25	SW	N.A.
MGT3-4	confidentialityOffset	O	10.7.25	SW	N.A.
MGT3-5	SAKs	O	10.7.26, 10.7.28	SW	N.A.
Can creation, control, or selection by management of the following be disabled individually?					
MGT3-6	Receive SCs and SAs	MGT3-1:M	10.7.11	SW	N.A.
MGT3-7	Transmit SAs	MGT3-2:M	10.7.22, 10.7.24	SW	N.A.
MGT3-8	The current CipherSuite	MGT3-3:M	10.7.25	SW	N.A.
MGT3-9	confidentialityOffset	MGT3-4:M	10.7.25	SW	N.A.
MGT3-10	SAKs	MGT3-5:M	10.7.25	SW	N.A.

Table 14. Management - statistics

Item	Feature	Status	References	Support	Features
Are each of the following interface statistics provided for the Controlled Port as specified in 10.7.6?					
MGT4-1	ifInOctets	M	10.7.6	HW+SW	Section 2.15
MGT4-2	ifInUcastPkts, ifInMulticastPkts, ifInBroadcast Pkts	M	10.7.6	HW+SW	Section 2.17, Section 2.18, Section 2.19
MGT4-3	ifInDiscards	M	10.7.6	HW+SW	Section 2.15
MGT4-4	ifInErrors	M	10.7.6	HW+SW	Section 2.15
MGT4-5	ifOutOctets	M	10.7.6	HW+SW	Section 2.57
MGT4-6	ifOutUcastPkts, ifOutMulticastPkts, ifOut BroadcastPkts	M	10.7.6	HW+SW	Section 2.59, Section 2.61, Section 2.62
MGT4-7	ifOutErrors	M	10.7.6	HW+SW	Section 2.15
Are each of the following frame verification statistics recorded as specified in 10.6 and maintained for the frame verification process as a whole?					
MGT4-8	InPktsUntagged	M	10.7.9, 10.6, Figure 10-5	HW+SW	Section 2.105
MGT4-9	InPktsNoTag	M	10.7.9, 10.6, Figure 10-5	HW+SW	Section 2.105
MGT4-10	InPktsBadTag	M	10.7.9, 10.6, Figure 10-5	HW+SW	Section 2.107
MGT4-11	InPktsUnknownSCI	M	10.7.9, 10.6.1	HW+SW	Section 2.108

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Table 14. Management - statistics...*continued*

Item	Feature	Status	References	Support	Features
MGT4-12	InPktsNoSCI	M	10.7.9, 10.6.1	HW+SW	Section 2.108
MGT4-13	InPktsOverrun	M	10.7.9, 10.6.3	SW	HW runs at line rate. SW can set this stat to 0.
Are each of the following frame verification statistics recorded as specified in 10.6 and maintained for each receive SC?					
MGT4-14	InPktsUnchecked	M	10.7.9, 10.6.5	HW+SW	Section 2.26
MGT4-15	InPktsDelayed	M	10.7.9, 10.6.5	HW+SW	Section 2.27
MGT4-16	InPktsLate	M	10.7.9, 10.6.2, 10.6.4	HW+SW	Section 2.28
Are each of the following frame verification statistics recorded as specified in 10.6 and maintained for each receive SC and for each of the four receive SAs corresponding to the last use of AN for that SC?					
MGT4-17	InPktsOK	M	10.7.9, 10.6.5	HW+SW	Section 2.36
MGT4-18	InPktsInvalid	M	10.7.9, 10.6.5	HW+SW	Section 2.41
MGT4-19	InPktsNotValid	M	10.7.9, 10.6.5	HW+SW	Section 2.42
MGT4-20	InPktsNotUsingSA	M	10.7.9, 10.6.1	HW+SW	Section 2.29
MGT4-21	InPktsUnusedSA	M	10.7.9, 10.6.1	HW+SW	Section 2.30
Are each of the following frame validation statistics recorded as specified in 10.7?					
MGT4-22	InOctetsValidated	M	10.7.10	HW+SW	Section 2.22
MGT4-23	InOctetsDecrypted	M	10.7.10	HW+SW	Section 2.24
Are each of the following frame generation statistics recorded as specified in 10.5 and maintained for the frame verification process as a whole?					
MGT4-24	OutPktsUntagged	M	10.7.18, 10.5	HW+SW	Section 2.60
MGT4-25	OutPktsTooLong	M	10.7.18, 10.5.5, Figure 10-4	HW+SW	Section 2.64
Are each of the following frame generation statistics recorded as specified in 10.5 and maintained for each of the four transmit SAs corresponding to the last use of AN for the transmit SC?					
MGT4-26	OutPktsProtected	M	10.7.18, 10.5.4	HW+SW	Section 2.72
MGT4-27	OutPktsEncrypted	M	10.7.18, 10.5.4	HW+SW	Section 2.73
Are each of the following frame protection statistics recorded as specified in 10.7?					
MGT4-28	OutOctetsProtected	M	10.7.19	HW+SW	Section 2.65
MGT4-29	OutOctetsEncrypted	M	10.7.19	HW+SW	Section 2.67

Table 15. Additional fully conforming Cipher Suite capabilities

Item	Feature	Status	References	Support	Features
No additional fully conforming cipher suites are supported. Only the default suite is supported.					

Table 16. Additional variant Cipher Suite capabilities

Item	Feature	Status	References	Support	Features
No additional variant cipher suites are supported. Only the default suite is supported.					

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6 Revision history

Table 17. Revision history

Document ID	Release date	Description
UM12059 v.1.0	20240606	• Initial version

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